

A

filaments. A-band is one of the two kinds of band visible under certain conditions in a skeletal muscle fibre. The other kind is the I-band (isotropic band), which contains the myosin filaments. The alternation of the two types produces the cross-striations on striated muscle. Striated muscles are attached mainly to the bones and skin and are responsible for the mobility of the body and limbs. The H-band is situated on either side of the M line, where the thick and thin filaments do not overlap and it contains only thick filaments but no thin filaments.

A-ESU (STATAMPERE; statA; stA): The unit of electric current in the centimetre–gram–second (cgs) system of electrostatic units (esu). It is equivalent to one statcoulomb (1 statC), also called one electrostatic unit (1 esu), of charge carriers moving past a specific point in one second. The statampere is a small unit of current, equivalent to approximately 3.3356×10^{-10} ampere (A) or 0.33356 nanoampere (nA). The International System of Units (SI) is the currently used system of units, and the ampere, the unit for electric current has replaced the statampere.

A-LAW: The standard codec (compression/decompression) algorithm for pulse code modulation (PCM) from the Telecommunication Standardization Sector of the International Telecommunications Union (ITU-T). A-Law is the type of PCM used in most parts of the world. The other type, mu-Law, is used in the United States and Japan.

A-SCAN ULTRASONOGRAPHY (AMPLITUDE MODULATED SCAN): A type of medical diagnostic technique that uses a single-beam, linear sound wave to detect abnormalities by returning an echo when interference disrupts its straight path. It is most commonly used to measure the axial length of the eye to determine the power requirement for a lens to replace the abnormal, opaque lens of the eye removed in cataract surgery.

A-WEIGHTED DECIBELS (dB): An expression of the relative loudness of sounds in air as perceived by the human ear. In the a-weighted system, the decibel values of sounds at low frequencies are reduced, compared with unweighted decibels, in which no correction is made for audio frequency. This correction is made because the human ear is less sensitive at low audio frequencies, especially below 1000 Hz, than at high audio frequencies. In computer systems, dB is often used to specify the loudness of the fan used to cool the microprocessor and associated components. Typical dB ratings are in the neighbourhood of 25 dB, representing 25 a-weighted decibels above the threshold of hearing. This is approximately the loudness of a person whispering in a quiet room.

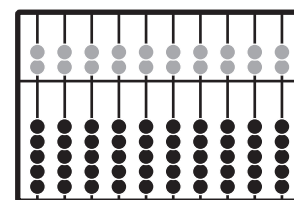
A+ CERTIFICATION (A-PLUS): An entry-level computer certification program, developed by the Computing Technology Industry Association (CompTIA), for knowledge and competency in installing, maintaining, customising and operating personal computers, as well as sponsoring specialised certification for document imaging and a network certification exam. The CompTIA certifications are in general more basic and less product-specific than other certifications, such as Microsoft Certified Systems Engineer (MCSE) or a Certified Novell Engineer (CNE).

AB-INITIO: 1. A Latin term which means "from the beginning" that is used in several contexts such as ab-initio calculation. 2. In chemistry, **ab initio** is an abbreviation referring to **ab initio quantum mechanical methods**, also known as **non-empirical quantum mechanical methods**, names that have replaced ab-initio calculation. It is defined by International Union of Pure and Applied Chemistry (IUPAC) as methods of quantum mechanical calculations independent of any experiment other than the determination of fundamental constants. The methods are based on the use of the full Schrodinger equation to treat all the electrons of a chemical system. It can be used, for example, to determine the bond lengths and bond angles of molecules by calculating the total energy of the molecule for a variety of molecular geometries and finding which conformation has the lowest energy. In practice, approximations are necessary to restrict the complexity of the electronic wave function and to make its calculation possible. For

example, in calculating the properties of an atom, the time required for the computation increases rapidly as the size of the atom or molecule increases. Calculations are now made easier by the use of computers to calculate the properties of both small and large molecules accurately. 3. In biophysics, **ab initio** is a method for predicting protein structures in protein folding. 4. In computing, **ab initio** is an extract, transform or load tool used to manipulate data.

A/B SWITCH: A switch with one input and two outputs (A and B), used to send signals to any of the two output devices. For example, it can be used to select and send signal to either of two printers, A or B attached to a computer; or in telephony, to use a wireless phone to switch from A to B frequency bands. Sometimes, an A/B switch can be used in reverse order, to select between two inputs to feed one output.

ABACUS: 1. An ancient, simple mechanical device that is still in use today for doing simple calculations or used as a teaching aid for counting and computation of numbers such as addition, subtraction, multiplication and division and can calculate roots up to the cubic degree. It consists of a frame with a crossbeam, made from many different materials, such as wood or brass with beads or small balls that slide to and fro on straight rods or wires. There are two types in use today: the Chinese suan pan and the Japanese soroban. The Chinese suan pan has rows, with the upper row above the beam having two beads on each rod and five beads below and uses a mixture of base 5 and 10. The Japanese soroban is based on the Chinese suan pan, but each rod carries one bead above and four below the beam. Each column of an abacus has beads that represent a sort of place value that can indicate numerals from 1 - 9, which could represent things such as units, tens, hundreds and thousands. The abacus is especially useful to the visually impaired. 2. In architecture, it is a slab on the top of the capital of a column. It is mainly used for providing a large supporting surface, which is wider than the capital (top part of a pillar or column) to receive the weight of the arch or the architrave (horizontal member above two columns spanning the interval between them) above.



Abacus

ABALONE (EAR SHELLS; ORMER): Any of several edible marine univalve molluscs of the class *Gastropoda* and the genus *Haliotis* having ear shaped shells, with two to three whorls. It is found mostly in the Pacific and Indian Oceans and also in warm seas worldwide. The outer surface of the single shell has a row of small pores, which are outlets for waste products. The colourful, pearly interior of the shell is used for jewellery and ornaments and articles such as buttons.

ABAMPERE (abA; BIOT, Bi): The unit of current in the centimetre–gram–second (cgs) system of electromagnetic units. It is the equivalent of one abcoulomb (1 abC) of charge carriers moving past a specific point in one second. The abampere is a moderately large unit of current, equivalent to 10 amperes (A). The International System of Units (SI) is the currently used system of units, and the unit for electric current has replaced the statampere.

ABANDONWARE: Computer software such as an operating system, word processor, interactive game or audio file that is no longer marketed or distributed by the company that created it, but is obtainable from some other source.

ABATEMENT: Any measure or set of measures designed to reduce or eliminate the degree or intensity of pollution.

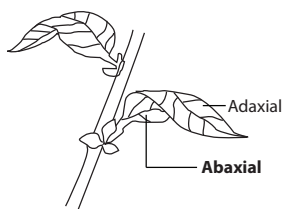
ABATTOIR (SLAUGHTERHOUSE): A place where animals are slaughtered and processed for meat. Abattoirs usually operate under government regulations. They have facilities for dressing, cutting, refrigerating and curing meat and for inspecting animals and meat. Abattoirs must also have water supply, drainage,

systems for disposal of animal waste and other refuse and also accommodation for animals awaiting slaughter.

ABAXIAL: A term used to describe lower surface of leaves and petals.

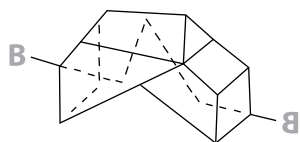
ABBE NUMBER (ABBE CONSTANT; ABBE-NUMBER; CONSTRICTENCE; NU-VALUE; RELATIVE DISPERSION; V-VALUE): The reciprocal of the dispersive power of a substance,

which is a measure of light dispersion in an optical medium. It is used to classify optically transparent materials. It is given by a function of the refractive index of a material at the Fraunhofer spectral lines f (486.1nm), d (587.6nm) and c (656.3nm) wavelengths. Its symbol is V and expressed as $V = n_d - 1/n_f - n_c$, where n_d , n_f and n_c are the refractive indices for the Fraunhofer spectral lines d (587.6 nm), f (486.1 nm) and c (656.3 nm). A material with a high Abbe number produces less chromatic aberration than one with a low number. Abbe numbers are used to classify optically transparent materials. For example, values of Abbe range from around 20 for very dense flint glass, around 30 for polycarbonate plastics and up to 58 for crown glass.



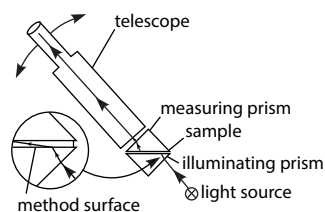
Abaxial of a leaf

ABBE PRISM: A type of roof prism, named after its inventor, the German physicist Ernst Abbe. It has 90°, 30° and 60° triangular faces that can be used to invert an image. Because of the perpendicular end faces, it can be inserted between the objective and eyepiece of a telescope. If the roof is replaced by a plane surface, it becomes a reversion or K prism.



Abbe prism

ABBE REFRACTOMETER: A high precision instrument in which the critical angle for total reflection at the interface of a film of liquid between two similar glass prisms is used to measure index of refraction. The main components of the Abbe refractometer are the illuminating prism and a refracting prism, the focusing telescope and a graduated glass circle with a reading microscope. The refracting prism is made of a glass with a high refractive index, for example, 1.75 and the refractometer is designed for use with samples having a refractive index smaller than that of the refractive prism. The focusing telescope is for observing the boundary line of total reflection. It also has two compensating Amici prisms to prevent dispersion, for eliminating the colour fringes along the boundary line and for measuring the mean dispersion. The eyepiece of the telescope is provided with crosshairs, so that in making a measurement, the prism angle is changed until the light-dark interface just coincides with the crosshairs. In Abbe's refractometer, the liquid sample is sandwiched into a thin layer between the illuminating and refracting prisms. The illuminating prism is matted, so light enters the sample at all possible angles, including those almost parallel to the surface. Light enters the sample from the illuminating prism, gets refracted at critical angle at the bottom surface of the measuring or refracting prism and enters the telescope. Bright and dark regions are visible in the telescope eyepiece. The boundary line between these two regions corresponds to the critical angle of total reflection. Illumination with white light



Abbe type critical angle refractometer

normally gives the boundary line a colour fringe in the beginning, which can be eliminated by the compensator. The now colourless boundary line is set to the intersecting point of the hairline cross. The refractive index n_d or the dry substance percentage of the investigated material can then be read. The mean dispersion $n_f - n_c$ and the Abbe number V can be determined at the same time from the reading of the drum graduation of the compensator and by means of a special chart or a nomogram.

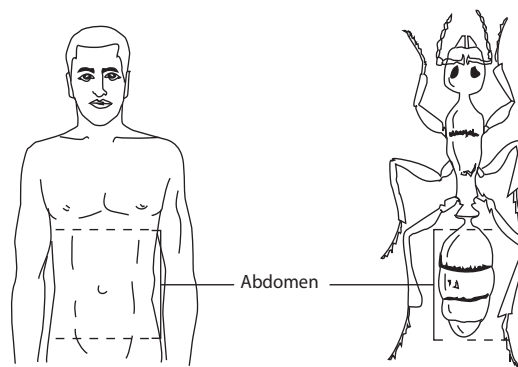
ABC CONJECTURE (OESTERLÉ–MASSER CONJECTURE): A conjecture that is, in number theory, stated in terms of three positive integers, a , b and c , which have no common factor and satisfy $a + b = c$. If d denotes the product of the distinct prime factors of abc , the conjecture essentially states that d cannot be much smaller than c . It was proposed in 1985 by Joseph Oesterle of the University of Paris and David Masser of the Mathematics Institute of the University of Basel in Switzerland. It is now considered one of the most important unsolved problems in number theory.

ABC MODEL: A model, which explains the genetic control of floral organ determination during flower development. According to the model, the four floral whorls, which include sepals, petals, carpels and stamens are specified by genes belonging to the three classes A, B and C. The organs are arranged in four concentric whorls and their identity is determined according to which combinations of genes are expressed in each whorl.

ABCOULOMB (abC): In the centimetre–gram–second (cgs) system of electromagnetic units, it is the charge that passes in one second through a section of conductor carrying a steady current of one ampere. The force with which two electrically charged bodies attract or repel one another depends on the product of the charges in abcoulombs in both objects and also on the distance between the objects. If the polarities are the same, the force is repulsive; if the polarities are opposite, the force is attractive. For any two charged bodies, the force decreases in proportion to the square of the distance between their charge centres. The International System of Units (SI) is the currently used system of units, and the coulomb (C) has replaced the abcoulomb (abC).

ABDOMEN (ABDOMINAL CAVITY): 1. In vertebrates, the frontal cavity of the trunk containing visceral organs such as the stomach and intestines. It is separated from the thorax by the diaphragm. Some examples of vertebrates with abdomens include mammals like humans, goats, sheep, dogs and cats. 2. In arthropods, it is the posterior portion of the body after the head and thorax or cephalothorax, which is made up of several repeating similar segments and containing the reproductive organs, respiratory system and digestive tracts. Some examples of arthropod organisms with abdomens include insects, crustaceans and arachnids.

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Abdomen of a human and an insect

ABDUCENS NERVE (ABDUCENT NERVE; SIXTH CRANIAL NERVE, CN6): Either of the sixth pair of cranial nerves, which innervates the movement of the lateral rectus muscle and is

responsible for lateral gaze or moving the eyeball outwards of the ipsilateral orbit. In other words, it contracts the lateral rectus which results in abduction of the eye or it pulls the eye's pupil away from the body's midline or axis.

ABDUCTOR: A muscle, which pulls a part of the body such as fingers, arms or toes away from the axis or midline of the body in the coronal plane. Abductors work in opposition to adductors, which pull the limbs towards the mid-section of the body.

ABEL SUMMATION (ABEL'S SUMMATION): The method of summability theory that computes the sum of a possibly divergent series of complex numbers as the limit, z approaches 1 from below, of the power series with the given sequence as coefficients. It is assumed that the series has a radius of convergence that is equal to 1.

ABEL TEST: An infinite series test used in testing its convergence. The test states that if $\sum d_n$ is a convergent sequence and $\{c_n\}$ is monotonically decreasing, that is $c_n + 1 \leq c_{n+1}$ for all n , then $\sum d_n c_n$ is also convergent.

ABEL TEST FOR UNIFORM CONVERGENCE: A uniform convergence test of an infinite series, stating that if $\{a_n(z)\}$ and $\{b_n(z)\}$ are sequences of complex functions on a compact set K , such that $\{a_n(z)\}$ is bounded in K , $\sum |a_n(z) - a_{n+1}(z)|$ is convergent and has a sum that is bounded in K , $\sum b_n(z)$ is uniformly convergent in K then $\sum a_n(z)b_n(z)$ is uniformly convergent on K .

ABEL, NIELS HENRIK (1802-1829): Norwegian mathematician, who at the age 16, proved that the binomial theorem is valid for all numbers. This extended Euler's result which had only held for rational numbers. At the age of 19, he proved that the general equation of a degree greater than four cannot be solved algebraically. In other words, there can be no formula for the roots of such an equation similar to the familiar formula for a quadratic equation. He was also an innovator in the field of elliptic functions and discovered Abelian functions.

ABEL'S LIMIT THEOREM: The result that the Abel summation method is regular in that the limit it assigns to a convergent series by Abel Summation agrees with the sum.

ABEL'S PARTIAL SUMMATION FORMULA: The formula of summability theory that for two arbitrary sequences $\{a_n\}$ and $\{b_n\}$ and with $\sum_{k=1}^n a_k = A_n$, has $\sum_{k=m}^n a_k b_k = \sum_{k=m}^n A_k (b_k - b_{k+1}) + A_n b_{n+1} - A_{m-1} b_m$.

ABELIAN GROUP (COMMUTATIVE GROUP): A group in which the binary operation is commutative, that is, the result of applying the group operation to two group elements does not depend on their order. It was named after a Norwegian mathematician called Niels Henrik Abel (1802 - 1829). Therefore, $ab = ba$ or $a + b = b + a$, for all elements a and b in the group. Abelian groups generalise the arithmetic of addition of integers.

ABELIAN THEOREM: A theorem stating that the convergence of series or integral implies summability, with respect to some summability method. It is named after Norwegian mathematician Niels Henrik Abel (1802 - 1829). Abel's theorem states that if a power series in z converges to $f(z)$ for $|z| < 1$ and to a for $z = 1$, then the limit of $f(z)$ as z approaches 1 equals a . In other words, it is a theorem stating that if a power series in z converges for $z = a$, it converges absolutely for $|z| < |a|$. It can also be defined as a theorem, which states that if the three series with n th term a_n , b_n and $c_n = a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0$, respectively, converge, then the third series equals the product of the first two series.

ABEND (ABNORMAL END): In computer science, it refers to software crashes. It is an abnormal termination of a software causing the computer to stop responding, commonly as a result of erroneous software logic in the application or operating system. It may be caused by, for example, a program that has attempted to address some

computer memory space that it has not got the right to address or when the computer is fed with instructions or data it cannot recognise.

ABEO-: The italic prefix of the form $x(y \rightarrow z)$ -abeo-. It is used to denote bond migration or the formal transfer of one end of a skeletal bond to another position, with a compensating transfer of a hydrogen atom. In this prefix, x is the locant of the stationary or the unchanged end of the migrating bond; y is the locant of the original position of the moving end of the migrating bond in the fundamental parent structure; and z is the locant of the position of the moving end of the migrating bond in the final structure. The numbering of the fundamental parent structure is retained in the new structure.

ABERRATION: 1. A departure from the normal or typical. 2. The apparent displacement of the position of a celestial body in the direction of motion of an observer on Earth, caused by the motion of Earth and the finite velocity of light. 3. A deviation in the normal structure or number of chromosomes in an organism. 4. A disorder or abnormal alteration in one's mental state. 5. In optics, also known as **optical aberrations**, it is a defect which causes an imperfection or blurring in the image seen in an optical instrument, caused by minute variations in lenses and mirrors and also because different parts of the light spectrum are reflected or refracted to varying degrees. The various types of **optical aberrations** include the following: (i) **astigmatism**, which is a type of optical aberration caused by a lens which has different focal lengths for rays of different orientations, resulting in a distortion of the image. (ii) **Chromatic aberration**, also called **achromatism** or **chromatic distortion** occurs in lenses because lenses bring different colours of light to a focus at different points. The image formed by a lens but not a mirror has a series of coloured images, even though only white light enters the system, as a result of the different extent to which light of different colours is refracted by the lens. It is present when using more than one wavelength of light. Chromatic aberrations are classified as transverse or lateral and longitudinal or axial. (iii) **Coma** is caused because off-axis rays do not quite converge at the focal plane. Coma is positive when off-axis rays focus furthest from the axis and negative when they are closest. (iv) **Distortion** is caused because the transverse magnification may be a function of the off-axis image distance. (v) **Field curvature** or **Petzval field curvature** is the result of the focal plane being not planar, but spherical. (vi) **Spherical aberration** occurs in a spherical lens or mirror because these do not focus parallel rays to a point, but instead along a line.

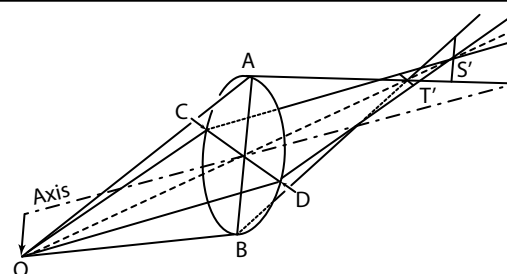


Fig. 1: Astigmatism (lens)

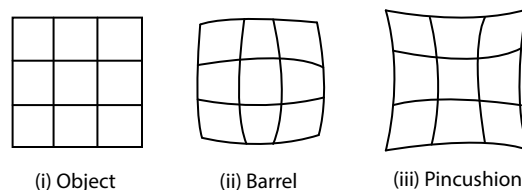


Fig. II: Types of Distortion

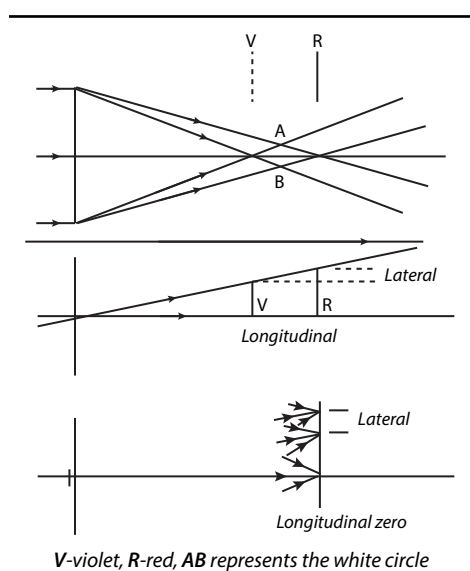


Fig. III: Chromatic aberration

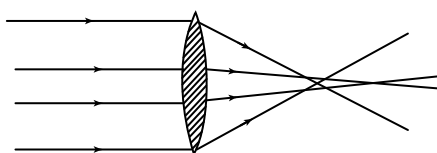


Fig. IV: Spherical aberration (lens)

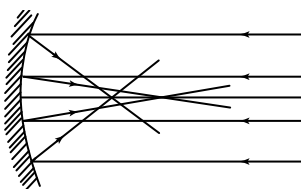


Fig. V: Spherical aberration (mirror)

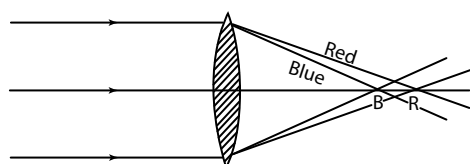


Fig. VI: Blue light focuses at B whereas red light focuses at R

ABFARAD (abF): The unit of electromagnetic capacitance in the centimetre–gram–second (cgs) system of units that if a charge of one abcoulomb (1 abC) produces a potential difference of one abvolt (1 abV) in a capacitance, then that capacitance is 1 abF. It is equivalent to 10^9 farads. A capacitance as large as 1 abF is unknown. The International System of Units (SI) is the currently used system of units, and the farad (F) has replaced abfarad.

ABHENRY (abH): The unit of electromagnetic inductance in the centimetre–gram–second (cgs) system of units. It is equivalent to 10^{-9} Henry. In a 1-abH inductance, a current that increases or decreases at one abampere per second (1 abA/s) will produce an electromotive force (EMF) of one abvolt (1 abV). The International System of Units (SI) is the currently used system of units, the henry (H) has replaced abhenry.

ABHESIVE (ABHERENT): A substance used as a slippery surface coating or releasing agent that prevents a material from adhering to itself or another material. An example is teflon, used to make frying pans non-sticky.

ABIOTIC (ABIOTIC FACTOR): A hypothesis that some organisms originated from inanimate material or the study of how life on Earth could have arisen from inanimate matter.

ABIOTIC (ABIOTIC FACTOR): A term used to describe the non-biological, non-living or the physical and chemical aspects such as soil composition, atmospheric gases, water, temperature, humidity and light, which influence the survival and distribution of organisms in an ecosystem. **Abiotic stress** is the negative environmental factors such as severe cold, drought, strong wind or extreme heat that may cause harm to living organisms such as plants and animals in the affected area. It is the most harmful factor affecting the growth and productivity of crops worldwide. **Abiotic stress resistance** is the resistance in organism to stress caused by non-biological factors such as extreme cold or heat.

ABIWORD (ABIWORD PROCESSOR): A free and open-source processing program that is similar to Microsoft Word that was first released on December 1, 1998 and provides all the major features found in other major processors. It is capable of saving and opening other word processor files created in other programs such as Microsoft Word and Corel WordPerfect (a word processing application owned by Corel).

ABLATION: 1. The process of getting rid of diseased tissue from the body by surgical or other means. 2. The melting or erosion of the protective outer surface of a spacecraft during re-entry through the Earth's atmosphere. 3. The loss of snow and ice from a glacier by melting and evaporation.

ABLATION TILL (MELT-OUT TILL): Debris, in the form of unstratified and unsorted accumulation of glacial sediment that has been deposited directly on the surface by melting of stagnant ice that does not flow.

ABNEY EFFECT: The shift in the apparent hue of a light that occurs when white light is added to a monochromatic light source. In other words, it is the apparent change in hue of spectral colours by mixing with white light.

ABNORMAL END (ABEND): In computer science, it refers to software crashes, an abnormal termination of a software causing the computer to stop responding, commonly as a result of erroneous software logic in the application or operating system. It may be caused by, for example, a program that attempts to address a computer memory space that it has not got the right to address or when the computer is fed with instructions or data it cannot recognise.

ABNORMAL HAEMOGLOBIN: Haemoglobin molecule with a different shape due to an altered amino acid sequence ultimately caused by an altered DNA base sequence, such as in the inheritable disease, sickle-cell anaemia. Haemoglobin E (HbE) is abnormal haemoglobin with a single point mutation in the β chain. At position 26 there is a change from a glutamate to a lysine. People with the HbE trait have a mild haemolytic anaemia, in which bone marrow activity cannot compensate for the increased loss of red blood cells (RBCs) and mild splenomegaly (an enlargement of the spleen). HbE trait is asymptomatic. In combination with certain thalassemia mutations, it provides an increased resistance to malaria.

ABO BLOOD GROUP SYSTEM: The classification of human blood based on the inherited properties of red blood cells (erythrocytes) as determined by the presence or absence of antigen A and antigen B, which are carried on the surface of red blood cells. Persons or individuals may therefore, have type A, type B, type O or type AB blood group. Austrian immunologist, Karl Landsteiner first identified the A, B and O blood groups in 1901. The antigens in ABO blood group exhibit unique characteristics, which are inherited according to a specific pattern. It is important for blood transfusion and also used in

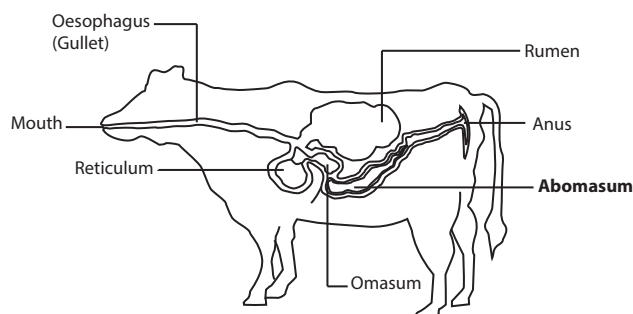
determining paternity suits, in forensic science and by anthropologists in the study of different populations.

A

BLOOD GROUP TYPE	ANTIGENS ON RED BLOOD CELL SURFACE	ANTIBODIES IN SERUM	BLOOD GROUP TYPE CAN RECEIVE BLOOD FROM	BLOOD GROUP TYPE CAN DONATE BLOOD TO
A	A	anti-b	A, O	A, AB
B	B	anti-a	B, O	B, AB
AB	A and B	-	A, B, AB, O	AB
O	-	anti-a/anti-b	O	A, B, AB, O

ABO Blood Group System

ABOMASUM (TRUE STOMACH): The fourth stomach of mammals with four-chambered stomach (ruminants) such as cattle and goats where mainly protein digestion takes place in acidic medium and which fills from the third stomach (omasum) and empties into the duodenum of the small intestine. The walls of the stomach of humans secrete dilute hydrochloric acid, mucous and the enzymes pepsin and rennin. Hydrochloric acid provides the acidic medium for the action of pepsin and rennin. Hydrochloric acid also kills bacteria that are subsequently digested, while mucous protects the walls of the abomasum from the action of pepsin. Rennin coagulates protein in milk into curds for retaining the milk for longer periods for further digestion.



Abomasum (true stomach) of a cow

ABORT: 1. To bring to a premature end due to a problem or fault. An example is terminating a flight because of an engine problem. 2. In biology, to terminate a pregnancy deliberately by causing the death of the foetus or to have a miscarriage, with loss of the foetus. 3. In computer science, it is to terminate an active computer process or command. For example, by pressing the keys ALT+CTRL+DEL, an active program that may have stopped responding is terminated. The keys, ALT+F4 could also be used to terminate a program that is not responding.

ABORTIFACIENT: Intended to produce or cause an abortion, for example, a drug that can induce abortion of the foetus.

ABORTION: 1. The loss of foetus before it is viable or at earlier stages of pregnancy before birth. **Spontaneous abortion**, also known as **miscarriage** is a natural biological process that ends a pregnancy. **Induced abortion** is a deliberate termination of pregnancy through human intervention such as an operation or other medical intervention by removing an embryo or foetus from the womb. 2. In botany, it refers to an embryonic fruit fall-off from flowering plants. 3. The stoppage of normal growth, especially of an organ or other body part, before its full development or maturation. 4. The premature and sudden ending of a mission such as a flight due to a problem that significantly affects the mission's chances of

success. 5. In computer science, it is a term used to describe the cancellation or termination of a computer process or command that is active. For example, CTRL+ALT+DELETE can be used to terminate a program that is no more responding.

ABORTIVE: 1. A structure in its rudimentary stage or not fully formed. 2. Abortifacient or intended to produce or cause an abortion.

ABOVE: The one-sided limit where x is restricted to values greater than the limit of a function from above. It is the right-hand limit written as $\lim_{x \downarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a^+)$.

ABRASION: 1. Using friction and rough texture to smoothen or clean a surface. 2. A type of erosion in which rock fragments, which may be carried in ice, called **glacial abrasion**, in water (**marine and fluvial abrasion**) or wind (**aeolian abrasion**) scrape and grind away a surface. 3. An injury or wound caused by rubbing or scraping the skin or mucous membrane.

ABRASIVE: A material of extreme hardness with refractory properties such as diamond, aluminium oxide and sand paper used for grinding, cutting and polishing or for removing small amounts of the surface of softer materials.

ABREAST PARLOUR: A type of milking parlour, such as rotary parlour and herringbone parlour, where cows stand side by side with their heads facing away from the milker.

ABRIKOSOV LATTICE (ABRIKOSOV VORTEX): In superconductivity, it is the arrangement of magnetic field lines or flux vortices or a vortex of supercurrent in a type II superconductor. The supercurrent circulates around the normal (i.e. non-superconducting) core of the vortex.

ABRUPTLY PINNATE (EVEN PINNATE): In biology, it refers to a pinnate compound leaf without an odd leaflet at the tip or a pinnate compound leaf with leaflets attached along an extension of the petiole called a rachis and a pair of leaflets at the apex or tip. Examples are found in mahogany, candle bush and tamarind. Pinnate in botany refers to a compound leaf having leaflets arranged on either side of the midrib, typically in pairs opposite each other.



Abruptly pinnate leaf

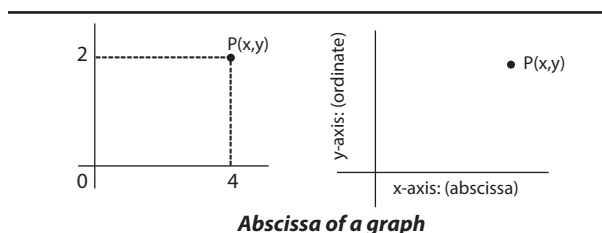
ABS PLASTICS: Any of a class of plastics based on acrylonitrile-butadiene-styrene copolymers. ABS plastics have good resistance to impact, heat and chemicals, as well as electrical insulating properties and ease of fabrication. They are produced in a wide range of grades including medium and high-impact, heat-resistant, plateable fire-retardant and both low and high gloss varieties. They are used in making pipes, car bumpers and are available as compounds for injection moulding, blow moulding, extrusion and calendaring, as sheet for thermoforming or cold forming and in expandable grades for foam moulding.

ABSCCESS: A painful swollen area on the body where pus forms. It is usually caused by bacteria, parasites or other foreign materials such as splinters. It is a collection of dead neutrophils or the build-up of fluid, living and dead white blood cells, dead tissue and bacteria or other foreign substances that have accumulated in a cavity formed by the tissue in which the pus resides. Abscess is a defensive reaction of the tissue to prevent the spread of infectious materials to other parts of the body. Abscesses most frequently occur on skin surface, in the lungs, brain, teeth, kidneys and tonsils.

ABSCISIC ACID, ABA (ABSCISSIN): A colourless plant hormone, with the chemical formula $C_{15}H_{20}O_4$, molar mass 264.32 g/mol, density 1.193 g/ml, melting point 186-188 °C (459-461 K) and boiling point 120 °C (393 K). It sublimates at this temperature. It is found in

all higher plants such as ferns, club mosses, flowering plants and conifers and is involved in the abscission of leaves, flowers and fruits and inhibits stem elongation, germination of seeds and the sprouting of buds.

ABSCISSA: The x -coordinate in the Cartesian plane that indicates the distance of a point from the y -axis. A point in the plane has coordinates (x,y) , where x is the abscissa and y is the ordinate. For example, for a point (2,3) in the plane, 2 is the abscissa and is 2 units distance away from the y -axis and 3 is the ordinate.



ABSCISSION: 1. Shedding of plant parts such as leaves, seed, fruit and unfertilised flowers, which occurs through the formation of a layer of cork cells, which seal the point of attachment and eventually cut off water and food from the part to be shed. 2. The shedding of a claw by an animal. 3. The separation of daughter cells at the end of cytokinesis, a process that generally begins immediately, following mitotic telophase.

ABSCISSION LAYER: In botany, it is the layer of specialised parenchymatous cells forming at the bases of fruits, flowers and leaves before they become separated from the plant to cut off conduction of water, mineral salts and food to avoid the plant from losing them.

ABSIEMENS (abS; ABMHO): The unit of electrical conductance in the centimetre–gram–second (cgs) electromagnetic system of units. It is equivalent to 10^9 siemens (S). When a current of one abampere (1 abA) flows through a conductance of 1 abS, the resulting potential difference across the component is one abvolt (1 abV). If the conductance is doubled while the current remains the same, the resulting voltage across the component is cut in half. Conversely, if the conductance is cut in half while the current remains the same, the resulting voltage across the component is doubled. The conductance in absiemens is the reciprocal of the resistance in abohms (abW). The International System of Units (SI) is the currently used system of units, and siemens (S) has replaced absiemens (abS).

ABSOLUTE: 1. Denoting a number or measurement that does not depend on a standard reference value or relating to a scale that has as its lowest temperature, absolute zero, which is zero Kelvin (0°K), equivalent to -273.16°C . It is the point at which all molecular motion ceases. 2. Also known as **numerical** of a value, it is a term used in mathematics to refer to the distance of a value from zero or the magnitude or value of numbers, without regard to their sign. For example, the numbers -3 and 3 have the same absolute value, written as $|3| = |-3| = 3$. However, the absolute value of 0 , $|0|$ is 0 . 3. True for all values of a variable in an algebraic expression. 4. Relating to or using basic units of length, time, mass and charge. 5. Involving or relating to measurements made relative to the vacuum state. 6. Not containing an algebraic variable. 7. Not changing in value, in varying mathematical expressions.

ABSOLUTE ACTIVATION ANALYSIS: Activation analysis in which the elemental concentrations in the material are calculated from known nuclear constants, irradiation and measurement parameters, rather than by comparing with known standards.

ABSOLUTE ADDRESS (EXPLICIT ADDRESS; SPECIFIC ADDRESS): A term used in computing to refer to the unique numerical identification of a memory location, peripheral device or location within a device. This is not the same as relative address, which is the common numerical difference between a desired address and a known reference address. Examples of absolute addresses are memory byte 107,443, disk drive 2 and sector 238.

ABSOLUTE AGE: The actual age of an object such as a fossil or rock expressed in units of time, usually years. This can be determined by measuring the amount of certain radioactive elements in the object.

ABSOLUTE ALCOHOL (ETHANOL; ETHYL ALCOHOL; PURE ALCOHOL; GRAIN ALCOHOL): A volatile, flammable, colourless chemical compound, with molecular formula $\text{C}_2\text{H}_5\text{O}$, density 0.789 g/cm^3 (at 25°C), molar mass 46 g/mol , melting point -114°C (159 K) and boiling point 78°C (351 K) that contains not more than 1% water. It is used in making alcoholic beverages.

ABSOLUTE CELL: In computing, it is a single fixed cell in a formula in spreadsheet display that does not change when the formula is copied or moved.

ABSOLUTE CELL REFERENCE (ACR): In Microsoft Excel and other spreadsheet programs, it is a spreadsheet cell that does not change regardless of its attributes.

ABSOLUTE CONFIGURATION (ABSOLUTE STEREOCHEMISTRY): The three-dimensional or spatial arrangement of the atoms of a chiral molecular group and its stereochemical description such as R or S in a molecule. The absolute configuration of a molecule can be determined by x-ray analysis or relating the configuration of a molecule to another molecule whose absolute configuration is known, using reactions of known stereochemistry.

ABSOLUTE CONTINUITY: 1. For a real valued function $f(x)$ on an interval $[a,b]$, the property that, for every $\epsilon > 0$, there is a $\sigma > 0$ such that if $\{(a_j, b_j)\}$ are intervals contained in $[a,b]$ with $\sum(b_j - a_j) < \sigma$ then $\sum|f(b_j) - f(a_j)| < \epsilon$. 2. For two measures μ and ν , absolute continuity of μ with respect to ν (written $\mu < \nu$) means that whenever E is a ν -measurable set with $\nu(E) = 0$, E is μ -measurable and $\mu(E) = 0$.

ABSOLUTE CONVERGENCE (ABSOLUTELY SUMMABLE): A series $\sum|s_n| = |s_1| + |s_2| + |s_3| + \dots + |s_n| + \dots$ that converges when all terms are replaced by their absolute values, i.e., a series that is mixed up with positive and negative terms, called absolutely convergent. If a series is absolutely convergent, then the sum is independent of the order in which the terms are summed. If the series is multiplied by another absolutely convergent series, the product series will also converge absolutely. Any series that converges absolutely is itself convergent.

For example, the series: $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2} \dots$ is absolutely convergent because

$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k^2} \right| = \sum_{k=1}^{\infty} \frac{1}{k^2}$ is also convergent. The ratio test can be used to test for absolute convergence. For an infinite integral $\int_a^b f(x) dx$ of real or complex valued function is said to be absolute convergence if $\int_a^b |f(x)| dx < \infty$. A series with mixed absolute terms is absolutely

convergent if; $\lim_{n \rightarrow \infty} \left| \frac{s_{n+1}}{s_n} \right| < 1$ and divergent if $\lim_{n \rightarrow \infty} \left| \frac{s_{n+1}}{s_n} \right| > 1$.

ABSOLUTE CONVEX SET: A subset of a vector space R or C that is both convex and balanced.

ABSOLUTE DATING: Any method of determining the actual age of an event or object in years. Some of the methods used in absolute dating include radiometric dating to determine the precise age of a rock or fossil, including isotope systems from very young (radiocarbon dating with ^{14}C) to systems such as uranium-lead dating that allow acquisition of absolute ages for some of the oldest rocks on earth. In archaeology, absolute dating methods are usually based on the physical, chemical, and life properties of the materials of artifacts, buildings, or other items that have been modified by humans and by historical associations with materials with known dates. Methods used include tree rings, radiocarbon dating of wood or bones, and trapped-charge dating methods such as thermoluminescence dating of glazed ceramics.

ABSOLUTE DIFFERENCE: The absolute value of the difference between two numbers.

A

ABSOLUTE ENTROPY: The increase in the entropy of a substance as it moves from a perfectly ordered crystalline form at -273°C (0 K), where its entropy is zero, to the temperature in question. Entropy is a measure of the energy in a system that is unavailable for work. Absolute entropy values are different at different temperatures and all compounds have an absolute entropy level that is an indication of the level of randomness of the molecules. For example, at 20°C (293K), liquid water has an absolute entropy level of $70\text{ J mol}^{-1}\text{ K}^{-1}$ and water in the vapour phase has an absolute entropy value of $189\text{ J mol}^{-1}\text{ K}^{-1}$.

ABSOLUTE ERROR: The actual error that exists when an estimate or approximation is made. For example, when a rope is measured as having a length of 99.95m when its length is actually 100m, then the absolute error is 0.05m.

ABSOLUTE EXPANSION: The true expansion of a liquid with temperature, taking into account the expansion of the container in which the volume of the liquid is measured. This contrasts with **apparent expansion**, which is the expansion of a liquid with temperature, as measured in a graduated container without taking into account the container's expansion.

ABSOLUTE FITNESS: Being one of the two ways by which fitness can be measured; it is a measure of biological fitness expressed as the total number of gene copies transmitted to the subsequent generation or the total number of surviving offspring that an individual produces during its lifetime. The other way of measuring fitness is called relative fitness.

ABSOLUTE FREQUENCY: In statistics, **absolute frequency** is the number of times an event occurs. For example, if a coin is tossed twice and two heads are observed, the absolute frequency of heads is one and the relative frequency is $1/4$.

ABSOLUTE HUMIDITY: The quantity of water vapour in a given volume of atmosphere. It does not fluctuate with the temperature of the air. It is usually expressed as the number of grams of water contained in 1 cubic metre of air.

ABSOLUTE LETHAL CONCENTRATION (LC_{100}): The lowest concentration of a chemical or a toxic substance in an environmental medium which kills 100% of test organisms under defined conditions. This value is dependent on the number of organisms used in its assessment.

Absolute lethal dose (LD_{100}) is the lowest amount of a chemical or a toxic substance which kills 100% of test animals under defined conditions. This value is dependent on the number of organisms used in its assessment.

ABSOLUTE LOCATION: The exact location of a place on the surface of Earth or a map, using a coordinate system, for example, latitude and longitude.

ABSOLUTE MAGNITUDE: The brightness of a star or other luminous object, when seen from a standard distance as if viewed from a distance of 10 parsecs or 32.6 light-years.

ABSOLUTE MAXIMUM (RELATIVE EXTREMUM OF THE FUNCTION): A number M , in the image of a function $f(x)$ on a set S , such that $f(x) \leq M$ for all $x \in S$. Suppose c is a number in the domain of a function f , and if and only if $f(c)$ is the maximum value of f on the domain of f , then $f(c)$ is called the absolute maximum value of f .

ABSOLUTE MINIMUM (RELATIVE EXTREMUM OF THE FUNCTION): A number M , in the image of a function $f(x)$ on a set S , such that $f(x) \geq M$ for all $x \in S$. Suppose c is a number in the domain of a function f , and if and only if $f(c)$ is the minimum value of f on the domain of f , then $f(c)$ is called the absolute minimum value of f .

ABSOLUTE PATH (ABSOLUTE PATHNAME; FULL PATH): A location, used in websites and operating systems that contains the root directory (the top-level directory) and all other sub directories. In Windows, it is designated (/).

ABSOLUTE PERMEABILITY: 1. The ratio of the magnetic flux density to the intensity of the magnetic field in a medium. Its unit is webers per square metre in the metre-kilogram-second system. 2. A measurement of the capacity for the flow of a single fluid such

as water, gas or oil through a rock formation when the formation is completely saturated with the fluid. Its symbol is k and the units are darcies or millidarcies.

ABSOLUTE PITCH (PERFECT PITCH): The ability to recognise and reproduce a tone without reference to a tuned musical instrument or an external standard.

ABSOLUTE REFRACTORY PERIOD (ARP): A point of time after a nerve or muscle has produced an action potential during which another action potential cannot be produced, no matter how strong the stimulus is. This occurs because the voltage-gated sodium channels which must open to produce an action potential have been inactivated.

ABSOLUTE SPACE: The space that exists as a background to events and processes and is not affected by objects or other entities in the universe or space, when matter and energy are not considered. Absolute space, in its own nature, without regard to anything external, remains always similar and all bodies are completely immovable. It is a concept underlying Newtonian physics, although many physicists have always regarded absolute space as an undesirable concept and suggested, as in Mach's principle, that fundamental physics should be described by rational theories.

ABSOLUTE TEMPERATURE: The absolute measure of temperature, using a scale that uses absolute zero as its lowest temperature that is independent of the properties of thermometric substance. An example is the Kelvin scale, whose zero is equivalent to -273.15°C . The standard measure of temperature in the International System of Units (S.I.) is the Kelvin scale, with a zero point (0 K) equivalent to -273.15°C . The Kelvin scale is equivalent to the Celsius ($^{\circ}\text{C}$) temperature scale with the points shifted by 273.15 degrees. An example of absolute temperature is the Kelvin scale, whose zero is equivalent to -273.15°C .

ABSOLUTE TIME: The geologic time measured in years, as determined by the decay of radioactive elements. The other geologic time is the **relative time**, which uses rock sequences and the fossils contained within the sequences to determine age. Ages may also be obtained by counting tree rings, by decay of a specific type of atom or annual sedimentary layers, such as varves in lakes or layers in a glacier.

ABSOLUTE UNITS: A unit of measurement defined in terms of the basic units of a particular measurement system (length, time, mass, current) and not based on arbitrary definitions. The units in which length is measured can be either relative or absolute.

ABSOLUTE VALUE (MODULUS): 1. Also known as **numerical value**, it is a term used in mathematics to refer to the magnitude or value of numbers, without regard to their sign. For example, the numbers -3 and 3 have the same absolute value, written as $|3| = |-3| = 3$. However, the absolute value of 0 , $|0|$ is 0 . In other words, for a real number x the absolute value is $|x|=x$ if $x \geq 0$ and $|x|=-x$ if $x < 0$. 2. The square root of the sum of the squares of the real and imaginary parts of a given complex number $z = x + iy$, $|z| = \sqrt{x^2 + y^2}$. Geometrically, it represents the distance from $0 \in \mathbb{C}$.

ABSOLUTE VALUE FUNCTION: A function that contains an algebraic expression within absolute value symbols. It is written as $f(x) = |x|$. It is a function in which the outputs of non-negative (i.e., $x \geq 0$) inputs equal the inputs, for example, $|3| = 3$, $|6| = 6$. If the outputs are negative (i.e., $x < 0$) inputs are the opposites of the inputs, for example, $|-3| = 3$, $|-6| = 6$.

ABSOLUTE WEIGHT: 1. The weight of a body measured without modifying influences such as the atmosphere and therefore measured in vacuum. In other words, it is the ratio of the mass and weight (mass/weight) of a body in vacuum or allowance must be made for buoyancy. 2. The weight of a specific number of seeds used in determining the average weight of a single seed.

ABSOLUTE ZERO: The lowest temperature theoretically possible. It is zero on the Kelvin scale and it is equal to -273.15°C . At or near this temperature molecules are theoretically motionless and the physical properties of some materials change considerably. For example, some metals lose their electrical resistance and become superconducting. The third law of thermodynamics shows that

absolute zero temperature cannot be reached, nonetheless in 1994, scientists at the National Institute of Standards and Technology (NIST) achieved a temperature of 700 nK ($1 \text{ nK} = 10^{-9} \text{ K}$), using optical lattice laser equipment to adiabatically cool caesium atoms; and in 2003, scientists at Massachusetts Institute of Technology (MIT) reached 450 pK (0.45 nK).

ABSOLUTELY NORMAL NUMBER: A real number that is a normal number in any scale or with respect to every base. A number is normal to base q if every sequence of n consecutive digits in its q -base expansion appears with limiting probability q^{-n} . A number is absolutely normal if it is normal to every base $q \geq 2$.

ABSOLUTELY SUMMABLE: An infinite series that has a convergent series of absolute values.

ABSOLUTELY SYMMETRIC: A function of two or more variables that is unchanged by any permutation of its variables.

ABSOLUTISM: In physics, the theories by English physicist and mathematician, Sir Isaac Newton (1642-1727) that space and time are absolutes. In other words, space is an independently existing thing and what facts about the universe there may be do not necessarily coincide with what can in principle be established by measurement. This is in contrast to **relationism**, which asserts that space is not an independently existing thing but merely a mathematical representation of the infinity of different spatial relations that particles may have to each other.

ABSORB: 1. The act of taking in or sucking up fluid, nutrients or chemicals gradually. For example, a liquid such as ink can be absorbed by a blotting paper or a gas such as ammonia by water. 2. Taking up light, noise or energy and not transmitting it. 3. For two subsets A , B of a topological vector space X , A is said to absorb B if, for some non-zero scalar α , $B \subset \alpha A = (\alpha x : x \in A)$.

ABSORBANCE (OPTICAL DENSITY): In physics, it is the capacity of a substance to absorb radiation. Its symbol is A_λ and expressed as $A_\lambda = \log(\frac{I_0}{I})$, where I_0 is the intensity of the incident ray and I is the intensity of the transmitted ray.

ABSORBED DOSE (D): 1. The amount of energy conveyed by ionising particles to a particular unit mass. The SI unit of absorbed dose is J kg^{-1} known as the gray with the symbol Gy. 2. The part of an administered medication, which is not excreted by the recipient's body.

ABSORBED ELECTRONS: The excess electrons present in a specimen under electron bombardment that are led to ground and measured as specimen current. The number of absorbed electrons per unit time or the specimen current equals the number of primary electrons minus the number of back scattered, secondary and transmitted electrons per unit time. Therefore, the fraction of electrons being absorbed depends on many parameters, including the composition and thickness of the specimen, the primary electron energy, the electron incidence angle and local electrostatic fields when present.

ABSORBED WATER: Water held mechanically in a soil mass and having physical properties similar to ordinary water at the same temperature and pressure.

ABSORBENT: A substance that has the capacity to absorb other substances, especially liquids.

ABSORBER: 1. A device used commonly for sampling by absorption in which a gaseous or liquid material is removed from another gas or liquid by selective absorption. An example is the impinge, which is used to sample dust in the air by drawing in a measured volume of dusty air and directing it through a jet to impact on a wetted glass plate; the dust particles adhering to the plate are counted. 2. In electronics, it is a material or device that takes up and dissipates radiated energy. It may be used to shield an object from the energy, prevent reflection of the energy, determine the nature of the radiation or selectively transmit one or more components of the radiation. 3. In mechanical engineering, it is a device that holds

liquid for the absorption of refrigerant vapour or other vapours. It is the part of the low-pressure side of an absorption system used for absorbing refrigerant vapour. 4. Any material that prevents ionising radiation from passing through. Examples include a sheet of paper, which can prevent alpha particles from passing through; a few centimetres of plastic material or 1 centimetre of aluminium that can absorb beta particles; materials such as lead, steel and concrete with a high atomic number and high density, which are used as absorbers of gamma rays; and boron, hafnium and cadmium, used as neutron absorbers in control rods for reactors.

ABSORBING ROD (CONTROL ROD): One of the numbers of rods of a material such as boron or cadmium that are used in nuclear reactors to absorb neutrons.

ABSORBING SET: A subset of a vector space over a field of numbers with the property that, for any point x in the space, tx belongs to the set whenever t is sufficiently small and positive. For example, the unit disk is an absorbing set in the Cartesian plane.

ABSORBING STATE: A state of a Markov chain from which the probability of exit is zero: a state that cannot be exited once entered. Expressed mathematically, state i is absorbing when $P(i, j) = 1$, hence $P(i, j) = 0$ for all $j \neq i$.

ABSORPTANCE: A measure of the ability of an object or substance to absorb radiant energy. Its symbol is α and it is given by the ratio of the radiant or luminous flux absorbed by a body to the incident radiation falling on it.

ABSORPTIOMETER: An instrument, equipped with a filter system or other simple dispersing system, used for measuring the absorption of nearly monochromatic radiation in the visible range by a gas or a liquid, to determine the concentration of the absorbing constituents in the gas or liquid.

ABSORPTION: 1. In chemistry, it is the up-take of a gas by a solid or liquid or the up-take of a liquid by a solid. Absorption differs from adsorption in that the molecules undergoing absorption are taken up by the volume, not by the surface. 2. In physics, it is the transformation of the energy of electromagnetic radiation such as sound and streams of particles into other forms of energy when passing through a medium. For example, in passing through a medium, the energy of electromagnetic radiation is taken up by matter, notably electrons, to transform it to heat. 3. In biology, it is the movement of dissolved substances, molecules or fluid such as water, gases or mineral ions across the membrane of a cell by diffusion, osmosis and/or by active transport to different parts of the body of an organism where they are required to build complex substances for normal growth and development. For example, in animals, soluble food materials digested are absorbed into cells lining the gut and to the blood system for transport to all body organs. In plants, water and mineral ions are absorbed from the soil by the roots.

ABSORPTION BAND: A series of very closely spaced absorption lines caused by the absorption of light by molecules. In other words it is a dark band in the spectrum of white light that has been transmitted through a substance that exhibits absorption at selective wavelengths. Alternatively, it is a range of wavelengths, frequencies or energies in an electromagnetic spectrum which are characteristic of particular transition from a lower to final energy state.

ABSORPTION COEFFICIENT: 1. A quantity that characterises the absorption of light or any other type of electromagnetic radiation as it passes through the absorbing material. Its symbol is ϵ and it has the dimensions of $1/(\text{concentration} \times \text{length})$. The absorption coefficient depends on the absorbing material and also on the wavelength of light being absorbed. Its highest value occurs where the absorption is most intense. 2. The volume of a given gas measured at a standard temperature and pressure that will dissolve in a unit volume of a given liquid.

ABSORPTION FIELD (SEPTIC DRAIN FIELDS): Part of a septic tank disposal system which typically consists of narrow trenches, which

are partially filled with gravel or crushed stone, into which perforated pipes are placed to allow liquid contaminated waste from the septic tank to seep into the soil.

A

ABSORPTION LAW: The law appearing in the definition of Boolean algebra and lattice which is stated as $A \cap (A \cup B) = A$ and $A \cup (A \cap B) = A$. The two parts of the absorption law are sometimes called the **absorption identities**.

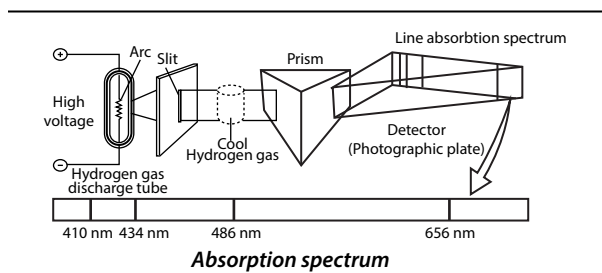
ABSORPTION LINE (ATOMIC FINGERPRINT): A narrow range of wavelengths or frequencies in the electromagnetic spectrum within which radiant energy is absorbed by the medium through which it is passing.

ABSORPTION OF COLOUR: The absorption of certain wavelengths of light by a filter. For example, pure red filter absorbs all light radiation other than shades of red and so only red light passes through. The intensity of the output light is, therefore, less than that of the input light.

ABSORPTION OF RADIATION: The loss of energy caused by the transfer of radiation from one medium to another.

ABSORPTION SPECTROSCOPY (ABSORPTIOMETRY): A technique for determining the identity and concentration of a chemical present in a substance. It is done by measuring the amount of electromagnetic radiation that the substance absorbs at specific wavelength.

ABSORPTION SPECTRUM: A pattern of dark lines or bands that is observed with a spectroscope, when electromagnetic radiation is absorbed by a medium. It is the result of unequal absorption of the radiation as it passes through the medium. A material's absorption spectrum shows the fraction of incident electromagnetic radiation absorbed by the material over a range of frequencies. Every chemical element has absorption lines at several particular wavelengths corresponding to the differences between the energy levels of its atomic orbitals. For example, an object that absorbs blue, green and yellow light will appear red when viewed under white light. Absorption spectra can therefore be used to identify elements present in a gas or liquid. This method is used in deducing the presence of elements in stars and other gaseous objects which cannot be measured directly.



ABSORPTION TOWER (ABSORPTION COLUMN): A long vertical tube used in industry to separate components of a rising gas with the use of a falling liquid to trap the gas. The gas is pumped in at the bottom of the column and the absorbing liquid, often water, is at the top and falls down against the concurrent of gas. The gas along with any impurities it contains floats to the top. As it moves toward the top, aerosolised liquid is sprayed into the tower. The droplets catch impurities in the gas and carry them to the bottom of the tower for collection. Some towers can have multiple points where liquid is sprayed out to capture different impurities or maximise the amount of material trapped.

ABSORPTIVE POWER (ABSORBENCY INDEX; ABSORPTION CONSTANT; ABSORPTIVITY): Any measure of the extent to which the intensity of a beam of light is reduced by passing through a distance of a solution having a molar concentration of a dissolved substance. It is also called **molar absorptivity** in the Beer's Law (Beer's law, Beer-Lambert law, Lambert-Beer law or the Beer-Lambert-Bouguer law), which relates the absorption of light to the properties of the

material through which the light is travelling). It is the constant, a , in the Beer's law relation $A = abc$, where A is absorbance (no units, since $A = \log_{10} P_0 / P$), e is the molar absorptivity with units of $L \text{ mol}^{-1} \text{ cm}^{-1}$, b is the path length of the sample, i.e., the path length of the cuvette in which the sample is contained. This measurement in centimetres c is the concentration of the compound in solution, expressed in mol L^{-1} . Therefore, if all the light passes through a solution without any absorption, then absorbance is zero and percent transmittance is 100%. If all the light is absorbed, then percent transmittance is zero and absorption is infinite. The reason for stating the law with this equation is because absorbance is directly proportional to the other parameters, as long as the law is obeyed.

ABSORPTIVE FEEDERS: Animals such as tapeworms that ingest food through the body wall.

ABSTRACT: 1. Theoretical; not applied or practical. 2. In logic, it is an expression, usually denoting a class or property formed by abstraction.

ABSTRACT ALGEBRA: The branch of mathematics concerned with algebraic structures such as groups, rings and fields, involving sets of elements with particular operations satisfying certain axioms. The purpose is to derive, from the set of axioms, general results that are then applicable to any particular example of the algebraic structure in question.

ABSTRACT MACHINE: Any hypothetical computing machine concerned with the detailed and precise analysis of how the computer system works rather than by its internal physical structure. It may also be defined as a procedure for executing a set of instructions in some formal language, possibly also taking in input data and producing output. An example is a Turing machine.

ABSTRACT NUMBER: A number regarded simply as a number in its entity, without reference to any material objects, units or specific examples. For example, the number 7, when it does not refer to 7 objects or quantities is called an **abstract number**. If the number refers to objects such as 5 oranges, it is called a **concrete number**. A **denominate number** is a type of concrete number with a unit of measure attached to it, for example, 10 metres.

ABSTRACT SYNTAX NOTATION ONE (ASN.1): A standard way of describing a message or a unit of application data that can be sent or received in a network. ASN.1 is divided into two parts: (i) the rules of syntax for describing the contents of a message in terms of data type and content sequence or structure; and (ii) how to actually encode each data item in a message.

ABSTRACT WINDOW TOOLKIT (AWT): A set of application program interfaces (APIs) used by Java programmers to create graphical user interface (GUI) objects such as buttons, scroll bars and windows. AWT is part of the Java Foundation Classes (JFC) from Sun Microsystems, the company that originated Java. The JFC are a comprehensive set of GUI class libraries that make it easier to develop the user interface part of an application program.

ABSTRACTION: 1. A generalised idea or theory developed from concrete examples of events or the formation of general ideas or concepts from concrete examples. It can also be defined as the process of making a general statement, which summarises what can be observed in particular instances. 2. In chemistry, it is the chemical reaction that involves bimolecular removal of an atom or ion from a molecule. An example is the abstraction of hydrogen from methane by reaction with a radical: $\text{CH}_4 + \text{X}^\cdot \rightarrow \text{H}_3\text{C}^\cdot + \text{HX}$. 3. The removal of water from a river or other source for industrial use. 4. In computer science and software engineering, it is the separation of essential concepts from background details so as to focus on a few concepts at a time rather than in terms of their details. For example, it can be used to identify software artifacts (objects) to model the problem domain in order to reduce the complexity of the design and implementation process of the software.

ABUNDANCE: 1. The measurement of how relatively common a chemical element is or how much of the element there is by comparison to all other elements. For example, the ratio of the total

mass of aluminium in the Earth's crust to the total mass of the Earth's crust is 8%. 2. The ratio of the number of atoms of a particular isotope of an element to the total number of atoms of all the isotopes present, often expressed as a percentage. For example, the abundance of uranium-235 in natural uranium is 0.71%.

ABUNDANT NUMBER: A positive integer that is smaller than the sum of its proper divisors. For example, 12 is an abundant number because it is smaller than the sum (16) of its proper divisors $1 + 2 + 3 + 4 + 6 = 16$.

ABVOLT (abV): The unit of electromotive force (emf) or potential difference in the centimetre-gram-second (cgs) electromagnetic system of units. When an emf of 1 abV exists between two points, then 1 erg of energy is needed to move one abcoulomb (1 abC) of charge carriers between those two points. The International System of Units (SI) is the currently used system of units, and the Volt (V) has replaced the abvolt.

ABWATT (abW): The unit of power in the centimetre-gram-second (cgs) electromagnetic system of units. In a direct-current (DC) circuit, 1 abW is the power dissipated, radiated or expended when one abvolt (1 abV) of potential difference drives a current of one abampere (1 abA) through a component. The International System of Units (SI) is the currently used system of units, and the watt (w) has replaced the abwatt.

ABYSSAL ZONE: That dark zone in the ocean, approximately 2,000 metres in depth having a temperature of about 4°C (277 K), where there is effectively no light penetration. Abyssal organisms are adapted for living under high pressures in cold dark conditions and may be blind or have their own light sources. The region above it is the **bathyal zone** and that below, is the **hadal zone**.

ABZYME: An antibody that is created by combining an antibody with an enzyme and used to accelerate chemical reactions.

AC: In physics, it is the abbreviation for alternating current.

AC-3 (DOLBY DIGITAL): A perceptual digital audio coding technique that reduces the amount of data needed to produce high-quality sound, developed by Dolby Laboratories. Perceptual digital audio coding takes advantage of the fact that the human ear screens out a certain amount of sound that is perceived as noise. Reducing, eliminating or masking this noise significantly reduces the amount of data that needs to be provided. Dolby Laboratories developed two other perceptual coding systems, AC-1 and AC-2. AC-3 was the first coding system designed specifically for multichannel digital audio. It is the sound format for digital television (DTV), digital versatile discs (DVDs), high definition television (HDTV) and digital cable and satellite transmissions.

AC'97 (AUDIO CODEC'97): A type of Intel Audio Codec that is integrated into Intel chips that provides six channels (5.1) of 20-bit, 48 kHz audio. Audio Codec is a digital electronic device or software that compresses and decompresses a digital audio signal, for example, MP3, Windows Media Audio (WMA), and Dolby Digital.

ACACIA: A genus of woody plants of the family Papilionaceae. The tree bears yellow or white flowers, typically bipinnate leaves and dark fruit pods. Some yield gum and some are good timber trees.

ACANTHION: The point at the base of the anterior nasal spine, which is formed where projecting processes of the maxillae (upper jaw bones) meet at the front of the face.

ACANTHOCEPHALA (SPINY/THORNY-HEADED WORMS): A phylum of worms living parasitically in the intestines of vertebrates such as fish and birds, causing fatal infections. Each one is characterised by retractable proboscis, which bears tiny hooks that anchor it to the intestinal walls of its host. They lack a mouth or alimentary canal and live in the intestines of their host and uptake digested nutrients of the host, directly, through their body surface. Acanthocephalans require two hosts to complete the life cycle. Juveniles are parasitic in arthropods and the adults, which are between 1-2 cm long and with an elongated, cylindrical body, live in the digestive tract of vertebrates, especially fish.

ACANTHUS: A spiny-leaved bush or perennial plant, which belongs to the genus *Acanthus*, of the Acanthaceae family and native to the Mediterranean. It is mostly a xerophytic species with white, purple flowers. One species in the Acanthaceae family, *Acanthus montanus*, also known as bear's breech or mountain thistle is effective in treating conditions such as gonorrhoea, syphilis, staphylococcus aureus, hepatitis, jaundice and typhoid.

ACARIASIS: A skin disease such as scabies caused by ticks or mites.

ACARICIDE: A name given to a group of pesticides capable of killing members of the class Arachnida and order Acari such as mites and ticks.

ACARINA: An order of small arthropods belonging to the class Arachnida and comprising the mites and ticks. Mites are parasitic, causing scabies in humans and mange in domesticated animals and can also provoke allergies or dermatitis. In plants they may infest arable and greenhouse crops. Ticks also transmit diseases such as encephalitis and Lyme disease.

ACARPOUS: In botany, it refers to a plant with flowers without pistils or gynoecia (gynoecium) or a sterile flower and, therefore, does not produce fruit.

ACAULESCENT: A term used in botany to describe plants such as tufted or rosette plants that have no stem or an extremely short stem concealed in the ground.

ACCELERANT: A flammable material such as petrol and paraffin, used as a catalyst or to start and spread a fire intentionally.

ACCELERATED AGING: A set of laboratory conditions or a testing method using factors such as temperature, vibration, light, oxygen, water and other environments designed to produce normal aging results in a short time. It is used to estimate the useful lifespan of a product or its shelf life when actual lifespan data is unavailable.

ACCELERATED FREEZE DRYING (AFD; LYOPHILISATION; CRYODESICCATION): A dehydration process used for food preservation. It is usually done in vacuum, where food is rapidly frozen with liquid nitrogen and then dehydrated by sublimation from the solid phase to the gas phase.

ACCELERATED GRAPHICS PORT (AGP): An interface specification that enables 3-D graphics to display quickly on personal computers. AGP is designed to convey 3-D images, for example, from Web sites or CD-ROMs much more quickly and smoothly than is possible today on any computer other than an expensive graphics workstation. It is especially useful in conjunction with gaming, three-dimensional (3D) video and sophisticated scientific/engineering graphics programs.

ACCELERATED HUB ARCHITECTURE (AHA; INTEL HUB ARCHITECTURE): An Intel 800-series chipset design that uses a dedicated bus to transfer data between the two main processor chips instead of using the Peripheral Component Interconnect (PCI) bus. AHA provides twice the bandwidth of the traditional PCI bus architecture at 266 MB per second. It consists of a memory controller hub and an input/output (I/O) controller hub.

ACCELERATING VOLTAGE SCAN (HIGH VOLTAGE SCAN): An alternative method of producing a momentum (mass) spectrum in magnetic deflection instruments by varying the accelerating voltage. This scan can also be used, in conjunction with a fixed radial electric field to produce an ion kinetic energy spectrum.

ACCELERATION: The rate at which the velocity of a body changes with time. Its unit is metre per second per second (m/s^2 or ms^{-2}) or kilometre per hour per hour (km/hr^2). It is a vector quantity denoted by a and expressed as $a = \text{change in velocity/time interval } (\Delta v/t)$. The slope of the velocity-time graph gives the acceleration of the body.

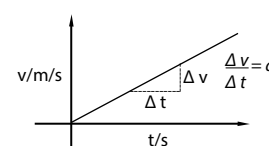


Fig i) Velocity-Time Graph

Since it is a vector quantity, the direction of the acceleration vector depends on two things: whether the object is speeding up or slowing down and whether the object is moving in the positive or negative direction. When an accelerating object changes its velocity by the same amount each second, for example, by 10 m/s in each consecutive second, it is referred to as a **constant acceleration**. A body travelling at constant speed may be said to be accelerating if its direction keeps changing as in a body moving in a circular path. There are two kinds of acceleration: average and instantaneous. **Average acceleration** is determined over a long time interval. The velocity at the beginning of this interval is called the **initial velocity** (v_0) and the velocity at the end is called the **final velocity** (v). It is calculated as follows:

$$a_v = \frac{(v - v_0)}{\Delta t}$$

Instantaneous acceleration is the limiting value of the acceleration calculated over a very short time interval at the appropriate instant of time. It can also be defined as the limit of average acceleration as the time interval approaches zero. It is the slope of the tangent of the velocity-time graph at the same instant of time. Instantaneous acceleration is calculated as follows:

$$\lim_{t \rightarrow 0} \frac{\delta v}{\delta t} = \frac{dv}{dt}$$

ACCELERATION DUE TO GRAVITY (ACCELERATION OF FREE FALL):

The acceleration of a body towards the earth's surface when the body is acted on by its weight only. In other words, it is the acceleration of a body falling freely under the influence of the Earth's gravitational pull or the acceleration experienced by any massive object falling freely in the earth's gravitational field. Its symbol is g . In a vacuum, all bodies accelerate in a gravitational field at the same rate relative to the centre of mass and this is true regardless of the mass or composition of the body. In a medium such as air or sea, the actual acceleration of a body is reduced by its buoyancy and the viscosity of the atmosphere. On the surface of the Earth, all objects fall in a vacuum with acceleration between 9.78 ms^{-2} and 9.82 ms^{-2} depending on latitude. Acceleration due to gravity has a typical value of 9.81 ms^{-2} close to the earth's surface. The acceleration due to gravity, g , can be calculated from a distance-time graph by plotting a graph of distance, s against the square of the time, t^2 . Two times the slope yields the acceleration due to gravity, g . Referring to the diagram, $y = mx + c$ can be compared to $s = \frac{1}{2}gt^2$, hence the slope $m = \frac{1}{2}g$. This implies that acceleration due

to gravity, g equals twice the slope of a displacement-time graph.

ACCELERATION HARDWARE: A device that speeds up data communications, storage and retrieval, encryption and decryption, mathematical operations, graphics and Web page viewing. An acceleration hardware may consist of an individual integrated circuit (an IC or chip), a printed circuit card or a self-contained system.

ACCELERATION SERVER (HIGH-SPEED DIALUP): An internet service provider (ISP) feature that speeds up data transfer by using a special server, called an acceleration server, to act as a bridge between

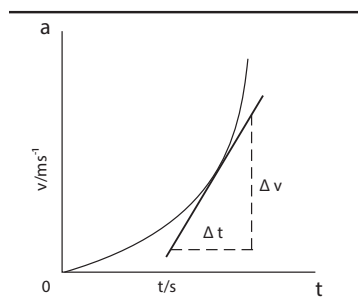
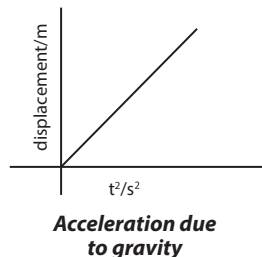


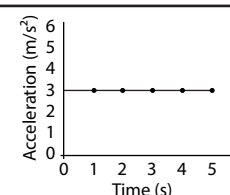
Fig (ii) Instantaneous Acceleration Graph



the user's dial up connection and a Web page. The acceleration server uses a fast broadband connection to request the Web page on behalf of the dial-up user. When the acceleration server receives the data it requested, it caches the page, compresses the data and filters out any pop up ads before sending the Web page on to the dialup user. To the average user browsing the internet, the caching, compression and filtering techniques can make a dial up connection seem up to five times faster than normal. Broadband has largely replaced dialup internet connections.

ACCELERATION STRESS: The total effects on human physiology and behaviour of increased gravitational forces, such as those experienced in a fighter aircraft or spacecraft, especially during take-off and re-entry.

ACCELERATION-TIME-GRAPH (A-T GRAPH): A graph that shows acceleration plotted against time for an object moving in a straight line. In case of the acceleration-time graph, acceleration of the object is generally plotted on the vertical axis and the time is plotted on the horizontal axis. For example, if an object at the start (0 second) accelerates at 3 m/s^2 and remains same at various time intervals of 1 second, the acceleration-time graph is as shown in the graph.



Acceleration-Time Graph

ACCELERATOR: 1. In physics, it is a device that uses electromagnetic fields to propel charged particles to high speeds and energies at which they can be of use in industry, medicine and nuclear and pure physics. The two basic types are electrostatic and oscillating field. An example is a cyclotron, that produces high-energy particles and focuses them on a target. 2. In automobiles, a device, usually operated by the foot, for controlling the speed of an engine. 3. In chemistry, it is a substance used to increase the rate of a chemical reaction. An example is the chemical used to speed up cross-linking reactions in polymers. Alternatively, it is a substance that increases the rate of chemical reactions, for example, a catalyst, especially in vulcanisation of rubber.

ACCELERATOR BOARD: A printed circuit board that enhances a computer's performance by substituting a faster microprocessor without replacing the entire motherboard and associated components. It accelerates transmission or processing beyond the CPU's capabilities and transmits faster floating-point units (FPUs) by assisting in mathematics calculations or by increasing speed. Without an expansion board, the CPU relies on software to implement certain tasks such as transferring images.

ACCELERATOR TUBE: An evacuated tube along which ions such as protons or alpha particles can be accelerated to high energy levels by applying a very high potential difference across its ends.

ACCELEROMETER: A mechanical or electromechanical instrument for measuring acceleration (rate of change of velocity) of a moving object, especially one in which a sensor converts acceleration into an electrical signal.

ACCENT: 1. Distinctive feature or quality, that accentuates, contrasts with or complements a decorative style or which makes an image stand out. For example, **accent colour** is the colour used as a complementary colour in quite small quantities to lift a colour scheme. It works best if it is a bright, vibrant colour. 2. In mathematics, it is a mark used as a superscript to distinguish among variables represented by the same symbol or a mark used as a superscript to indicate the first derivative of a variable. 3. A mark or one of several marks used as a superscript to indicate a unit, such as feet (') and inches (") in linear measurement.

ACCEPTABLE DAILY INTAKE (ADI): It is the estimate of the amount of a food additive or a substance in food or drinking water, expressed on a body mass basis, which can be ingested daily over a lifetime without appreciable health risk. It is usually expressed in mg or kg of body weight. Ingestion that exceeds this amount may

cause toxic effects. For calculation of the daily intake per person, a standard body mass of 60 kg is used.

ACCEPTABLE USE POLICY (AUP; INTERNET USE POLICY; AUP/IUP): A policy that a user must agree to follow in order to be provided with access to a network or to the Internet. It is common practice for many businesses and educational facilities to require that employees or students sign an acceptable use policy before being granted a network ID.

ACCEPTANCE REGION: A term used in statistics to represent a region, where the null hypothesis, H_0 , is accepted when the test statistic falls within it. The acceptance region is defined by the critical values of the test.

ACCEPTANCE SAMPLING: It is a procedure that is applied in quality control in the selection of sample from a batch and a decision to accept the batch made on the basis of the quality of the sample. In most cases, the simple procedure of accept or reject criterion is applied. The main advantage of this method is the reduction in cost of taking samples to satisfy quality control criteria.

ACCEPTOR: 1. In physics, it is a substance that is added as an impurity to a semiconductor because of its ability to accept electrons from the valence bands. 2. In chemistry, it is an atom or a group of atoms that accepts electrons to form a coordinate bond during the formation of a chemical compound. 3. In biochemistry, it is the receptor that binds a hormone without any apparent biological response.

ACCEPTOR IMPURITIES: Trivalent impurity atoms used to increase the number of holes in a pure or intrinsic semiconductor. Aluminium (Al), indium (In), boron (B), and gallium (Ga) are examples of trivalent impurities.

ACCEPTOR LEVEL: The energy level of an acceptor atom in a semiconductor, such as boron in silicon, that results from the presence of acceptor atoms. The acceptor impurity introduces an additional energy level just above the edge of the valence band. Acceptor level is very close to the top of the valence band and hence causes a *p*-type conduction.

ACCESS: 1. It refers to the right or permission to use a computer, a software or service you have authorised access to, usually granted by entering your login details. 2. In computer science, it is to retrieve data from or write data to a storage device. 3. **Microsoft Access (MS ACCESS)** is a computer software program. It is a database management program that allows users to create custom databases that store information.

ACCESS ARM (HEAD ARM/ACTUATOR ARM): A mechanical arm which moves the read/write head to a specific track on the hard drive where information needs to be read or written to. It is found inside all computer hard disk drives.

ACCESS CONTROL: Any mechanism by which a system grants or revokes the right to access some data or perform some action. The mechanism controls what operations the user may or may not make by comparing the User ID to an Access Control database. It normally requests a user to Login to a system, using some authentication system such as User ID. The system includes the following: file permissions, such as create, read, edit or delete on a file server; program permissions, such as the right to execute a program on an application server; and Data rights, such as the right to retrieve or update information in a database.

ACCESS CONTROL FACILITY (ACF2; CA-ACF2): A set of programs from Computer Associates that enables security on mainframes. ACF2 prevents accidental or deliberate modification, corruption, mutilation, deletion or viral infection of files. With ACF2, access to a system is denied to unauthorised personnel. Any authorised or unauthorised attempt to gain access is logged.

ACCESS LIGHT (ACTIVITY LIGHT): A LED (light emitting diode) on the front of a computer and/or disk drive that illuminates to indicate that the computer or disk is engaged.

ACCESS CONTROL LIST (ACL): A list of user permissions for a file, folder or other object. It defines what users and groups can access and

what operations they can perform. These operations typically include read, write and execute. For example, if an ACL specifies read-only access for a specific user of a file, that user will be able to open the file, but cannot write to it or run the file.

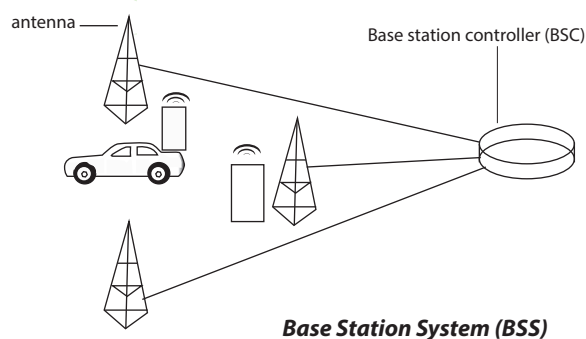
ACCESS KEY (ACCESS CODE): A series of numbers and/or letters that allow access to a particular system. It may be a password, although passwords are generally used in conjunction with usernames. Access keys need not be attached to a specific user. Many users could use the same access key for a specific system or object without being identified as a specific user.

ACCESS LOG: A list of all the requests for individual files that people have requested from a Web site. These files will include the HTML files and their imbedded graphic images and any other associated files that get transmitted.

ACCESS METHOD: 1. A program or a hardware mechanism that moves data between the computer and an outlying device such as a hard disk (or other form of storage) or a display terminal. The term is sometimes used to refer to the mechanics of placing or locating specific data at a particular place on a storage medium and then writing the data or reading it. It is also used to describe the way that data is located within a larger unit of data such as a data set or file. 2. An Application Programming Interface (API) that a programmer uses to create or access data sets or to read from or write to a display terminal or other output device. Examples are the Virtual Sequential Access Method (VSAM) and the Virtual Telecommunication Access Method (VTAM).

ACCESS PATH: A directory that tells the computer running Microsoft DOS/Windows where to look if a command is not part of an internal command.

ACCESS POINT (BASE STATION): A device, such as a wireless router that allows wireless devices to connect to a network. An access point connects users to other users within the network and also can serve as the point of interconnection between the Wireless Local Area Network (WLAN) and a fixed wire network.



ACCESS POINT NAME (APN): The name assigned to a mobile or wireless device network and the public Internet in order to determine the correct IP address and connect to the correct secure gateway or to a network like virtual private network (VPN).

ACCESS PROVIDER (INTERNET ACCESS PROVIDER, IAP; INTERNET SERVICE PROVIDER, ISP): Any organisation that arranges for an individual or an organisation to have access to the Internet. Access providers are generally divided into two classes: Internet Access Providers (IAPs) and Online Service Providers (OSPs). ISPs can be local businesses that pay for a high-speed connection to one of the companies that are part of the Internet. They can also be national or international companies that have their own networks such as AT&T's WorldNet or IBM's Global Services. **AccessChk** is a free Sysinternals command-line utility that shows what access a user or group of users has to a particular service, file, folder or registry key.

A

ACCESS PRIVILEGE: In computer networking, authorised access to files and other shared resources. The ability to authorise or restrict access selectively to files or directories, including separate privileges such as reading, writing or changing data, is a key element in computer security systems. This kind of system ensures that, for example, a company's employees cannot read its personnel files or alter payroll data unless they work for the appropriate departments or that freelance or temporary staff can be given access to some areas of the computer system but not others. Access privileges may also exist for shared printers and devices such as tape drives and modems.

ACCESS RATE: In computer networking, it is the maximum data rate of a channel between a user site and a network, as defined by the bandwidth of the access link available for data transmission.

ACCESS RIGHTS: The authorisation granted by an administrator to a user to read, write and modify files in the computer.

ACCESS TIME (REACTION TIME): The time taken by a computer to read from or write to memory after an instruction has been given.

ACCESSENUM: A Sysinternals tool that offers administrators a view of the full file system and registry security settings to ensure that users have appropriate permissions to access files and directories. It uses standard Windows security APIs to show permissions information, offering more granular results than standard access control lists.

ACCESSORY: Any device added to a computer that performs an additional feature, but which the computer does not require to function. An example is a printer, which gives the computer the ability to print, but the computer functions even without a printer.

ACCESSORY BUD (SUPERNUMERARY BUD): In botany, it refers to any bud of a plant, which is positioned at the side or above an axillary bud.

ACCESSORY CELL (SUBSIDIARY CELL): 1. Any of a group of morphologically differentiated epidermal cells immediately surrounding the guard cells on the leaves of many plants. Stomatal complexes lacking subsidiary cells are termed **anomocytic**. The number, arrangement and development of accessory cells are often useful taxonomic characters. 2. Any nonlymphocytic cell that helps in the induction of the immune response by presenting antigen to a helper T lymphocyte.

ACCESSORY CHROMOSOME (B CHROMOSOME; SUPERNUMERARY CHROMOSOME; ODD CHROMOSOME; UNPAIRED ALLOSOME; UNPAIRED CHROMOSOME): Any chromosome additional to the regular karyotype of a species, which typically varies in size and composition and may or may not affect the phenotype. It is an excess chromosome resulting from unsuccessful meiotic divisions where resultant cells have extra chromosomes. In humans, about one person in a thousand carries an accessory chromosome and some of these are associated with mental or physical abnormalities. For example, cat-eye syndrome occurs in people who carry an extra chromosome that partly duplicates chromosome 22.

ACCESSORY FRUIT (FALSE FRUIT; PSEUDOCARP): A fruit that develops from ripened ovaries and some other parts of the flower and contains much fleshy tissue derived from part of the plant outside the ovary. They are usually seedless. Examples are apple, strawberry, cashew and pineapple.

ACCESSORY ORGAN: An organ or lump of tissue adjacent to an organ that assists with the functioning of some other organ within a system. Examples are teeth, tongue, salivary glands, gallbladder, liver, pancreas, ocular muscles and eyelids.

ACCESSORY PIGMENTS: Light-absorbing compounds other than chlorophyll, found in photosynthetic organisms such as plants, algae and some bacteria that capture light energy and transfer it to the principal active pigment responsible for photosynthesis. They

are usually a type of chlorophyll. Examples are carotenoids and phycobiliproteins.

ACCESSORY TRANSFUSION TISSUE: A type of tissue in gymnosperms, such as Cycads, that extends through the mesophyll from the midrib to the margin. It is composed of three kinds of cells: thick walled tracheids, parenchyma and albuminous cells.

ACCLIMATION: The natural physiological changes occurring in an organism in response to a change in a particular environmental factor such as altitude, oxygen concentration or temperature that equips the organism to adapt to living in the environment. Acclimation, like acclimatisation is a phenotypic change that occurs within an individual organism and results from an environmental change with the changes occurring within a lifetime, is reversible and is an ecological response. The main difference between acclimation and acclimatisation is that while acclimation can be achieved in a laboratory or controlled field settings, acclimatisation is due to natural conditions only.

ACCLIMATISATION: 1. The progressive physiological, biochemical or anatomical adaptation of an organism to any change in its natural environment that subjects it to physiological stress. For example, a man who migrates to live on a very high altitude must acclimatise to his new environment by developing an increase in the number of red blood cells to increase the oxygen carrying capacity of the blood, in order to compensate for lower levels of oxygen in the air. This explains why athletes from Kenya, who live on high altitudes, are very good long distance runners. Also, before major tournaments such as the world cup, teams arrive several days early to get used to the climatic conditions so that it does not affect them substantially. Acclimatisation, like acclimation is a phenotypic change that occurs within an individual organism and results from an environmental change with the changes occurring within a lifetime, is reversible and has an ecological response. The main difference between acclimation and acclimatisation is that while acclimation can be achieved in a laboratory or controlled field settings, acclimatisation is due to natural conditions only. 2. The overall sum of processes by which an organism attempts to compensate for conditions that would significantly lower the amount of oxygen delivered to its cells.

ACCOMMODATION: 1. The process by which the vertebrate eye focuses or maintains a clear image of near or far objects by changing the shape of the lens. 2. The psychological adjustment made by animals in response to continuously changing environmental conditions.

ACCOMMODATION COEFFICIENT (STICKING COEFFICIENT): The ratio of the average energy actually transferred between a surface and impinging gas molecules such as aerosol particles and cloud droplets scattered by the surface, to the average energy which would theoretically be transferred if the impinging molecules reached complete thermal equilibrium with the surface.

ACCOUNT: In computer science, it is a unique record for each type of asset, liability, equity, revenue and expense.

ACCOUNT LOCKOUT: A Windows 2000 and later security feature that locks a user account if a certain number of logon attempts fail within a specified interval of time.

ACCOUNTING MACHINE: An earlier non-electrical computer system that used magnetic stripe ledger cards or punch cards to perform functions such as to calculate, prepare invoices, payrolls, and various financial data.

ACCOUNTING MANAGEMENT: In information technology, it is the process of keeping track of user activities on network resources usage, based on call detail recording.

ACCRESCENT: In botany, it is the enlargement or increasing in size of the calyx after the flower is fully developed and functioning.

ACCRETION: A process of gradual growth in the size of an object by accumulation of matter due to gravity. As small bits of matter are

attracted toward an object due to gravity and stick to the object, it gets bigger and bigger. Most galaxies, stars, and planets, are formed by accretion. 3. In geology, it is the slow addition to land by deposition of water-borne sediment or an increase of land along the shores of a body of water, for example, by alluvial deposit. 4. In biology, it is the growing together or adherence of parts that are normally separate.

ACCRETION DISC (ACCRETION DISK): A disc-shaped rotating structure formed by gravitational attraction. In other words, it is a disc of material being drawn in by gravity to a centre of mass, such as a star or black hole. As the material comes in, it spins round the central body and radiates energy. As the material moves in, gravitational energy released in that process is transformed into heat and rotational movement.

ACCRETIONARY LAPILLI: Rounded tephra balls between 2 mm and 64 mm in diameter consisting of tiny volcanic ash particles. They are formed in an eruption column or cloud by moisture or electrostatic forces, with the volcanic ash nucleating on some object and then accreting to it in layers before the accretionary lapillus falls from the cloud.

ACCRETIONARY WEDGE: A body of deformed sediments, wedge-shaped in two dimensions or prism-shaped in three dimensions that accumulate and deform where oceanic and continental plates collide. These sediments are scraped off the top of the down going oceanic crustal plate and are appended to the edge of the continental plate.

ACCUMBENT COTYLEDON: Cotyledons of an embryo lying against one edge of the radical.

ACCUMULATION: 1. Also known as **biological accumulation** or **bioaccumulation**, it is the exposure to toxic substances in the environment. Examples of such toxic substances are pesticides or other chemicals that may concentrate within the tissues or organs of a living organism. 2. In ecology, it is the process in which iceberg and glaciers are formed through snowfall and avalanches, which can store frozen water for many years. This is in contrast to ablation, which removes snow and ice from a glacier or iceberg by melting or evaporation.

ACCUMULATION POINT (CLUSTER POINT; LIMIT POINT): In mathematics, a point where neighbourhoods intersect with the set other than at the point itself.

ACCUMULATOR: 1. In computing, it is an intermediate register or memory location in the arithmetic and logic unit of the computer processor for temporary storage of arithmetic and logic data before being transferred to the main memory or to another register. 2. Any device in which energy in the form of electrical, hydraulic, mechanical and pneumatic is stored and used to boost the system's energy supply into a short surge of energy or vice versa. Examples include capacitors, compulsators, steam accumulator, wave energy machines and pumped-storage hydroelectric plants. 3. A group of rechargeable cells connected to store electrical energy in the form of chemical energy. Examples are the 12 volt lead-acid battery consisting of lead electrodes immersed in an electrolyte of dilute sulfuric acid and commonly used in motor vehicles; and nickel-iron (NiFe) accumulator, also known as Edison accumulator, which is a type of secondary or rechargeable battery having a positive plate of nickel oxide and a negative plate of iron immersed in an electrolyte of potassium hydroxide.

ACCURACY (EXACTNESS): An indication of how close a measurement is to its true value or the closeness of a measurement to the true value of the quantity that is measured. In physics, the degree of accuracy of a measured value depends on the least count of the instrument used. The term is also used for anything that is precise or error free or conforming exactly to a given standard. For example, an **accurate** or **exact calculation** is one which is free from error and conforms exactly to standard.

ACCURATE (CORRECT) TO *n* DECIMAL PLACES: Approximating a number with accuracy indicated by the number of decimal places or

correctly representing the first *n* digits of the given number following the first non-zero digit but approximating to nearest digit in the final position. For example, 3.142631 to four decimal places is given by 3.1426.

ACCURATE (CORRECT) TO *n* SIGNIFICANT FIGURES:

Approximating a number with the accuracy indicated by the number of significant figures stated in the rounded number. In other words, giving the first *n* digits after the decimal point without further approximation. For example, 5.369872 to six significant figures is given by 5.36987. Usually significant figures are counted from the none zero figures, for example, 0.00063401 to two significant figures is 0.00063. In this example, the zeros in front of the 6 are not significant. Also, 2003.443 to five significant figures gives 2003.4, with the zeros between the 2 and 3 being significant.

ACD CANVAS: A drawing software used for creating and editing technical illustrations. It is especially used for editing vector-based images and other graphics. The program supports more than 100 input file formats, which include CAD and page layout file types. The software also contains several tools used for enhancing images and visualising scientific data formats.

ACE INHIBITOR (ANGIOTENSIN-CONVERTING ENZYME INHIBITOR): A group of medicines that are used in the treatment

of various disorders such as hypertension and heart failure. They are often prescribed with other drugs, for example, diuretics or beta-blockers. ACE inhibitor drugs work by blocking the action of the enzyme that converts angiotensin (a protein present in the blood) from an inactive form, angiotensin I, to an active form, angiotensin II, which constricts blood vessels. By reducing production of angiotensin II, ACE inhibitors reduce constriction of blood vessels and relax blood vessels, which makes it easier for the blood to flow through them and thus reduces blood pressure.

ACELLULAR: 1. Not containing any cells, non-cellular or a living entity without cells, such as viroids and plasmids. 2. A unicellular or single-cell organism such as bacteria, protozoans and ciliates. 3. Tissues or organisms that are not made up of separate cells but often have more than one nucleus. An example of acellular structures is muscle fibres.

ACENAPHTHENE (1,2-DIHYDROACENAPHTHYLENE; 1,8-ETHYLENENAPHTHALENE; PERI-ETHYLENENAPHTHALENE; NAPHTHYLENEETHYLENE): A white or

pale yellow crystalline aromatic compound, with chemical formula $C_{12}H_{10}$, molar mass $154.21 \text{ g mol}^{-1}$, density 1.222 g/ml , boiling point 279°C (552 K) and melting point 93.4°C (367K). It is an intermediate in the production of some dyes, soaps, pigments, pharmaceuticals, insecticides, fungicides, herbicides and plant growth hormones.

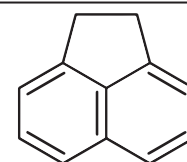
ACENTRIC: Without a centre, a term used to describe an aberrant chromosome fragment that lacks a centromere because it is unable to orientate properly during cell division. In other words, it lacks the structure at which the two arms of a chromosome join the centromere.

ACEPHALOUS: Headless or lacking a clearly defined head.

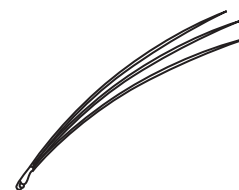
ACEROSE: Needle-shaped as occurs, for example, in the leaves of spruce and pine.

ACERVULUS: A small disc-shaped asexual fruiting body that erupts through the epidermis of plants infected by fungi of the order Melanconiales and bears conidiophores (specialised filaments or hyphae) that form conidia (spores).

ACESULFAME-K: A white crystalline powder, with chemical formula $C_4H_4KNO_4S$, molecular weight 201.24 g/mol , density 1.81 g/cm^3 and



Chemical structure of acenaphthene

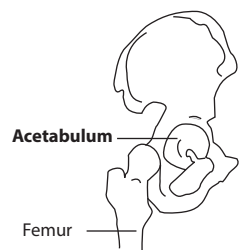


Acerose-shaped leaves

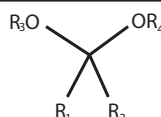
melting point 225 °C (498 K). A non-carbohydrate sweetener, it is about 200 times as sweet as sugar and composed of carbon, nitrogen, oxygen, hydrogen, sulfur and potassium atoms. Unlike aspartame, it retains its sweetness even when heated, making it suitable for cooking and baking. It is used in soft drinks and desserts, baked goods and preferred by those on low carbohydrate diets such as diabetic patients.

ACETABULUM (COTYLOID CAVITY):

Either of the two semi circular cavities, on each side of the pelvic girdle that houses the ball shaped head of the femur.



Pelvic girdle showing the acetabulum

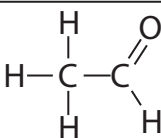


Chemical structure of an acetal compound

ACETAL: An organic functional group where two separate oxygen atoms are single-bonded to a central carbon atom; or organic compounds formed by addition of alcohol molecules to aldehyde molecules. Its structure is R₂C(OR')₂. If one molecule of aldehyde (RCHO) reacts with one molecule of alcohol (R'OH), a hemiacetal is formed (RCH(OH)OR'). The rings of aldose sugars are hemiacetals. Further reaction with a second alcohol molecule produces a full acetal R₂C(OR')₂. Examples of acetal are dimethoxymethane, also known as methylal, a colourless flammable liquid with a low boiling point, low viscosity and excellent dissolving power, used as a solvent and in the manufacture of perfumes, resins, adhesives, paint strippers and protective coatings; dioxolane (1,3-dioxolane) with the chemical formula (CH₂)₂O₂CH₂; and metaldehyde, an organic compound with the formula (CH₃CHO)₄, used as a pesticide against slugs, snails and other gastropods.

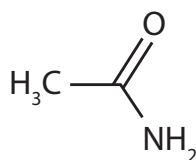
ACETALDEHYDE (ACETIC ALDEHYDE; ETHANAL; ETHYL ALDEHYDE):

A colourless, flammable liquid organic compound with the chemical formula C₂H₄O, melting point -123 °C, (150 K) and a boiling point of 20.8 °C (293.95 K). It is one of a group of chemicals known as the volatile organic compounds (VOCs). It is miscible in water, alcohol, acetone, gasoline, toluene, xylene, benzene, ether, paraldehyde and organic solvents. It is an intermediate product of higher plant respiration and also as an intermediate in the metabolism of alcohol and formed as a product of incomplete wood combustion in fireplaces and woodstoves, coffee roasting, burning of tobacco, vehicle exhaust fumes and coal refining and waste processing. Many individuals are exposed to acetaldehyde by breathing ambient air. It is highly reactive and a strong reducing agent which undergoes numerous condensation, addition and polymerisation reactions. Acetaldehyde can react violently with acid anhydrides, alcohols, ketones, phenols, ammonia, hydrogen cyanide, hydrogen sulfide, halogens, phosphorus, isocyanates, strong alkalis and amines. It is used mainly as an intermediate in the synthesis of other chemicals; and also in the manufacture of acetic acid, perfumes, polyester resins, basic dyes and drugs.



The two-dimensional molecular structure of acetaldehyde

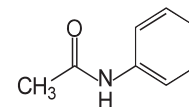
ACETAMIDE (ETHANAMIDE): A colourless, deliquescent, crystalline organic compound with chemical formula CH₃CONH₂, melting point 82.3 °C (355.45 K) and boiling point 221.25 °C (494.4 K) which crystallises in the form of long white crystals, with a characteristic smell of a mouse. It is used in organic synthesis as solvent.



Chemical structure of acetamide

ACETAMINOPHEN [PARACETAMOL; 4-ACETAMIDOPHENOL; N-(4-HYDROXYPHENYL)ETHANAMIDE; N-(4-HYDROXYPHENYL)ACETAMIDE]: A synthetic analgesic drug similar to aspirin having the chemical formula C₈H₉NO₂, molecular mass 151.163 g/mol, density 1.263 g/cm³, melting point 169 °C and boiling point of 420 °C. It is effective for relieving pain such as headache, muscle pain and lower back pain; and also used in reducing fever. It has weak anti-inflammatory properties. An overdose can cause severe, often irreversible or even fatal, liver and kidney damage.

ACETANILIDE (N-PHENYLETHANAMIDE): A white crystalline primary amide of ethanoic acid with chemical formula CH₃CONHC₆H₅, melting point 114.3 °C (387.45 K) and boiling point 304 °C (577.15 K). It is made by reacting phenylamine (aniline) with excess ethanoic acid or ethanoic anhydride; and used in the manufacture of dyestuffs and rubber.



Chemical structure of acetanilide

ACETATE: An ester or salt of acetic acid (CH₃COOH) in which the terminal hydrogen atom has been substituted by a radical or by a metal atom. Examples are ethyl acetate (CH₃COOC₂H₅) and copper acetate Cu(CH₃COO)₂. Acetates are useful in the biochemical synthesis of fats from carbohydrates in plants and animals. Vinyl acetate is used in plastic production, cellulose acetate in photographic films and textiles and volatile organic esters are used as solvents. It is also used in printing.

ACETIC ACID (ETHANOIC ACID): A colourless, corrosive liquid, with pungent odour, chemical formula CH₃COOH and boiling point 118 °C (391 K). It solidifies at low temperature, 18 °C (291 K), to glacial acetic acid. It occurs naturally as a metabolic intermediate in body fluids and plant juices and it is made industrially by the oxidation of naphtha. It plays an important role in energy metabolism. It is used for making vinegar (2 to 5%); and also used to produce chemicals used in printing and as plastics, photographic films, textiles and solvents.

ACETIC ANHYDRIDE (ETHANOIC ANHYDRIDE): A colourless liquid with a sharp, pungent odour. Its molecular formula is C₄H₆O₃, boiling point 139.8 °C (412.9 K) and a melting point of -73.1 °C (200.1 K). It is produced by carbonylation of methyl acetate. It is mainly used in converting cellulose to cellulose acetate, a coating material used in the production of plastics, photographic film and fabrics. It is also used in the production of aspirin (acetylsalicylic acid) and as a wood preservative. It can be safely used to substitute ethanoyl (acetyl) group (CH₃CO-) into organic compounds.

ACETO-ORCEIN: A stain used in the preparation of root-tip or anther squashes for chromosome examination. The material to be stained is placed in the aceto-orcein (which is acidified with hydrochloric acid) and heated at 60 °C for 15 minutes. The material is then removed and mounted in acetic acid.

ACETOGENS (ACETOGENIC BACTERIA): Rod-shaped aerobic, gram negative bacteria that reduce CO₂ to acetate via the acetyl coenzyme A (acetyl-CoA) or Wood-Ljungdahl pathway. The Wood-Ljungdahl pathway is common in anaerobic bacteria and archaea and is used for both anabolism and catabolism. Acetogens use hydrogen gas for energy and derive carbon for building organic molecules from inorganic carbon dioxide. They can inhabit very diverse ecosystems with respect to temperature, pH and salinity.

ACETONAEMIA (KETOSIS): A disease of cows, in which acetone or ketone bodies accumulate in relatively large amounts in the blood. It is characterised by loss of appetite, rapid weight loss and severe ketonuria and the smell of acetone in breath, milk and urine.

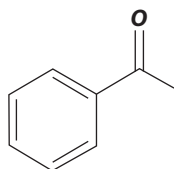
ACETONE (PROPANONE): A colourless, volatile, extremely flammable liquid ketone having a pleasant odour with the chemical formula CH₃COCH₃, molar mass 58.08 g mol⁻¹, density 0.791 g cm⁻³,

boiling point 56.5 °C (329.65 K) and melting point -94.8 °C (178.35). Acetone is produced mainly directly or indirectly from propylene mainly through the cumene process, therefore, acetone production is tied to phenol production. A small quantity is also produced and disposed of in the human body as one of the ketone bodies produced during ketoacidosis through normal metabolic processes and is normally present in blood and urine. It is used as solvent for paint, plastics and nail polish and also for the manufacture of organic chemicals, artificial fibres, explosives, resins, paints, inks, cosmetics, coatings and adhesives.

ACETONE BODY: Ketone substance produced metabolically during fasting, when body fat is broken down. The concentration of ketone bodies in blood and urine increases in starvation, diabetes and pregnancy and may be clinically useful in cases of diabetes, for detecting ketosis.

ACETONE-CHLOR-HAEMIN TEST (WAGENAAR TEST): A presumptive test for blood in which a small amount of acetone is added to the bloodstain, followed by a drop of hydrochloric acid. This forms distinctive crystals with haemoglobin derivatives haematin, haemin and haemochromogen. If haemoglobin is present it is identified under the microscope.

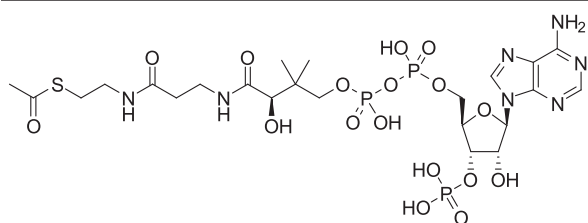
ACETOPHENONE (PHENYL METHYL KETONE): A colourless liquid organic compound having a sweet pungent odour, with the chemical formula $C_6H_5COCH_3$, density 1.028 g/cm³ and a melting point of 19.6 °C (292.75 K). It occurs naturally in many foods such as apple, cheese, apricot, banana, beef and cauliflower. Acetophenone can be obtained as a by-product of the oxidation of ethylbenzene, which mainly gives ethylbenzene hydroperoxide used in the production of propylene oxide. It is the simplest aromatic ketone and used, for example, as a chemical intermediate, as a precursor to useful resins and fragrances, and as a raw material for the synthesis of some pharmaceuticals.



Chemical structure of acetophenone

ACETYL CHLORIDE (ETHANOYL CHLORIDE; ACYL CHLORIDE): A highly refractive colourless liquid, with a strong odour, chemical formula CH_3COCl , density 1.104 g/ml (liquid) and boiling point 52 °C (325 K). It is an acid chloride derived from acetic acid and used in preparing esters and in the manufacture of dyestuffs and pharmaceuticals.

ACETYL COENZYME A (ACETYL CoA): A compound, with the chemical formula $C_{25}H_{38}N_7O_{17}P_3S$, which is formed in the mitochondria when an acetyl group (CH_3CO-) derived from the breakdown of fats, proteins or carbohydrate combines with thiol group ($-SH$) of coenzyme A. It acts to convey the carbon atoms within the acetyl group to the citric acid cycle (Krebs cycle). It is essential for the oxidative metabolism of glucose and other simple sugars. It also plays a role in the synthesis and oxidation of fatty acids.



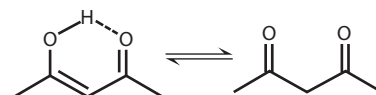
Chemical structure of Acetyl-CoA

ACETYL GROUP (ETHANOYL): A two carbon organic radical containing a methyl group (CH_3) and a carbonyl (CO) group, with the chemical formula CH_3CO- .

ACETYLACETONATO [acac; TRIS(ACETYLACETONATO) IRON(III), $Fe(acac)_3$]: An ion, which is red solid in appearance with

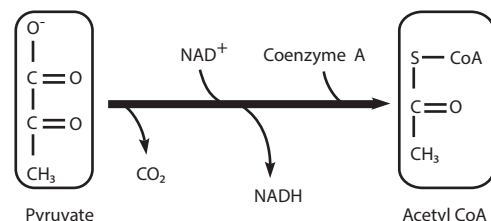
the chemical formula $Fe(C_5H_7O_2)_3$ or $FeC_5H_7O_2$; molar mass 353.17 g/mol, density 1.348 g/cm³ which functions as a bidentate ligand in which two ligands can bind to a central atom. In formulae, the symbol **acac** is used. Acetylacetonato ligand can be either a hydrogen atom donor or acceptor. A bidentate ligand is a chelating agent having two groups capable of attachment to a metal ion. Examples of bidentate ligands are ethylenediamine and oxalate ion.

ACETYLACETONE: A colourless liquid with a pleasant odour, chemical formula $CH_3COCH_2OCCH_3$, density 0.98 g/ml and boiling point 140.5 °C (413.65 K). It is soluble in water and is a precursor to acetylacetonate (acac). It exists in two tautomeric forms which rapidly interconvert. It is used as a solvent, lubricant additive, paint drier and pesticide.



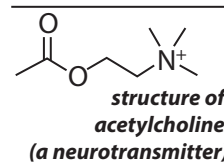
Chemical structure of acetylacetone

ACETYLATION (ETHANOYLATION): The introduction of an acetyl group (CH_3CO-) to an organic compound. For example, the acetylation of coenzyme A to form acetyl coenzyme A is an important stage in the Krebs cycle. The acetyl group is derived from pyruvate (pyruvic acid) after the removal of a molecule of CO_2 and two hydrogen atoms (2H) during respiration, which is essential for oxidative metabolism of glucose and other simple sugars.



Acetylation

ACETYLCHOLINE (ACh): A white crystalline derivative of choline, with chemical formula $C_7H_{17}NO_3$ and found in the body of animals. It is released at some nerve endings and serves as a neurotransmitter communicating nerve impulses between the cells of the nervous system and from motor neurons to muscle cells, causing the muscle cells to contract.



structure of acetylcholine (a neurotransmitter)

ACETYLENE (ETHYNE): A colourless, flammable, explosive organic gas with an ether odour in its pure state. Its chemical formula is C_2H_2 and it is used in organic synthesis and as a fuel, especially by welders because its flame temperature reaches 3,300 °C (3,573 K). It is the simplest alkyne, produced by the reaction of water with calcium carbide.

ACETYLENE BLACK: A special type of carbon black with high electrical conductivity. It is formed by decomposing acetylene by heat. It is characterised by the highest degree of aggregation and crystalline orientation when compared with all types of carbon black.

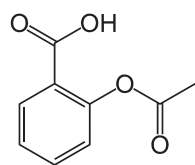
ACETYLENE BLOCK ASSAY: A process that determines the release of nitrous oxide gas from acetylene treated soil, which is used to estimate denitrification.

ACETYLENE REDUCTION ASSAY: A process that is used to estimate nitrogenase activity by measuring the rate of reduction of ethylene to acetylene.

ACETYLENE TRICHLORIDE (1,1,2-TRICHLOROETHENE; 1,1-DICHLORO-2-CHLOROETHYLENE; 1-CHLORO-2,2-DICHLOROETHYLENE; TRICHLOROETHYLENE; TRICHLOROETHENE): A colourless, toxic and non-flammable liquid with the chemical formula C_2HCl_3 , density 1.46 g/cm³ (20 °C),

melting point $-73\text{ }^{\circ}\text{C}$ (200 K) and boiling point $87.2\text{ }^{\circ}\text{C}$ (360 K). It has a smell resembling that of chloroform (trichloromethane). It is widely used as a solvent in dry cleaning and degreasing. It is also used to extract oils from nuts and fruit, as an anaesthetic and fire extinguisher.

ACETYSALICYCLIC ACID (ASPIRIN): A white crystalline, weakly acidic substance with chemical formula $\text{C}_9\text{H}_8\text{O}_4$, density 1.40 g/cm^3 , melting point $135\text{ }^{\circ}\text{C}$ (408 K), boiling point of $140\text{ }^{\circ}\text{C}$ (413 K) and decomposes. It was first synthesised by the German chemist Felix Hoffman in 1893. It is part of a group of medications called nonsteroidal anti-inflammatory drugs (NSAIDs). NSAIDs work by blocking the production of prostaglandins, which are body chemicals that cause inflammation and contribute to the brain's perception of pain. Acetylsalicylic acid is used to treat pain, rheumatic fever and inflammatory diseases, such as rheumatoid arthritis and pericarditis; and also used to reduce fever. It has anti blood clotting effect by inhibiting the production of thromboxane (a substance in platelets that causes blood clotting and constriction of blood vessels). It is used in low doses and on long-term basis to help prevent heart attacks, strokes and blood clot formation in people at high risk of developing blood clots. It may also be given at low doses immediately to people who suffer heart attack to reduce the risk of another heart attack or of the death of cardiac tissue.



Chemical Structure of
Acetylsalicylic Acid
(Aspirin)

ACHATINA: Edible snail belonging to the phylum, Mollusca. They are common in rain forests, particularly in moist litter and soil or in gardens. They are herbivores. Currently many people have gone into snail farming on commercial bases.

ACHENE (AKENE): A dry, single seeded fruit such as sunflowers and dandelions that forms from a single ovary. It is indehiscent and does not split open to disperse the matured seed.

ACHESON PROCESS: An industrial process for the manufacture of graphite by heating coke mixed with clay, patented in 1896 by the US inventor Edward Goodrich Acheson (1856-1931). The reaction involves the production of silicon carbide, which loses silicon at $4150\text{ }^{\circ}\text{C}$ (4423 K) to leave graphite.

ACHILLES TENDON: A powerful, cordlike band of connective tissue in the legs of mammals. It is one of the largest tendons in the human body, joining the calf muscle to the bone in the heel. In people who live sedentary lifestyle, the Achilles tendon may become weakened and may rupture if the calf muscle contracts suddenly, which may result in immediate loss of the normal use of the leg and foot.

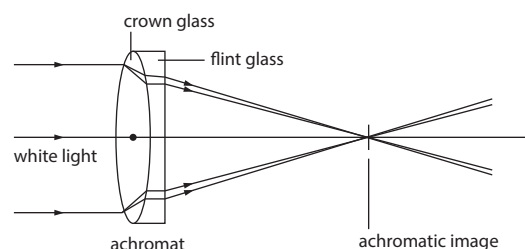
ACHIRAL: It refers to a molecule whose arrangement of atoms is such that it cannot be superimposed on its mirror image or to a molecule that does not contain a chirality element. **Chirality element** is the part of a molecule that makes it exhibit left-handed and right-handed forms.

ACHLAMYDEOUS: A flower lacking both calyx and corolla. Example is found in the willow flower.

ACHLOROPHYLLOUS: Having no chlorophyll; appearing without green colour. **Achlorophyllous plants** cannot assimilate carbohydrates through photosynthesis and must have another source of obtaining nutrients. They are two groups of these plants, depending on how they obtain nutrients: **parasitic plants** such as mistletoe or dodder, which develop special organs called haustoria (haustorium) that penetrate the tissues of a host and absorb nutrients and water; and **mycoheterotrophic** plants obtain carbon from their mycorrhizal fungi, which live inside their roots.

ACHROMATIC: 1. In optics, **achromatic lens** refers to a lens that corrects chromatic aberration by using a combination of two lenses, made of different kinds of glass, such that their dispersions neutralise each other although their refractions do not. It was discovered in 1729 by Chester Hall, an English inventor. 2. In printing, the term is used for "black and white". A black and white print has shades of grey, but greyscale is also considered to be achromatic because it lacks hue,

which means it cannot be classified as a subset of the colours red, green, blue (RGB) or yellow.



Achromatic Lens

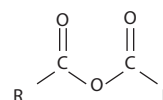
ACICULAR: 1. In botany, also known as **acerose**, it is a slender or needle-shaped leaf, as found in the leaves of Pine. 2. In mineralogy, it refers to a crystal habit composed of a radiating mass of slender, needle-like crystals. Minerals with this habit tend to be fragile and complete, undamaged specimens can be uncommon. Examples are malachite, gypsum and rutile.

ACID: 1. It is generally defined as a sour-tasting compound that releases excess hydrogen ions (H^+) in aqueous solution with a pH of less than 7, reacts with a base to form a salt and turns blue litmus red or a compound that can donate a proton or accept a pair of electrons to form a covalent bond with a base. Specifically, there are three common definitions for acids: the Arrhenius definition, the Brønsted-Lowry definition and the Lewis definition. The Arrhenius definition is a sour-tasting compound that releases excess hydrogen ions in an aqueous medium to form a solution with a pH less than 7. The Brønsted-Lowry definition is a compound that can donate a proton. The Lewis definition is a compound that can donate a proton or accept a pair of electrons to form a covalent bond with a base. Acids turn blue litmus paper red. Strong acids ionise completely (complete dissociation) in solution. Examples are hydrogen chloride acid (HCl), tetraoxosulfate(VI) acid (H_2SO_4) and trioxonitrate(V) acid (HNO_3). On the other hand, weak acids only slightly ionise in solution. An example is ethanoic acid (CH_3COOH). Acids can be solutions, liquids or solids.

Acidification is the addition of acid to a solution to make it distinctly acidic or to make the pH to fall below 7. It also refers to the fall in the pH of less than 7 of water in lakes, rivers, wells and in the ground due to contamination by acidic gases such as sulfur dioxide (SO_2), nitrogen oxides (NO_x) and ammonia (NH_3). 2. In computer science, it is also known as **atomicity, consistency, isolation, durability** and is a set of properties that guarantee that database transactions are processed reliably. In the context of databases, a single logical operation on the data is called a transaction. For example, a transfer of funds from one bank account to another, even involving multiple changes such as debiting one account and crediting another, is a single transaction.

ACID ANHYDRIDE: Chemical compounds that react with water to form acidic solution.

Anhydrides of inorganic acids are usually oxides of nonmetallic elements. For example, SO_3 is the acid anhydride of H_2SO_4 (sulfuric acid) and carbon dioxide reacts with water



Chemical Structure of Acid Anhydride

to form carbonic acid. Examples of anhydrides of organic acids are ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$ and benzoic anhydride, $(\text{C}_6\text{H}_5\text{CO})_2\text{O}$.

ACID CHLORIDE (ACYL CHLORIDE): A member of a class of organic compounds with the general formula RCOCl , for example benzoyl chloride. They cannot form hydrogen bonds and therefore have lower boiling and melting points than similar carboxylic acids. For example, acetic acid boils at $118\text{ }^{\circ}\text{C}$, whereas acetyl chloride boils at $51\text{ }^{\circ}\text{C}$. They are prepared by reacting carboxylic acids with

phosphorus (III or V) chloride or thionyl chloride (SOCl₂). They are volatile, fuming, pungent liquids, corrosive and very reactive. They react with alcohols to give esters, with ammonia and amines to give amides and with water to give carboxylic acids.

ACID DEPOSITION (ACID PRECIPITATION): The wet or dry deposition from the atmosphere containing higher than normal amounts of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) on the surface of the Earth. Its formation is caused by natural sources such as volcanoes and decaying vegetation and man-made sources such as the emissions of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) resulting from fossil fuel combustion that have been blown into the atmosphere and react with water, oxygen and other chemicals to form various acidic compounds and fall back to Earth. There are two types of acid deposition: wet deposition and dry deposition. **Wet Deposition** is the precipitation of acidic rain, fog, mist, sleet and snow on the surface of the Earth with pH of less than 5.6, typically 4.0 – 4.4. In contrast, normal, clean rainwater has a pH value of between 5.0 and 5.5, which is slightly acidic and pure water has a pH value of 7. It results from acidic chemical pollution caused mainly by industrial burning of coal and other fossil fuels combining with atmospheric water that falls to the ground in the form of rain, snow, fog or mist. **Dry Deposition** results, when acidic chemicals mix with dust or smoke and fall to the ground, sticking on vegetation and other objects and surfaces. The acid chemicals can be washed from these objects; making the runoff water mixture more acidic. This can eventually be carried as pollutants into streams, lakes and rivers, making it dangerous for types of fish and other living things that rely on the water. Acid deposition has many ecological effects, especially on lakes, streams, wetlands and other aquatic environments. It also damages forests and vegetation, especially those at higher elevations. Acid deposition also causes clay soils to release aluminium and magnesium further lowering the pH in some areas; the leaching of minerals and other nutrients from the soil; and the inhibition of nitrogen fixation and photosynthesis in plants.

ACID DISSOCIATION CONSTANT (ACIDITY CONSTANT; ACID IONISING CONSTANT): A quantitative measure of the strength of an acid in solution. Its symbol is K_a and it is the equilibrium constant for the dissociation reaction $HA \rightleftharpoons A^- + H^+$, where HA is a generic acid that dissociates into A⁻, known as the conjugate base of the acid and the hydrogen ion or proton, H⁺. In aqueous solution the H⁺ exists as a solvated hydronium ion H₃O⁺. The dissociation constant is usually written as a quotient of the equilibrium concentrations (in mol/L), as follows:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

ACID DYE: A class of chemical colourants that contain acidic groups, usually in the form of sodium or potassium salts that are applied in weak acid solutions to stain wool, silk and polyamides (nylon); and also used in staining cytoplasm and various acidophilic structures of cells or tissues. Acid dyes require the use of an acid such as vinegar, acetic or sulfuric acid to set the colour.

ACID GROWTH RATE (AGR): A process that describes how auxin stimulates cell expansion in certain plant tissues such as coleoptiles of cereals and other grasses.

ACID GROWTH THEORY: A theory governing how auxins stimulate cell expansion in certain plant tissues, notably in coleoptiles of cereals and other grasses, such as oat (Avena). It asserts that the auxins induce acidification of the immediate cell-wall environment, thereby activating enzymes that loosen load-bearing bonds within the cell wall.

ACID PROTEASE: A protein-digesting enzyme that exhibits maximum activity and stability in acidic conditions for pH values between 2.0 and 5.0 and becomes inactive at pH values above 6.0. Acid proteases have a low isoelectric point and are low in basic amino acids. Those from *Aspergillus* and *Mucor* are used in the food and beverage industries.

ACID RAIN: Rainfall on the surface of the Earth with pH of less than 5.6, typically 4.0 – 4.4. In contrast, normal, clean rainwater has

a pH value of between 5.0 and 5.5, which is slightly acidic and pure water has a pH value of 7. It results from acidic chemical pollution caused mainly by industrial burning of coal and other fossil fuels, which contain sulfur and nitrogen oxides combining with atmospheric water that falls as rain.

ACID ROCK: Any igneous rock that contains more than 66% of silica. The main silica mineral is quartz, but some feldspars are also quite rich in silica. Acid igneous rocks are usually light in colour. The most common is granite. When they are weathered to form soil, they are generally infertile.

ACID SALT: Salt in which only some of the replaceable hydrogen atoms in a polyprotic acid or polyprotic acids molecule have been partially replaced by a metal or cation. Examples are sodium hydrogen sulfate (NaHSO₄), monopotassium phosphate (KH₂PO₄) and sodium bicarbonate (NaHCO₃). Although a salt, it still contains replaceable hydrogen, so it may undergo the typical reactions of an acid. It is formed when the acidic hydrogen of an acid is replaced partially by a metal or cation.

ACID SOIL: A soil with a pH of less than 7.0. Some plants tolerate acidic soils, examples are azalea, rhododendron, blueberry, strawberry, holly; blackberry, raspberry, potato, sweet potato, apple, carrot, cauliflower, corn, cucumber, eggplant, garlic, pepper, pumpkin and melon.

ACID TEST: A chemical or metallurgical test that uses acid to test a mineral or rock for calcite or calcium in its composition.

ACID VALUE (ACID NUMBER): A measure of the amount of free fatty acid present in fat or oil, given as the number of milligrams of potassium hydroxide required to neutralise the fatty acids in 1g of the substance being tested.

ACID-BASE BALANCE: The regulation of the concentrations of acids and bases in blood and other body fluids in the range of between pH 7.35 and 7.43 in man so that it is physiologically acceptable. This is essential for the normal functioning of the body because even a small deviation from the normal range can severely affect many organs. The body uses different mechanisms to control the blood's acid-base balance. Some of the mechanisms are: (i) using the lungs to control blood pH by expelling carbon dioxide from the lungs; (ii) using the kidneys to control blood pH by the amount of acid or base that is excreted. Improper control of acid-base balance may result in **acidosis**, in which the blood has too much acid or too little base, resulting in a decrease in blood pH. Another is **alkalosis** in which the blood has too much base or too little acid, resulting in an increase in blood pH.

ACID-BASE EQUILIBRIUM: A condition in which acidic and basic ions in a solution neutralise each other exactly.

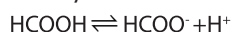
ACID-BASE INDICATOR: An organic compound that reveals through characteristic colour changes, the degree of acidity or basicity of solutions; or a weak complex organic acid or base that gives different colour in its molecular or ionised form depending on the pH of the solution. Examples are phenolphthalein, which is colourless in acid and neutral solutions, pink at about pH 8.5 and purplish at pH 10; methyl orange, which is yellow in basic and neutral solutions and reddish below pH 3.7; and p-nitrophenol, which is colourless from pH 1 to 5 and yellow from pH 5 to 9.

ACID-BASE REACTION: A chemical reaction that occurs between an acid and a base to form a salt and water. When a strong acid and a weak base are mixed, the resulting salt will be acidic; likewise, if a strong base and a weak acid are mixed, the resulting salt will be basic. However, when a strong acid and a strong base solution are mixed, a neutralisation reaction occurs and the products do not have characteristics of either acids or bases. Instead, a neutral salt and water are formed. The positive ion from a base forms a salt with the negative ion from an acid. For example, two moles of sodium ion from the base sodium hydroxide (NaOH) combine with one mole of sulfate ion from sulfuric acid (H₂SO₄) to form one mole of sodium sulfate (Na₂SO₄). Two moles of water are also

formed. That is, $2\text{NaOH}_{(\text{aq})} + \text{H}_2\text{SO}_{4(\text{aq})} \rightarrow \text{Na}_2\text{SO}_{4(\text{aq})} + 2\text{H}_2\text{O}_{(\text{l})}$. Also, $\text{NaOH}_{(\text{aq})} + \text{HCl}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$. The anion from the acid (Cl^-) reacts with the cation from the base (Na^+) to give a salt. A salt is defined as any compound formed whose anion comes from an acid and whose cation comes from a base. To determine if a salt formed in a particular reaction is neutral, acidic or basic, one needs to know which acid reacted with which base to form this salt and also if the acid is strong or weak and then, also whether the base is strong or weak. K_2CO_3 is formed when the base, potassium hydroxide (which is strong since potassium is a 1A metal), reacts with the acid, H_2CO_3 (which is weak since it is not one of the six strong acids). This reaction is a combination of a strong base and a weak acid, therefore, the salt formed will be basic.

ACIDIC: Describing a compound that is an acid or a solution that has an excess of hydrogen ions or a compound that forms an acid when dissolved in water. Carbon dioxide, for example, is an acidic oxide.

ACIDIC HYDROGEN (ACID HYDROGEN): A hydrogen atom in a compound that forms a positive ion when the acid dissociates in a solution. For example, in methanoic acid (HCOOH) the hydrogen atom on the carboxylate group is the acidic hydrogen (the one bound directly to the carbon atom does not dissociate).



ACIDIC OXIDE: A non-metallic oxide, normally an anhydride, which reacts with water to form an acid and a base, to form a salt and water. Examples include sulfur dioxide (SO_2) which reacts with water to form sulfurous acid (H_2SO_3), dinitrogen pentoxide (N_2O_5), boron trioxide (B_2O_3) and carbon dioxide (CO_2). An example of those that do not dissolve in water include silicon dioxide (SiO_2).

ACIDIC SOLUTION: A solution, which has a hydrogen index (pH) of less than 7. This is an indication that the solution has a higher concentration of acid than base. It neutralises alkalis and turns blue litmus paper red.

ACIDIMETRY (ACIDOMETRY): The volumetric analysis that uses standard solutions of acids to determine the amount of base present in a solution. The measurement of the strength of acids and bases, especially by such a chemical process, is based on the law of chemical combinations or the fact that, to produce a complete reaction, a certain definite weight of reagent is required. An **acidimeter** is an instrument used for measuring the concentration of acids in a solution.

ACIDITY: 1. The extent to which something is acidic or the concentration of acid in a substance, measured by the dissociate ion constant K value. Higher K values mean stronger acidity. For Brønsted acids, it is the tendency of a compound to act as a hydrogen donor. It can be quantitatively expressed by the acid dissociation constant of the compound in water or some other specified medium. For Lewis acids it relates to the association constants of Lewis adducts and π -adducts. 2. The excess production of acid by the gastric glands of the stomach, which results in symptoms such as dyspepsia, heartburn and the formation of ulcers. The stomach normally secretes acid that helps in breaking down food during digestion.

ACIDITY CONSTANT: The equilibrium constant with the symbol K_{an} for splitting off the n th proton from a charged or uncharged acid. It can also be defined as the relative capacity of different compounds or functional groups to donate a proton to a common base under identical conditions. Its symbol is K_{a} .

ACIDITY FUNCTION: Any function that measures the thermodynamic hydron-donating or hydron-accepting ability of a solvent system or a closely related thermodynamic property, such as the tendency of the lyate ion of the solvent system to form Lewis adducts. It is usually expressed in terms of its ability to donate protons to or accept protons from a solute.

ACIDOPHILUS: 1. Organisms, which prefer or love acidic medium. They include many plants, also known as **calciphobe plants** that

strive in soil with a pH of less than 7 and usually cannot tolerate calcite in the substratum. An example is the cacti. 2. A species of gram-positive, rod-shaped bacteria in the genus *Lactobacillus*, present naturally in the acidic environment of gastrointestinal tract, mouth and vagina of humans and animals. It is used to make the fermented product, acidophilus milk such as yoghurt and also available in tablet, capsule, powder and suppository forms. Acidophilus has got many health benefits to the body including: (i) helping the body to fight cancer; (ii) helping the body to fight against various digestive disorders and vaginal yeast infections; (iii) helping in lowering the serum cholesterol level; (iv) helping in the synthesis of various vitamins such as vitamin B12 and K and also thiamine and folic acid in the intestine; (v) reducing the recurrence of urinary tract infections. 3. General term for a number of probiotics that are taken to help digestion.

ACIDOSIS: A condition in which the blood and tissue fluids become abnormally highly acidic due to the reduction in the alkali reserve of the blood. There are two types of acidosis: (i) metabolic acidosis that occurs when the kidneys do not efficiently eliminate acid buildup in the body or when the body gets rid of too much base; and (ii) respiratory acidosis that occurs when excessive CO_2 builds up in the body. Metabolic acidosis include the following: (i) diabetic acidosis that occurs in uncontrolled diabetes due to acidification of the blood caused by ketones building up in the body; (ii) hyperchloremic acidosis due to loss of sodium bicarbonate in the body, which may be caused by diarrhoea and vomiting; and (iii) lactic acidosis that occurs when there is too much lactic acid in the body, which may be caused by, for example, chronic alcohol use, heart failure and cancer. Examples of general symptoms of acidosis include rapid and shallow breathing, thirst, tiredness, confusion, fatigue, headache and sleepiness.

ACINOSE: 1. Resembling a bunch of grapes. 2. Pertaining to one of the small sacs (as in a compound gland).

ACINOUS: 1. A rounded sac containing secretory cells, which is found at the ends of the ducts in an exocrine gland or the smallest unit of a multicellular gland, such as the pancreas. Each acinus in the pancreas is made up of a hollow cluster of acinar cells, which produce the digestive enzymes secreted in pancreatic juice. 2. A drupe in a multiple fruit or any of the small globes drupelets that make up an aggregate fruit. Examples are the raspberry and blackberry.

ACK: An abbreviation for **acknowledgement**, an answer given and sent by the destination station back to the sending station after the receipt of a recognisable block of data of specific size.

ACKEE (ACHEE; AKEE APPLE; *Blighia sapida*): An evergreen tree with a dense crown of spreading branches, native to West Africa. It grows to about 7-25 metres in height with a short trunk, 1.8 metres in circumference. It has compound bright-green evergreen leaves, which alternate with 3 to 5 pairs of oblong, obovate-oblong or elliptic leaflets, rounded at the base, short-pointed at the apex. The fragrant inflorescence is about 20 centimetres long and bears unisexual five-petalled, white and hairy flowers in simple racemes. The pear-shaped fruit grow in bunches on powerful, strong-rooted trees; it is green when young, becoming yellow or red when mature. At full maturity, it splits open revealing three large shiny black seeds surrounded by cream-coloured flesh called aril. The tree is used as an ornamental shade tree and planted in villages and along streets. The fleshy arils around the seeds are edible but other parts of the fruit and seeds are poisonous. The fruit is very popular in Jamaican cuisine and is carefully prepared before eating. The dried seeds, fruit, bark and leaves are used medicinally.

ACLINIC LINE (THE MAGNETIC EQUATOR): An imaginary line near the geographic equator where the compass needle balances horizontally, with the attraction of the north and south magnetic poles being equal. It passes through all points where a magnetic needle has no dip.

ACNE: A common inflammatory disorder of the sebaceous glands occurring mostly in the face and upper trunk and most prominent in adolescence. It is characterised by the presence of blackheads with

papules, pustules and in severe cases by cysts and scars. Acne may be aggravated by oily foods such as chocolate and nuts, by hormone imbalance, by greasy skin or by poor hygiene. Methods of treatment include degreasing the skin, removing the blackheads, controlling diet and exposure to ultraviolet radiation.

ACNODE (ISOLATED POINT): A point not lying on a curve, but whose coordinates satisfy its equation.

ACOELOMATE: Any bilaterally symmetrical animal that does not possess a coelom. Examples are the flatworm, fluke and tapeworm, which possess one internal space, the digestive cavity. Most belong to the subkingdom Eumetazoa.

ACORN: A hard, dry, indehiscent fruit with a single large seed and a cup-like base. Example is found in the fruits of Oak.

ACOS: Abbreviation and symbol for the inverse cosine function.

ACOSH: Symbol for the inverse hyperbolic cosine function.

ACOSEC: Abbreviation and symbol for the inverse cosecant function.

ACOSECH: Symbol for the inverse hyperbolic cosecant function.

ACOT: Abbreviation and symbol for the inverse cotangent function.

ACOTH: Symbol for the inverse hyperbolic cotangent function.

ACOTYLEDONOUS: A plant without cotyledon. Examples are the Dodder, Ferns and Mosses.

ACOUSTIC COUPLER: A device used between the modem of a computer terminal and a standard telephone line to permit transmission of digital data in either direction without making direct connections.

ACOUSTIC IMPEDANCE (SOUND IMPEDANCE): The complex ratio of the sound pressure on a given surface to the sound flux through that surface with the symbol Z , defined as the product of the density of the medium (ρ) and acoustic velocity within the medium (V): $Z = \rho V$. Its unit is rayls per square metre. Acoustic resistance is the real component of acoustic impedance and acoustic reactance is the imaginary component. Acoustic impedance is important in the determination of acoustic transmission and reflection at the boundary of two materials having different acoustic impedances, the design of ultrasonic transducers and assessing absorption of sound in a medium.

ACOUSTIC INTENSITY (SOUND INTENSITY): Amount of energy that a sound wave brings to a unit area every second in a direction perpendicular to that area. Its unit is ergs per second per square centimetre, or in units of power, as microwatts (10^{-6} watt) per square centimetre.

ACOUSTIC LEVITATION: A method, which uses acoustic radiation pressure from intense sound waves in the medium to suspend matter such as solids, liquids and heavy gases. Acoustic levitation is possible because of the non-linear effects of intense sound waves. There is no known theoretical limit to what acoustic levitation can lift given enough vibratory sound, but in practice current technology limits the amount that can be lifted by this force to at most a few kilograms. Acoustic levitators are used mostly in industry and for researchers of anti-gravity effects such as NASA.

ACOUSTIC NERVE (AUDITORY NERVE; VESTIBULOCOCHLEAR): The eighth cranial nerve that transmits information from the inner ear to the hindbrain, which it enters between the pons and medulla. It serves two distinct functions: The acoustic portion carries sensations of sound and pitch from the organ of hearing, the cochlea; these are relayed to the auditory part of the cerebral cortex; The vestibular portion carries postural sensation from the semicircular canals of the inner ear to the cerebellum. It provides information about position.

ACOUSTIC OHM: Centimetre-gram-second (cgs) unit of acoustic impedance, which is the ratio of the sound pressure on a surface to the sound flux through the surface. It is similar to the ohm as the unit of electrical impedance. The International System of Units (SI) is the currently used system of units, and the pascals has replaced acoustic ohm.

ACOUSTIC RADIATION FORCE: A physical phenomenon resulting from the interaction of an acoustic wave with an obstacle placed along its path. Acoustic radiation force is currently being utilised in medical ultrasonic imaging to generate images based on the mechanical properties.

ACOUSTIC RADIATION FORCE IMPULSE IMAGING (ARFI IMAGING): The use of acoustic radiation force to generate images of the mechanical properties of soft tissue. It uses brief, high-intensity, focused ultrasound to generate radiation force and conventional, ultrasonic, correlation-based methods to track tissue displacement.

ACOUSTIC RADIATIONS: 1. Infrasonic, sonic or ultrasonic waves propagating through a solid, liquid or gaseous medium. 2. Also known as **auditory radiations**, they are radiation of fibres found in the brain, in the ventral cochlear pathway, a part of the auditory system. Lesions to the auditory radiations could be a cause of cortical deafness.

ACOUSTIC SHIELDING: A sound barrier that prevents the transmission of unwanted sounds, usually by introducing sound-absorbing materials in the path of the sound waves. Sound enters a room through the air, through the structure or by the diaphragm action of floors, walls and ceilings. In buildings, acoustic shielding can be achieved by using a combination of conventional building materials such as fibreglass panels, mineral board fillers, mass loaded vinyl, sealants and absorbent linings for ducts. Acoustic shielding is used in, for example, Magnetic Resonance Imaging (MRI) installations and in hospitals and clinics, where noise suppression specifications often exist.

ACOUSTICS: 1. The study of sound and sound waves, especially of its generation, transmission and reception. 2. The characteristics of a building or space with respect to its ability to enable speech and music to be heard clearly and distinctly within it, as it responds to sound.

ACOUSTO-ELECTRONIC DEVICE (ELECTROACOUSTIC DEVICE): An electro-optical device that functions as an electronically tunable excitation filter to simultaneously modulate the intensity and wavelength of multiple laser lines from one or more sources. It has optical properties that are altered in the presence of an acoustic wave. It can be used to select and transmit a single wavelength from the incoming light. They can also replace radio/infrared filter banks by adding a piezoelectric transducer at one end of the cell to convert a downmixed radio signal into an acoustic vibration. It has a fast response time, is accurate and exhibits a high extinction ratio.

ACQUIRED CHARACTER (ACQUIRED CHARACTERISTICS): A non-hereditary feature of the body that develops during the lifetime of an individual plant or animal, usually as a result of disease, use and disuse or any distinctly environmental influence and cannot be inherited or transferred through genes to offspring. An example is the enlarged muscles of a weightlifter or boxer.

ACQUIRED IMMUNE DEFICIENCY SYNDROME (ACQUIRED IMMUNO DEFICIENCY SYNDROME; AIDS): It is a viral disease caused by human immunodeficiency virus. It is transmitted through sexual contact with infected persons without protection or sharing of needles with infected individuals as well as exchange of other body fluids from infected persons. Two types of the virus have been identified (Type-1 and Type-2) which differ in some of their genes and sequences of their genome. These slight differences change the extent of their virulence with Type-1 being more virulent than Type-2. While the Type-2 is the commonest in West Africa and parts of Europe, the Type-1 is the most common in the world. AIDS is the end stage of the HIV virus infection. Because the virus selectively infects the immune cells bearing the CD4 ($CD4^+$ cells) such as helper T cells, macrophages and monocytes, chronic infection leads to decreased ability to fight other infections since these immune cells are depleted by the viral infection.

A

ACQUIRED IMMUNITY (ADAPTIVE IMMUNITY): An immunity to a particular disease that has been obtained by the development of antibodies after an attack of an infectious disease, for example, a pregnant mother passing antibodies through the placenta to a foetus or through breast milk, called **natural passive immunity**; or by vaccination or the injection of antiserum, called **artificial passive immunity**.

ACRASIOMYCETES: A class of the *Myxomycota* containing organisms that feed on soil bacteria. It includes some 25 species divided among 10 genera. They are found in, for example, soil and decomposing plant materials. The feeding stage generally consists of individual amoeboid cells (*myxamoebae*). Fruiting bodies or the differentiated, spore-bearing structure are formed by aggregation of *myxamoebae* and differentiation of the resulting pseudoplasmodium.

ACRE: A unit of area, which is equivalent to 4,047 square metres or 0.405 hectare. 640 acres is equal to one square mile. It is used to measure tracts of land. **Acreage** is the total area of land measured in acres.

ACRE-FOOT: A unit of volume sometimes used to measure large quantities of water, such as reservoir, canals and sewer flow that is equal to its area in acres multiplied by its average depth in feet. One acre-foot equals 30.48 centimetres in depth 1,233.5 cubic metres.

ACRID: An unpleasantly sharp smell, bitter to taste or pungent.

ACRIDINE: A colourless, crystalline, heterocyclic organic compound with chemical formula, $C_{13}H_9N$, melting point 110°C (383K), boiling point 107°C (380K). It occurs in coal tar from which it is extracted using dilute acids but can also be obtained synthetically. Most derivatives of acridine are used to make dyes, drugs and biological stains. It is capable of causing frameshift mutation in the DNA sequence.

ACRILAN: A synthetic fibre produced by copolymerisation of acrylonitrile and vinyl acetate. Acrilan fibre is characterised by strength, softness and crease-resistance and used for making things such as clothing, upholstery, carpets, socks, hats, gloves, scarves and sweaters.

ACROCARPOUS: A term used to describe mosses of the subclass Eubrya, in which the main axis is terminated by the development of the reproductive organs so subsequent growth is sympodial. In such mosses the main axis is almost always erect.

ACROCENTRIC: A term used to describe a chromosome in which the centromere is not centrally placed, so that during the anaphase stage of cell division, movement of an acrocentric chromosome towards one end results in two arms of different lengths.

ACRODONT: Having teeth that are rootless and without sockets and fused to the bones of the upper jaw, as in some reptiles.

ACRODROMOUS: A leaf with two or more veins running in convergent arches towards the apex.

ACROGEN: A non-flowering plant growing and producing its reproductive structures at the apex only, as found in ferns.

ACROGENOUS: In botany, it refers to flowerless plants such as ferns or mosses in which growth occurs only at the tip of the stem.

ACROMEGALY: Enlargement of the body's extremities as a result of the overproduction of the growth hormone somatotropin. It is usually caused by a tumour in the pituitary gland. Some of the characteristics include gradual enlargement of the hands, feet, face, nose, lips, tongue and abdominal organs, widely spaced teeth and enlargement of the bones of the extremities.

ACROMION (ACROMION PROCESS): The lateral end of the spine of the scapula (shoulder blade), which forms the highest point of the shoulder and connects with the clavicle at a small oval surface in the middle of the spine. Part of the acromion articulates with the clavicle (collar bone) to form the acromioclavicular joint. From the acromion a portion of the deltoid muscle arises.

ACRONEMATIC FLAGELLUM (WHIPLASH FLAGELLUM): A type of zoospore of the oomycota with straight threadlike projection and a smooth surface terminating with a fine fibril. It is used by

some algae, bacteria and fungi to propagate themselves. It has an acute bend at the end and as the flagellar movement propels the cell, the bent end whips back and forth.

ACRONYM EXPANSION: A feature in some content management systems (CMS) and Web-based search systems that provides automatic expansion of the first letters of words within phrases. When a user enters an acronym or abbreviation the CMS will spell out the phrase or provide a list of matching phrases for the user to choose from. Acronym expansion is used in data queries and provides suggestions relative to whatever was entered into the search function box.

ACROPETAL: Describing movement, differentiation, development or maturing from the base toward the apex, as in those plant organs, which occur from the base or point of attachment, so that the oldest parts are at the base and the youngest are at the tip. For example, the development of flowers is usually acropetal with the calyx being formed first and the gynoecium last. Movement of water through the plant is also acropetal.

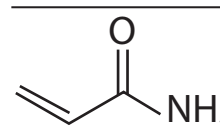
ACROPHOBIA: An abnormal fear of heights or being at high places.

ACROSCOPIC: A term that describes the nature of facing or on the side towards the apex. For example, in plants, leaves are described as acroscopic when their branches grow and face the apex.

ACROSOME: A membrane bound structure found at the anterior end of the sperm cell, which releases digestive juices (lytic enzymes) just before fertilisation to digest and break down the egg layers to permit the entry of the sperm to cause fertilisation.

ACROTONIC: 1. It refers to flowering seasonal shoots that produce leaves below the inflorescence. 2. The development of branches, in which those near the apex of the plant are the most developed. 3. The development of new organs on or near the apex of existing organs.

ACRYLAMIDE: A readily polymerised amide with the chemical formula $\text{CH}_2=\text{CHCONH}_2$, density 1.13 g/cm^3 and melting point 84.5°C (358.65 K). It is derived from acrylic acid and is used in synthetic fibres and sewage treatment. It is a carcinogen and is present in some foods, especially starches and cereals that are cooked at high temperatures.



Chemical structure of acrylamide

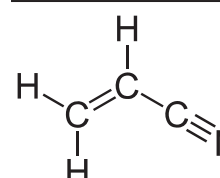
ACRYLIC: In information technology, it refers to a code name for Microsoft's Expression Design product, a professional design tool used to create sophisticated vector graphics for web applications.

ACRYLIC ACID (PROP-2-ENOIC ACID; PROPENOIC ACID): An unsaturated aliphatic acid with the chemical formula $\text{C}_3\text{H}_4\text{O}_2$ with a pungent odour. It has a melting point of 13°C (286 K), a boiling point of 141.6°C (414.75 K) and it is used in making acrylic resins.

ACRYLIC RESINS: Synthetic resins made by polymerising esters or other derivatives of acrylic acid (propenoic acid). Examples are polypropenenitrile such as acrilan and polymethylmethacrylate such as perspex.

ACRYLONITRILE (2-PROPENENITRILE; CYANOETHENE; VINYL CYANIDE): An

explosive, which is poisonous, flammable and has a pungent-smell. It is a colourless liquid and partly soluble in water. Its chemical formula is $\text{C}_3\text{H}_3\text{N}$ and it has a density of 0.81 g/cm^3 , a melting point of -84°C (189 K) and a boiling point of 77°C (350 K). It is used in organic syntheses and in the manufacture of plastics.



Chemical Structure of Acrylonitrile

ACTH (ADRENOCORTICOTROPHIC HORMONE; CORTICOTROPHIN): A hormone produced by the anterior pituitary gland that stimulates secretion of glucocorticoids composed of cortisol, corticosterone and cortisone. They are involved in carbohydrate, protein and fat metabolism and have anti-inflammatory properties. A deficiency of ACTH causes secondary adrenal insufficiency and an excess causes Cushing's syndrome. Its secretion occurs in short bursts every few hours and is increased by stress.

ACTIN: A contractile protein found in all cells and in muscle tissue, occurring in the form of filaments, where it plays a role in contraction. It is found in the microfilaments that form part of the cytoskeleton of all cells.

ACTINIC FLUX: The quantity of light available to molecules at a particular point in the atmosphere and which, on absorption, drives photochemical processes in the atmosphere.

ACTINIC RADIATION: Electromagnetic radiation that is capable of producing a chemical change, especially ultra-violet radiation and radiation that will affect a photographic emission.

ACTINIDE (ACTINIDE SERIES; ACTINOID): Any of a series of 14 radioactive metallic chemical elements with atomic number 90 (thorium) to 103 (lawrencium). Actinides are grouped together because of their chemical similarities, with the properties differing only slightly with atomic number. Only the first four, ie, actinium, thorium, protactinium and uranium occur in nature in appreciable quantities.

ACTINIUM: A white radioactive metallic element belonging to Group III of the Periodic Table and the first of the actinide series. It was discovered by the French chemist André Louis Debierne in 1899. It has the symbol Ac, atomic number 89 and relative atomic mass 227 and an estimated boiling and melting points of 3200 °C (3473 K) and 1050 °C (1323 K) respectively. It is a weak emitter of high energy alpha particles. Actinium occurs with uranium and radium in pitchblende and other ores and can be synthesised by bombarding radium with neutrons.

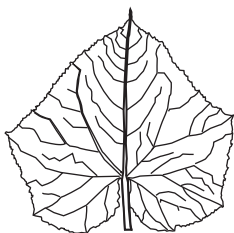
ACTINIUM SERIES: Natural radioactive decay series beginning with uranium-235 and ending with lead-207. They include actinium, thorium, protactinium, uranium, neptunium, plutonium, americium, curium, berkelium, californium, einsteinium, fermium, mendelevium and nobelium. The elements with atomic numbers of 93 and above are called **transuranium elements**.

ACTINOBRILLILOSIS (CRUELS; WOODEN TONGUE): A disease that affects the throat and tongue of animals especially cattle. In sheep, it appears as swellings on the lips, cheeks and jaws.

ACTINOBACTERIA (ACTINOMYCETES; ACTINOMYCOTA): A phylum of gram-positive, mostly anaerobic bacteria, which includes bacteria of the actinomycetes. Some species cause disease in animals and humans and others such as streptomycetes produce antibiotics such as streptomycin.

ACTINODROMOUS VENATION:

Palmately veined, with three or more veins radiating from the tip of the petiole or from a single point. It is seen, for example, in the leaves of sycamore (*Acer pseudoplatanus*).



Actinodromous venation

ACTINOID (ACTINIDE; ACTINIDE SERIES):

Any of a series of 14 radioactive metallic chemical elements with atomic number 90 (thorium) to 103 (lawrencium). Actinides are grouped together because of their chemical similarities, with the properties differing only slightly with atomic number. Only the first four, ie, actinium, thorium, protactinium and uranium occur in nature in appreciable quantities.

ACTINOID CONTRACTION: A smooth decrease in atomic or ionic radius with increasing proton or atomic number, as found in the actinoids.

ACTINOMETER: Any of the various instruments or physical devices, which determines the number of photons in a beam integrally or per unit time. Actinometer commonly refers to devices used in the ultraviolet and visible wavelength ranges. Examples are solutions of iron(III) oxalate used as a chemical actinometer, while bolometers, thermopiles and photodiodes are physical devices giving a reading that can be correlated. In meteorology, actinometer is used to measure the heating power of solar radiation as transmitted directly by the Sun, scattered by the atmosphere or reflected by the Earth.

ACTINOMORPHIC (RADIALY SYMMETRICAL): Having a cut along any plane through the centre that divides an object or specimen into two equal halves. Examples are seen in cotton, hibiscus, crocus, lily, okra flowers and starfish.

ACTINOMYCETES: A group of gram-positive mostly anaerobic bacteria identified by a branching growth pattern resulting in large threadlike structures. All species are fungus-like filaments producing spores on aerial branches similar to the spores of certain moulds.

ACTINOMYCOSIS (LUMPY JAW): An inflammatory disease of cattle and pigs, in which the animal is infected with bacteria, forming abscesses in the mouth and lungs.

ACTINORHIZA: A root nodule formed as a result of a symbiotic relationship between an actinomycete and fine plant roots.

ACTION: 1. In mathematics, a **group action** is an operation such that for a group G , on a non-empty set, S , it is a homomorphism ϕ from the group to the group of permutations on S . For g in G and s in S the image $\phi(g)(s)$ may be written gs , sg or $s(g\phi)$. Thus, there is an associative mapping with an identity from $G \times S$ into S , under which the product of any member of the group with any member of the set is a member of the set, such that for each g, h in G and s in S , $g(hs) = (gh)s$ and $1s = s$. 2. In physics, the definite integral of the Lagrangian of a given body or a discrete set of particles from some reference time to the current time. It is the product of energy and time and its SI unit is joule-second. It is a mathematical functional which takes the trajectory, of the system as its argument and has a real number as its result. The action is typically represented as an integral over time, taken along the path of the system between the initial time and the final time of the development of the system, $S = \int_{t_1}^{t_2} L dt$, where the integrand L is called the Lagrangian.

ACTION AND REACTION FORCES: Forces, which always come in pairs. According to Newton's third law, also known as the **weak law of action and reaction**, for every action force there is a corresponding reaction force, which is equal in magnitude and opposite in direction. According to the **strong law of action and reaction**, for every action force, there is a corresponding reaction force, which is equal in magnitude and opposite in direction and the forces are central forces, which act along the line joining the particles.

ACTION AT A DISTANCE: A direct interaction between bodies that are not in physical contact with each other, which is inconsistent with the special theory of relativity, which states that nothing can travel faster than the speed of light in vacuum. This kind of interaction is appropriately described with the field theory or by the exchange of virtual particles.

ACTION LIMIT: In the production process, it is the outer limit set on a control chart. If the observed value falls beyond these limits, then the action will be executed, often resetting the machine. For example, consider a sample of size n in a process with standard deviation σ with target mean μ , the action limit will be set at $\mu \pm 1.96 \frac{\sigma}{\sqrt{n}}$.

ACTION POTENTIAL: A short-term electrical activity across the membrane of a nerve cell by a change in the potential difference (voltage) when an impulse passes along it. It may be caused by electrical, chemical or mechanical stimulation, by temperature change, etc. A change in potential from about -60 to +45 millivolts accompanies the passage of sodium and potassium ions across the membrane.

A

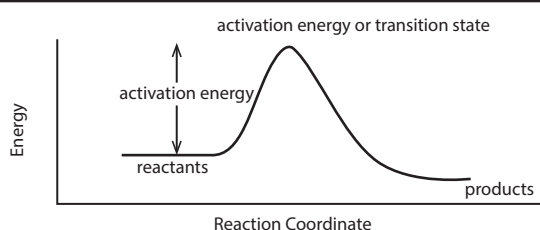
ACTION SPECTRUM: A plot of the rate of a physiological activity against the wavelength of light. It shows which wavelength of light is most effectively used in a specific chemical reaction. For example, chlorophyll is much more efficient at using the red and blue spectrums of light to carry out photosynthesis, thus the action spectrum graph would show spikes above the wavelengths representing the colours red and blue.

ACTIONSRIPT: An object-oriented scripting and programming language designed to provide rich interactive abilities to the Adobe Flash Player platform. ActionScript makes it possible for developers to create onscreen environments (such as games, tutorials and e-commerce applications) that can respond to user input through the keyboard or mouse.

ACTIVATED ADSORPTION: Adsorption that involves activation energy, which occurs in certain cases of chemisorption. If the temperature coefficient of the rate of adsorption is substantial, an adsorption process is said to be activated or have significant activation energy. In this case, the sticking coefficient is small. In general, the activation energy of activated adsorption is a function of coverage and it usually increases with increasing coverage.

ACTIVATED CARBON (ACTIVE CARBON; ACTIVATED CHARCOAL): A highly absorbent carbon with a very large surface area and high porosity which is obtained by special treatment including steaming or heating granulated charcoal in a vacuum to remove contained gases. An example is zinc chloride (ZnCl_2), before, during or after carbonisation in order to increase its adsorptive properties. It is used primarily for purifying gases by adsorption, solvent recovery or deodorisation and as an antidote to certain poisons.

ACTIVATED COMPLEX: The association of atoms of highest energy, formed in the transition state of a chemical reaction or the highest free energy state of a complex in going from reactants to products. In other words, it refers to a collection of intermediate structures in a chemical reaction that persist while bonds are breaking and new bonds are forming. It therefore, represents not one defined state, but rather a range of transient configurations that a collection of atoms passes through in between clearly defined products and reactants. In the potential energy diagram shown below, at the peak of energy (potential energy), the reactants are in the **transition state** or **activated complex**, halfway between being reactants and forming products. The **activation energy** is the minimum amount of energy needed to start a reaction. The **reaction coordinate** represents the progress (from reactants to products) along the reaction pathway.



Variation of energy along the reaction coordinate for the bimolecular nucleophilic substitution (SN_2) reaction between bromomethane and the hydroxide anion

ACTIVATED SLUDGE PROCESS: A sewage and waste water treatment process in which the sludge produced after primary treatment is pumped into aeration tanks, where aerobic decomposition by microorganisms occurs. The sludge and effluent are separated in a settling tank and the effluent is further treated by chlorination or oxidation.

ACTIVATED-COMPLEX-THEORY: A theory that enables the rate constants in chemical reactions to be calculated using statistical thermodynamics. The events assumed to be taking place can be shown in a diagram, with the potential energy as the vertical axis and the horizontal axis, called the reaction coordinate, represents the course of the reaction.

ACTIVATION: 1. In electrochemical corrosion, it is the process of transition from the passive to the active state by removal of the

passivating film. A necessary condition for activation is an electrode potential negative to the equilibrium potential of formation of the passivating film. Activation is achieved by cathodic currents, by a reduced substance in the adjacent solution or by contact with an electronic conductor having a suitably negative corrosion potential. 2. The process of inducing radioactivity by irradiation.

ACTIVATION ANALYSIS: A specific, highly sensitive technique, applicable to almost every element in the Periodic Table, in which a neutron, charged particle such as alpha particle or gamma photon, is captured by a stable nuclide through nuclear reaction to produce a different radioactive nuclide whose radioactivity is then measured and used to quantify the analyte elements. Neutron activation is the most popular activation method and uses neutrons for the activation.

ACTIVATION ENERGY: The minimum amount of energy such as heat, electromagnetic radiation or electrical energy needed to start a chemical reaction or the minimum amount of energy which reactant particles (atoms, ions or molecules) must possess before a reaction can occur. The lower the energy, the more likely it is for a reaction to take place.

ACTIVATOR: 1. A type of transcription factor that enhances the transcription of a gene by binding to a region of DNA called an enhancer. 2. Any substance which, when bonded to an allosteric site on an enzyme, enables the active site of the enzyme to bind to the substrate. 3. Any compound that makes effective the activity of a drug or other foreign substance in the body. It is also any substance which, when bonded to an allosteric site on an enzyme enables the active site of the enzyme to bind to the substrate. 4. A type of dopant used in phosphors and scintillators.

ACTIVE ACIDITY: The activity of hydrogen ion in the aqueous phase of a soil. It is measured and expressed as a pH value.

ACTIVE ATTACK: In computing security, it is an attack characterised by the attacker attempting to break into the system. During an active attack, the intruder will introduce data into the system as well as potentially change data within the system.

ACTIVE BEAM: 1. A bundle of rays emitted by a radiation source, such as x-ray tube. Normally, the active beam is limited to the required size by a diaphragm arrangement. 2. A type of chilled beam system that contains enclosed space in buildings and make heat transfer more effective due to forced convection. In an active beam system, air duct and nozzle blowers provide ventilation air, which helps draw the warm room air up through the heat exchange coils where it cools and then falls to the floor and the cycle starts over again.

ACTIVE CARRIER: An infected person with no signs or symptoms of an infectious disease, but is capable of transmitting the disease to other individuals.

ACTIVE CELL (CURRENT CELL; SELECTED CELL): It is a rectangular box in a spreadsheet where data can be entered. It is bounded by a heavy border around it. Active cell helps identify which cell is being worked on. Only one cell can be active at a time.

ACTIVE CONSTRAINT: An inequality that is active at a point on a boundary. For example, $3x + 2y \geq 16$, the points (4, 2) satisfies the inequality.

ACTIVE CONTINENTAL MARGIN: Plate boundary found on the leading edge of a continent. Active margins are tectonically active marked by earthquakes, volcanoes, and formation of mountains. They are found at West Coast of North America and South America.

ACTIVE DESKTOP: An optional feature of Microsoft Windows introduced with Microsoft's Internet Explorer 4 Web browser that began in Windows 98 allowing a Desktop to be used as a web page. In other words, it allows the storage of Web information on the desktop that is updated as needed. It enables programs to be run with a single click of the mouse instead of the usual double-clicking. It is an integral component of Windows XP.

ACTIVE DEVICE: 1. An electronic component that is capable of amplification and requires an energy source for its operation. Examples are LEDs, transmitters, controlled power supplies and transistors. 2. An artificial satellite that receives information and retransmits it after amplification. 3. A radar device that emits microwave radiation and provides information about a distant body by receiving a reflection of this radiation.

ACTIVE DIRECTORY: Microsoft's trademarked directory service, an integral part of the Windows 2000 architecture. Like other directory services, such as Novell Directory Services (NDS), Active Directory is a centralised and standardised system that automates network management of user data, security and distributed resources and enables interoperability with other directories. Active Directory is designed especially for distributed networking environments.

ACTIVE FAULT: A fault that is likely to move and cause another earthquake sometime in the future. Faults are commonly considered to be active if they have moved one or more times in the last 10,000 years. Faults are cracks in the earth's crust with lengths of up to many hundreds of kilometres.

ACTIVE GALACTIC NUCLEI (AGN): The small very bright and luminous region at the centre of some galaxies. They emit very great amount of energy in the form of radio, optical, X-ray, or gamma radiation or high-speed particle jets. Examples of such galaxies include quasars, radio galaxies, and Seyfert galaxies. Quasars are the most luminous AGN.

ACTIVE GALAXY: A galaxy that contains an active galactic nucleus (AGN) whose central region exhibits strong emission activity, from radio to x-ray frequencies. This may be due to the gravitational collapse, resulting into a massive black hole at the centre of the galaxy. Examples are M82 galaxies, Seyfert galaxies, N galaxies and possibly quasars.

ACTIVE IMMUNITY: The body's natural resistance to disease due to the immune system's response to foreign bodies such as microorganisms, an antigen or a vaccine. During this process, lymphocytes in the blood multiply rapidly and produce plasma cells and memory cells. The plasma cells produce antibodies, which fight the foreign body, while the memory cells remain in the blood after the original infection is gone but would respond rapidly if the body encounters the same antigen or microorganism again. Response to a second infection is therefore, much more rapid providing immunity immediately.

ACTIVE INGREDIENT (AI; ACTIVE PHARMACEUTICAL INGREDIENT; API; ACTIVE SUBSTANCE; BULK ACTIVE): The substance in a pharmaceutical drug or in a pesticide, that is responsible for the observed biological activity.

ACTIVE MATRIX LCD (THIN FILM TRANSISTOR LCD; TFT LCD): A type of colour liquid crystal display commonly used in laptop computers. TFT screens have one or more transistors to control each pixel on the screen, as opposed to the less expensive passive matrix displays where each pixel is not matched with a transistor. TFT LCDs have a faster refresh rate than other types of LCD, reducing the lag between computer and display.

ACTIVE MATRIX ORGANIC LIGHT-EMITTING DIODE (AMOLED): A type of organic light-emitting diode (OLED) used in screen displays consisting of a thin film transistor of an organic substance, often thin films of polymer sandwiched between two conductors that emits light when an electric current is applied. AMOLED has a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes. The screen is lighter, thinner, and durable. AMOLED screen technology has been used in smartphones, television screens, computer monitors, etc. Another type of organic light-emitting diode (OLED) is the passive-matrix organic light-emitting diode (PMOLED) that does not contain a storage capacitor and so the pixels in each line are actually off most of the time and needs more power to make them brighter.

ACTIVE METAL (ALKALI METAL): A collective name for metals such as lithium, sodium, potassium, rubidium and caesium that react strongly or quickly with other substances. Active metal has low ionisation energy and loses electrons readily to form cations.

ACTIVE NETWORK: A network in which the nodes are programmed to perform custom operations on the messages that pass through the node. For example, a node could be programmed or customised to handle packets on an individual user basis or to handle multicast packets differently than other packets. Active network approaches are expected to be especially important in networks of mobile users.

ACTIVE OPTICS: A computer-controlled system for reflecting telescopes that prevents the distortion of a telescope's mirrors caused by external influences such as gravity, wind, temperature, mechanical stress and thus allows a thinner mirror and a lighter support structure to be used. It works by monitoring the image of a guide star and sending appropriate signals to actuators behind the mirror, which in turn control movable supports to correct the mirror's shape and alignment.

ACTIVE PARTITION: A partition on a hard drive that contains the operating system that a computer attempts to load into memory by default when it is started or restarted. A partition is a logically independent section of a hard disk drive (HDD). When booting or starting a computer, the master boot record (MBR) looks at the partition table and selects the primary partition that is marked active in that table. Only one partition on each hard drive can be set as an active partition or bootable partition. For example, in Microsoft Windows, the partition that contains Windows is the active partition.

ACTIVE PROGRAM: Any program that is loaded and ready to be executed or any program that is currently running or being displayed.

ACTIVE REGION: A localised volume of the Sun's outer atmosphere where powerful magnetic fields, emerging from subsurface layers, give rise to various short-lived features. These features may include: (i) sunspots and faculae in the photosphere; (ii) plagues, fibrils and filaments in the chromospheres; (iii) coronal condensations in the corona. Solar flares are also associated with active regions.

ACTIVE SATELLITE: A satellite that carries equipment, including onboard power supplies, for collecting, transmitting or relaying data. An active satellite may perform signal processing functions such as amplification, regeneration, frequency translation and link switching, to make the signals suitable for retransmission.

ACTIVE SERVER PAGE (ASP): An HTML (Hypertext Markup Language) page that includes one or more scripts (small embedded programs) that are processed on a Microsoft Web server before the page is sent to the user. Typically, the script in the Web page at the server uses input received as the result of the user's request for the page to access data from a database and then builds or customises the page on the fly before sending it to the requestor.

ACTIVE SITE (FUNCTIONAL SITE; CATALYTIC SITE): 1. The site on the surface of an enzyme or a receptor to which the substrate molecules bind. Its properties are determined by the three dimensional arrangement of the polypeptide chains of the enzymes and their constituent amino acids. 2. Part of an enzyme formed by specific amino acid side chains at which catalysis occurs.

ACTIVE SOLAR HEATING: The use of solar collectors (the device in which a fluid is heated by the Sun) to convert solar energy into heat energy. There are two basic types of active solar heating systems based on the type of fluid that is heated in the solar energy collectors. Liquid-based systems heat water or an antifreeze solution in a collector and air-based systems heat air in an air collector. Both systems collect and absorb solar radiation, then transfer the solar heat directly to the interior space or to a storage system, from which the heat is distributed.

ACTIVE SOLID: A porous solid possessing adsorptive properties and used for chromatographic separations. The separations resulting from this action follow laws different from those deriving from the partitioning action of the liquid phase.

ACTIVE SUN: The condition of the Sun during the portion of its 11-year cycle characterised by unusually large numbers and size of spots, flares, prominences and variations in radio-frequency emission.

ACTIVE TRANSPORT: Movement of molecules or ions across a cell membrane through carrier proteins in living cells against a concentration gradient using metabolic energy in the form of adenosine triphosphate (ATP) produced by mitochondria of the cell. In other words, it is the movement of molecules or ions from a less concentrated region to a more concentrated region using energy from the cell across a selectively permeable membrane through carrier proteins. Cells in which active transport occur contain numerous mitochondria. Active transport is extremely important in living organisms because most cells take substances that are only present in low concentrations. Examples are found in the uptake or absorption of mineral ions from the soil by roots; in loading of sugars into the phloem sieve tubes; and in the reabsorption of useful filtrate from the renal tubule into the peritubular capillaries in the kidney during urine formation.

ACTIVE VOLUME: A volcano that has had at least one eruption during the past 10,000 years.

ACTIVE WINDOW: The window that is receiving input from the mouse and keyboard. It is distinguishable by the unique colour of its title bar and window border.

ACTIVE ZONE: A relatively dense and thickened region of presynaptic membrane within a synapse where synaptic vesicles fuse with the membrane to release neurotransmitter from the neuron into synaptic cleft.

ACTIVEX: A technology introduced by Microsoft in 1996 for Internet Explorer to facilitate rich media playback. It is an extension to Microsoft's OLE and COM technologies, and includes a collection of prewritten software components that developers can implement within an application or webpage. This provides a simple way for programmers to add extra functionality to their software or website without needing to write code from scratch. Software add-ons created with ActiveX, called ActiveX controls, for example, might display a clock on a webpage. ActiveX controls, may also allow users to scan their computer for viruses in their browser using free online antivirus scanners or scan their computer for any Windows updates.

ACTIVEX CONTROL: A component program object that can be re-used by many application programs within a computer or among computers in a network. ActiveX controls can be downloaded as small programs or animations for Web pages, but they can also be used for any commonly-needed task by an application program in the latest Windows and Macintosh environments. In general, ActiveX controls replace the earlier OCX (Object Linking and Embedding custom controls).

ACTIVEX DATA OBJECTS (ADO): An easy-to-use application programming interface (API) that wraps OLE DB for use in languages such as Visual Basic, Visual Basic for Applications, Active Server Pages (ASP) and Microsoft Internet Explorer Visual Basic Scripting.

ACTIVITY: 1. Also known as **disintegration rate** or **decay rate**, it is the number of nuclear decays, which occur in a given quantity of material in a small time interval, divided by that time interval. Its symbol is A and expressed as follows: $A = dN/dt$. The SI unit of activity is the becquerel (Bq), named after a French physicist, Henri Becquerel (1852-1908), who discovered radioactivity. One Bq is defined as one transformation or decay per second. 2. A thermodynamic parameter used in place of concentration in equilibrium constants for reactions involving non-ideal gases and solutions. It is a measure of the "effective concentration" of a species in a mixture.

ACTIVITY SERIES (REACTIVITY SERIES; ELECTROCHEMICAL SERIES): A series of chemical elements arranged in descending order of their chemical reactivity or electrode potentials, with the most active

at the top and the less reactive or more noble metals at the bottom. The series shows the order in which metals replace one another from their salts. For example, electropositive metals will replace hydrogen in acids; zinc will displace hydrogen from hydrogen chloride. Basically, such an activity series need not be limited to the metals but may be carried on through the electronegative or nonmetallic elements as well.

METALS	OXIDATION REACTION
Lithium	$\text{Li(s)} \rightarrow \text{Li}^+(\text{aq}) + \text{e}^-$
Potassium	$\text{K(s)} \rightarrow \text{K}^+(\text{aq}) + \text{e}^-$
Barium	$\text{Ba(s)} \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{e}^-$
Calcium	$\text{Ca(s)} \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{e}^-$
Sodium	$\text{Na(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{e}^-$
Magnesium	$\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$
Aluminium	$\text{Al(s)} \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^-$
Manganese	$\text{Mn(s)} \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{e}^-$
Zinc	$\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$
Chromium	$\text{Cr(s)} \rightarrow \text{Cr}^{3+}(\text{aq}) + 3\text{e}^-$
Iron	$\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$
Cobalt	$\text{Co(s)} \rightarrow \text{Co}^{2+}(\text{aq}) + 2\text{e}^-$
Nickel	$\text{Ni(s)} \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{e}^-$
Tin	$\text{Sn(s)} \rightarrow \text{Sn}^{2+}(\text{aq}) + 2\text{e}^-$
Lead	$\text{Pb(s)} \rightarrow \text{Pb}^{2+}(\text{aq}) + 2\text{e}^-$
Hydrogen	$\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
Copper	$\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$
Silver	$\text{Ag(s)} \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$
Mercury	$\text{Hg(l)} \rightarrow \text{Hg}^{2+}(\text{aq}) + 2\text{e}^-$
Platinum	$\text{Pt(s)} \rightarrow \text{Pt}^{2+}(\text{aq}) + 2\text{e}^-$
Gold	$\text{Au(s)} \rightarrow \text{Au}^{3+}(\text{aq}) + 3\text{e}^-$

Activity Series of Metals in Aqueous Solution
(arranged in descending order of easier oxidation)

ACTIVITY SPACE: The entire range of climatic and environmental conditions suitable to normal functions, process and activities of a living organism.

ACTOMYOSIN: The complex formed from the interaction of the proteins actin and myosin during the process of muscle contraction. It forms the basic unit of contraction in muscle cells.

ACTUAL YIELD: The amount of product actually obtained in a reaction.

ACTUARIAL STATISTICS: The branch of statistics that is concerned with insurance and loss, including reinsurance, ruin theory and run-off triangles.

ACTUARY: An expert in statistics who calculates insurance risks and relates them to the premiums to be charged.

ACTUATOR: 1. A device that translates signals into mechanical actions. It is a type of motor for mechanically controlling a system and it is operated by a source of energy, usually in the form of an electric current, hydraulic fluid pressure, pneumatic pressure or their combinations, which converts that energy into some kind of motion. An example is a stepper motor which acts on an electrical signal received from a computer instructing it to turn its shaft a certain number of degrees or rotations. 2. A mechanism by which an agent acts upon an environment, with the agent being either an artificial intelligence agent or any other autonomous being such as humans, other animals, etc.

ACULEATE: 1. In botany, having or being covered with sharp-pointed structures or prickles, for example, bougainvillea. 2. In zoology, it refers to organisms such as hymenopterans that are armed with a sting.

ACULEOLATE: In biology, it refers to a structure having or covered with minute or tiny prickles.

ACUMEN: In biology, it refers to a tapering or sharp point such as the point of an acuminate leaf.

ACUMINATE: In biology, it refers to a plant or an animal structure such as a leaf or teeth that tapers gradually to a sharp point and forming concave sides along the tip.

ACUPUNCTURE: A type of complementary medicine that originated in China and now used in other parts of the world, which involves using needles to prick the skin or tissues to alleviate pain and to treat various physical, mental and emotional conditions.

ACUTE: 1. Sensing or perceiving accurately, clearly, effectively or sensitively, for example, acute vision. 2. Characterised by sharpness or severity, for example, acute pain. 3. In mathematics, an **acute angle** is an angle which is more than 0° and less than 90° or an amount of turn that is less than a quarter of a circle. An acute triangle is a triangle with three acute angles. 4. In botany, it is any part of a plant that is sharply pointed or tapers to a pointed apex with more or less straight sides. 5. In medicine, an **acute disease** is a disease with a rapid onset that lasts for a short period, often accompanied by distinct symptoms.

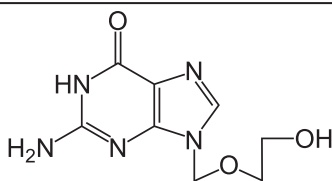
ACUTE PHASE RESPONSE: A mechanism of innate immunity in which the liver markedly increases its synthesis of certain immune proteins (the acute-phase proteins) in response to infection, physical trauma or malignancy to prevent ongoing tissue damage, isolate and destroy the infective organism and activate the repair processes necessary to restore an organism's normal function.

ACUTE TOXICITY: A measurement of the harmful effects of a single exposure or multiple exposures to a chemical in a short period of time.

ACYCLIC: 1. A term describing a chemical compound having a molecular structure in which the atoms are arranged in a string whose ends do not meet an open chain or a chemical compound that does not contain a ring of atoms in its molecular structure. 2. A term describing flowers that have their parts arranged in a spiral, not a whorl.

ACYCLIC MOTION (IRROTATIONAL FLOW; IRROTATIONAL MOTION): In physics, it is a fluid flow in which the curl of the velocity function is zero everywhere, so that the circulation of the velocity about any closed curve vanishes. The velocity is given by the gradient of a scalar field.

ACYCLOVIR (ACYCLOGUANINE; 2-AMINO-1,9-DIHYDRO-9-(2-HYDROXYETHOXY) METHYL)-6H-PURIN-6-ONE): A compound with the chemical formula $C_8H_{11}N_5O_3$, molar mass 225.21 g/mol, derived from guanine, which is used for the treatment of cold sores, genital blisters, shingles and other diseases caused by herpesvirus guanine. It acts by interfering with DNA replication virus.



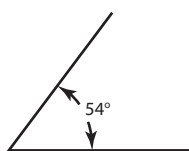
Chemical Structure of Acyclovir

ACYL CARRIER PROTEIN (ACP): A low-molecular-weight conjugated protein found in organisms such as animals, bacteria and plants. It is an important component in both fatty acid and polypeptide biosynthesis that picks up acetyl and malonyl groups from acetyl coenzyme A and malonyl coenzyme A and links them by condensation to form β -keto acid acyl carrier protein, releasing carbon dioxide and the sulfhydryl form of acyl carrier protein.

ACYL FISSION: The breaking of the carbon-oxygen bond in an acyl group. It occurs in the hydrolysis of an ester to produce an alcohol and a carboxylic acid.



Structure of an acuminate leaf



Acute angle

ACYL GROUP: A radical formed from an organic acid by the removal of a hydroxyl group or a compound derived from a carbonic acid by replacing the -OH group with a halogen, which is usually Cl. An example is the acetyl group CH_3CO- .

ACYL HALIDES (ACID HALIDE): A chemical compound, with the general formula $RCOCl$ derived from a carboxylic acid by replacing a hydroxyl group with a halide group, using a halogenating agent such as PCl_5 . An example is ethanoyl chloride (CH_3COCl). Acyl halides react readily with water, alcohols, phenols and amines and are used in acylation reactions.

ACYLATION: The process of introducing an acyl group ($RCO-$) into a compound. The usual method is to react an alcohol with an acyl halide or a carboxylic acid anhydride. An example is $RCOCl + R'OH \rightarrow RCOOR' + HCl$. The introduction of an acetyl group (CH_3CO-) is acetylation, a process used for protecting -OH groups in organic synthesis.

ACYLGLYCEROL (GLYCERIDE): An ester of glycerol (propane-1,2,3-triol) in which a hydroxyl group is replaced by a fatty acid. Acyl-glycerols are one of the four main types of complex lipids. They are subdivided into triglycerides, 1,2- or 1,3-diglycerides and 1- or 2-monoglycerides, according to the number and position of acyl. The recommended method for naming individual glycerides is mono-, di- or tri-Oacylglycerol, as appropriate. Soaps are formed from the reaction of glycerides with sodium hydroxide.

ACYLOINS (KETOLS): A class of organic compounds, which share common functional group consisting of a hydroxyl group placed on the α -position of a carbonyl group. Acyloins include alpha-ketols, which have the hydroxyl group adjacent to the keto group; beta-ketols, which have the hydroxyl group at the second carbon from the keto group; and gamma-ketols, which have the hydroxyl group at the third carbon from the keto group.

ACYLOXY RADICALS: Oxygen-centred radicals consisting of an acyl radical bonded to an oxygen atom. They are valuable intermediates in the Barton and related reactions, where they serve as versatile precursors of alkyl radicals used in synthesis of complex natural products such as interiorins and kadsulignans and mitomycins.

AD BLOCKER: A software that removes different kinds of advertising from a Web user's experience online. These programs target certain kinds of ads, such as pop-ups, banner ads and other common forms of online advertisement, allowing a user to surf the Web without annoying distractions or interruptions.

AD HOC NETWORK: A temporary network that allows individual wireless devices within range of each other to locate and communicate without involving central access points such as a router or a switch.

AD INFINITUM: A Latin phrase for "to infinity" or "forever more" and used in mathematics to refer to an infinite number of times. It is normally abbreviated as ad inf.

AD LIBITUM (FREE FEEDING): The feeding management through which farm animals are offered as much feed as they want; that is, they always have feed in the trough at their disposal.

AD-AWARE: A free anti-virus and anti-spyware program that helps to combat the most complex forms of computer malware, spyware and adware. The anti-spyware software program was developed by Lavasoft in 1999 and is licensed under proprietary software. Ad-Aware software can be downloaded free of charge from the official Lavasoft Ad-Aware website.

ADA PROGRAMMING LANGUAGE: A high-level computer programming language, used for military and other complex applications. It was developed for the US military in the 1970s and named after Augusta Ada Byron (1815-1852), Countess of Lovelace and an English mathematician.

ADAMS CATALYST (PLATINUM DIOXIDE): A dark brown powder, which is a hydrated form of platinum (IV) oxide (PtO_2), produced by heating chloroplatinic acid (H_2PtCl_6) with sodium nitrate ($NaNO_3$). The product, platinum nitrate, decomposes to platinum (IV) oxide

with evolution of nitrogen dioxide (NO₂) and oxygen. It is used in hydrogenations of alkenes to alkanes, nitro compounds to amines and ketones to alcohols. The actual catalyst is not the oxide but finely divided platinum black, which forms during the hydrogenation reaction.

ADAPTATION: 1. Any change in the structure or functioning of an organism that makes it better suited more perfectly for existence in its environment. Natural selection of inheritable adaptations ultimately leads to the development of new species. Increasing adaptation of a species to a particular environment tends to diminish its ability to adapt to any sudden change in that environment. 2. The alteration in the degree of sensitivity, either an increase or a decrease, of a sense organ to suit conditions more extreme than normally encountered. An example is the adjustment of the eye to vision in very bright or very dim light.

ADAPTATION ASSESSMENT: The practice of identifying options to adapt to climate change and evaluating them in terms of criteria such as availability, benefits, costs, effectiveness, efficiency, and feasibility.

ADAPTATION BENEFITS: The avoided damage costs or the accrued benefits following the adoption and implementation of adaptation measures.

ADAPTATION COSTS: Costs of planning, preparing for, facilitating, and implementing adaptation measures, including transition costs.

ADAPTER: It is a term used to refer to any additional circuit board that is built onto the computer motherboard or that comes in the form of a peripheral device that gives the computer additional functionality or compatibility. Examples are a video card and a sound card.

ADAPTIVE CAPACITY: The ability of a system to adjust to climate change (including climate variability and extremes) to moderate potential damages, to take advantage of opportunities, or to cope with the consequences.

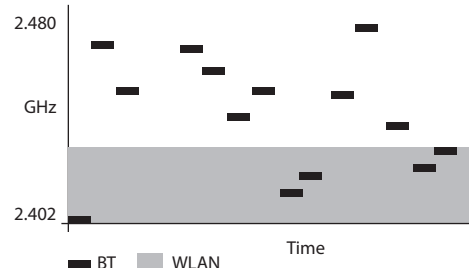
ADAPTIVE DIFFERENTIAL PULSE-CODE MODULATION (ADPCM): A technique for converting sound or analogue information to binary information by taking frequent samples of the sound and expressing the value of the sampled sound modulation in binary terms. ADPCM is used to send sound on fibre-optic long-distance lines as well as to store sound along with text, images and code on a CD-ROM.

ADAPTIVE ENZYME (INDUCED ENZYME): An enzyme that is normally present in minute quantities within a cell and synthesised in response to high concentrations of its substrate compound. In other words, it is an enzyme produced by a cell or an organism in response to the accumulation or addition of a particular substance. When the substrate is absent the genes governing the enzyme's synthesis are repressed.

ADAPTIVE FEATURES: Any feature of an organism that allows it to survive in its environment. These may be structural, behavioural or physiological. Structural adaptations are physical features of an organism such as shape, body covering, etc. For example, a leaf may have stomata to regulate water loss and permit gas exchanges, drip tips to guide water off, toothed edges to promote leaf opening from winter dormancy or leaf hairs as both mechanical protection from chewing insects and to control air movement across a leaf's surface thus slowing water loss. In animals such as dogs, adaptive features include the fact that they can coordinate themselves (hunt in packs), their acute sense of smell enables them to find food, potential mates and potential rivals and their strong jaw muscles are important for hunting, while in fishes, the most obvious would be the streamlined structure which makes it easier to move by reducing friction and water resistance. Also the gills are useful in allowing the fish to breathe. The fact that some fishes swim in schools decreases their chance of being selected by predators as food.

ADAPTIVE FREQUENCY HOPPING (AFH): A process that reassigns the transmission of packets on frequency channels that have interference to other channels that have lesser interference levels. Devices communicating by means of Bluetooth wireless technology run the risk of causing and encountering interference in environments where other wireless technologies are in use. AFH allows Bluetooth to

adapt to the environment by identifying fixed sources of interference and excluding them from the list of available channels. This process of remapping also involves reducing the number of channels to be used by Bluetooth.



Collisions resulting from random frequency hopping adapting to the environment

ADAPTIVE IMMUNITY (ACQUIRED IMMUNITY): A type of immunity to a particular disease that has been obtained by the development of antibodies after an attack of an infectious disease. Examples of immunity obtained by the development of antibodies are a pregnant mother passing antibodies through the placenta to a foetus and by vaccination.

ADAPTIVE LANDSCAPE (ADAPTIVE SURFACE): The surface plotted in a three-dimensional graph; having all possible combinations of allele frequencies for different loci plotted in the plane; and mean fitness for each combination plotted in the third dimension.

ADAPTIVE OPTICS: An optical technique that allows rapid fluctuations in a ground-based telescope's image quality, caused by Earth's atmospheric turbulence, to be corrected in fractions of a second. Adaptive optics measures the atmospheric distortions in the incoming light from a star or other object and sends electronic signals to a deformable mirror that can change its shape rapidly to correct for distortions.

ADAPTIVE PACKET ASSEMBLY (APA): An error control technique applied to modems to increase line fault tolerance and facilitate split data packet transmission at short intervals. APA works with other modem error control techniques to improve modem transmission quality.

ADAPTIVE PEAK: An equilibrium state in a population when the gene pool has allele frequencies that maximise the average fitness of a population's members. In other words, it is a high point (perhaps one of several) on an adaptive landscape or fitness landscape, used to visualise the relationship between reproductive success (fitness) and an organism's phenotype, which is determined by its genotype. Increasing fitness is represented on the landscape as a steeper peak.

ADAPTIVE RADIATION (BRANCHING EVOLUTION; CLADOGENESIS; DIVERGENT EVOLUTION): The formation of several species with adaptations to different ways of life or evolution of different species in different directions from one from a parent species. This is likely to occur whenever members of a species migrate to exploit new habitats and food sources and new populations emerge, each adapted to its particular environment. Eventually, these become sufficiently distinct to be recognised as new species. Examples are the 13 species of Darwin's finches on the Galapagos Islands, each filling a different niche on different islands but evolved from one ancestral species. Also, early placental mammals, for example, gave rise to modern burrowing, climbing, flying, running and swimming organisms. The vertebrate limb is another example of divergent evolution. The limb in many different species has a common origin, but has diverged somewhat in overall structure and function.

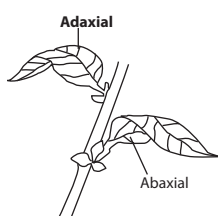
ADAPTIVE ROUTE: An optimal and efficient routing path that is selected when routing priorities change or failures occur with routing

devices, nodes or other network components. An adaptive route ensures continuous network connectivity and operations.

ADAPTIVE SOFTWARE (ASSISTIVE SOFTWARE): A computer program designed for specialised hardware used by physically challenged people. Examples include programs for screen magnification, screen reading, speech recognition, text-to-speech, Braille printers, Braille scanners, touch screen displays, oversized mice and oversized joysticks. Adaptive software and associated hardware are the computer-related components of a larger category of products known as assistive technology or adaptive technology.

ADAPTOR PROTEIN: A protein that takes part in intracellular signalling pathways by recruiting specific components of the pathway in response to activation of the pathway's receptor molecule. Adaptors lack enzymatic activity and possess at least two types of binding sites: one type to bind to the cytoplasmic region of the activated receptor and others to bind specific intracellular molecules in such a way that they become activated.

ADAXIAL: 1. Located on the side nearest to the axis or central line of an organ or organism. 2. A term used to describe the side or surface facing or nearest to the axis of an organ, such as the upper surface of a leaf; ventral, as shown in the diagram.



Adaxial of a leaf

ADD: Combining numbers or quantities by finding the total number of units contained in them. The addition of two natural numbers is defined by $a + 0 = a$;

$a + (n + 1) = (a + n) + 1$. Also the definition applies in terms of the cardinality of the disjoint union of sets whose cardinalities are the given numbers.

ADD-DROP MULTIPLEXER (ADM): A critical element of an optical fibre network used for inserting or extracting packets of data. It can combine (i.e. multiplex) several low-bandwidth streams of data into a single light beam; and simultaneously, it can drop or remove other low-bandwidth signals from the stream of data and direct them to other network routes. The filtering technology used with ADM is called Fabry-Pérot etalon.

ADD-IN (ADD-ON; PLUG-IN): A software utility or other program that can be added to a primary or existing program to increase memory or add graphics or extend the capabilities of larger programs in a computer. They can come in the form of chips, circuit boards, cartridges, video adapters, graphics accelerators, sound cards, accelerator boards and internal modems. For example, there are many Excel add-ins designed to complement the basic functionality offered by Excel. Microsoft Bookshelf is an add-in for Word and Analysis Toolpak is an add-in for Microsoft Excel. Also, Mozilla Firefox, a popular Web browser, supports add-ons such as the Google toolbar, ad blockers and Web developer tools. Some computer games support add-ons that provide extra maps, new characters or give the player game-editing capabilities.

ADDEND: In algebra, any of a set of numbers to be added. For example, in the expression $x + y + z$, the addends are x , y , z .

ADDER: 1. An electronic circuit in a computer or calculator that carries out the process of adding two binary numbers. 2. Any of highly venomous snakes of the family Viperidae having haematotoxic venom and movable fangs. A typical example in Africa is the puff adder (*Bitis arietans*).

ADDICTION: A state of physiological or psychological dependence on a potentially harmful drug such as morphine, heroin and alcohol; or a behavioural practice, which is considered harmful.

ADDING MACHINE: A mechanical device for doing simple additions and other arithmetical operations like subtraction, multiplication and division and usually does not have provision for a print out. These machines have largely been replaced in most applications by electronic calculators.

ADDISON'S DISEASE (ADRENOCORTICAL INSUFFICIENCY; HYPOCORTISOLISM): A rare metabolic disease, named after the British physician Thomas Addison (1793–1860), who first identified it. It occurs when the adrenal glands do not produce enough cortisol and other cortical steroids. The disease is characterised by weight loss, increasing weakness, fatigue, hypotension or low blood pressure and abnormal pigmentation of the skin.

ADDITION: A mathematical operation, process or action of calculating the sum of two or more numbers or quantities with an operation represented by the addition sign $+$ (usually read plus). Addition can follow the **commutative law** and expressed, for example, as $a + b = b + a$, in which case the order it follows does not matter. It can also follow the **associative law**, in which case in the addition of more than two numbers, the order in which it is performed does not matter, expressed as

$$(a + b) + c = a + (b + c) \text{ or,}$$

$$(a + b) + c = a + (b + c) = (a + c) + b.$$

For example, the numbers 2, 3 and 4, may be performed as follows: $(2 + 3) + 4 = 5 + 4 = 9$ or $2 + (3 + 4) = 2 + 7 = 9$. If addition is done in combination with multiplication, the **distributive law** is applied, which follows the rule that the same product results in multiplication, when performed on a set of numbers as when performed on members of the set individually. For example,

$$a \times (b + c) = a \times b + a \times c, \text{ then multiplication is distributive over addition.}$$

ADDITION FACTS: The basic facts as we know them in addition. For example, the addition facts of zero are:

$$0 + 0 = 0, 0 + 1 = 1, 0 + 2 = 2, 0 + 3 = 3, 0 + 4 = 4, \text{ etc.}$$

ADDITION FORMULA: 1. Any analogous formula for a function f that gives the value of $f(x + y)$ in terms of $f(x)$, $f(y)$ and related functions. 2. Any equation that expresses trigonometric functions of the sum or difference of two angles in terms of separate functions of the angles. They are used to simplify trigonometric expressions, for example, in solving an equation. The fundamental formulae of angle addition in trigonometry are given by the following identities:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B;$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B;$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B;$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B;$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B};$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}.$$

ADDITION LAW OF PROBABILITY: A statistical property that states the probability of one and/or two events occurring at the same time is equal to the probability of the first event occurring, plus the probability of the second event occurring, minus the probability that both events occur at the same time. For example, if A and B are two events then the addition law states that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \text{ If the events } A \text{ and } B \text{ are mutually exclusive events this reduces to}$$

$$P(A \cup B) = P(A) + P(B), \text{ where } P(A \cap B) = 0.$$

ADDITION MUTATION: A type of gene mutation in which the deletion or addition of one or more nucleotides causes a shift in the reading frame of the codons in the mRNA. It may eventually lead to the alteration in the amino acid sequence at protein synthesis (translation).

ADDITION OF BINARY NUMBERS: Addition of numbers based on base 2 (consisting of the numbers 0 and 1) instead of base 10 for decimal addition (ordinary addition of numbers). For example, decimal (denary) numbers 2 and 3 occur as 10 and 11 in a binary system. The rules of binary addition of numbers are:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10_{\text{two}}$$

ADDITION OF FRACTIONS: Fractions can be added only if they are expressed with the same denominator. For example, the lowest common denominator of $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{15}$ is 30 and fractions can be stated as $\frac{15}{30}$, $\frac{10}{30}$ and $\frac{2}{30}$. They can now be added as

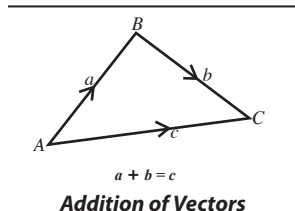
$$\frac{1}{2} + \frac{1}{3} + \frac{1}{15} = \frac{15+10+2}{30} = \frac{27}{30} = \frac{9}{10}.$$

ADDITION OF MATRICES: If two or more matrices are of the same order they can be added. The order that should be followed is the number of rows, followed by the number of columns. For example,

if $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is added to $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ this becomes

$$A+B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}.$$

ADDITION OF VECTORS: A vector is a quantity that has both magnitude and direction and is usually symbolised by a letter with an arrow above it, for example, $\vec{a}$. When a lower case letter is used for a vector, for example, vector a , it can also be denoted by $\vec{a}$. The vector $\overrightarrow{AB}$ is the movement from point A to B , vector $\overrightarrow{AC}$ is the movement from point A to C , and vector $\overrightarrow{BC}$ is the movement from point B to C . $\overrightarrow{AC}$ is the vector sum or resultant of $\overrightarrow{AB}$ and $\overrightarrow{BC}$, i.e., $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$. Addition of vectors obey the commutative and associative laws. It is done by adding the head of one to the tail of the other vector. This is known as head-to-tail addition.



ADDITION POLYMER: A polymer such as plastics which is formed from simple monomers, usually containing C=C by addition reaction.

ADDITION POLYMERISATION: The condensation of two or more molecules (monomers) of the same kind to form a complex molecule (polymers) without elimination of any small molecule. An example is the free-radical polymerisation of propylene to polypropylene.

ADDITION REACTION: The chemical reaction of two or more reacting molecular entities, resulting in a single reaction product containing all atoms of all components, with formation of two chemical bonds and a net reduction in bond multiplicity in at least one of the reactants. In other words, it is a reaction in which two molecules, one of which is unsaturated combine to give a single product. An example is the reaction between hydrogen bromide (HBr) and ethylene ($\text{CH}_2=\text{CH}_2$) to form ethyl bromide ($\text{C}_2\text{H}_5\text{Br}$). The reverse process is called an elimination reaction. The addition may occur at only one site called alpha-addition (α -addition) or 1/1/addition. It can occur at two adjacent sites (1/2/addition) or at two non-adjacent sites (1/3/- or 1/4/addition, etc.).

ADDITION STRATEGIES: Methods that help beginners to add two or more numbers. For example, counting all, forward counting, making groups of tens and ones, doubling a number, etc.

ADDITION WORD PROBLEM: A mathematical problem that is expressed in words rather than in mathematical notation and requires beginners to find an unknown quantity using an addition operation or strategy.

ADDITION-DELETION MUTATION: A type of gene mutation in which the addition or deletion of one or more nucleotides causes a shift in the reading frame of the codons in the mRNA, thus, may eventually lead to the alteration in the amino acid sequence at protein synthesis (translation).

ADDITIVE: 1. In mathematics, of a function between semigroups, distributing over addition, so that $f(x+y) = f(x)+f(y)$. The only continuous or measurable additive functions on the real line are of

the form $f(x) = cx$. 2. Of a set function, having the property that the function of the union or sum of two quantities is equal to the sum of the functional values of each quantity: $f(A \cup B) = f(A) + f(B)$. 3. A small amount of substance that is added to another to alter or improve the general quality or to counteract undesirable properties. For example, food additive is added to food during processing to give it particular properties such as to prolong the shelf life, change the colour or flavour or to improve its food value. Examples of food additives are flavour enhancers such as artificial sweeteners including aspartame, sucralose and saccharin; preservatives such as calcium propionate, sodium nitrate, sulfites, nitrites and nitrates; stabilisers and emulsifiers, which include tricalcium phosphate, an anticaking agent; lecithin, an emulsifier; alginic acid and xanthan gum, bulking and thickening agents; and stearic acid, a glaze enhancer.

ADDITIVE COMPARISON: A numerical comparison that indicates the relation between two quantities such as numbers or length measures, by how much more or less one quantity is compared to the other.

ADDITIVE FUNCTION: A function in which $f(a+b) = f(a) + f(b)$. For example, $f(a) = 4a$ is an additive function since $f(a+b) = 4(a+b)$, but $g(a) = \sqrt{a}$ is not additive since $\sqrt{a+b}$ is not equal to $\sqrt{a} + \sqrt{b}$.

ADDITIVE GROUP: A group that has addition (+) as its operation. The operation is usually represented by only addition; if it is commutative it is referred to as Abelian group.

ADDITIVE IDENTITY: An identity element under the operation addition (+). The number zero is the additive identity element because it satisfies the property that the addition of zero does not change a number: $x + 0 = 0 + x = x$ for all x .

ADDITIVE INVERSE: The inverse of an element under the addition operation. For example the inverse of 2 under additive operation is -2. It is a number that gives zero, when added to a given number. The additive inverse of b ($b \in R$) is also called the negative or opposite of b (written as $-b$), thus $b + (-b) = b - b = 0$. For example, -2 is the additive inverse of 2 and 6 is the additive inverse of -6.

ADDITIVE NAME: The formal assembly of names for the components of a compound without loss of atoms or groups of atoms from any component.

ADDITIVE PROCESS: A process of colour photography in which colours are formed by mixing lights of additive primary colours consisting of red, green and blue-violet in various proportions.

ADDITIVITY: The process by which each chemical species present in the ion source at a certain partial pressure makes a contribution to the total mass spectrum which is the same as that it would give if that chemical species alone were present in the ion source at a pressure equal to this certain partial pressure.

ADDITIVITY PRINCIPLE: The hypothesis that each of several structural features of a molecular entity makes a separate and additive contribution to a property of the substance concerned. More specifically, it is the hypothesis that each of the several substituent groups in a parent molecule makes a separate and additive contribution to the standard Gibbs energy change (or Gibbs energy of activation) corresponding to a particular equilibrium (or rate of reaction).

ADDRESS: 1. A letter-number pair used to locate a spreadsheet cell. For example, A5 is the fifth cell in column A. **Address box** is a place where the address of a spreadsheet cell is shown when the cell is selected. 2. In computer science, a number indicating a specific location in a computer memory. At each address, a single piece of data can be stored. The term also refers to a device or storage

location by an identifying number, character or group of characters. Computer address can be classified as one of the following: (i) **Physical address** is a unique hexadecimal number used to identify the network interface card. (ii) **IP address** is a unique number (a 32-bit long binary number represented as 4-octets) in LAN/WAN used to logically identify a host in the network. (iii) **Port address** is used to identify application software accessing the network.

ADDRESS BAR (ADDRESS BOX; LOCATION BAR; URL BAR):

A text field near the top of a Web browser window that displays the URL of the current webpage. The URL or web address, reflects the address of the current page and automatically changes whenever you visit a new webpage.

ADDRESS BOOK: In computer science, a database that stores names, addresses and other contact information for a computer user.

ADDRESS BUS: The electrical pathway or bus used to select the route for any particular data item, as it is moved from one part of a computer to another.

ADDRESS RESOLUTION PROTOCOL (ARP): A protocol for mapping an Internet Protocol address (IP address) to a physical machine address that is recognised in the local network. For example, in IP Version 4, the most common level of IP in use today, an address is 32 bits long. In an Ethernet local area network, however, addresses for attached devices are 48 bits long.

ADDRESS SPACE: The amount of memory allocated for all possible addresses for a computational entity, such as a device, a file, a server or a networked computer. Address space may refer to a range of either physical or virtual addresses accessible to a processor or reserved for a process. A processor's address space is always limited by the width of its address bus and registers.

ADDUCT: 1. A compound formed by addition reaction or a product of two atoms or molecules chemically combined as in an addition reaction. The term is used particularly for compounds formed by coordination between a Lewis acid and a Lewis base. 2. To draw towards or past the median axis of the body or bring together or forward.

ADDUCTION: The movement of a body part toward the axis or midline of the body in the coronal plane. In the case of the fingers and toes, the term adduction refers to the drawing together of these structures, i.e., their movement toward the axis of the limb. The opposite movement is called **abduction**. An example of the action of an abductor muscle is raising the arm out from the body.

ADDUCTOR: 1. A muscle whose contraction pulls down a part of the body, such as a finger, arm or toe, towards the mid-line of the body. Examples include *musculus adductor brevis*, the short adductor muscle of the thigh; *musculus adductor longus*, the long adductor muscle of the thigh; great adductor muscle or *musculus adductor magnus*, the muscle that adducts and extends the thigh; and *musculus adductor hallucis*, the adductor muscle of the great toe. 2. A muscle that closes the valves of a bivalve mollusc such as clams, scallops, mussels and oysters.

ADDUCTOR MAGNUS: A large, powerful triangular muscle in the thigh. It consists of two parts, the adductor and hamstring. It is one of the two long adductor muscles, the other being the gracilis.

ADELPHOUS: A term used to describe an androecium or the stamens of a flower with fused filaments, instead of growing singly.

ADENINE (9H-PURIN-6-AMINE; 6-AMINOPURINE): One of the major component bases of nucleotides, DNA and RNA, with the chemical formula $C_5H_5N_5$. It is a purine derivative which occurs freely in tea and in combined form in nucleic acids, adenosine triphosphate (ATP), vitamin B_{12} and several coenzymes. In DNA, its complementary base is thymine.

ADENOIDS (PHARYNGEAL TONSILS): Masses of lymphoid tissue, similar to tonsils, located in the upper part of the throat, behind the nose. They are part of a child's defences against the entry of germs but usually shrink and disappear by the age of ten.

ADENOSINE: A nucleotide with the molecular formula $C_{10}H_{13}N_5O_4$, present in all human cells comprising one adenine molecule linked to a D-ribose sugar molecule. It is a purine nucleoside found in DNA, RNA and many co-factors. It readily combines with phosphate to form various chemical compounds including adenosine monophosphate (AMP) and adenosine triphosphate (ATP).

ADENOSINE DIPHOSPHATE (ADP; ADENOSINE 5'-(TRIHYDROGEN DIPHOSPHATE); ADENOSINE 5'-DIPHOSPHATE):

The nucleotide consisting of the pyrophosphate group, the pentose sugar ribose and the nucleobase adenine, having the molecular formula $C_{10}H_{15}N_5O_{10}P_2$. It is formed by the removal of one phosphate from an adenosine triphosphate (ATP) molecule during metabolic reactions in living cells and stored in dense bodies inside blood platelets and is released upon platelet activation. It is an important intermediate in cellular metabolism and plays a critical role in supplying energy for many life processes such as necessary growth and cell maintenance. Animals use the energy released in the breakdown of glucose and other molecules to convert adenosine diphosphate (ADP) to ATP. ADP can be converted back to adenosine triphosphate through the process of releasing the chemical energy available in food. In humans, this is performed via aerobic respiration in the mitochondria. Plants use photosynthetic pathways to convert and store the energy from sunlight, via conversion of ADP to ATP.

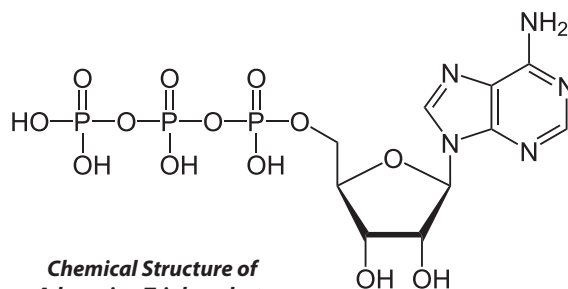
ADENOSINE MONOPHOSPHATE (AMP; 5'-ADENYLIC ACID):

A phosphorylated nucleoside consisting of phosphate group, the sugar ribose and the nucleobase adenine, having molecular formula of $C_{10}H_{14}N_5O_7P$. It can be produced during adenosine triphosphate (ATP) synthesis by the enzyme adenylate kinase by combining two ADP molecules: $2ADP \rightarrow ATP + AMP$. It may also be produced by the hydrolysis of one high energy phosphate bond of ADP:

$ADP \rightarrow AMP + P_i$. Another process is by hydrolysis of ATP into AMP and pyrophosphate: $ATP \rightarrow AMP + PP_i$. AMP is involved in the regulation of glycolysis and gluconeogenesis, promoting the formation of fructose biphosphate from fructose 6-phosphate (i.e. promoting glycolysis), while inhibiting the back reaction to fructose biphosphate.

ADENOSINE TRIPHOSPHATE, ATP [ADENOSINE 5'-(TETRAHYDROGEN TRIPHOSPHATE)]:

A nucleotide molecule with the molecular formula $C_{10}H_{16}N_5O_{13}P_3$. It is formed by adding yet another phosphate group to the pyrophosphate group of ADP. It is the major energy-transferring molecule in all biological systems and found in all cells that are fundamentally important to all living organisms. Adenosine triphosphate is involved in the regulation of sugar oxidation, being an inhibitor of the conversions of pyruvic acid to acetyl CoA and of fructose 6-phosphate to fructose biphosphate. It is also essential for the synthesis of the nucleic acids DNA and RNA.



ADENOVIRUS: One of a group of DNA-containing viruses found in rodents, fowls, cattle, monkeys and man. In man, they produce active respiratory-tract infections resembling those of common cold.

ADENYLATE: Any anion of adenylic acid (the phosphoric acid ester of adenosine) or any salt or ester of adenylic acid, especially either of the nucleotides adenosine triphosphate (ATP) or adenosine diphosphate (ADP).

ADENYLATE CYCLASE: The enzyme that catalyses the formation of cyclic adenosine monophosphate (AMP) from adenosine triphosphate (ATP) and acts as a signal transducer, coupling hormone binding to changes in cytoplasmic cyclic AMP. Many hormones exert their physiological effects by increased synthesis of cyclic AMP through the activation of adenylate cyclase.

AERATED WATER: Water that has had air or oxygen pumped into it to remove some of the sulfur contents. This may be done to polluted water bodies to help support aquatic life and prevent the growth of putrefying bacteria.

AGENDA 21: A non-binding programme adopted by representatives of 178 countries in June 1992, at the United Nations Conference on Environment and Development in Rio de Janeiro, Brazil, to achieve sustainable development and sets out a framework of recommendations designed to protect the environment. It includes Social and Economic Dimensions; Conservation and Management of Resources for Development; Strengthening the Role of Major Groups; and Means of Implementation that includes science, technology transfer, education, international institutions and financial mechanisms.

ADH: Abbreviation for **antidiuretic hormone**, a hormone produced in some mammals by specialised nerve cells in the hypothalamus, which is transported to, stored and secreted by the posterior pituitary gland.

ADHERENS JUNCTION (ZONULA ADHERENS): A type of cell junction, especially found in epithelial cells that forms a strengthening and interlocking belt encircling the exterior of adjacent cells.

ADHERENT: In biology, it is the joining of dissimilar parts or organs which are not united. An example is the attachment of the anthers to the style.

ADHERENT POINT: A point of the closure of a set. If X is a topological space and $A \subset X$ is a subset, a point $x \in X$ is an adherent point for A if every open set containing x contains at least one point of A . A point x is an adherent point for A if and only if x is in the closure of A .

ADHESION: 1. The tendency of dissimilar particles or surfaces to cling to one another. The various kinds of adhesion include chemical adhesion, dispersive adhesion and diffusive adhesion. **Chemical adhesion** occurs when the surface atoms of two separate surfaces form ionic, covalent or hydrogen bonds. **Dispersive adhesion**, also called **adsorptive adhesion**, is a mechanism for adhesion which attributes attractive forces between two materials to intermolecular interactions between molecules of each material. This mechanism is widely viewed as the most important of the five mechanisms of adhesion due to its presence in every type of adhesive system and relative strength. **Electrostatic adhesion** occurs when some types of conducting materials pass electrons to form a difference in electrical charge at the join. This results in a structure similar to a capacitor and creates an attractive electrostatic force between the materials. **Diffusive adhesion** occurs, when two materials merge at the joint by diffusion. This may occur when the molecules of both materials are mobile and soluble in each other. **Cohesion** is the tendency of similar or identical particles or surfaces to cling to one another. 2. In medicine, adhesions are bands of scar-like tissue that form between two surfaces inside the body and cause them to stick together. The

causes include inflammation, surgery or injury. Adhesions can form anywhere in the body including joints such as the shoulder, in the eyes and inside the abdomen or pelvis.

ADHESIVE: A substance such as glue or epoxy used to stick one material to another.

ADHESIVE FORCES: Forces of attraction between a dissimilar or unlike molecules. Adhesive force explains why water clings to the wall of a glass vessel, because there is a force of attraction between the molecules of water and molecules of the glass vessel.

ADIABATIC (ADIABATIC PROCESS): 1. A term used to describe any thermodynamic process that occurs with neither a gain nor loss of energy between the system and its surroundings; or any thermodynamic process that occurs without heat entering or leaving a system. Since no heat enters nor leaves the system, heat (Q) is zero, hence the first law of thermodynamic becomes: $\Delta E_{in} = -W$, where ΔE_{in} is the internal energy of the system and W is the work done. This process generally involves a fall or rise in temperature of the system. The term can be applied in several contexts as follows: (i) **Electronically adiabatic** is a reaction in which there is no change of electronic state or multiplicity. (ii) **Vibrational adiabaticity** is a reaction in which there is no change of vibrational state during the course of reaction. The term also applies to a process in which excess vibrational energy in the reactants appears as vibrational energy in the products or in which ground-state vibration in the reactants leads to ground state vibration in the products. (iii) **Rotational adiabatic** is a reaction in which excess rotational energy in the reactants appears as rotational energy in the products or in which ground-state rotation in the reactants leads to ground-state rotation in the products. 2. In the Rice-Ramsperger-Kassel-Marcus (RRKM) theory of unimolecular reactions, a degree of freedom whose quantum number is more or less preserved during energisation.

ADIABATIC APPROXIMATION: An approximation used in quantum mechanics, when the time dependence of parameters, such as the inter-nuclear distance between atoms in a molecule is slowly varying.

ADIABATIC DEMAGNETISATION (GIAQUE-DEBYE METHOD; MAGNETIC COOLING; PARAMAGNETIC COOLING): A technique for cooling a paramagnetic salt such as potassium chrome alum to a temperature near absolute zero.

ADIABATIC ELECTRON TRANSFER: Electron transfer process in which the reacting system remains on a single electronic surface in passing from reactants to products. For adiabatic electron transfer, the electronic transmission factor is close to unity.

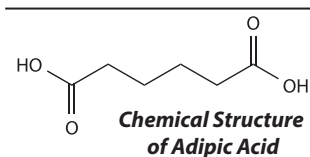
ADIABATIC IONISATION: A process in which an electron is either removed from or added to an atom, ion or molecule in the gas phase. This produces an ion in its ground state. **Adiabatic ionisation energy** is the lowest energy required to remove an electron from an atom, ion or molecule in the gas phase. It is the difference between the ground state energy of the ion formed and the energy of the original atom, molecule or ion.

ADIABATIC LAPSE RATE (ADIABATIC RATE): The rate of decrease in temperature with increase in altitude of an air parcel which is expanding slowly to a lower atmospheric pressure without exchange of heat. For a descending parcel it is the rate of increase in temperature with decrease in altitude. **Dry adiabatic lapse rate** is when unsaturated air loses about 1°C per 100 m. **Saturated adiabatic lapse rate** is when saturated air loses an average 0.5°C per 100 m.

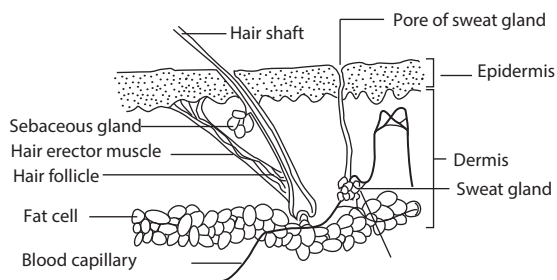
ADIABATIC TRANSITION-STATE THEORY: A form of transition-state theory in which the system is assumed to preserve its internal quantum states as it moves over the potential-energy surface.

ADIPIC ACID (HEXANE-1,6-DIOIC ACID; HEXANEDIOIC ACID): A colourless crystalline dicarboxylic acid with chemical formula $C_6H_{10}O_4$.

density 1.36 g/cm³, molar mass 146.14g/mol, melting point 152.1 °C (425 K) and boiling point 337.5 °C (611.65 K) which is a major constituent of rosin. It is derived from oxidation of various fats, slightly soluble in water but soluble in alcohol and acetone. It is used in the manufacture of nylon and polyurethane foams.



ADIPOCYTE (FAT CELL): Any of the cells of subcutaneous layer or the layer beneath the dermis of the skin in which fats are stored.



Adipocyte (Fat Cell)

ADIPOSE TISSUE (SUBCUTANEOUS TISSUE; ADIPOSE LAYER; FATTY LAYER): A layer of cells found chiefly below the skin and around major organs such as the kidneys, heart and the gut made up of fat and oils which act as energy reserve and also provide insulation against heat loss.

ADJACENCY MATRIX (CONNECTION MATRIX): An $n \times n$ square matrix with rows and columns corresponding to graph vertices, with a 1 or 0 in position of which the i, j^{th} entry is 1 if vertex i is adjacent to vertex j and 0 if not. For a simple graph with no self-loops, the adjacency matrix must have 0s on the diagonal. For an undirected graph, the adjacency matrix is symmetric.

ADJACENT: 1. The side of a right-angled triangle other than the hypotenuse that forms an arm for a given angle. 2. **Adjacent angles** are two angles that have a common arm or side and a common vertex (corner point) that do not overlap. In the diagram Fig. I, adjacent angles are A and B , B and D , C and D , A and C , E and F , F and H , G and H , G and E . Angles such as A and D and E and H are not adjacent as they do not share a side. 3. **Adjacent edges** are pair of edges in a graph joined by a common vertex. 4. In a right-angled triangle, the **adjacent sides**, also called **consecutive sides** are the sides between the angle θ and the right angle, as shown in Fig II. In a polygon, they are two sides having a common vertex; and in a polyhedron, two sides with a common edge. The hypotenuse of a right-angled triangle is the longest side, which is the one opposite the right angle. The opposite side is opposite the angle θ . 5. **Adjacent vertices** are a pair of vertices in a graph that have common edges or a pair of edges in a graph that have a common vertex.

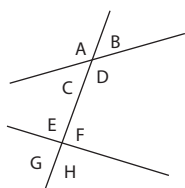


Fig I:
Adjacent angles

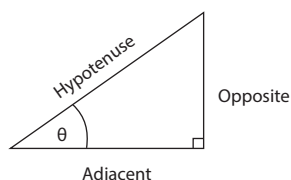


Fig II: Adjacent of a right-angled triangle

ADJOINT: 1. Relating to or denoting a function or quantity related to a given function or quantity by a particular process of transposition. 2. Also known as **Hermitian conjugate** and often denoted

A^* , A' or $A^\perp$, it is the matrix that is the transpose of the matrix of the complex conjugates of the entries of a given matrix. Alternatively, it is also known as **dual** and is the the operator A^* that is conjugate to a given linear operator A between normed spaces X and Y . 3. **Hilbert space adjoint** is the operator A^* that is conjugate to a given linear operator A . It is defined on Hilbert space by $\langle Ax, y \rangle = \langle x, A^*y \rangle$. 4. Also known as **adjugate**, the adjoint of a square matrix A , denoted by $\text{adj } A$, is the transpose of the matrix of cofactors of A . For $A = (a_{ij})$, A_{ij} denote the cofactor of the entry a_{ij} . The adjoint of a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\text{Adj. } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

ADJOINT EQUATION: A linear differential equation, usually derived from a given differential equation using integration by parts that any solution of one equation is an integrating factor of the other.

ADJUVANT: A non-antigenic substance that in combination with an antigen enhances antibody production by inducing an inflammatory response, which leads to a local influx of antibody-forming cells. It is often used in the therapeutic preparation of vaccines to enhance the recipient's immune response to a supplied antigen while keeping the injected foreign material at a minimum.

ADMISSIBLE HYPOTHESIS: A hypothesis that can be true or has not been over-ruled.

ADMISSIBLE VARIATION: The differentiable function v that vanishes at the endpoints of an interval so that for all scalars t , $x + tv$ agrees with x on the boundary.

ADMITTANCE: The reciprocal of impedance, which is a measure of the ability of an electrical current to flow. It is expressed as $1/Z$, where Z is the impedance. Its symbol is $\mathcal{Y}$ and its unit is siemens.

ADMIXTURE: 1. The proportion of a seed crop which is made up of weed, seeds or other crop species. 2. A material other than water, aggregates, lime or cement, used as an ingredient of concrete or mortar and added to the batch immediately before or during its mixing. It is used as a water repellent, colouring agent, retarder or accelerator to modify its setting rate.

ADNATE: The fusion of different parts of unlike parts or organs. An example is the fusion of the corolla to stamens.

ADOBE ACROBAT: A family of application software that allows the viewing, printing and management of portable document files (PDF). It is distributed as commercial software under proprietary license by Adobe Inc.

ADOBE FLASH: A proprietary application development platform developed by Adobe Systems. The primary focus of the Flash platform is the creation of Rich Internet Applications (RIA), which combine graphics, animation, video and sound for an enhanced Web user experience.

ADOBE INTEGRATED RUNTIME (AIR): Often referred to as **Adobe AIR**, it is an environment developed by Adobe that can be used on different operating systems. This allows Web developers to use their existing Web development skills, code and tools to build and deploy rich Web applications and a content desktop. Adobe AIR has a set of features that support building applications using HTML, JavaScript and Adobe Flash. It also allows developers to build Rich Internet Applications (RIA) across a variety of devices and platforms, including PCs, smartphones and TVs.

ADOLESCENCE: The period between the beginning of puberty and adulthood in the human lifecycle.

ADOPTIVE IMMUNITY (TRANSFER IMMUNITY): Immunity produced by transferring lymphoid cells such as T cells from an immune donor to a genetically identical recipient. An example is in cancer patients when adoptive immunisation is performed to boost immunity, following intensive radiotherapy, using T cells of the patient harvested prior to treatment.

ADP: Abbreviation for **adenosine diphosphate**, the nucleotide consisting of the pyrophosphate group, the pentose sugar ribose and the nucleobase adenine, having the molecular formula $C_{10}H_{15}N_5O_{10}P_2$.

ADRESSED: In biology, it refers to organs lying close to one another but not fused together.

ADRENAL: Relating to or on the kidneys.

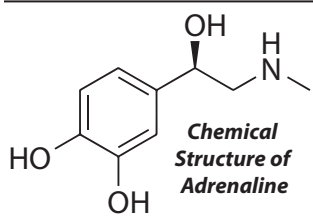
ADRENAL CORTEX: The outer layer of the adrenal gland located on top of each kidney which produces several steroid hormones such as mineralocorticoids which conserve sodium in the body; glucocorticoids which increase blood glucose levels; and gonadocorticoids which regulate sex hormones such as oestrogen.

ADRENAL GLANDS (SUPRARENAL GLANDS): A pair of two small triangular endocrine glands, weighing about 5 grammes in humans which are situated immediately above the kidneys. It lies anterior to the mammalian kidney. The inner portions of the adrenals secrete the hormone adrenaline, which causes increased heart beat on breathing, in response to conditions of stress.

ADRENAL MEDULLA: The inner reddish-brown part of the adrenal gland, in which adrenaline is produced.

ADRENALINE (EPINEPHRINE; ADRENALIN; 4,5-B-TRIHIDROXY-N-METHYLPHENETHYLAMINE):

A hormone and a neurotransmitter, one of a group of monoamines called catecholamines. Its molecular formula is $C_9H_{13}NO_3$ and molar mass 183.204g/mol. It is synthesised in some neurons of the central nervous system and in the chromaffin cells of the adrenal medulla from the amine acids phelalamine and tyrosine. It is released during fight or flight response of the sympathetic nervous system. Its functions include stimulation of glucose production and regulating heart beats.



ADRENERGIC: A term applied to nerve fibres that release adrenaline or noradrenaline when stimulated.

ADRENOCEPTOR (ADRENORECEPTOR; ADRENERGIC RECEPTOR): Any cell receptor that binds and is activated by the catecholamines adrenaline or noradrenaline. The two principal types are the alpha and beta adrenoceptors with differing sensitivities to certain drugs.

ADRIAN, EDGAR DOUGLAS (1889-1977): A British neurophysiologist, who became famous for his work on nerve impulses, establishing that messages are conveyed by changes in the frequency of the impulses. He held numerous positions, which include Professor of Physiology in the University of Cambridge (1937-1951); President of the Royal Society (1950-1955); President of the Royal Society of Medicine (1960-1962); and Chancellor of the University of Cambridge (1967-1975). He shared the 1932 Nobel Prize for Physiology with Sir Charles Sherrington for their work on the function of neurons. In 1955, he was made Baron Adrian of Cambridge.

ADSL: Abbreviation for asymmetric digital subscriber line, a form of digital subscriber line, loop or link, standard for transmitting data through existing copper telephone wires. technology that enables faster data transmission or broadband communication via the internet by using pre-existing copper telephone lines, while allowing simultaneous use of line for normal telephone calls by utilising a DSL filter or splitter.

ADSORBED WATER: 1. Water held in a soil mass or rock by electrochemical forces and having physical properties that are different from absorbed water or chemically combined water at the same temperature and pressure. 2. Water that is held on the surface of materials by electrochemical forces. This water, such as that on the surfaces of aggregate in a concrete mix, has a higher density and thus different physical properties from those of the free water in the mix.

ADSORPTION: The process by which a substance, especially a solid, called the **adsorbent** attracts to its surface very thin layer of molecules of gases, liquids, or solutes, called the **adsorbate** with which it is in contact. For example, activated charcoal adsorbs coloured impurities in a solution during recrystallisation; and iron adsorbs oxygen to its surface, the oxygen being an **adsorbate**. There are two types of adsorption: **physical adsorption**, which depends on van der Waals forces of attraction between molecules; and **chemical adsorption** often called **chemisorptions**, in which the molecules are held to the surface by chemical forces specific to the chemicals involved. In terms of soil, it is the adhesion of ions such as potassium (K^+) or calcium (Ca^{++}) or molecules to the surface of soil particles.

ADSORPTION COMPLEX: The group of substances in the soil capable of adsorbing other materials. Organic and inorganic colloidal substances form the greater part of the adsorption complex. The noncolloidal materials, such as silt and sand, exhibit adsorption to a much lesser extent than the colloidal materials.

ADSORPTION INDICATOR: A type of indicator used in titration reactions that involve precipitation. The yellow dye fluorescein is a common example, used for the reaction

$NaCl_{(aq)} + AgNO_{3(aq)} \rightarrow AgCl_{(s)} + NaNO_{3(aq)}$. As silver nitrate solution is added to the sodium chloride, silver chloride precipitates. As long as Cl^- ions are in excess, they adsorb on the precipitate particles. At the end point, no Cl^- ions are left in solution. Fluorescein ions are then adsorbed, giving a pink colour to the precipitate.

ADSORPTION ISOBAR: The function relating the amount, mass or volume or corresponding excess of substance adsorbed by a given amount of solid, to the temperature at constant pressure. It is also a graph showing how adsorption varies with some parameter, such as temperature, while holding pressure constant.

ADSORPTION ISOSTERE: The function relating the equilibrium pressure to the temperature at a constant value of the amount or excess amount, of substance adsorbed by a given amount of solid.

ADSORPTION ISOTHERM (SORPTION ISOTHERM): The mathematical relationship between the amount of a substance adsorbed onto a surface and its pressure, if a gas or its concentration is a solution, at a constant temperature. It is a curve obtained by plotting at constant temperature the quantity of adsorbate against the concentration of the substance in the original gas or solution.

ADULT: A human or other organism that has grown physically and reaches sexual maturity. In humans, it also includes full intellectual, emotional and psychological development. Examples are a fully matured person, animal, plant or other form of life.

ADVANCED BIOFUEL (SECOND GENERATION BIOFUEL): Fuel that can be derived from plant materials that do not have an alternative use as food. Such materials include agricultural or forestry residues or wastes, and dedicated energy crops, using a different technology. Second generation biofuel that has been developed as first generation biofuel is derived from the main crop and cannot produce enough biofuel without threatening food supplies and biodiversity. The end product second generation biofuel such as advanced bioethanol or advanced biodiesel is the same as that produced by first generation technology.

ADVANCED BUSINESS APPLICATION PROGRAMMING (ABAP): A programming language for developing applications for the SAP R/3 system, a widely-installed business application subsystem. The latest version, ABAP Objects, is object-oriented programming. SAP will run applications written using ABAP/4, the earlier ABAP version, as well as applications using ABAP Objects.

ADVANCED CONFIGURATION AND POWER INTERFACE (ACPI): An industry specification for the efficient handling of power consumption in desktop and mobile computers. ACPI specifies how a computer's basic input/output system, operating system and peripheral devices communicate with each other about power usage.

ACPI is a key component of Intel's "Instantly Available" technology. Microsoft refers to its support as "Always On."

ADVANCED ENCRYPTION STANDARD (AES): An encryption algorithm that uses block ciphers and supports 128-bit blocks and can operate with keys that are 128, 192 and 256 bits long for data encryption. It is used in the implementation of WIFI security application called WIFI Protected Access 2 (WAP 2).

ADVANCED GAS-COOLED REACTOR (AGR): A type of nuclear reactor widely used in Western Europe, which uses fuel of enriched uranium dioxide in stainless-steel cladding and has graphite as the moderator and carbon dioxide as coolant.

ADVANCED INTELLIGENT NETWORK (AIN): A telephone network architecture that separates service logic from switching equipment, allowing new services to be added without having to redesign switches to support new services. It encourages competition among service providers since it makes it easier for a provider to add services and it offers customers more service choices.

ADVANCED INTELLIGENT TAPE (AIT): A magnetic tape and drive system used for computer data storage and archiving. The tapes measure eight millimetres across. A helical scanning technique, similar to that used in Mammoth drives, optimises the data transfer rate and the storage capacity.

ADVANCED PROGRAM-TO-PROGRAM COMMUNICATION (APPC): A communication protocol and programming interface standard used by applications to communicate with each other over a network. APPC works in the application layer of the OSI model. It enables communication between programs from desktop computers and other devices such as mobile devices and midrange computers.

ADVANCED TECHNOLOGY ATTACHMENT PACKET INTERFACE (ATAPI): A type of hardware connected between hard disks, CD-ROMs and tape drives and other computer peripherals to a PC, to support the additional commands needed for controlling a CD-ROM player or tape backup so that the computer can use the IDE interface and controllers to control these relatively newer device types.

ADVANTAGE: A factor or combination of factors that gives a better chance of success. Advantage enables an organism to survive and reproduce better than other organisms in a population in a given environment.

ADVECTION: The transport of a substance or conserved property by a fluid due to the fluid's bulk motion. An example of advection is the transport of pollutants or silt in a river by bulk water flow downstream. In meteorology, it is the transport of some property of the atmosphere or ocean, such as heat, humidity or salinity. Advection is important for the formation of orographic clouds and the precipitation of water from clouds, as part of the hydrological cycle. **Advection fog** is the fog formed by warm air meeting a colder current or flowing over a cold surface.

ADVECTIVE DERIVATIVE (SUBSTANTIVE DERIVATIVE): A form of derivative which takes into account the fact that a quantity changes both as it moves to a different region and as the overall field is changing. It is most commonly used in fluid mechanics.

ADVENTITIOUS: Organs or other structures that arise in unusual position of a plant. For example, an **Adventitious Bud** is a bud that develops from unusual parts such as the root tuber of sweet potato whiles an **Adventitious Root** is a root developing in an unusual position and does not develop from the radicle. An example of an adventitious root is roots growing from the stem of maize plant. Another example is the banyan (*Ficus benghalensis*), where roots grow out of the branches forming, supporting pillars. Adventitious root system occurs commonly in monocots and pteridophytes, but also in many dicots, such as clover (*Trifolium*), ivy (*Hedera*), strawberry (*Fragaria*) and willow (*Salix*). Adventitious roots and buds are very important when propagating plants via cuttings, layering and tissue culture. These roots often grow from leaf and/or stem cuttings and can develop with or without the assistance of growth hormones. Adventitious plant buds can be attached to a variety of root stock via

grafting and in some plants, such as raspberries, apples and cabbage are a means of vegetative propagation.

ADVENTIVE: Something introduced and beginning to spread in the new region; not native. **Adventive species** is a species that is not native to and not fully established in a new habitat or environment and has arrived in a new habitat from somewhere else by means such as through imported plants.

ADVENTIVE CONE (LATERAL CONE; PARASITIC CONE): A secondary cone that is on the flank of and subsidiary to a larger volcano. It is fed by conduits branching from the main feeder or directly from the magma chamber.

ADVENTIVE EMBRYONY (SPOROPHYTIC APOMIXIS; ADVENTITIOUS EMBRYOGENESIS): A form of apomixis in which the embryo forms in some position other than within an embryo sac or embryonic development without fertilisation from cells that are not egg cells. Adventive embryos may develop from somatic cells of the nucellus or chalaza (apospory) and may occur together with a normal zygotic embryo. One seed can contain several embryos, as is often the case in Citrus seeds. Alternatively, an embryo may develop from the unreduced egg cell.

ADVERSE CLIMATIC EVENT: Weather conditions, such as frost, storm and hail, ice, heavy rain or severe drought, which can be considered as a natural disaster.

ADVERSE EFFECT: In biology, it is the change in biochemistry, morphology, physiology, growth, development or lifespan of an organism which results in impairment of functional capacity or impairment of capacity to compensate for additional stress or increase in susceptibility to other environmental influences. In medicine, an **adverse effect** or **side effect** is a harmful and undesired effect resulting from a medication or other intervention such as surgery.

ADWARE: A software program that is designed to run once a web page has been accessed. This is usually in the form of banner or popup advertisements. **Spyware** is a type of adware designed to be installed on a system without the operator's consent or knowledge and used to monitor surfing habits so that their software can deliver better targeted advertisements. **Malware** is a type of adware designed to monitor keyboard keystrokes so that the author of the software can gain access to password protected accounts.

AECIOSPORE (AECIDIOSPORE): One of the five types of spores formed by rust fungi in a sorus called aecium. Each spore has two nuclei and is part of a chain of spores. The aeciospores cannot again infect barberry but will germinate on certain grass or cereal leaves where the subsequent infection shortly gives rise to uredospores.

AEDES: A mosquito, with the botanical name, *Aedes aegypti*, which is native to the tropics and subtropics and can transmit diseases such as yellow fever and dengue fever.

AEOLIAN (EOLIAN; AEOLIAN SOIL): Sediments carried, formed, eroded or deposited by the wind or soil that is transported from one place to another by the wind.

AEON (EON): 1. A large division of geological time consisting of several eras. 2. A term for a unit of geologic time, which is equal to a thousand million (10^9) years.

AERATION (AERIFICATION): The act or process of circulating air through a medium. Some examples of aeration include the following: blowing compressed air into a reagent tank to provide oxygen to promote bacterial action in the treatment of sewage. Similarly, one can add carbon dioxide, for example, to a liquid under pressure. Aeration is also used in the fermentation in soft drinks industries, as well as in the manufacture of penicillin and other antibiotic. Soil aeration is the process by which air in the soil is replaced by air from the atmosphere. Soil can be aerated by digging it or by the burrowing activity of soil living organisms. In a well-aerated soil, the soil air is similar in composition to the atmosphere above the soil. Poorly aerated soils usually contain a much higher percentage of carbon dioxide and a correspondingly lower percentage of oxygen than the atmosphere.

The rate of aeration depends largely on the volume and continuity of pores in the soil. **Aerate** is to circulate air through a medium.

AERENCHYMA: A specialised parenchyma plant tissue with numerous air-filled spaces between the cells. It occurs in the stems and roots of many aquatic plants where it aids buoyancy and facilitates a low-resistance internal pathway for the exchange of gases such as oxygen and ethylene between the plant above the water and the submerged tissues.

AERIAL: 1. Existing or suspended in the air. 2. Also called **antenna**, it is wires, rods or dish for sending or receiving radio waves or signals.

AERIAL BEHAVIOUR: A type of communicative or playful behaviour, mostly exhibited by certain animals such as whales and dolphins when they surface above water to either jump, leap, or just flit across.

AERIAL OXIDATION: A reaction in which air is used to oxidise another substance. An example is the contact process for the manufacture of sulfuric acid and in the souring of wine.

AERIAL ROOT: A root that develops from the shoot of plants above the ground, absorbing water from humid air. It is produced mostly by plants which live as epiphytes in tropical forests. Aerial roots may be of enormous lengths and on reaching the soil produce a normal branching root system. Among the epiphytic orchids, aerial roots are short and hang loosely in the damp part of the forest and absorb rain and condensed moisture directly from the air with a special tissue, called **velamen**, surrounding the root.

AERO GENERATOR: An electricity generator powered by the wind. Aerogenerators are installed either in stand-alone mode or along with solar photovoltaic (SPV) systems to form a wind-solar hybrid system for decentralised power generation. The aerogenerator rotor starts moving at a wind speed of 9-12 km per hour. It produces optimum power at the rated wind speed of 40-45 km per hour. The power generated can be stored in a battery bank for use, when needed.

AEROBALLISTICS: The study of the motion of bodies such as of rockets, guided missiles and other projectiles through the atmosphere by applying the principles of both aerodynamics and ballistics.

AEROBE (AEROBIUM): Microorganism, especially certain bacteria and certain yeasts that survive only if there is a reasonable supply of oxygen. They require the oxygen for metabolism. This is in contrast to anaerobes which do not require oxygen for metabolism. Those organisms that are able to survive with or without the presence of oxygen are called **facultative anaerobes**.

AEROBIC: 1. Having molecular oxygen as a part of the environment. 2. Growing only in the presence of molecular oxygen, such as aerobic organisms. 3. Occurring only in the presence of molecular oxygen, as applied to certain chemical or biochemical processes such as aerobic decomposition. **Aerobic respiration** is the metabolic process occurring in the mitochondrion inside cells that absorbs free oxygen to break down food molecules such as glucose and releases energy. The oxygen is used as the terminal electron acceptor and the ultimate products are water and carbon dioxide. Approximately 700 kcals are released when a molecule of glucose is completely oxidised in aerobic respiration. It occurs in the mitochondrion inside cells.

AEROBIC ANOXYGENIC PHOTOSYNTHESIS: Photosynthetic process in which light energy is captured and stored as ATP (adenosine triphosphate). This takes place under aerobic conditions and does not result in the formation of oxygen and water is not used as an electron donor.

AEROBRAKING (ATMOSPHERIC BREAKING): The action of atmospheric drag (the friction the spacecraft experiences as it passes through the top layer of the atmosphere) in slowing down an object that is approaching a planet or some other body with an atmosphere. The friction of the atmosphere against the surface of the spacecraft slows down and lowers the craft's orbital altitude.

AEROCAPTURE: A technique used to reduce the velocity of a spacecraft, flying by a planet so as to place the spacecraft in orbit around the planet with a single atmospheric pass. It is similar to

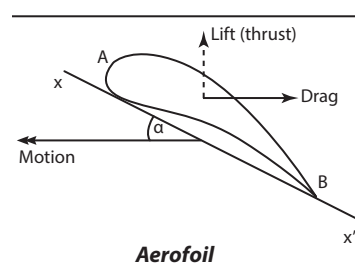
aerobraking but it occurs at higher speeds and digs deeper into a planet's atmosphere.

AERODYNAMIC DRAG: It is the force experienced by a body moving in the air as a result of the flow of air over the surface of the body. An example is an aeroplane flying in the atmosphere. The force is the summation of the aerodynamic drag, which is tangential to the flight path and the lift, which is perpendicular to flight path.

AERODYNAMICS: A branch of physics that studies the forces exerted by the air on other objects in motion such as the airflow around bodies of land vehicles, bullets, rockets and aircrafts moving at various speeds through the atmosphere. It is also involved in the design of automobiles, trains and ships and even stationary structures such as bridges and tall buildings, which must withstand high winds.

AEROEMBOLISM (BENDS): The presence of air in the blood vessels of the body. Direct entry of air into the veins may occur through trauma, cannulation, surgery or by the rapid decrease of the atmospheric pressure such as with flying to high altitude or sudden surfacing after deep diving. Symptoms include bubbles or swellings under the skin and extreme pain, usually spread over many parts of the body. Temporary paralysis of some parts of the body may also occur along with the possibilities of permanent damage or even death.

AEROFOIL: An object with a curved surface like an aeroplane wing or propeller blade that experiences an upward thrust when a stream of air moves across it.



AEROGEL: A microporous solid in which the dispersed phase is a gas or a low-density extremely light and porous transparent material that consists of more than 90% air. Examples are microporous silica, microporous glass and zeolites. Aerogels are produced from certain gels such as silica by heating the gel under pressure, which causes the liquid in the gel to become supercritical (in a state between a liquid and a gas) and lose its surface tension. Aerogels are used as drying agents, insulators and filters.

AEROMEDICINE: A branch of medicine that deals with physiological and psychological disorders associated with atmospheric or space flight. It has two sub-branches: (i) aviation medicine, concerned with flight in Earth's atmosphere and under at least normal Earth gravity; (ii) space medicine, which is concerned with flight beyond the atmosphere in which humans are typically exposed to a fraction of normal Earth gravity.

AEROMETER: An instrument for determining the weight and density of air or another gas. **Aerometric measurement** is the measurement of the temperature, pressure, air motion (velocity vectors) or other physical measurement to characterise an air mass.

AERONAUTICS: The science of travel through the Earth's atmosphere. This includes aerial navigation, aerodynamics and jet and rocket propulsion. The science of aeronautics also studies design, construction and operation of aircraft within the Earth's atmosphere. Studies involving travel in space is called **astronautics**.

AERONOMY: The study of the upper atmosphere of the Earth and other planets where the ionisation of gas takes place. It also includes the study of atmospheres around comets and satellites.

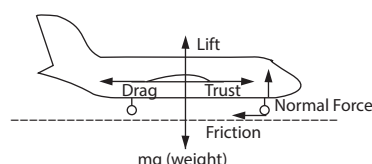
AEROPAUSE: A region of indeterminate limits in the upper atmosphere, in which the air is too thin for aircraft to operate properly. It is considered to be the boundary or transition layer between the denser portion of the atmosphere and space.

AEROPHORE: 1. Also known as **pneumatophores**, **breathing root** or **respiratory root**, it is a specialised erect respiratory root structure that protrudes some distance above soil. The surface of the aerophore is perforated by numerous lenticels. Because these

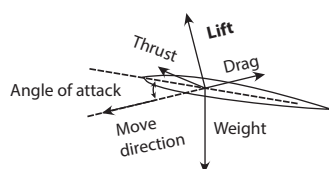
roots are exposed at least part of the day and not always submerged underwater, the root system can obtain oxygen that promotes gaseous exchange. It is formed in large numbers by certain plants such as the bald cypress aquatic plants and some mangrove species, growing in areas with waterlogged badly aerated soils. 2. A portable device for supplying air to the lungs of newborn babies who fail to breathe at birth or to a person in an environment, where there is no oxygen, such as mines and under water.

AEROPLANE (AIRPLANE): A heavier than air streamlined flying vehicle with fixed metallic wings or propellers and at least one engine. It is kept aloft by the upward thrust exerted by the passing air on its fixed wings and driven by propellers or jet propulsion. The modern aeroplane has five basic structural components: fuselage, wings, empennage or tail structures, power plant (propulsion system) and the undercarriage. The fuselage is the main body structure to which all other components are attached, containing the cockpit or flight deck, passenger compartment and cargo compartment. The wings are the most important lift-producing part of the aircraft. The wings generate most of the lift to hold the plane in the air. Wings vary in design depending upon the aircraft type and its purpose. The empennage or tail assembly provides stability and control for the aircraft. It is composed of two main parts: the vertical stabiliser (fin) to which the rudder is attached; and the horizontal stabiliser to which the elevators are attached. The stabilisers' job is to provide stability for the aircraft, to keep it flying straight. The vertical stabiliser keeps the nose of the plane from swinging from side to side. The horizontal stabiliser prevents an up-and-down motion of the nose, which is called **pitch**. The undercarriage or landing gear consists of struts, wheels and brakes. The landing gear can be fixed in place or retractable. Smaller, low-speed aeroplanes use propellers for the propulsion system. They use reciprocating engines or piston engines to turn a propeller to create thrust. Modern, high speed passenger and military aircrafts are powered by gas turbine engines or jet engines that create forward thrust by taking in a large amount of air and discharging it as a high-speed jet of gas. There are several types of jet engine: **ramjet, pulsejet, turbojet, turbofan**. A gas turbine engine sucks in air at the front with a fan. A compressor raises the pressure of the air and compresses the air. The compressed air is then sprayed with fuel and an electric spark lights the mixture. The burning gases expand and blast out through the nozzle, at the back of the engine. As the jets of gas shoot backward, the engine and the aircraft are thrust forward. Gas turbine engines have a great power-to-weight ratio compared to reciprocating engines and are smaller than their reciprocating counterparts of the same power. Most aeroplanes are flown by a pilot on board the aircraft, but some are designed to be remotely or computer-controlled.

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A free-body force of an aeroplane in a horizontal flight at constant speed



A free-body force of an aeroplane in an inclined plane

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AEROPONICS: The process of growing plants without soil. The plant roots hang in the air and are misted regularly with a nutrient solution. The system can grow many different plant species and cultivars including nursery stock, vegetables, bedding and houseplants. Unlike hydroponics, which uses water as growing medium and essential minerals to sustain plant growth, aeroponics is conducted without a growing medium.

AEROSOL: Liquid or solid particles, with a typical size between 0.01 and 10 mm suspended in a gas or in the atmosphere for at least several hours. Its source may be either natural or anthropogenic (man-made). Fog, smoke or dust originating from sources such as volcanoes, dust storms, and forest and grassland fires are common natural sources examples. Emissions from engine exhaust, burning of firewood, coal, agricultural waste, and biomass for energy are examples of anthropogenic sources. An **aerosol can** is a small container holding a substance that can be dispensed under pressure by a propellant as a spray. Aerosol cans are used to deliver antiperspirants, spray paint, hair spray, insecticides, deodorants, etc. Aerosol cans make people sick with heart and lung disease, and also play a major role in smog problems and contribute to global warming. In general, aerosols may influence climate directly through scattering and absorbing radiation, and indirectly through acting as condensation nuclei for cloud formation or modifying the optical properties and lifetime of clouds.

AEROSPACE: 1. The Earth's atmosphere and the space beyond it. It is considered as a zone for the flight of aircraft and spacecraft. It extends from the Earth's surface outward through the atmosphere to the farthest part of the observable universe. 2. The industry associated with air and space travel, including research and manufacturing toward these aims. Examples are National Aeronautics and Space Administration (NASA) in the United States, European Space Agency (ESA) in Europe and the China National Space Administration in China.

AEROTAXIS: A form of chemotaxis in which an organism, especially a bacterium, moves toward or away from air or oxygen.

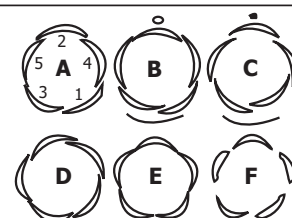
AEROTHERMODYNAMIC BORDER: An altitude at about 160 km, above which the atmosphere is so thin that an object moving through it at high speed generates virtually no surface heat.

AEROTOLERANT ANAEROBES: Microbes such as Streptococcus, Clostridium, and Actinomyces that cannot use oxygen for growth, but can survive in both, aerobic and anaerobic conditions, because they obtain their energy by fermentation.

AEROTROPISM: A behavioural response in aerobic organisms such as a plant in which it grows towards or away from a region of higher oxygen content. When the plant grows away from the presence of oxygen, it is known as **negative aerotropism**.

AESTIVAL: Appearing in the summer or flowering.

AESTIVATION (ESTIVATION): 1. A state of inactivity or dormancy characterised by reduced metabolic activity and usually encystment in response to high temperature and dry arid conditions. It occurs in some animals, notably lungfish, during the summer or dry season. 2. Also known as **prefoliation**, it is the arrangement of the parts of a flower bud especially of the sepals and petals before it has opened. It can be an important taxonomic diagnostic.



Examples of petal or sepal aestivation in flower buds.
A: quincuncial; B,C: cochleate;
D: contorted; E: valvate; F: open

AETIOLOGICAL

(ETIOLOGICAL): In botany, it is the abnormal growth observed when plants grow in darkness or severely reduced light. It is characterised by long, weak stems, small leaves and a pale yellowish colour owing to lack of chlorophyll.

AETIOLOGY (ETIOLOGY): The branch of medicine that studies the causes and origins of diseases or the set of factors that contributes to the occurrence of diseases.

AFFECTIVE COMPUTING: A human-computer interaction in which a device has the ability to detect and appropriately respond to its user's emotions and other stimuli. Facial expressions, posture, gestures, speech, the force or rhythm of key strokes and the temperature

changes of the hand on a mouse can all signify changes in the user's emotional state and these can all be detected and interpreted by a computer. A built-in camera captures images of the user and algorithms are used to process the data to yield meaningful information.

AFFERENT: A term used to describe nerves that carry impulses from the body toward the brain or spinal cord or blood vessels that carry blood to an organ. For example, afferent nerves also called sensory nerves carry impulses towards the brain and the afferent arteriole of the nephron carry blood towards the glomerulus.

AFFILIATE PROGRAM: An automated electronic program that directs customers to another business or website and receives a percentage of the fees users pay if they sign up. This is typically done by including display advertisements on the websites of website users. Common types of affiliate programs include pay-per-click, pay-per-lead and pay-per-sale.

AFFINE: Of, pertain to or involving affine transformation or to the geometry of affine spaces.

AFFINE CHAIN BEHAVIOUR: Behaviour of a polymer network in which the junction points deform uniformly with the macroscopic deformation of the network.

AFFINE GEOMETRY: Geometry that preserves parallelism and the ratios between intervals on any line segment. In mathematics, affine geometry is the study of geometric properties which remain unchanged by affine transformations, i.e. non-singular linear transformations and translations.

AFFINE HULL: In mathematics, the affine hull as defined in a set is the set of all affine combinations of points in the set. This is equivalent to the intersection of all affine sets containing the set.

AFFINE PLANE: A two-dimensional affine geometry constructed over a finite field. For a field F of size n , the affine plane consists of the set of points that are ordered pairs of elements in F and a set of lines that are themselves a set of points. Adding a point at infinity and line at infinity allows a projective plane to be constructed from an affine plane. An affine plane of order n is a block design of the form $(n^2, n, 1)$. An affine plane of order n exists if and only if a projective plane of order n exists.

AFFINE SPAN: The smallest affine manifold containing a given subset of a vector space.

AFFINE SUBSPACE (AFFINE MANIFOLD; FLAT): A subset of a vector space, usually over the field of real numbers that contains all lines between points of the subset. In a vector space, when the origin of a linear subspace is translated into another point, the result is an affine subspace. A subset $U \subset V$ of a vector space V is an affine space if there exists a $u \in U$ such that $U - u = \{x - u | x \in U\}$ is a linear subspace of V . A non-trivial affine manifold in three-space must be a point, line or plane.

AFFINE TRANSFORMATION (AFFINITY): A function or transformation of a plane which preserves points, straight lines and parallel nature of lines as well as the ratios of their distances. In other words, it is a change of transformation that is a combination of single transformations such as translation or rotation or reflection on an axis.

AFFINELY INDEPENDENT SET: A minimal subset with a given affine span.

AFFINITY: 1. A chemical attraction that is a measure of the likelihood of a chemical reaction taking place between two substances. In other words, it is the force by which atoms are held together in chemical compounds. 2. A similarity in structure, such as in species or languages that may suggest a common origin. 3. Also known as **affine transformation**, it is a transformation of a plane, a function or transformation of a plane which preserves points, straight lines and parallel nature of lines as well as the ratios of their distances.

AFFINITY CHROMATOGRAPHY: A type of chromatography that is dependent on the affinity between specific molecules such as that between antigen and antibody, enzyme and substrate or receptor and ligand. It is used to separate and purify a biological molecule from a mixture, based on the attraction of the molecule of interest

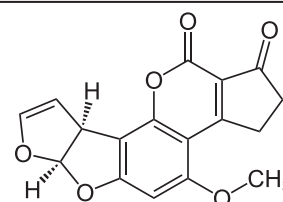
to a particular ligand which has been previously attached to a solid, inert substance. Some applications include nucleic acid purification, protein purification from cell free extracts and purification from blood.

AFFINITY OF REACTION: The negative partial derivative of Gibbs energy with respect to extent of reaction (ξ) at constant pressure and temperature. It is positive for spontaneous reactions. Its symbol is A .

AFFIRMATIVE: 1. In logic, of a categorical proposition, affirming a predicate of a subject (the predicate affirms something about the subject), for example, "All men are happy". 2. Not containing negation. 3. An affirmative proposition.

AFFORESTATION: Areas of land planted with trees to form a forest or planting of trees in an area that has not previously had trees.

AFLATOXIN: A toxin produced by some fungi strains of *Aspergillus flavus* and *Aspergillus parasiticus* in crops, especially peanuts, cotton seeds, various kinds of tree nuts, maize grains, wheat grains and various kinds of cereals. Aflatoxins are found most abundantly in foods derived directly from plants are the B and G types. B and G represent their blue or green fluorescence in ultra violet light. They are among the most potent mutagenic and carcinogenic compounds produced in nature.



Chemical structure of Aflatoxin B₁

AFRICAN LETTUCE (YANRIN; WILD LETTUCE; *Launaea taraxacifolia*): A perennial herb having an erect woody stem and a creeping root system found in West Africa. It has a woody rhizome and an erect stem and grows up to a height of about 1.3 metres tall. The plant can spread by means of underground rhizomes. It possesses rosette leaves at the base, which alternate and are sessile. The leaves are simple to pinnatifid and are spatulate or elliptical in shape. The plant produces branched synflorescences with each flower being ligulate. The petals of the flowers are fused, forming tubes and are golden yellow in colour. The anthers of the stamens are fused and the ovary is epigynous. It produces fruits whose shapes range from cylindrical to fusiform achene. The leaves are used either fresh as a salad or cooked in sauces or soup. It is also used as a treatment against vomiting, teeth pain, diabetes, etc.

AFRICAN MISTLETOE (*Tapinanthus dodoneifolius*): An evergreen partial parasitic shrub growing on trees such as cocoa and citrus. The stem is smooth, grey to brown and dense, with inconspicuous lenticels. The leaves are elliptical and oppositely arranged. It produces red flowers which resemble a bundle of vertical matches whose corolla tubes are red with white spots. The parasitic plant attaches itself to the host by means of a root-like structure called haustorium for support and absorption of water and mineral salts. It produces fruits which are berries and sticky seeds dispersed by birds to branches and stems of host plants where they germinate quickly. In Northern Nigeria, it is used by the Fulanis for treatment of many human and animal ailments that include stomach ache, diarrhoea, dysentery, wound and cancer. It has also got antimicrobial activities against certain multiple drug resistant bacterial and fungal isolates of farm animals.

AFTER-RIPENING: The process of further development undergone by the seeds of some plants before germination can occur. This is a case in which the embryo is not fully mature at the time of dispersal and must develop further before germination can take place.

AFTER-SHAFT FEATHER (AFTERFEATHER): A small secondary feather, growing from the underside of the base of the shaft of a contour feather. It resembles a smaller down feather. It is found in many birds and used to keep warm.

AFTERBIRTH: The special tissues associated with the development of a foetus in the uterus such as placenta, umbilical cord, foetal membrane and blood which are expelled or discharged from the womb or uterus soon after a mammalian foetus is born.

AFTERBURNING: Method of increasing the thrust of gas turbine (jet) aeroplane engine by spraying additional fuel into the hot exhaust duct between the turbojet and the tail pine where it ignites.

AFTERGLOW: 1. A broad glowing arc, sometimes seen high in the western sky at twilight. It is caused by fine particles of dust suspended in the high regions of the atmosphere. 2. The atmospheric glow that remains for a short time after sunset. 3. The glow in a material after the removal of an external source of fire to which it is exposed or after the cessation of natural or induced flames. 4. An effect seen in x-ray intensifying screens and fluoroscopic screens, in which small amounts of light are emitted by a phosphor after the stimulating radiation has been removed.

AFTERHEAT: Heat produced by a nuclear reactor after it has been shut down.

AFTERIMAGE (GHOST IMAGE): Persistence of an image on the retina of the eye after the object producing it has been removed. This leads to persistence of vision, a necessary phenomenon, which allows a rapid series of pictures to portray motion, which is the basis of animation and cinema.

AFTERMATH: The part of a grass that grows quickly after cutting for hay and which can be ready for another hay.

AFTERSHOCK: A smaller earthquake that occurs after the mainshock (largest earthquake in a sequence) of an earthquake sequence, in the same area of the mainshock. Aftershocks can continue over a period of weeks, months or years. In general, the larger the mainshock, the larger and more numerous the aftershocks and the longer they will continue. If an aftershock is larger than the main shock, the aftershock becomes the mainshock and the original main shock is known as a foreshock.

AFZELIA: A genus of about 13 species of trees in the family Leguminosae or Fabaceae (legumes) in the major group Angiosperms (Flowering plants), native to tropical Africa and Asia. The leaves are paripinnate with the leaflets opposite or subopposite. The inflorescence is a simple raceme or racemes arranged in panicles. Flowers are arranged spirally. The seeds are embedded in a white pith with a basal, brightly coloured, aril. *Azelia* species have durable, good quality wood and do not require treatment with preservatives. Some of the species include *Azelia africana*, an evergreen, small to fairly large tree, growing up to 40 m tall. It occurs from Senegal east to Uganda and south to Democratic Republic of Congo. Its wood is characterised by an excellent stability with little susceptibility to variations in humidity, small shrinkage rates during drying and a good natural durability. It is used for boat keels, stems and panels, for bridges, panelling, doors, frames, stairs, furniture, sporting goods, etc. The nitrogen-rich leaves enrich the soil and valued in agroforestry systems. *Azelia bipindensis* is a large tree growing from 15 - 40 metres tall, with a straight cylindrical bole that can be up to 20 metres tall distributed in Nigeria to Congo, south to Angola and Zambia, used for exterior joinery, flooring, heavy construction, furniture, vats and tanks. *Azelia xylocarpa* can reach up to 30 m tall with a trunk up to 2 m in diameter and found in Southeast Asia in deciduous forests. The wood is used for ornamental woodturning, pens, knife handles, carvings and musical instruments. *Azelia quanzensis* is a very attractive, medium-sized, deciduous tree, growing up to 35 m high, with a large spreading crown. It is distributed from Northern KwaZulu-Natal, through to Limpopo, Zimbabwe and other neighbouring countries and grows in low altitude woodland and dry forests, usually in deep sand. Its hard wood is light red-brown with a good grain. It has been used for building, making plywood, furniture, panelling and for flooring and used as railway sleepers.

AGAMOSPECIES: A species of organism that reproduces only by asexual means. Examples are many bacteria and some plants and fungi.

AGAMOSPERMY: A form of reproduction in flowering plants with the formation of embryos and seeds without fertilisation. It is a form of apomixes.

AGAR: Jellylike substance obtained from certain red algae, notably Gelidium species. It is used mainly in experiments as solid culture

medium for growing bacteria, fungi and micro-algae and also as a stabiliser and thickener in many food products. Agar is widely used because it is resistant to attack by almost all microorganisms and, as it does not melt until 100 °C, incubation of the selected culture can take place at high temperatures. When agar is used in a Petri dish it is known as an agar plate.

AGARICALES (GILLED MUSHROOMS): An order of fungi in the class Agaricomycetes with the phylum Basidiomycota, containing the mushrooms and toadstools. The best known families are Agaricaceae, in which the hymenium is borne on the surface of gills; and the Boletaceae, in which the hymenium lines fleshy pores. Both families contain many edible species, including the cultivated mushroom *Agaricus bisporus*.

AGAROSE: A carbohydrate polymer that is a component of agar. It is used in chromatography and electrophoresis.

AGATE: A bonded or cloudy type of chalcedony, a form of silica (SiO₂) that forms in rock cavities and has a pattern of concentrically arranged bands or layers that lie parallel to the cavity walls. It is used as ornamental stones and for art objects.

AGE OF EARTH: The time since the Earth emerged as a planet of the sun, formed from a giant cloud of gas and dust known as the solar nebula. It is estimated, based on evidence from radiometric age dating of meteorite material and consistent with the ages of the oldest-known terrestrial and lunar samples, to be about 4.54 billion (4.6 × 10⁹) years old. The solar nebula consisted mainly of hydrogen, the lightest element, but the nebula was also seeded with a smaller percentage of heavier elements, such as carbon and oxygen. The oldest minerals analysed to date are small crystals of zircon from the Jack Hills of Western Australia, estimated to be at least 4.404 (4.4 × 10⁹) billion years old.

AGE OF UNIVERSE: A time determined by the reciprocal value (1/H) of the Hubble, about 13.75 (13.75 × 10⁹) billion years. The actual age of the universe depends not only on Hubble's constant but also on how much the gravitational pull of the mass in the universe slows the universe's expansion. The age of the universe is usually calculated by assuming that the expansion of the universe can be described by the big-bang theory. Expansion of the universe measurements give accurate data to calculate the age of the universe.

AGE POLYETHISM: The phenomenon whereby an animal shows different forms of behaviour at different ages. This is best observed in colonies of social insects, where certain members of the colony may perform different tasks as they get older. An example is the worker honeybees which generally perform duties within the hive up to the age of 2-3 weeks and then switch to foraging outside the hive for the rest of their lives.

AGE STRUCTURE (POPULATION AGE STRUCTURE): The relative number of individuals of each age in a population. It is used in understanding and predicting population growth. Graphs called **age pyramids** are used to depict the age structure of populations. Age groups are plotted along the vertical axis (y-axis) and population size along the horizontal axis (x-axis). Each age group is broken into males and females, with a bar graph for each running horizontally to the left and to the right. Age pyramid has a relatively broad base, which indicates the number of children, while the peak indicates the increasing likelihood of death as people age. A young population has an age structure with a very broad base and a sharp narrow peak. An old population has a structure whose base is not much wider than the rest of the pyramid. See population age structure for diagram.

AGE-SPECIFIC DEATH RATE: The rate calculated within specific age group. For example, the age specific death rate among 40-50 year (yr.) olds can be calculated as:

$$\text{Death Rate}_{40-50} = \frac{\text{no. of deaths of 50 yr olds in a yr.}}{\text{average population size in 40 - 50 yr. olds in a yr.}}$$

AGEING (AGING): 1. In chemistry, it is the time-dependent physical and/or chemical changes of a material such as loss of water, growth

A

of crystals, recrystallisation, decrease of the specific surface, loss of coprecipitated substances, under defined environmental conditions, leading to improvement or deterioration of properties. 2. In biology, it is also known as senescence, which is the gradual changing processes in organisms, leading to degenerative changes and a decrease in functional capacity of all parts of the body including organs, tissues, skin, bones, heart, blood vessels, lungs, nerves and cells. Older organisms are increasingly vulnerable to a wide variety of age-related diseases, eventually ending in death.

AGENT: 1. A force or substance that produces an effect or change. For example, yeast is the raising agent in bread. 2. In computer systems management, a user that, for a particular interaction, has assumed an agent role. 3. An entity that represents one or more managed objects by the following: (i) emitting notifications regarding the objects; and (ii) handling requests from managers for management operations to modify or query the objects. 4. A system that assumes an agent role. 5. Software that acts on behalf of a user as it performs tasks within an application program. An agent may run on both the client and the server.

AGENT ORANGE: Also called **orange** from the identifying colour strip on its container, it is a very poisonous mixture of two weed-killers 2,4-D (2,4-dichlorophenoxyacetic acid) and 2,4,5-T (2,4,5-trichlorophenoxyacetic acid) with trace amounts of dioxin, used as an herbicide to destroy enemy crops and defoliate trees. It was widely used as a defoliant in the Vietnam War.

AGERATUM: A low-growing garden plant or a herb, with blue, white or purplish flowers in thick clusters that grow as an annual. Several species abound such as *Ageratum conoides*, *Ageratum houstonianum* and *Ageratum conyzoides*.

AGGLOMERATION: The process in which dispersed molecules or particles assemble rather than remain as isolated single molecules or particles.

AGGLUTINATION: 1. The process by which red blood cells clump together when the antigens on their surfaces react with complementary antibodies. 2. The coalescing of small particles that are suspended in solution. These larger masses are then (usually) precipitated. 3. An allergic reaction, in which cells become more compacted together to prevent foreign materials entering them, usually as a result of an antigen in the vicinity of the cells.

AGGLUTININ: A substance, such as an antibody found in blood plasma which causes antigens such as foreign red blood cells and bacteria to aggregate by binding to the antigen-binding sites of antibodies. Agglutinins work by clumping on particles causing the particles to change from fluid-like state to thickened-mass state. Each agglutinin acts on a specific antigen, removing it from the blood. An agglutinin is produced in large quantities after immunisation with its particular antigen. Agglutinins which agglutinate erythrocytes (red blood cells) are called **isohemagglutinins** and the blood group of an individual is determined by which of these are present in his or her blood. Group O blood contains isohemagglutinins anti-A and anti-B; group A contains anti-B; group B contains anti-A and group AB contains neither.

AGGLUTINOGEN: Any of the antigens present on the outer surface of red blood cells (erythrocytes) that stimulates the production of a particular agglutinin, such as an antibody. Antibodies in the plasma, known as agglutinins, react with the agglutinogens in the blood of an incompatible blood group.

AGGRADATION: Filling in or levelling by deposition.

AGGREGATE: 1. A mixture of minerals or rock fragments that resembles rock or any material, which can be cemented together to form a solid material. They are used in road building, laying foundation of buildings and in producing building materials as bricks, blocks and cement. 2. **Aggregates** are soil units containing mostly clay, silt and sand particles held together by a gel-type substance formed by organic matter. There are three main types of aggregates: (i) **platelike**, in which the aggregates are flat and mostly horizontal; (ii) **prismlike**, which forms vertical than horizontal dimension; and (iii) **blocklike**,

which is roughly equal in horizontal and vertical dimensions and either angular or rounded. 3. In botany, **aggregates** are random associations of animals with little or no internal organisation. They form in response to a single stimulus and disperse when the stimulus is removed. It is one of the three broad classes of social organisation. 4. In computer programming languages, a structured collection of data objects that form a data type.

AGGREGATE CULTURE: 1. The use of a solid material such as sand, gravel or marbles to grow plants with the additional use of a nutrient water-soluble solution. 2. A cell culture technique that has been used widely to provide a three-dimensional (3-D) environment for differentiating cells, particularly to study in vitro chondrogenesis.

AGGREGATE FRUIT: A fruit such as raspberry formed from one flower which has many free carpels.

AGGREGATOR: Any device that serves multiple other devices or users either with its own capabilities or by forwarding transmissions in a more concentrated and economical way. An aggregator may be able to handle up to 100 dial-up modem calls, support a certain number of Integrated Services Digital Network connections and support leased line and frame relay traffic while also functioning as a router.

AGGRESSIN: A toxic substance that is secreted by certain parasitic micro-organisms to inhibit the natural defence mechanisms of host organisms.

AGGRESSION: Behaviour used to frighten or injure another organism usually for purposes of gaining a territory, a mate or food. It often involves series of threats without having to engage in potentially dangerous physical contact. Examples include roaring by red deer, snarling by dogs, fluffing up of feathers by birds or raising of fins by some species of fish.

AGISTMENT: Money paid for grazing stock on land owned by another person. The owner of the land is responsible for the feeding and care of the livestock.

AGITATOR: The part of a root crop harvesting machine that shakes the earth off the crop after it has been lifted.

AGLET (AGILE APPLET): In computer technology, it refers to a small application program or applet with the capability to serve as a mobile agent of services in a computer network.

AGLYCON (AGLYCONE): The non-sugar compound remaining after replacement of the glycosyl group from a glycoside by a hydrogen atom.

AGNATHAN: A type of fishlike vertebrates such as lampreys and hagfish that lack jaws, have slimy scaleless skins, skeletons made of cartilage and well-developed sucking mouthparts, with horny teeth. They are found in marine and freshwater bodies.

AGONISTIC BEHAVIOUR: Any form of social behaviour, often associated with defence of a territory. It is characterised by aggression, including threat, attack, appeasement or avoidance. For example, a threat display by the defending individual is often met with an appeasement display from the intruder, thus avoiding harmful conflict.

AGONIST: 1. A drug, hormone, neurotransmitter or other signal molecule that binds to a receptor site and triggers an active response from a cell. 2. In biology, it is a muscle that contracts and causes a movement of a part of the body directly. When an agonist contracts, it causes its antagonist to relax. For example, the biceps in the front of the upper arm bends the elbow while the triceps lying behind the biceps straightens the arm.

AGONOMYCETALES (*Mycelia sterilia*): An order of imperfect fungi that do not produce spores, although some develop sclerotia. It contains about 200 species and 28 genera including *Rhizoc-tonia*, certain species of which cause-off disease and *Sclerotium*.

AGORAPHOBIA: An anxiety disorder characterised by extreme or abnormal fear of open spaces, public places or being in crowds such as stadiums or using public transportation. People with such conditions feel trapped, helpless, uncomfortable or embarrassed in such places.

AGOSTIC: A term used to describe structures in which a hydrogen atom is bonded to both a carbon atom and a metal atom. Examples are the interaction of a carbon-hydrogen bond (C–H bond) and an unsaturated transition-metal compound; or the bonding of a transition metal with silicon-hydrogen (Si– compounds).

AGOSTIC INTERACTION: The interaction of a coordinatively-unsaturated transition metal with a carbon-hydrogen bond (C–H bond), when the two electrons involved in the C–H bond enter the empty d-orbital of a transition metal, resulting in a three-centre two-electron bond. Agostic interactions are observed in alkyl, alkylidene and polyenyl ligands.

AGRANULAR CARBON: A monogranular or monolithic carbon material with homogeneous microstructure which does not exhibit any structural components distinguishable by optical microscopy.

AGRANULAR ENDOPLASMIC RETICULUM (SMOOTH ENDOPLASMIC RETICULUM): A part of endoplasmic reticulum that is tubular in form and lacks ribosomes. Its functions include lipid synthesis, carbohydrate metabolism, calcium concentration, drug detoxification and attachment of receptors on cell membrane proteins.

AGRANULOCYTE: Any white blood cell such as lymphocytes and monocytes with a large spherical nucleus and characterised by the absence of granules in its cytoplasm. Lymphocytes are much more common in the lymphatic system and include natural killer T-cells (NKT cells).

AGRANULOCYTOSIS (GRANULOCYTOPENIA): A rare blood disease in which the bone marrow produces inadequate numbers of granulocytes (white blood cells that eliminate bacteria). It is caused by some drugs and chemicals or by the exposure to radiation. Symptoms include high fever, severe throat infections and sepsis.

AGRAVIC: Of or pertaining to a state of no gravitation or weightlessness.

AGREE: When two functions have the same values for the same argument, we say that the functions agree on the set S if $f(x) = g(x)$ for all x in S . The sets of measurable functions that agree almost everywhere form equivalence classes under this relation.

AGRI-ENVIRONMENT-CLIMATE MEASURES: In the European Union (EU), they are practices, undertaken voluntarily by farmers, over a set period. Support may be provided through rural development programmes. The practices bring environmental benefits and /or help to mitigate and adapt to climate change. The payments compensate farmers for the extra costs that they incur and the income that they forego when they undertake these practices. The practices must go beyond a number of obligations which apply to farmers in any case – including (but not limited to) cross-compliance and relevant national legislation.

AGRIBUSINESS: Commercial firms that have developed with or stemmed out of agriculture. **Agribusiness management** is the human element that carries out a plan to meet goals and objectives in an agricultural science business.

AGRICULTURAL ACTIVITY: The 2013 reform of the Common Agricultural Policy which introduced the stipulation that, for the purpose of receiving direct payments, farmers shall have an agricultural activity which means the following: (i) the production, rearing or growing of agricultural products including harvesting, milking, breeding animals and keeping animals for farming purposes; or (ii) maintaining the agricultural area in a state which makes it suitable for grazing or cultivation without any particular preparatory action going beyond usual agricultural methods and machinery based on criteria to be defined by member states on the basis of a framework established by the Commission; or (iii) carrying out a minimum activity to be established by member states on agricultural areas naturally kept in a state suitable for grazing or cultivation.

AGRICULTURAL ADVISORY SERVICES (FARM ADVISORY SYSTEM, FAS): In agriculture, services that support and facilitate people engaged in agricultural production to improve the economic

and environmental performance of their holdings. These services provide tailor-made advice, taking into consideration the specificities of the farm, to contribute to the sustainability and climate friendliness of the holding, and to obtain information, skills and technologies to improve their livelihoods.

AGRICULTURAL AREA (AGRICULTURAL LAND): Any area taken up by arable land, permanent grassland or permanent crops, especially the rearing of livestock and production of crops, in order to produce food for humans.

AGRICULTURAL ECONOMICS: Management of agricultural resources, including farms and agribusinesses.

AGRICULTURAL ENGINEERING: Application of engineering principles, including the disciplines of animal biology, plant biology and mechanical, civil, electrical and chemical engineering principles, with a knowledge of agricultural principles in agricultural settings.

AGRICULTURAL EXTENSION: The process whereby the beneficial products of research are taken to farmers and the problems of farmers taken to research institutions for solution.

AGRICULTURAL HOLDING (HOLDING): An economic unit engaged in agricultural production activities under a single management and comprises all the livestock kept and all the land used, wholly or partly, for agricultural production purposes, without regard to title, legal form or size.

AGRICULTURAL LIME (AG LIME): Solid limestone comprising primarily of calcium carbonate (CaCO_3) as the active ingredient that is crushed to varying degrees of fineness and used as soil additive. Dolomite is also a source of lime. Its effects on soil include the following: (i) it lowers the acidity of the soil and alkalinity increases; (ii) it improves the physical, chemical and biological properties of the soil; (iii) it provides calcium, magnesium and other minerals to plants; (iv) it permits improved water penetration for acidic soils.

AGRICULTURAL SCIENCE (AGRISCIENCE): The application of scientific principles and new technologies to agriculture. It includes all jobs related to plants, animals and renewable natural resources. It is a multidisciplinary field of biology that includes the parts of the natural, exact, economic and social sciences that are used in the understanding and practice of agriculture. It includes animal science in its definition but often excludes veterinary science.

AGRICULTURE: Activities concerned with the production of plants and animals and the related supplies, services, mechanics, products, processing and marketing. An **agriculturist** is a person engaged in the occupation of agriculture.

AGRICULTURE EXTENSION SERVICE: A governmental agency with specialists that do research, and education for rural and urban producer and consumer groups. Agriculture extension can bring farmers new technologies and provide them with knowledge, skill and information which they do not have. Agriculture extension helps people to solve problems and encourages farmers to make decisions.

AGRIN: A component of the synaptic basal lamina that causes aggregation of acetylcholine receptors and acetylcholinesterase on the surface of muscle fibres of the neuromuscular junction. It is also important in the natural regeneration of damaged neuro-muscular junctions and seems to play a role in the differentiation of presynaptic nerve terminals. It occurs in embryonic nervous tissue and muscle, especially during early development. At least five different forms arise by alternative splicing and differ in their acetylcholine receptor clustering activity.

AGRISCIENCE MECHANICS (AGRICULTURAL SCIENCE MECHANICS): Design, operation, maintenance, service, selling and use of power units, machinery, equipment, structures and utilities in agricultural science.

AGRISCIENCE PROCESSING AND DISTRIBUTION: Industry which hauls, grades, processes, packages and markets commodities from production sources.

AGRISCIENCE SUPPLIES AND SERVICES: Businesses that sell supplies and agencies that provide services for people in agricultural science.

AGROBACTERIUM TUMEFACIENS (RHIZOBIUM RADIOBACTER): A gram-negative bacterium that has its name updated as *Rhizobium radiobacter* which is of significance for its use in genetic engineering because of its possession of tumour inducing plasmid (Ti plasmid) used in transferring cells of host plants. It also infects a wide range of plants and causes tumorous growth known as crown gall especially at the root-stem junction. It is found in soil and parasitic to plants unlike other *Rhizobium* species that dwell in root nodules of legumes and are mutualistic.

AGROCHEMICAL: A natural or artificially produced chemical used in modern, intensive agricultural industry. Examples are herbicides, insecticides, fungicides, fertilisers, pesticides, some animal feed additives and pharmaceuticals.

AGRONOMY: The science of soil management, land cultivation and crop production or the study of the cultivation of crops with various physical and biological factors, including soil management, tillage, crop rotation, breeding, weed control, climate-related issues and crop production. An **Agronomist** is a person who specialises in agronomy. Some of the various disciplines include soil microbiology, soil conservation, soil physics, soil fertility and plant nutrition, chemistry, biochemistry and mineralogy. Specialties in the crops area relate primarily to plant genetics and breeding, crop physiology and management, crop ecology, turf-grass management, seed production and physiology.

AHARONOV-BOHM EFFECT: A quantum mechanical effect that occurs in small disordered metallic conductors when a beam of electrons is split into two beams that travel in opposite directions around a region containing magnetic flux and are then recombined, whereby the intensity of the resulting beam oscillates periodically as the enclosed magnetic field is changed.

AIDS (ACQUIRED IMMUNE DEFICIENCY SYNDROME): The advanced stage or the final, life-threatening stage of infection with the human immunodeficiency viruses (HIV) infection. HIV is transmitted through body fluids such as blood, during blood transfusion or sharing syringes and contaminated fluid released during sexual contact. The virus destroys a type of white blood cells, which makes antibodies, thus weakening the body's immune responses. This leads to various unusual and severe infections, cancers and debilitating illness. An AIDS sufferer is open to diseases (opportunistic diseases) that uninfected people normally fight off without difficulty.

AIGRETTE: A plume or tuft of long and loose feathers, arranged as a head ornament used by breeding herons and egrets, during courtship displays.

AILERON: The hinged part of the wing of an aircraft used to control its balance while it is flying.

AIM: 1. The reason for a scientific investigation. 2. To intend to do something or plan to achieve an objective.

AIR: Colourless, odourless and tasteless mixture of gases including nitrogen (78.1%), oxygen (20.9%), argon (0.9%) and other gases (0.1%) surrounding the earth and making up the Earth's atmosphere.

AIR BAG (AIRBAG): 1. A safety device fitted in motor vehicles consisting of an inflatable plastic bag mounted under the dashboard or on the back of the front seat and also sometimes in the roof and doors. It inflates automatically in an accident and prevents the passengers from being thrown forwards. A person in the moving car has a momentum. This momentum must be brought to zero when the car suddenly stops as a result of collision. Since the person's momentum must be reduced to zero in a very short time, it implies that the force on the person's body must be very high (Newton's second law of Motion: $F = \Delta mv / \Delta t$). The air bag extends the interval of time over which the person's momentum is reduced to zero and as a result reduces the rate of change of momentum and hence, the force. 2. Any of a number of large inflatable bags comprising part of a spacecraft's landing system, enabling the craft to come to rest

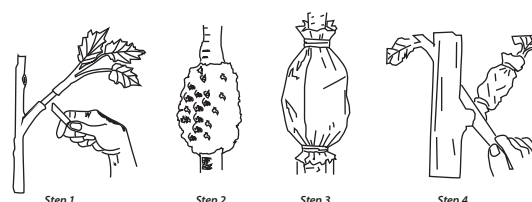
through a series of bounces to provide a softer, protective surface for a person to strike on in case of collision.

AIR BLADDER (SWIM BLADDER): A hydrostatic organ of thin-walled air sac, containing gas, usually oxygen, lying above the alimentary canal in bony fishes that enables them to adjust their density and thereby control their buoyancy and also as a resonating chamber to produce or receive sound; in certain primitive fishes it functions as a lung or respiratory aid. It is missing in some bottom-dwelling and deep-sea bony fish (teleosts) and in all cartilaginous fish such as sharks, skates and rays.

AIR CHAMBER: 1. Any of the air-filled cavities beneath the upper epidermis of the gametophyte thallus in many liverworts of the order Marchantiales. The air chambers increase the surface area of the internal photosynthetic cells and so facilitate gaseous exchange and also maintain a humid atmosphere around these cells, thus reducing transpiration. Connection of the air chambers with the external atmosphere is facilitated by pores in the upper surface of the thallus. 2. A chamber, especially in a hydraulic system, in which air elastically compresses and expands to regulate the flow of a fluid. 3. A vertical pipe stub which is sealed at the top and contains air in a water piping system near a valve or faucet. The entrapped air provides a cushion when the valve is closed suddenly, thereby eliminating the noise of water hammer.

AIR CONDITIONER (AIR CONDITIONING): A system for controlling, especially lowering, the temperature and humidity of the air circulating in an enclosed space such as a room or building.

AIR LAYERING (MARCOTTING): Plant propagation by girdling a plant stem, wrapping with peat moss and protecting it with plastic to encourage root formation.



Air layering (marcotting)

AIR MASS: A qualitative term to describe a widespread body of air with approximately uniform characteristics which had been identified at a given time over a particular region of the earth's surface. Sometimes an air mass is marked by inert tracers such as SF₆ which may be added to it. The composition of a given air mass undergoes alteration as it migrates, chemical changes occur, compounds are removed by dry and wet deposition and new impurities are added to the mass. An air mass forms when air rests over an area long enough to pick up the conditions of that area. It has distinct boundaries and may extend hundreds or thousands of miles horizontally and sometimes as high as the top of the troposphere. In astronomy, air mass is the optical path length through the Earth's atmosphere for light from a celestial source. As it passes through the atmosphere, light is attenuated by scattering and absorption. The more atmosphere through which it passes, the greater the attenuation. Consequently, celestial bodies at the horizon appear less bright than when at the zenith. The attenuation, known as atmospheric extinction, is described quantitatively by the Beer–Lambert–Bouguer law. Air mass normally indicates **relative air mass**, the path length relative to that at the zenith at sea level, so by definition, the sea-level air mass at the zenith is 1. Air mass increases as the angle between the source and the zenith increases, reaching a value of approximately 38 at the horizon. Air mass can be less than one at an elevation greater than sea level; however, most closed-form expressions for air mass do not include the effects of elevation, so adjustment must usually be accomplished by other means. In some fields, such as solar energy and photovoltaics,

air mass is indicated by the acronym AM. The value of the air mass is often given by appending its value to AM, so that AM1 indicates an air mass of 1, AM2 indicates an air mass of 2, etc. The region above the Earth's atmosphere, where there is no atmospheric attenuation of solar radiation, is considered to have "air mass zero" (AM0).

AIR PLANT (EPIPHYTE; AEROPHYTE): A non-parasitic plant usually growing on another plant for support, which lacks roots in the soil and lives above the ground surface. It obtains its nutrients from the air, rainwater and from organic debris on its support. Their aerial roots form a tangled network that catches falling leaves and other organic material, which provide a source of mineral salts for the plant. In addition, the mesh of roots and organic debris act as a sponge to collect and hold water. Examples of air plants are some orchids and ferns. Epiphytes growing on the leaves of another plant are termed **epiphyllous** while those growing on rock outcrops are called **lithophytes**. Epiphytes are also a group of plants in the Raunkiaer system of classification.

AIR POLLUTANT: A gaseous substance or aerosol which has been introduced into the air either by human activity or by natural processes in sufficient concentration to produce a measurable effect on humans, animals, vegetation or materials. Examples of air pollutants are sulfur dioxide (SO₂), nitrogen dioxide (NO₂), particulate matter (PM₁₀) or suspended particulates, carbon monoxide (CO) and ozone (O₃). Air pollution also leads to effects such as increased UV radiation due to the depletion of the ozone layer, acid rain, greenhouse effect and photochemical smog.

AIR POLLUTION: It is the release into the atmosphere substances such as particulates, biological molecules or various harmful gases such as carbon dioxide (CO₂), carbon monoxide (CO), nitrogen dioxide (NO₂) and sulphur dioxide (SO₂), which are harmful to all living organisms. Air pollution leads to effects such as increased UV radiation due to the depletion of the ozone layer, acid rain, greenhouse effect and photochemical smog.

AIR POLLUTION INDEX (AIR QUALITY INDEX; AQI): A mathematical combination of the concentrations of air pollutants which gives an approximate numerical measure of the quality of the air at a given time. It is a non-linear index that runs from 0 to 500 for reporting daily air quality to the public. The higher the value, the greater the level of air pollution and the greater the health concern. For example, a value of 50 represents good air quality with little potential to affect public health, while a value over 300 represents hazardous air quality. The API level is based on the level of five atmospheric pollutants, which are sulfur dioxide (SO₂), nitrogen dioxide (NO₂), particulate matter (PM₁₀) or suspended particulates, carbon monoxide (CO) and ozone (O₃) measured at the monitoring stations throughout each city.

AIR POROSITY: The portion of the bulk volume of soil that is filled with air at any given time or under a given condition such as a specified soil water potential. Usually, this portion is made up of large pores, that is those drained by a tension of less than about 100 em of water.

AIR PRESSURE: The force per unit area that the air exerts on any surface in contact with it, measured with a barometer. It varies from place to place and from hour to hour. Generally the standard value of air pressure at the earth's surface is about 100 kPa.

AIR QUALITY CHARACTERISTIC: One of the quantifiable properties relating to an air sample such as concentration of a constituent, wind speed and temperature. The quantity of air quality characteristic is the true value of the characteristic being investigated. It is recognised that in practice, this value can only be approximated by existing methods.

AIR RESISTANCE: The force acting against an object moving through air. It is caused by friction between the object and the air around it.

AIR RESOURCE MANAGEMENT: The detailed planning and the implementation of air pollution control programs. It is designed to give guidance and facilitation to manage air resource by mitigating

the air pollution in order to preserve the health and welfare of the people in the region, the plant and animal life, physical property, good visibility and other factors which determine the air quality and the maintenance of an aesthetically acceptable environment.

AIR SAC: 1. A bag-shaped part of an animal or plant, surrounded by a membrane. It may contain a gas or a liquid or an organ. 2. An air-filled space in the body of a bird that forms a connection between the lungs and bone cavities and aids in breathing and temperature regulation. 3. Another name for the **alveoli** in the lungs of mammals where exchange of gases takes place.

AIR SAMPLING NETWORK: A globally distributed network of air sampling stations from which air samples are collected for measurements of both pollutant concentrations and meteorological quantities such as wind speed, direction, rain fall and humidity. Measurement data are used to determine the extent and the nature of the air pollution and to identify long-term trends, seasonal variability and spatial distribution of carbon cycle gases.

AIR STREAM: A high altitude wind that blows at much the same speed and in much the same direction for days.

AIR-BREATHING ENGINE: An engine that draws in continuous stream of air via an inlet duct in order to burn fuel. The air is compressed, mixed with fuel, ignited and expelled as the exhaust gas. The maximum velocity of the air-breathing engine vehicle is limited to 2 km/s due to extreme temperature and dissociation of the exhaust gas. Examples of such engine include the ramjet, scramjet, turbojet, turbofan and pulse-jet. Some vehicles, such as space planes, may be fitted with both air-breathing and rocket engines for efficient operation both within and beyond the atmosphere.

AIR-COMPRESSOR: A machine that draws in air at atmospheric pressure, compresses it and then delivers it at higher pressure.

AIR-DRY: 1. To remove moisture from a substance or object by exposing it to, or using dry air. This can be used to prevent rusting and corrosion. 2. The state of dryness of a soil at equilibrium with the moisture content of the surrounding atmosphere. The moisture content depends on the relative humidity and the temperature of the surrounding atmosphere.

AIR-LIFT BIOREACTOR: A type of fermenter, for large-scale bioprocesses, for animal and plant cell culture in particular. In this type of bioreactor, the reaction medium is kept mixed and gassed by introduction of air or another gas (mixture) at the base of a column-like reactor. It is equipped either with a draught tube or an external tube which improves circulation and oxygen transfer.

AIR, SOIL ATMOSPHERE: The soil atmosphere, the gaseous phase of the soil, is the volume not occupied by solid or liquid.

AIRBORNE: 1. Transported by air. For example, diseases or seeds transported by air are said to be airborne. **Airborne disease** is ill-health or unhealthy condition caused by disease-causing agents that are transmitted through the air. 2. In flight or flying such as an air plane or the state of an aircraft from the moment it becomes entirely sustained by air or has gained flying speed until it lands.

AIRBORNE PARTICULATES (PARTICLE POLLUTION; PARTICULATE MATTER; PM): Total suspended particulate matter composed of a mixture of solid particles and liquid droplets in the atmosphere. Some particles, such as dust, dirt, soot or smoke, known as PM₁₀ are large or dark enough to be seen with the naked eye and come from sources such as dust from the desert or agricultural fields and dust kicked up on unpaved roads by vehicles. Fine particulates known as PM_{2.5} are so small, they can only be detected using an electron microscope and are generally emitted from activities such as industrial and residential combustion and from vehicle exhaust. They are also formed in the atmosphere when gases such as sulfur dioxide, nitrogen oxides and volatile organic compounds, emitted by combustion activities, are transformed by chemical reactions in the air. Chemical composition of particulates varies widely, depending on location and time of year.

AIRBORNE TRANSMISSION: The transmission of certain infectious bacteria or viruses that reside in its host's respiratory tract, which become aerosolised when an infected host, for example, a human sneezes, coughs, etc., and is transmitted to another host.

AIRCRAFT: Any machine or structure that can fly in the air and is regarded as a vehicle or carrier. For example, balloons, airships, aeroplane and helicopter.

AIRCRAFT CARRIER: A warship with a long flat deck designed to carry aircraft. Aircrafts can land on the deck or take off.

AIRCUSHION VEHICLE (ACV; HOVERCRAFT): A vehicle that can travel over land and water, supported by a cushion of slowly moving air that is produced by fans, enclosed within a flexible skirt beneath the hull, which is blown downwards.

AIRFOIL: The cross section of a body that is designed to develop an aerodynamic force by reaction with air. Examples of airfoils are cross sections of wings, propeller blades, windmill blades, compressor and turbine blades in a jet engine and hydrofoils on a high-speed ship. Airfoils that are manipulated to produce variable forces are called **control surfaces**.

AIRGLOW: Faint infrared and ultraviolet emissions in the Earth's atmosphere, produced by chemical reactions in the ionosphere. It is usually observed in the night sky.

AIRSHIP (DERIGIBLE): A special lighter-than-air, self-propelled aircraft whose buoyancy is provided by balloons containing hydrogen or helium. The balloon forms the streamlined envelope or hull and has below it propulsion system, steering mechanism and space for crew, passengers and/or cargo.

AIRWAYS: 1. A passageway from the nose or mouth through which air enters and leaves the lungs. 2. A tube-like device used to maintain adequate, unobstructed respiration, for example, during general anaesthesia. 3. The band of frequencies, taken collectively and used by radio broadcasting stations. 4. Any passage in a mine used for the purposes of ventilation. 5. Also known as **air corridors**, it is an invisible three-dimensional network of roads that zigzag within controlled airspace. Most are 14 kilometres wide. Within these corridors, aircraft fly separated by at least 305 metres of distance above and below them when they are at altitudes of up to 8,839 metres. Above this distance, aircraft are separated by at least 610 metres because controllers have more difficulty tracking aircraft at this altitude and flying at the speeds for that altitude.

AIRY DISC (AIRY PATTERN; AIRY DISK): A term, which is used in optics to describe the best focused spot of light that a perfect lens with a circular aperture can make, limited by the diffraction of light. It was named after George Airy (1801–1892), British mathematician and astronomer.

AIRY FUNCTION: Any of the functions $Ai(z)$ and $Bi(z)$, which are the special solutions of the second-order differential equation

$$\phi'' = t\phi = 0; \text{ which is } \phi(t) = \frac{1}{\pi} \int_0^\infty \cos\left(tx + \frac{x^3}{3}\right) dx.$$

There are four varieties of Airy functions: $Ai(z)$, $Bi(z)$, $Gi(z)$ and $Hi(z)$. $Ai(z)$ and $Bi(z)$ are used more frequently than $Gi(z)$ and $Hi(z)$. Airy functions commonly appear in physics, especially in optics, quantum mechanics, electromagnetics and radiative transfer.

AITKEN PARTICLES: Aerosol particles, which are less than 0.1 μm in diameter and are small enough to get into the human respiratory system. These particles are the most abundant in the atmosphere when classifying atmospheric particles by size. Their concentrations can be determined with the Aitken counter which measures total particle number density. Owing to their small size, Aitken particles contribute little to the total mass concentration of all aerosol particles; this is determined primarily by particles less than 0.1 μm in diameter.

AITKEN'S METHOD: A numerical method used to accelerate the convergence of an equation. Normally, the formula $X_{i+1} = f(X_i)$ is used to solve the equation. The method uses initial value and the first two values

obtained by the formula, to calculate a good approximation than the results of the iterative formula.

AKINETE: A large thick walled dormant cell that is non-motile with food reserves that serves as the reproductive cell in certain filamentous cyanobacteria.

AL-KHWATIZMI, MOHAMMED IBN MUSA (780 AD - 850 AD): A Muslim mathematician whose works introduced the modern numerals (the Hindu-Arabic numerals) to Europe. The title of his book *Kitab al-jabr Waa Muqabalah* provided the source for the term algebra. His name is the source of the term algorithm.

ALA: 1. One of the two lateral or wing-like petals of a papilionaceous corolla as in *Crotalaria* or of certain flowers in the pea family; or the flattened border of some stems, fruits and seeds. 2. A flat, winglike anatomical process or part, especially of bone.

ALABASTER: A naturally-occurring, fine-grained, white or light coloured, translucent form of gypsum with chemical formula, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. It is used for decorative carving and the production of dry lining boards for the building industry. It is also used as a soil conditioner for heavy, poorly draining, soils where the included sulfur also aids plant growth by reducing the alkalinity. A small quantity is also included in Portland cement to control the set time.

ALANINE (2-AMINOPROPANOIC ACID): Any of the two non-essential amino acids with the chemical formula $\text{C}_3\text{H}_7\text{NO}_2$, molecular mass 89.09 g mol^{-1} , density 1.424 g cm^{-3} and melting point 258°C (531 K). Alpha-alanine is one of the nonessential amino acids, found in most proteins and particularly abundant in fibroin, the protein in silk. It is used in research and as a dietary supplement. Beta-alanine is a naturally occurring amino acid not found in proteins. It is an important constituent of the vitamin pantothenic acid and is used in its synthesis, as well as in biochemical research, electroplating and organic synthesis.

ALAOGLU THEOREM (BANACH-ALAOGLU THEOREM): The theorem that in a dual Banach space the unit ball is weak star compact and, more generally, that the polar of a neighbourhood of the origin in a topological vector space is also weak star compact.

ALARM RESPONSE: A sudden reaction to any stimulus that potentially threatens the well-being of an organism. It involves the release of adrenaline and noradrenaline from the adrenal glands, triggered by increased nervous activity. These hormones enhance the effects of the sympathetic nervous system, such as increasing heart and breathing rates and promoting glycogen breakdown, which supplies large amounts of glucose for increased respiration and energy release. Examples include sound calls such as roaring in lions and behaviours such as raising of hair in dogs, raising of feathers in birds and discharge of pheromones in ants.

ALARM SIGNAL: 1. A signal of caution given by an animal to other members of its population in response to perceived danger, usually the approach of a predator. For example, the alarm signal of a rabbit involves thumping the ground and then flashing the white of its tail as it runs, which alerts nearby rabbits. 2. A signal of caution given by an electrical/electronic system.

ALATE: Any structure or organ in plants which is winged as in combretum or having wing-like extensions or margins; winged.

ALBEDO: 1. The fraction of the energy of electromagnetic radiation reflected from a body (or surface) relative to the energy incident upon it. In other words, it is the fraction of solar energy (shortwave radiation) reflected from the Earth back into space. It is a measure of the reflectivity of the earth's surface. For example, ice, especially with snow on top of it, has a high albedo as most sunlight hitting the surface bounces back towards space. Water is much more absorbent and less reflective. So, if there is a lot of water, more solar radiation is absorbed by the ocean than when ice dominates. The reflection of light from a surface is dependent on the wavelength of the light, the nature of the surface and its angle of incidence with the surface. 2. The probability that a neutron entering a body of material will be reflected back through the same surface as it entered.

ALBEDO FEATURE: A marking on the surface of a planet or a celestial object in space which is significantly brighter or darker than its surroundings. It may not be an actual geological or topographical feature.

ALBINISM: A condition in organisms, such as humans and mice, in which the organism's melanin pigmentation fails to develop, giving rise to characteristics such as light skin, white hair and pink eyes. Albinism results when the recessive alleles are inherited from both parents. It is caused by the absence of the enzyme tyrosinase, which is necessary for controlling melanin production. Albinism affects both sexes, since it affects autosomal chromosomes. Although albinism is rare in animals, any organism that makes melanin is susceptible. Examples of other organisms affected by albinism are dogs, elephants, deer, rats, snakes, some birds and plants. In animals albinism is mainly characterised by pink/red eyes and white colour (although white colour may not be the main deciding factor, as many animals are naturally white). A person or animal having albinism is called an albino. Some human albinos are sensitive to sun exposure and are susceptible to skin cancer.

ALBITE: A mineral, with the chemical formula $\text{NaAlSi}_3\text{O}_8$, Mohs scale hardness of 6 - 6.5 and specific gravity of 2.60 - 2.65 that occurs in igneous rocks. It often forms vitreous crystals of various colours. It is a variety of plagioclase, one of the three end-members (pure compounds) of the feldspar group.

ALBIZA: A common name for a group of small trees and shrubs of the mimosa sub-family, of the pea family and native to warm regions.

ALBUMEN (ALBUMIN): A soluble protein found in the core of egg white and also present in blood, milk and plants or any of a group of sulfur containing protein. Albumen helps regulate osmotic pressure and hence plasma volume, binds and transports fatty acids.

ALBUMINOUS CELLS: 1. Parenchyma cells, adjacent to the sieve cells with which they are associated physiologically and anatomically. They are found only in seedless vascular plants (ferns, club mosses, horsetails) and gymnosperms such as conifers and Ginkgo. They are distinguished by staining deeply with cytoplasmic stains. 2. Also known as **serous cell**, it is a cell, especially of the salivary gland, that secretes a watery or thin albuminous fluid, as opposed to a mucous cell.

ALBURNUM (SAPWOOD): The thin white layer between the bark and wood of a tree. It is the outer functional part of the secondary xylem cylinder, where sap flows, as distinct from the nonfunctional heartwood, where it does not flow.

ALCIAN BLUE: A copper phthalocyanin stain, with the chemical formula $\text{C}_{56}\text{H}_{58}\text{Cl}_{14}\text{CuN}_{16}\text{S}_4$ that enhances the preservation of cell surface materials. It is usually used during fixation with glutaraldehyde for transmission electron microscopy and also used as a stain for connective tissue mucins and a number of epithelial mucins and as a gelling agent for lubricating fluids.

ALCOGEL: A gel in which the swelling agent consists predominantly of an alcohol or a mixture of alcohols or a gel formed by the coagulation of an alcisol in which the liquid is alcohol. At the gel point, the mixture forms a rigid substance that can stand on its own. The liquid and solid parts of an alcogel occupy the same volume.

ALCOHOL: 1. Common name for ethanol, ethyl alcohol or grain alcohol. Its chemical formula is $\text{C}_2\text{H}_5\text{OH}$. 2. Any organic compound whose molecules have one or more hydroxyl groups attached to saturated carbon atoms. The general formula is $\text{C}_n\text{H}_{2n+1}\text{OH}$. Examples include methanol, propanol and glycerol. They react with organic acids to form esters. They are used in industries as solvents and for medical substances in pharmacy.

ALCOHOLIC FERMENTATION: A form of anaerobic respiration in which glucose is broken down to form alcohol and carbon dioxide. It is carried out by yeasts, plants, various fungi and by certain bacteria. Fermentation takes place outside the organism and is catalysed by

enzymes of the zymase complex. Fermentation usually stops due to cell poisoning when the alcohol level reaches about 15%. The process is central to the brewing, wine-making and baking industries.

ALCOHOLIC SOLUTION: A solution produced when a solute is dissolved in ethanol.

ALCOHOLOMETRY: Measurement of the percentage of alcohol in a liquid using a kind of hydrometer called alcoholometer. For proof spirits such as whisky, an instrument called Sikes's hydrometer is used to determine the strength; and for fermented beer, Atkins's hydrometer is used.

ALCYONARIA: A subclass of Actinozoa (Anthozoa). Members are colonial coelenterates, with most of them being sedentary and littoral. Examples are sponges.

ALDARIC ACIDS: Polyhydroxy dicarboxylic acids having the general formula $\text{HOC}(=\text{O})[\text{CH}(\text{OH})]_n\text{C}(=\text{O})\text{OH}$. They are formed by oxidation of both terminal groups of an aldose to carboxyl groups.

ALDEHYDES: Organic compounds with the general formula $\text{RC}(=\text{O})\text{H}$, in which a carbon atom is bonded to one hydrogen atom with a single covalent bond and oxygen atom with a double covalent bond and single-bonded to another chemical group such as methane, benzene, another hydrogen, etc. The carbon oxygen double bond part is known as a carbonyl group ($\text{C}=\text{O}$), often referred to as a formyl or methanoyl group. Aldehydes are formed by oxidation of primary alcohols. Further oxidation yields carboxylic acids. Examples of aldehydes are methanol (formaldehyde) and ethanol (acetaldehyde). Ethanol is a carbonyl group single-bonded to a hydrogen and single-bonded to a methane. They are reducing agents. Some other types of organic compounds containing carbonyl groups are ketones and carboxylic acids.

ALDICARB (2-METHYL-2-(METHYLTHIO)PROPANAL O-(N-METHYLCARBAMOYL)OXIME): A crystalline, white solid having slightly sulfurous smell with the chemical formula $\text{C}_7\text{H}_{14}\text{N}_2\text{S}$, molar mass 190.26 gmol^{-1} , density 1.195 g/cm^3 , melting point 100°C (373 K) and decomposes before boiling point. It is barely soluble in water but readily dissolves in organic solvents and it is used as systemic insecticide, acaricide and nematocide.

ALDIMINE: Any imine derived from aldehydes. The general structure is $\text{RCH}=\text{NR}$, where R may be any organyl group or H.

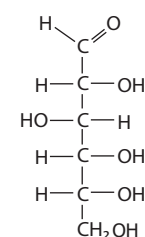
ALDITOLS: Formerly called glycol or sugar alcohol; they are acyclic polyols having the general formula $\text{HOCH}_2[\text{CH}(\text{OH})]_n\text{CH}_2\text{OH}$. They are derived from the acyclic form of a monosaccharide by reduction of the aldehyde or keto group to an alcoholic group.

ALDOHEXOSE: A hexose sugar with an aldehyde group attached to one end of the compound. Examples are glucose and galactose.

ALDOL: A compound (beta-hydroxy aldehyde or ketone) formed by the condensation of two molecules of aldehyde or ketone containing an aldehyde ($-\text{CHO}$) or ketone (RCOR) group and an alcohol ($-\text{OH}$) group, hence ald-ol reaction. Aldol reactions are catalysed by acids or bases. They are used in manufacturing rubber age resistors, accelerators and vulcanisers.

ALDONIC ACIDS: Polyhydroxy acids such as D-gluconic acid and D-gluconolactone having the general formula $\text{HOCH}_2[\text{CH}(\text{OH})]_n\text{C}(=\text{O})\text{OH}$. They are derived from an aldose by oxidation of the aldehydic group.

ALDOSE: Type of monosaccharide or simple sugar such as glucose and fructose with the chemical formula $\text{C}_n(\text{H}_2\text{O})_n$ and whose molecule

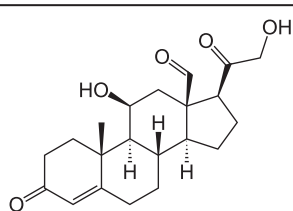


**Chemical Structure of Glucose
(an Aldohexose)**

contains an aldehyde (-CHO) group and one or more alcohol (-OH) groups.

A

ALDOSTERONE: A hormone secreted by the cortex of the adrenal glands that controls the reabsorption of sodium in the renal tubule of the nephron. Aldosterone acts directly on the kidney to decrease the rate of sodium-ion excretion with accompanying retention of water and to increase the rate of potassium-ion excretion.



Chemical structure of Aldosterone

ALDOXIMES: Compounds formed by reaction between hydroxylamine and aldehyde $RCHO + H_2NOH \rightarrow RCH:NOH + H_2O$. If R is an aliphatic group, the aldoxime is generally a liquid or low melting solid. If R is aromatic, the aldoxime is a crystalline solid. Aldoximes are useful in qualitative analysis of aldehydes.

ALDRIN: A white crystalline water-insoluble organo-chlorine pesticide with the chemical formula $C_{12}H_8Cl_6$. It is soluble in solvents such as acetone, benzene, ethanol and xylene and stable in strong alkali. It is used as soil insecticide mainly against termites. Aldrin is a persistent organic pollutant, which has been banned.

ALEPH (ALEPH NUMBERS): In set theory, they are a sequence of infinite numbers used to represent the cardinality or size of infinite sets. They are represented by the symbol $\aleph$, the first Hebrew letter.

Aleph-null, also known as **aleph-nought** is the smallest infinite cardinal number and is a set of cardinal numbers that can be put into a bijection or one-to-one correspondence between it and the natural numbers, which implies that it is countably infinite; the sets are countable or denumerable. It is represented by $\aleph_0$. Aleph-one, with the symbol $\aleph_1$, is the cardinality of the set of all countable ordinal numbers, called ω_1 or Ω . It is an ordinal number larger than all countable ones; therefore it is an uncountable set.

ALEPH-NULL (ALEPH-NOUGHT; ALEPH-ZERO): It is the smallest cardinal number or a set of cardinal numbers that can be put into one-to-one relation. These sets are said to countable or denumerable. It is denoted by $\aleph_0$.

ALERT: 1. In computer science, a message sent to a management services focal point in a network to identify a problem or an impending problem. 2. In SNA/MS (Systems Network Architecture/Microsoft System) management services, it is a high priority event that warrants immediate attention.

ALETHIC: 1. In logic, it is used to represent such philosophical concepts such as truth, necessity, possibility and contingency. Alethic modality, expresses modalities of truth and falsehood, such as possibly true and necessarily true. 2. Designating the branch of modal logic that deals with the formalisation of these concepts.

ALEURONE: Granules of protein present in some plants, especially in seeds and grains.

ALEURONE LAYER (PROTEINACEOUS ENDO-SPERM LAYER): The single outer layer of living cells of endosperm as in maize and other grain species containing granules of protein which secretes enzymes into the endosperm to initiate digestion of stored food reserves in the endosperm during germination.

ALEUROPLAST: A colourless plastid, which stores protein.

ALEXANDER'S SUB-BASE THEOREM: The theorem put forward by James Waddle Alexander (1888-1971), US mathematician and topologist that if every open cover of a topological space, by elements of a sub-base has a finite sub-cover, then the space is compact. More formally, if S is a sub-base for the topology of a space X and any cover of X by elements of S has a finite sub-cover, then X is compact.

ALEXANDRITE: Rare gemstone variety of the mineral chrysoberyl (beryllium aluminate), which is green in daylight but appears red in artificial light.

ALEXANDROFF ONE-POINT COMPACTIFICATION (ONE-POINT COMPACTIFICATION): A compactification that adds a single point, designated ∞ , to a Hausdorff space and adds complements of closed subsets of the original space as neighbourhoods of this point. The real numbers can be compactified in this manner.

ALEXIN: A substance present in blood plasma, which upon combining with antiserum destroys bacteria.

ALFALFA: A perennial leguminous flowering crop (*Medicago sativa*) in the pea family, Fabaceae that is widely grown as fodder for livestock.

ALGA: A diverse group of simple plants that contain chlorophyll and live in aquatic, damp or moist habitats on land with relatively little differentiation of tissues and organs as compared to bryophytes and tracheophytes. The group includes prokaryotic and eukaryotic organisms with single-celled to multi-cellular forms. For example, *Spirogyra* and *Euglena* are single-celled algae while sea weeds such as *Fucus* and *Sargassum* are multi-cellular algae.

ALGAL BLOOM: A population explosion of algae and other phytoplankton, which is coloured green or red. It is caused, when sewage introduces nitrates and industries introduce phosphate from detergents and run-offs, which carry excess fertiliser into water bodies such as lakes and lagoons. The density of the algae may prevent light from passing to lower depths in the water body. Sustained algal blooms may lead to the eutrophication of a lake.

ALGEBRA: 1. A system of arithmetic that applies to any set of non-numerical symbols, usually letters, axioms and rules by which they are combined or operated upon. In other words, it is the branch of mathematics in which symbols, usually letters of alphabets, represent numbers or quantities and express general relationship that hold for all members of a specified set, for example $x + y = y + x$. 2. Also called **abstract algebra**, it is the study of structures that deal with algebraic systems rather than the usual number systems and includes groups, rings and fields. 3. Any formal calculus used to model and study the properties of the entities that are the intended interpretation of their symbols, such as the algebra of logic and the algebra of classes. 4. A Boolean algebra or sigma-algebra (σ -algebra) and in particular, the algebra of subsets or algebra of propositions. 5. Any formal system with only functions and constants, but no relations except possibly identity.

ALGEBRA OF PROPOSITIONS: The Boolean algebra that interprets propositional calculus.

ALGEBRA OF SETS (ALGEBRA OF SUBSETS; FIELD OF SETS): The properties and laws of sets, which includes the set-theoretic operations of union ($\cup$), intersection ($\cap$) and complementation and the relations of set equality and set inclusion. It also provides systematic procedures for evaluating expressions and performing calculations, involving these operations and relations. The binary operations of set union ($\cup$) and intersection ($\cap$) satisfy many identities. Several of these identities or laws have the following names:

(i) **Commutative laws:** $A \cup B = B \cup A$ and $A \cap B = B \cap A$;

(ii) **Associative laws:** $A \cup (B \cap C) = (A \cup B) \cap C$ and $(A \cap B) \cup C = A \cap (B \cup C)$.

(iii) **Distributive laws:** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. There are also some more important laws of set algebra. For example, Proposition 3 states that for any subsets A and B of a universal set U, the following identities hold: idempotent laws given by

(i) $A \cup A = A$ and (ii) $A \cap A = A$.

The de Morgan's laws are given by (i) $(A \cup B)' = A' \cap B'$ and

(ii) $(A \cap B)' = A' \cup B'$.

ALGEBRA OVER A FIELD: A ring that is also a vector space in which the scalars are members of a field that satisfies the condition that if $\mathbf{x}$ and $\mathbf{y}$ are any members of the ring and a and b are any scalars, then $(a\mathbf{x})(b\mathbf{y}) = (ab)(\mathbf{xy})$.

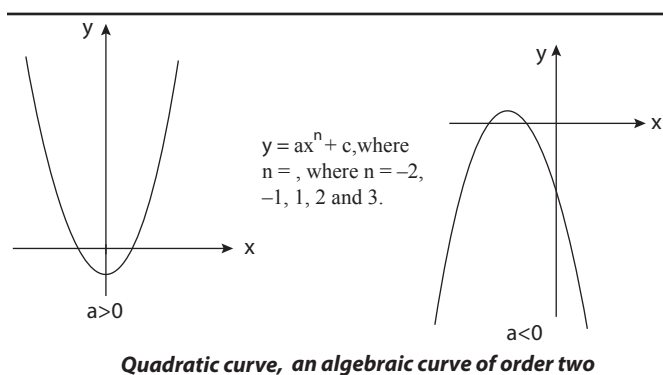
ALGEBRA TILES: Objects in the form of small squares, large squares, and rectangles which represent numbers and variables used to develop algebraic thinking in young people. Different colours can be used to represent positive and negative variables.

ALGEBRAIC: 1. Of, relating to or involving algebra or using only finite numbers, expressions and only the operations of addition, subtraction, multiplication, division and exponentiation with constant rational exponents. 2. Consisting of or relating to the roots of a polynomial equation with rational coefficients. Also, **algebraic over a field** is about the roots of a polynomial equation with coefficients that are members of the given field.

ALGEBRAIC ADDITION THEOREM: 1. Any theorem or identity that gives an addition formula for a function f in terms of a polynomial P in three complex variables such that $P(f(x), f(y), f(x+y)) = 0$ for all complex arguments. 2. A classical result in elliptic function theory that for every elliptic function there is an algebraic addition theorem. A meromorphic function has such an addition theorem if and only if it is rational, trigonometric or elliptic. For example, $\exp(x+y) = \exp(x)\exp(y)$ is an addition theorem for the exponential with $P(x, y, z) = xy = z$.

ALGEBRAIC CLOSURE: It is the inclusion of all the roots of polynomials with coefficients of a given set, when the given set is extended. An example is $x^2 + 1 = 0$, which has the smallest algebraic closed set with complex solution.

ALGEBRAIC CURVE: A plane curve satisfying a polynomial equation in two variables x and y , for example, $p(x, y) = 0$. These are functions that, in their most general form, can be written as a sum of polynomials in x multiplied by powers of y , equals to zero. A curve in the real plane that is algebraic is defined by $p(x, y) = 0$, where p is a real polynomial. For example, if $p = x^2 + y^2 - 1$, then $p(x, y) = 0$ defines a circle. Examples of algebraic curves are (i) **quadratic curve**, an algebraic curve of degree two and includes circle, ellipse, hyperbola, and parabola. In standard form, a quadratic function is written as $y = ax^2 + bx + c$. (ii) **Cubic curve**, an algebraic curve of degree three. Examples are the cissoid of Diocles, conchoid of de Sluze, folium of Descartes, Maclaurin trisectrix, Maltese cross curve, right strophoid, semicubical parabola and serpentine curve. (iii) **Quartic Curve**, an algebraic curve of degree four. Examples are ampersand curve, bean curve, bicorn, bicuspid curve, bifoliate, bifolium, bitangent-rich curve, bow, bullet nose, butterfly curve, capricornoid, cardioid and Cartesian ovals. (iv) **Quintic Curve**, an algebraic curve of degree five. Examples include the Burnside curve, butterfly catastrophe curve and stirrup curve. (v) **Sextic Curve**, an algebraic curve of degree six. Examples include the astroid, atriphtaloid, Cayley's sextic, cornoid, cycloid of Ceva, dumbbell curve, ellipse evolute, epicycloid, Freeth's nephroid and heart curve (first). (vi) **Octic Curve** is an algebraic curve of degree eight, for example, the pear curve. (vii) **Dodecic curve** is an algebraic curve of degree twelve, for example, the ranunculoid.



ALGEBRAIC EQUATION: An equation of the form $f(x) = 0$, where f is a polynomial. It shows the equality of two expressions formulated

by applying to a set of variables, the algebraic operations namely, addition, subtraction, multiplication, division, raising to power and extraction of root. An example is $4x^3 + 6x^2 - 7x + 4 = 0$.

ALGEBRAIC EXPRESSION: An expression containing a collection of numbers, variables and signs, positive or negative and arithmetic operations that must make mathematical and logical sense, for example, $5x^3 - 3y^2 + 2xy$. Two types exist: simple and compound. Simple algebraic expression has one term, called a monomial, for example, $3a$. Compound algebraic expression has two or more terms: the one with two terms, for example, $4x + 3y$ is a binomial; the one with three terms, for example, $2x^2 - 3x + 1$, $3m - 2n + 5z$, is a trinomial; and the one with more than three terms is a multinomial. Alternatively, an algebraic expression may not contain any individual quantifiers and is expressed only in terms of variables and intended to be interpreted as being true for all members of some set; for example, $x(y+z) = xy + xz$.

ALGEBRAIC EXTENSION (ALGEBRAIC EXTENSION FIELD): An extension field of a base field with the property that every element of the extension is algebraic over the base. Let L/K be an extension of fields. L/K is said to be an algebraic extension field if every element of L is algebraic over K . If L/K is not algebraic and contains transcendental elements then it is a transcendental extension field. An algebraic extension of the rationals is an algebraic number field. For example, the field extension R/Q , that is the field of real numbers as an extension of the field of rational numbers, is transcendental, while the field extensions C/R and $Q(\sqrt{2})/Q$ are algebraic, where C is the field of complex numbers.

ALGEBRAIC FUNCTION: A function that may be expressed in a finite number of terms, of only the elementary algebraic operations of addition, subtraction, multiplication, division and root extraction, for example, $f(x) = mx^3 + x^{\frac{1}{2}} - \frac{2}{x}$. They are distinguished from transcendental functions, such as the exponential, logarithmic and trigonometric functions, which can be expressed only by an infinite series. Another way of saying this is that the function $f(x)$ is algebraic if $y = f(x)$ is a solution of an equation of the form $p_n(x)y^n + \dots + p_1(x)y + p_0(x) = 0$, where $p_0(x), p_1(x), \dots, p_n(x)$ are polynomials in x . All polynomials are algebraic. A function that satisfies no such equation is transcendental. There are various types of algebraic functions. A rational function does not have any fractional powers of the variable or variables. Integral functions do not include the operation of division in any of their terms. A homogeneous function is one in which the terms are all of the same degree, i.e., the sum of the indices of the variables in each term is the same for every term. For example, $u^4 + u^3v + u^2v^2 + uv^3 + v^4$ is a rational, integral, homogeneous function of the fourth degree in u and v .

ALGEBRAIC GEOMETRY: The study of geometries that come from algebra, in particular, from rings. In classical algebraic geometry, the algebra is the ring of polynomials and the geometry is the set of zeros of polynomials, called an **algebraic variety**. For instance, the unit circle is the set of zeros of $x^2 + y^2 = 1$ is an algebraic variety, as in all the conic sections.

ALGEBRAIC INDEPENDENCE: Let $f_1, \dots, f_n$ be polynomials in $F[x_1, \dots, x_n]$; $\{f_1, \dots, f_n\}$ are algebraically independent if and only if there is no non-zero polynomial $A \in F[y_1, \dots, y_n]$ such that $A(f_1, \dots, f_n) = 0$. If the polynomials do not satisfy this, they are said to be algebraically dependent and A is their annihilating polynomial. For example, $\{x_1, x_2\}$ are algebraically independent, but $\{x_1, x_2, x_1^3 + x_2^2\}$ are not.

ALGEBRAIC INTEGER: The root of a polynomial with integer coefficients, where the coefficient of the highest power is 1. The polynomial is of the form $x^n + a_{n-1}x^{n-1} + \dots + a_n = 0$, where $a_1, \dots, a_n$ are integers; For example, $x_1 = 2 + \sqrt{3}$ is an algebraic integer, since

$x_1^2 + 4x_1 + 1 = 0$. The set of all algebraic integers is closed under addition and multiplication and therefore, a subring of complex numbers denoted by A .

ALGEBRAIC NUMBER: A number that is the root of an algebraic polynomial. For example, in the expression $x^2 = 2$, the square root of 2 ($\sqrt{2}$) is an algebraic number because it is a solution of the equation $x^2 = 2$. Another source defines it as any number that is the root of a polynomial equation with coefficients drawn from a given field, in particular the rationals; in that case, $\sqrt{2}$ is an algebraic number, but π is not. A number which is not algebraic is called transcendental. The algebraic numbers form a field.

ALGEBRAIC NUMBER FIELD: A subfield of the complex numbers that arises as a finite degree algebraic extension field of the rational field.

ALGEBRAIC NUMBER THEORY: That part of number theory that studies the arithmetic of algebraic number fields such as the ring of integers in the number field, the ideals and units in the ring of integers and the extent to which unique factorisation holds.

ALGEBRAIC OPERATION: Any of the operations of addition, subtraction, multiplication, division, raising to a power or extraction of roots. Operations that involve infinite series and functions such as $\log_x$ are not algebraic (they depend on the use of limits). The term algebraic is also applied to arithmetic operations carried out with due regard to sign. Thus, the algebraic sum of a and $-b$ is $a - b$, the algebraic product of $-a$ and $-b$ is ab , etc.

ALGEBRAIC STRUCTURE: A term used in algebra to describe concepts that are abstract and have elements with operations fulfilling certain given axioms. Examples include groups, rings and fields.

ALGEBRAIC SUM: The total of a set of quantities, with regard to their signs, as $+$ or $-$, according to the rules of addition in algebra. For example, the algebraic sum of -2 , 8 and -1 is 5 .

ALGEBRAIC SYSTEM: A set including the operations and relations expressed on that set.

ALGEBRAIC TOPOLOGY: The study of topological spaces, especially homology and homotopy theories, using abstract algebra. In other words, it is the branch of mathematics that uses tools from abstract algebra in the study of topological spaces.

ALGEBRAIC VARIETY: The set of solutions of a system of polynomial equations, over the real or complex numbers. In other words, it is a subset of an n -dimensional algebraic geometry, consisting of all points $(x_1, \dots, x_{n+1})$ satisfying a system of polynomial equations $P_\alpha(x_1, \dots, x_{n+1}) = 0, \alpha \in A$. It is the main ingredients of the study of algebraic geometry.

ALGEBRAICALLY SOLUBLE: Of a first-order ordinary differential equation, such that, where u and t are respectively the dependent and independent variables, either it can be solved for $p = du/dt$ and the resulting first-degree equation in p can be solved; or else it can be solved for u , giving an equation that can be differentiated to give a first-order equation in the dependent variable t and independent variable p ; or else it can be solved for t ; giving an equation that can be differentiated with respect to u to give a linear first-order equation in the independent variable p , after writing $dt/dp = 1/p$.

ALGIN (ALGINIC ACID): 1. A complex polysaccharide occurring in the cell walls of the brown algae. It is produced commercially from species of *Laminaria* and *Macrocystis* in the form of alginates which are used as stabilisers and texturing agents in the food industry. 2. A yellowish-white organic solid, with the chemical formula $(C_6H_8O_6)_n$.

It is a polymer of mannuronic acid in the pyranose ring form that occurs in brown seaweeds. It has many industrial applications due to the high level of viscosity of its dilute solutions.

ALGINATE: Salt or ester of alginic acid. It is obtained from brown seaweeds and used as thickener or emulsifier in textiles, paper, food products, cosmetics and pharmaceuticals.

ALGOL (ALGORITHMIC LANGUAGE): One of several high-level programming languages, used for programming mathematical and scientific data. Most programming languages are algorithmic languages.

ALGORITHM: 1. Procedure systematic steps for executing an operation procedure or series of steps, for example, by mechanical means, that can be used to effect a particular calculation or to manipulate data in a certain way, usually to solve a particular problem in mathematics and computer science. In elementary arithmetic, simple algorithms are those used in the extraction of square roots and in long division. In computer science, the term is commonly used in the context of programming. The word is derived from the name of the 9th-century mathematician, Ibn-Masa al-Khwarizmi. 2. A recursive definition enabling any member of an infinite sequence of terms to be generated by its repeated application.

ALGORITHMIC COMPLEXITY: The size of the smallest program that can produce a particular sequence of numbers. Algorithmic complexity quantifies how complex a system is in terms of the length of the shortest computer program or set of algorithms, needed to completely describe the system. In other words, it is concerned about how fast or slow particular algorithm performs. Regular patterns have low algorithmic complexity and random sequences have high algorithmic complexity.

ALIAS: In an internet, a name assigned to a server that makes the server independent of the name of its host machine. The alias must be defined in the domain name server.

ALIASING: The generation of a false frequency along with the correct one in frequency sampling, when converting analogue signals to digital in sound and image generation. For images, this produces a jagged edge or stair-step effect. For sound, it produces a buzz.

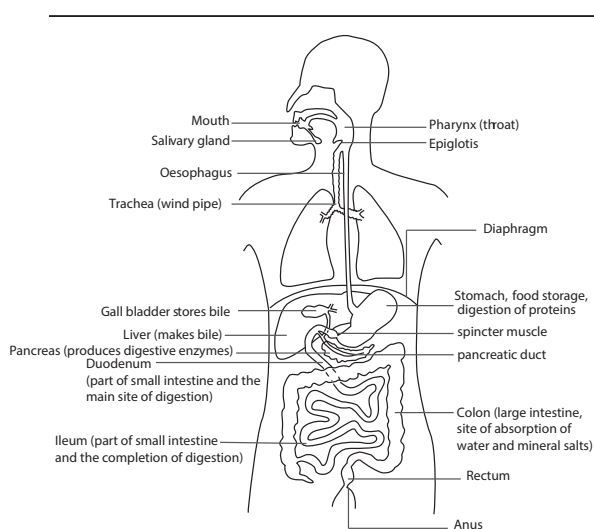
ALICYCLIC: A term used to describe a class of organic chemicals that possess properties of both aliphatic and cyclic compounds. **Alicyclic compounds** are organic compounds in which carbon atoms are linked to form one or more rings, which may be saturated or unsaturated, but may not be a benzenoid or other aromatic system. They are named according to the IUPAC system of nomenclature and are often prefixed by cyclo-. Examples are cyclopentane, cyclobutane cyclohexane.

ALIDADE: A device equipped with a telescopic sight and used for measuring optical heights and distances.

ALIEN (EXOTIC): An organism that is not native to a locality, having been moved there from its natural range by man or its agents. An example is rats introduced mainly by accidents in cargo or transport vessels.

ALIGNMENT: In bioinformatics, it is the process of matching up base sequences or amino acids sequences to reveal similarities and differences between them.

ALIMENTARY CANAL (DIGESTIVE TRACT): A tube-like organ in animals, extending from the mouth to the anus that is divided into a series of zones specialised for taking food into the body, digestion and absorption of food and for the elimination of indigestible material.



Pathway of food is the mouth, oesophagus, stomach, small intestine, large intestine, rectum and anus

Fig. I: The alimentary canal of a human

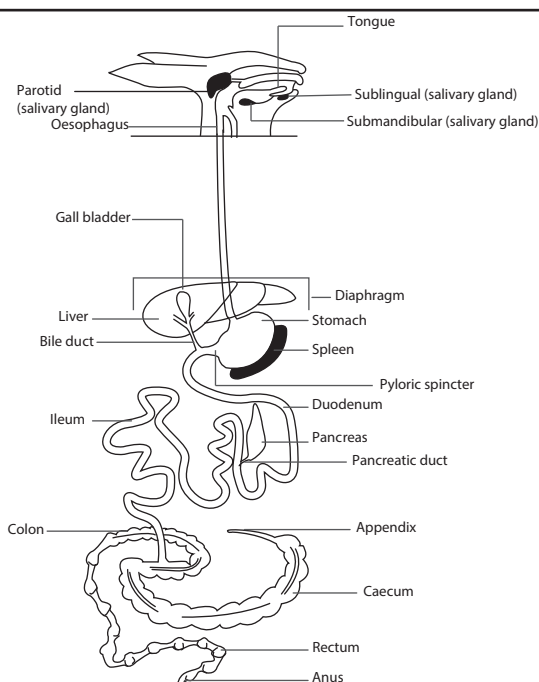


Fig II: The alimentary canal of a rabbit

ALIORELATIVE (IRREFLEXIVE): A relation R on a set S is aliorrelative or irreflexive provided that no element is related to itself or a relation that can never hold between itself and other relations. For example, the relation $<$ is an irreflexive relation on the set of natural numbers. An irreflexive relation is never reflexive on any subdomain whatever.

ALIPHATIC COMPOUND: Any organic chemical compound in which the carbon atoms are joined in straight chains, as in hexane or in branched chains, as in 2-dimethyl pentane. The chains can be straight or branched, can contain single or double or triple carbon-carbon bonds or be attached to other carbon atoms. Alkanes, alkenes, alkynes and their derivatives form this class.

ALIPHATIC HYDROCARBONS (OPEN CHAIN HYDROCARBONS): One of the two major groups into which hydrocarbons are classified.

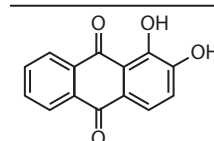
They have carbon atoms arranged in chains that have open ends and may be connected in single bonds (alkanes), double bonds (alkenes) and triple bonds (alkynes). Cyclic hydrocarbons are classified into alicyclic or aromatic hydrocarbons.

ALIQUANT PART: A number or an expression which is not a precise divisor of a given number or expression. For example, $x + 3$ is an aliquot of $x^2 + 9$ and 3 is an aliquant part of 10.

ALIQUOT: 1. In mathematics, it is a number or quantity that is an exact divisor of a given number or quantity. For example, 2 is an aliquot of 6. **Aliquot part** is any divisor of a number that isn't equal to the number itself. For example, the aliquot parts of 12 are 1, 2, 3, 4 and 6. 2. In chemistry, it is a portion of a known amount of a homogeneous solution, assumed to be taken with negligible sampling error. In simpler words, an aliquot is a solution to which a titrant is being added. The term is usually applied to fluids and usually used when the fractional part is an exact divisor of the whole. The term has also been used when the fractional part is not an exact divisor of the whole. For example, a 15 ml portion is an aliquant of 100 ml.

ALISMATIDAE (ALISMIDAE): A subclass of the monocotyledons containing aquatic or semiaquatic herbaceous flowering plants in the class Liliopsida, generally having apocarpous flowers and trinucleate pollen and lacking endosperm. They are characterised by unfused carpels, trinucleate pollen, absence of a starchy endosperm and absence of vessels in the stem and often also in the roots. There are usually two subsidiary cells associated with each stoma. Examples are *Ruppia* (ditch grasses), *Butomus umbellatus* (flowering rush) and Potamogetonaceae (pondweeds).

ALIZARIN: An orange-red dye, with the chemical formula $C_{14}H_8O_4$, which forms lakes when heavy metal salts are added to its alkaline solutions. It occurs naturally in madder (*Rubia tinctorum*), but is generally synthesised by heating anthraquinone with sodium hydroxide.



Chemical Structure of Alizarin

ALK: Resin obtained from *Pistacia terebinthus*, commonly known as the turpentine tree which grows chiefly in the Mediterranean region. It produces a yellow to green liquid, from which a type of turpentine called Chian or Chio turpentine is distilled.

ALKALI: A soluble base, which dissolves in water to give a pH greater than 7. It turns red litmus paper blue and neutralises acids. Examples are caustic soda or sodium hydroxide (NaOH) and soda ash or sodium carbonate.

ALKALI FLATS (SALT FLATS): Level, barren areas in dry regions covered with evaporites, mainly salts of the alkali metals and alkaline earth metals. They are formed by the evaporation of water in a depression and deposition of its fine sediment and dissolved minerals.

ALKALI METALS: Very reactive metals found in Group 1 (IA) of the Periodic Table of Elements. They are lithium, sodium, potassium, rubidium, caesium and francium. They react violently with water to form solutions, which are alkaline. They react readily with Group 17 (VIIA) elements such as fluorine, chlorine, bromine and iodine to form salts. Due to their reactivity they do not occur freely in nature and are stored in the laboratory in paraffin oil to prevent oxygen and water contact.

ALKALIMETRY (ALKIMETRY): Volumetric analysis using standard solutions of alkali to determine the amount of acid present in a solution or the specialised analytic use of acid-base titration to determine the concentration of a basic (alkaline) substance. The techniques used include titration and other analytical methods. An alkalimeter is an instrument used for measuring the concentration of alkalis in a solution.

ALKALINE: A term used to describe a solution with pH greater than 7.

ALKALINE BATTERY: A dry cell in which the electrolyte contains potassium hydroxide (KOH).

A

ALKALINE EARTH METALS: Metals found in Group 2 (IIA) of the Periodic Table. They are beryllium, magnesium, calcium, strontium, barium and radium. They are less reactive than the alkali metals, but like the alkali metals, produce alkaline solution when reacted with water.

ALKALINE PHOSPHATASE: An enzyme catalysing the hydrolysis of phosphoric esters under conditions of alkaline pH. In humans the blood levels of alkaline phosphatase are measured as part of the assessment of liver function. In the blood the enzyme also breaks down phosphates required for the mineralisation of the bone. There are two distinct forms of the enzyme, the alkaline phosphatase and bone alkaline phosphatase. It is found in bacteria, fungi and animals but not in higher plants.

ALKALINE SOIL (ALKALI SOIL): A type of clay soil such as a rendzina having a high degree of alkalinity (pH of 8.5 or higher). It occurs in arid and semi-arid regions. The natural cause of alkaline soil is the presence of soil minerals producing sodium carbonate (Na_2CO_3) and sodium bicarbonate (NaHCO_3) upon weathering. There are also many man-made causes such as using soft water containing relatively high proportion of sodium bicarbonates and less calcium and magnesium in irrigation. Alkaline soils are poor in nutrients, especially micronutrient and have iron deficiency.

ALKALINITY: The acid-base relationship of any solution that has a lower concentration of hydrogen ions or a higher concentration of hydroxide ions than pure water, which is an arbitrarily neutral standard with a pH of 7.0 at 25 °C. It can also be defined as a measure of the ability of a solution to neutralise acids to the equivalence point of carbonate or bicarbonate. It is equal to the stoichiometric (the relative quantities of reactants and products in chemical reactions) sum of the bases in solution.

ALKALIPHILES (ALKALOPHILES): Microbes, including some bacteria and archaea classified as extremophiles, that grow very well in highly alkaline environments, such as soda lakes and carbonate-rich soils, with pH values of between 9 - 11. Alkaliphiles cannot grow or grow only slowly at the near-neutral pH value of 6.5. They maintain a mildly alkaline intracellular pH of about 8 amid surroundings of much higher pH by continuously pumping hydrogen ions across their cell membranes into their cytoplasm. Examples include *Geoalkalibacter ferrihydriticus*, *Bacillus okhensis* and *Alkalibacterium iburiense*.

ALKALIZATION: The process of making alkaline in a solution. In soils, this results in the increase of the exchangeable sodium content.

ALKALOID: A group of organic compounds composed of carbon, hydrogen, nitrogen and usually oxygen that are physiologically active as poisons or drugs contained in some plants. Examples include cocaine, quinine, caffeine, morphine and nicotine.

ALKALOSIS: A pathological condition in which the hydrogen ion substance concentration of body fluids is below normal and hence the pH of blood rises above the reference interval. Excess alkali can make muscles overreact, causing them to go into cramp-like spasms.

ALKANAL: An aliphatic aldehyde or an organic compound containing a carbon atom connected to an oxygen atom by a double bond and to a hydrogen atom in a string open chain.

ALKANES: A group of saturated open-chain hydrocarbon containing only carbon-to-carbon or carbon-to-hydrogen single bonds and belonging to a series with general formula $\text{C}_n\text{H}_{2n+2}$ which is commonly known as paraffin. In nature they are found in natural gas and petroleum. They are chemically inert, inflammable and stable.

ALKANIUM ION: Any carbocation obtained by hydrogenation of an alkane and containing at least one pentacoordinate carbon atom. Examples are methanium ($^+\text{CH}_5$) and ethanium [C_2H_7] $^+$.

ALKANOL: An aliphatic alcohol similar in structure to the alkanes ($\text{C}_n\text{H}_{2n+2}$), except that one hydrogen has been replaced by an OH (hydroxyl) group. The three types are

Primary, Secondary and Tertiary Alkanols. **Primary Alkanols:**

the -OH is attached to a carbon atom which carries two

hydrogen atoms. **Secondary Alkanols:** the -OH is attached to a carbon atom which carries one hydrogen atom. **Tertiary**

Alkanols: the -OH is attached to a carbon atom which does not carry any hydrogen atom.

Unlike alkanes, alkenes and alkynes, alkanols are not hydrocarbons as, in addition to carbon and hydrogen atoms, they contain oxygen atoms.

Alkanols are flammable and burn in excess oxygen to form carbon dioxide and water and can be used to fuel cars. They react with sodium (Na) to produce sodium alkoxide and hydrogen gas.

ALKAPTONURIA (ALCAPTONURIA): An inherited metabolic disorder in which large amounts of homogentisic acid (alkapton) are excreted in the urine. This is caused by a deficiency of the enzyme homogentisic acid oxidase, which is required for the complete breakdown of the amino acids tyrosine and phenylalanine.

ALKENE PHOTOCYCLOADDITION: A photochemical process leading to formation of a cyclobutane upon addition of an electronically excited alkene to another ground-state alkene. When an electronically excited α,β -unsaturated carbonyl compound is added to a ground-state alkene, the process is called a **de Mayo reaction**.

ALKENE PHOTODIMERIZATION: A photochemical process, which results in the formation of a cyclobutane from two identical alkenes or from the photocyclo addition of an electronically excited alkene to a ground-state alkene.

ALKENE PHOTOREARRANGEMENT: A photochemical process, which results in the skeletal rearrangement of an alkene.

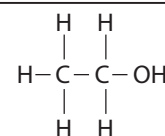
ALKENES (OLEFINS): Acyclic branched or unbranched hydrocarbons whose molecules contain one or more carbon-carbon double bonds having the general formula C_nH_{2n} . Examples are ethane and propene. Examples of acyclic branched or unbranched hydrocarbons having more than one double bond are alkadienes and alkatrienes.

ALKOXIDE (ALCOHOLATE): A compound formed by the reaction of an alcohol with an alkali metal by the replacement of the hydrogen atom of the hydroxyl group with a metal. Alkoxides are salt-like compounds containing the ion RO^- , where R is the organic substituent from the alcohol. An example is sodium methoxide (CH_3ONa), which is formed from methyl alcohol (CH_3OH) by the reaction with sodium.

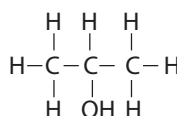
ALKYD: A synthetic resin, generally produced by copolymerising phthalic anhydride with glycerol, to give a brittle cross-linked polymer. The properties of such resins can be modified by adding monobasic acids or alcohols during the polymerisation. Alkyds can be moulded at high speed under low pressure and cured quickly. Alkyd resin is widely used as paint resins and other surface coating. They are also used in products where good insulation, high strength and good temperature, voltage and humidity stability are required, such as in car ignition systems.

ALKYD RESIN: A type of polyester resin used in paints and other surface coating. The original alkyd resins were made by copolymerising phthalic anhydride with glycerol, to give a brittle cross-linked polymer. The properties of such resins can be modified by adding monobasic acids or alcohols during the polymerisation.

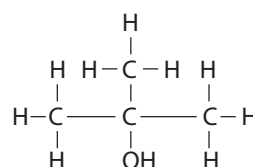
ALKYL GROUPS: Univalent groups derived from alkanes by removal of a hydrogen atom from any carbon atom $-\text{C}_n\text{H}_{2n+1}$. The groups



Primary alkanol



Secondary alkanol



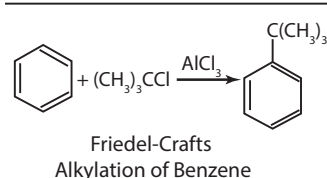
Tertiary alkanol

derived by removal of a hydrogen atom from a terminal carbon atom of unbranched alkanes form a subclass of normal alkyl (*n*-alkyl) groups $H(CH_2)_n$. The groups RCH_2 , R_2CH ($R \neq H$) and R_3C ($R \neq H$) are primary, secondary and tertiary alkyl groups, respectively.

ALKYL HALIDE (HALOALKANES; HALOGENOALKANES): An organic compound containing a halogen atom covalently bonded to an sp^3 hybridised carbon atom given the symbol R-X. They are prepared by direct halogenation of alkanes (except for the iodides), addition of hydrogen halides to alkenes or halogenation of alcohols by hydrogen halides, phosphorus (III) halides, etc. The halogen atom is readily replaced by other nucleophiles such as hydroxide or cyanide. Elimination of hydrogen halides (base-catalysed) yields alkenes. Examples of alkyl halides are chloroform, carbon tetrachloride and the chlorofluorocarbons (CFCs) which have been used in air conditioners, etc., as refrigerants. They are also used as solvents and as intermediates in chemical manufacture.

ALKYL RADICALS: Carbon-centred radicals derived formally by removal of one hydrogen atom from an alkane. An example is propyl (C_3H_7).

ALKYLATION: A chemical reaction that introduces an alkyl group into an organic molecule. The Friedel-Crafts reaction results in alkylation of aromatic compounds. In oil refining, it is defined as the process for converting light, gaseous olefins into high-octane benzene (petrol) components, which is the reverse process of cracking. It produces synthetic C_7 - C_8 alkylate, which is a premium blending stock for benzene (petrol). In medicine, alkylation of DNA is used in chemotherapy to damage the DNA of cancer cells. It is accomplished with the class of drugs called alkylating antineoplastic agents.



ALKYLBENZENE: A compound derived from benzene that contains an alkyl group bonded to a benzene ring. It is a product of alkylation in which one or more hydrogen atoms are replaced by alkyl groups of different sizes. An example is toluene, which is the simplest member of the group, with a hydrogen atom of the benzene replaced by a methyl group.

ALKYLIDENE GROUPS: The divalent groups formed from alkanes by removal of two hydrogen atoms from the same carbon atom, the free valencies of which are part of a double bond. An example is benzene.

ALKYNES (ACETYLENES): Acyclic branched or unbranched hydrocarbons having a carbon-carbon triple bond and the general formula C_nH_{2n-2} , $RC \equiv CR$. Acyclic branched or unbranched hydrocarbons having more than one triple bond are known as alkadiynes, alkatriynes, etc. The simplest member of the alkynes is ethyne also known as acetylene (C_2H_2). It is used for welding and cutting metals, as an illuminant and commercial chemical synthesis. Another simple member of the alkynes is propyne ($CH_3C \equiv CH$).

ALL-OR-NONE LAW: It states that no matter how strong a stimulus is, the size of an action potential is always the same. Therefore, information about the strength of a stimulus is carried along a nerve fibre not as variations in the size of nerve impulses, but by changes in their frequency.

ALL-OR-NONE RESPONSE: Response to a stimulus by an organism that comes with full intensity and is totally complete or totally absent. In other words, it is a response in which organisms decide to respond to a stimulus or not to respond to a stimulus as partial responses could aggravate the opponent. It is meant to overpower the enemy and if not possible then the response is not necessary.

ALLAMANDA: A genus of about 12 species of flowering evergreen trees, shrubs or vines in the dogbane family, Apocynaceae containing white latex. They are native to the tropical America and distributed from Mexico to Argentina. They do not grow well under shades and salty or alkaline soils. The plants have two cotyledons in the seed which usually appear at germination. The leaves are whorled and generally oval in shape with the margin being entire or smooth. Its

funnel-shaped flowers are borne in terminal cymes and have five lobed sepals with 5 large overlapping petals. The petals' colours vary from yellow, cherry, white, peach, chocolate, cream and golden-orange. All parts of the plant are poisonous. Some of the species produce colourful flowers and are cultivated as ornamental plants. Some species, for example, *Allamanda cathartica*, commonly known as golden trumpet, common trumpetvine and yellow allamanda have medicinal properties and are used to treat various ailments such as liver tumours, jaundice, splenomegaly and malaria.

ALLANTOIS: A membranous sac that grows from the lower gut in mammal, bird and reptile embryos. In mammals, it combines with the chorion to form the umbilical cord and placenta.

ALLEE EFFECT: A phenomenon in biology characterised by decline in individual fitness (often measured as per capita population growth rate) at low population size or density that can result in critical population thresholds below which populations crash to extinction. Allee effects are considered to have two manifestations: (i) component Allee effects, which are exhibited by a population in which there is a positive association between some fitness component such as viability, juvenile survivorship and fecundity and population size; and (ii) demographic Allee effects that occur when these component Allee effects produce a positive association between per capita population growth and population size. Allee effect was named after Warder Clyde Allee (1885–1955), an American Zoologist and Ecologist, who first described it.

ALLELE (ALLELOMORPH): One of the alternative forms of the same gene that occupies relative positions on homologous chromosomes producing contrasting characteristics. In most organisms, there are two alleles of any one gene (one from each parent). It determines which trait or character the organism will display, depending on whether it is dominant or recessive.

ALLELE FREQUENCY (GENE FREQUENCY): The occurrence of an allele in a population in relation to all alleles of that gene at the same locus, expressed as a fraction.

ALLELIC EXCLUSION: A process by which only one allele of a gene is expressed while the other is silenced. For most genes, the individual inherits one copy of each gene from each parent. Allelic exclusion occurs in certain cells of the immune system. For example, mature B cells belonging to a particular clone produce identical antibodies, despite the presence in each cell, of several genes determining antibody structure.

ALLELOCHEMICAL: A substance produced by members of one species that influences the behaviour or growth of members of another species.

ALLELOPATHY: The secretion of chemicals, such as phenolic and terpenoid compounds, by plants that inhibit the growth or germination of other plants with which they are competing. These allelochemicals can have beneficial (positive allelopathy) or detrimental (negative allelopathy) effects on the target organisms.

ALLENES: Compounds that contain cumulate double bonds, $>C=C=C<$, in which three carbon atoms are linked by two adjacent double bonds. The outer carbon atoms are each linked to two other atoms or groups by single bonds. The simplest example is 1,2-propadiene, $CH_2=C=CH_2$. Allenes are **dienes** (unsaturated hydrocarbon alkene containing two carbon-to-carbon double bonds) with typical reactions of alkenes. Under basic conditions, they often convert to alkynes. In an allene, the two double bonds lie in planes that are perpendicular to each other. Consequently, in an allene of the type $R_1R_2C=C=CR_3R_4$, the groups R_1 and R_2 lie in a plane perpendicular to the plane containing R_3 and R_4 . Under these circumstances, the molecule is chiral (molecule whose arrangement of atoms is such that it cannot be superimposed on its mirror image) and can show optical activity.

ALLERGIC REACTION: An inflammatory response triggered by the overreaction of the immune system to a harmless weak antigen, called **allergen**. When the immune system tries to protect the body,

it produces IgE antibodies to that allergen. The antibodies cause certain cells in the body to release large amounts of histamine from mast cells. The histamine acts on the eyes, nose, throat, lungs, skin or gastrointestinal tract and causes the symptoms of the allergic reaction. Allergens come into contact with the skin, nose, eyes, respiratory tract and gastrointestinal tract and can be breathed into the lungs, swallowed or injected. Most individuals do not react to allergens. An example of allergic reaction is asthma attack, which is triggered, when a susceptible person breathes an allergen such as pollen into the lungs. Some common allergens include animal dander, plants, pollens, bee stings or stings from other insects, some foods, insect bites and medications. **Allergy** is a condition of being usually reactive to something eaten, breathed in or touched that causes allergic reaction.

ALLEY CROPPING: A type of cropping where cereals such as millet or maize are planted between trees. The trees help the soil from being eroded.

ALLIACEOUS: Having a taste or smell of garlic.

ALLIGATOR: A broad snouted crocodilian species in the genus Alligator of the family Alligatoridae, subfamily Alligatorinae found in subtropical regions of America and China. The two living species are the American alligator (*A. mississippiensis*) and the Chinese alligator (*A. sinensis*). It is a large reptile with short legs, a long tail and very powerful jaws with very sharp teeth. An average adult American alligator's length is about 4 metres with a weight of about 360 kg. The Chinese alligator is smaller in size, about 2 metres in length and weighs about 45 kg. Crocodiles and alligators share many physical similarities, and it is difficult to distinguish between them. One distinguishing factor is the head; an alligator has a wider and U-shaped head, whilst that of a crocodile is longer and narrower, described as V-shaped. An alligator's diet consists mostly of fish, but also, insects, snails, crustaceans, and worms when young. More matured ones prefer larger fish such as gar, turtles, and various mammals and reptiles that they can kill. Female alligators reproduce by laying eggs in nests of vegetation, which provide warmth for them to hatch. Alligators have been raised or hunted by men for food and their skins that are used to make shoes, handbags, belts, and other leather items.

ALLOCHTHONOUS: This refers to materials or organisms that are foreign to where they are found. This is the opposite of autochthonous which describes an organism that is native to the place in which it is found.

ALLODYNIA: Pain due to a stimulus that does not normally provoke pain. A related term, **hyperalgesia** is the excessive sensitivity to pain or painful stimulus that produces pain. It may be caused by damage to nociceptors (a peripheral nerve organ that is a receptor for pain).

ALLOGAMY (CROSS-FERTILISATION): The fertilisation of an ovum from one individual with the spermatozoa of another. In plants, it is the use of pollen from one plant to fertilise the flower of another plant.

ALLOGENIC: The variation that exists in the genotypes of different individuals of the same species or a change in the environment of an individual organism brought about by some external factor.

ALLOMETRIC GROWTH: A phenomenon in which parts and organs of the same organism grow at different rates. In other words, it is the growth in part of an organism in relation to its whole body. An example is legs increasing in length faster than the arms. It is studied by measuring the mass and size of any organ or part of the body and expressed in relation to the total mass or size of the entire organism using the relation $Y = bx^\alpha$, where Y is the mass of the organ, x is the mass of the organism, α is the coefficient of the organ and b is a constant. It is important in studying relative growth rates.

ALLOMETRY: A regular relationship between a given feature and the size of the body as a whole, when the relationship is not a simple proportion of body size. Thus, an organ may increase in size proportionately faster or slower than body size does.

ALLOPARENTAL CARE (ALLOPARENTING): A behaviour in which other animals within a social group provide care through activities such as feeding, carrying, babysitting, etc., for young that are not their own offspring. It can also occur without any ongoing social interaction between the true parent and the alloparent. This behaviour is found in some birds, such as pigeons; and social mammals, such as meerkats. Usually both the alloparent and the young benefit.

ALLOPATHY: The treatment of disease by conventional means or orthodox medicine, using therapies designed to have effects opposite to the symptoms of the disease. This contrasts with homeopathy, which treats a disease by administering minute doses of natural substances that would in larger amounts produce in healthy persons symptoms similar to those of the disease.

ALLOPATRIC SPECIATION: Groups of similar organisms that could interbreed but do not because they are geographically separated. For example, when a barrier, such as a stretch of mountain range, river or sea, separates different populations of a particular species, the populations may no longer be able to cross the barrier to interbreed. Each species develops differently based on the demands of their unique habitat or the genetic characteristics of the group that are passed on to offspring. Example of allopatric speciation is found in the many different populations of pupfish that live in the Death Valley region of Nevada and California in the United States.

ALLOPHANE: An amorphous hydrated aluminosilicate of variable composition and various colours such as blue, green, or yellow, with the chemical formula $Al_2O_3 \cdot (SiO_2)_{1.3-2} \cdot (2.5-3)H_2O$.

ALLOPOLYPLOID: A polyploid (having more than twice the basic number of chromosomes) organism, that contains multiple sets of chromosomes derived from different species.

ALLOSTERIC EFFECT: A regulatory effect on macromolecules or enzymes that takes place at a site distinct from the enzyme's catalytic site because the macromolecules or enzymes have multiple ligand binding sites. For example, in haemoglobin, allosteric effects can generate co-operative behaviour. Both allosteric effects (which enhance activity) and allosteric inhibitors (which reduce activity) are utilised to control enzyme reactions.

ALLOSTERIC ENZYME: Enzymes, which contain regions to which small, regulatory molecules may bind in addition to and separate from substrate binding sites. On binding the effector, the catalytic activity of the enzyme towards the substrate may be enhanced, in which case the effector is an activator or reduced, in which case it is an inhibitor.

ALLOSTERIC SITE: A specific receptor or molecular site on a protein, usually an enzyme remote from the active site, where a molecule that is not a substrate may bind to a different molecule, called an **effector**. A protein molecule may also bind with the substrate at the active site. A molecule binds to change the shape of the active site, making it either more or less receptive.

ALLOSTERY: A phenomenon whereby the conformation of an enzyme or other protein is altered by combination, at a site other than the substrate-binding site, with a small molecule, referred to as an effector, which results in either increased or decreased activity by the enzyme.

ALLOSYNDESIS: Pairing of chromosomes derived from different species, seen in some diploid hybrids and in segmental allopolyploids. If the genetic composition of an allopolyploid is given by AABB, where AA represent the chromosomes derived from one parent species and BB the chromosomes derived from the other parent species, then during meiotic prophase, A undergoes allosyndetic pairing with B.

ALLOTETRAPLOID: A hybrid that has a chromosome set four times that of a haploid organism. Allotetraploids are created as a result of both chromosome sets of each parents being present in gametes. An example is Raphanobrassica, which is obtained by crossing radish (*Raphanus sativus*) with cabbage (*Brassica oleracea*).

ALLOTRANSPLANTATION (ALLOGENEIC TRANSPLANT; HOMOGRAFT): The transplantation of an organ or tissue from one individual to another of the same species with a different genotype. The transplant is called an allograft.

ALLOTROPE (ALLOTROPIC MODIFICATIONS): An element, which exists in two or more different forms in the same physical state but, has the same chemical properties. One allotrope is more stable than another at a particular temperature and pressure. For example, diamond and graphite are allotropes of carbon. Normal dioxygen (O_2) and ozone (O_3) are allotropes of oxygen. Other elements, which exist in different allotropic forms are monoclinic and rhombic sulfur, red and white phosphorous and α -tin (grey tin) and β -tin (white tin). The existence of an element in more than one crystalline or molecular form is called **allotropy**.

ALLOTYPE: 1. Any of various allelic variants of a protein, characterised by antigenic differences when introduced into the bodies of most other individuals of the same species. 2. An animal or plant specimen that shows physical characteristics, usually sexual ones, not shown in the holotype.

ALLOXAN (MESOXALYLUREA): A compound with molecular formula $C_4H_2O_4N_2$ that is an important decomposition product of uric acid. It is obtained by oxidising uric acid with nitric acid. It can react with alkali to take up two molecules of water to produce urea and mesoxalic acid. It can be used experimentally to induce diabetes mellitus in animals by selectively destroying the insulin-secreting islet cells of the pancreas and therefore used in medical research.

ALLOWED TRANSITION: A transition between two electronic states, which is permitted according to the selection rules. Allowed transitions have higher probability of occurring.

ALLOY: A metal mixed or blended with another metal or non-metal to give it special qualities such as resistance to corrosion and greater hardness, which none of the metals or non-metals alone has. The physical properties of an alloy depend on the metals chosen, the proportions combined and the process by which the alloy is formed. Examples of alloys are bronze, which is an alloy of copper and tin; steel, an alloy of iron and carbon and brass, which is an alloy of copper and zinc.

ALLOY	COMPOSITION (%)	USES
Bell Metals	Cu 80, Sn 20	Bells
Brass	Cu 60-80, Zn 40-20	Cooking utensils & knives
Bronze	Cu 75-90, Sn 10-25	Household utensils
Duralumin	Al 95, Cu 4, Mg 0.5, Mn 0.5	Aircraft construction
German Silver	Cu 60, Zn 25, Ni 15	Utensils
Gun Metal	Cu 85, Zn 10, Sn 5	In Engineering
Solder	Sn 50-75, Pb 50-25	Soldering of metal
Stainless Steel	Fe 82, Cr or Ni 18	Cooking utensils & knives
Steel	Fe 98, C 2	Nails, screws, construction

Important Alloys, Composition And Uses

ALLOY STEEL: Steel containing a variety of elements in certain percentages of vanadium, molybdenum or other elements, but not carbon and also larger amounts of manganese, silicon and copper to

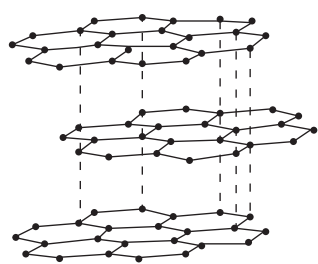


Fig I: Allotrope of graphite

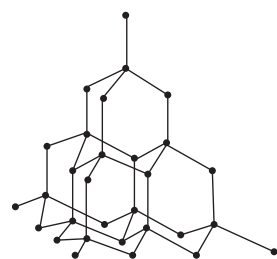


Fig II: Allotrope of diamond

give it certain physical, mechanical or chemical properties. Compared to carbon steel, alloy steel has some improved qualities such as strength, toughness, wear resistance, hardness and hot hardness. Some of the uses of alloy steel are as automobile gears and axles, roller skates, carving knives, etc.

ALLOZYME: Common biological enzyme that exhibits high levels of functional evolutionary conservation occurring in chromosomes as variants of enzymes that are coded and determined by different alleles at a single genetic locus. It is applied in many population genetics studies, including measurement of outcrossing rates, sub-population structure, population divergence and also used in concert with gene markers in fingerprinting.

ALLUVIAL: Relating to, consisting of or formed by sediment deposited by flowing water. Some alluvial deposits may contain rich deposits of minerals such as gold, platinum, gemstones and tin ore (cassiterite).

ALLUVIAL FAN: Roughly triangular sedimentary formation found at the base of slopes or a fan-shaped deposit of sediment comprising alluvium formed at the point where a stream enters a valley or plain or another larger stream.

ALLUVIAL SOIL (ALLUVIAL DEPOSIT): A fertile soil deposited by water flowing over flood plains or in river beds or clay, silt or gravel carried by rushing streams and deposited where the streams slow down. Its texture may be coarse to fine (drift sand and rich, loamy soil to silt clays) depending on proximity to water deposition source. New material is successively deposited on the surface when the land is subjected to flooding. Alluvial soil requires the least water for crops to flourish due to its high porosity and therefore, has the highest productivity compared to other types of soils. In tropical regions such areas may be used as paddy fields and in arid regions the extent of the alluvial soils may be artificially increased by irrigation methods.

ALLUVIUM: Unconsolidated materials such as gravel, sand, silt or organic matter, deposited by streams and rivers outside the water. Alluvium makes rich agricultural soil and the earliest civilisations originated as farming communities centred on alluvial flood plains. All rivers have some alluvium deposits, but large rivers such as the Mississippi or Amazon have alluvial plains which are several kilometres wide.

ALLYLIC GROUPS: An allyl and derivatives formed by substitution. The term 'allylic position' or 'allylic site' refers to the saturated carbon atom. A group, such as $-OH$, attached at an allylic site is sometimes described as 'allylic'. An allyl group is a substituent with the structural formula $H_2C=CH-CH_2R$, where R is the connection to the rest of the molecule. It is made up of a methylene ($-CH_2-$), attached to a vinyl group ($-CH=CH_2$).

ALLYLIC INTERMEDIATES: Carbanions, carbenium ions or radicals, formally derived by detachment of one hydron, hydride or hydrogen from the CH_3 group of propene or derivatives thereof. An example is allyl cation ($H_2C=CHCH_2^+$).

ALMANAC: An annual reference book composed of various lists, tables and often brief articles relating to a particular field or many general fields. An example is one that lists the future positions of the Moon, planets and other prominent celestial objects, together with other useful astronomical data.

ALMOST ALL (ALMOST EVERYWHERE): In mathematics, often written **a.e.** or **p.p.**, it is the truth set of all values with the exception of sets on zero measure. For example, if $f(x) = 1$ for all real x , $g(x) = 1$ for all irrational x and 0 for all rational x , then f and g agree almost everywhere, since the rationals have measure zero.

ALMOST DISJOINT: In mathematics, two sets are said to be almost disjoint if the intersections are negligible or very small.

ALMOST UNIVERSAL: A class is called almost universal if every subset of it is contained in some member of it.

A

ALMOST PERFECT NUMBER: The powers of 2, because the aliquot parts (proper divisors) of 2^n sum to $2^n - 1$. The only known almost perfect numbers are 1, 2, 4, 8, 16, 32, ..., which form the powers of 2.

ALMOST SURELY: A terminology used in probability measure to represent nearly everywhere.

ALN: Abbreviation and symbol for the antilogarithm of a natural logarithm.

ALNICO: A trade name for a series of alloys containing iron, aluminium, nickel, cobalt and copper used to make permanent magnets.

ALOG: Abbreviation and symbol for antilogarithm. If no base is specified, it is taken to be 10.

ALOPECIA: Loss or absence of hair on parts of the body or the whole body, especially from the scalp.

ALPHA: 1. The first letter of the Greek alphabet denoted by A for the upper case and α for the lower case. They are used in many scientific contexts. 2. In astronomy, it is the brightest star in a constellation. 3. In chemistry, relating to one of two or more isomeric forms of a chemical compound, especially one in which a group is attached to the carbon atom to which the principal group is attached. 4. In social animals, the highest ranked or most dominant individual in a group. 5. In computing, **alpha test** is the first version of a new software program that may not contain all of the features that are planned for the final version that is tested seriously by the developers of the program. The second version, known as the **beta version** is usually given to a limited number of external users for testing.

ALPHA ADRENOCEPTOR (ALPHA ADRENERGIC RECEPTOR; ALPHA RECEPTOR): 1. A molecule on the surface or within a cell that recognises and binds with specific molecules to produce a specific effect in the cell. Examples are the cell surface receptors for antigens or cytoplasmic receptors for steroid hormones. 2. A molecule that responds to norepinephrine and epinephrine but blocking agents such as phenoxybenzamine in cells so that these cells can initiate physiological responses such as vasoconstriction, pupil dilation, contraction of cardiac muscles and contraction of pilomotor muscles to raise the hair during cold weather. 3. A sensory nerve ending or effector that responds to various stimuli.

ALPHA CELLS: Cells found in the islet cells (islets of Langerhans) of the pancreas which produce glucagon, which raises glucose (type of sugar) levels in the blood. One of the causes of elevated levels of glucose in the blood is diabetes, when the body is unable to produce enough insulin to control the body's blood glucose or sugar and the diabetic does not control the glucose level through medication, exercises, etc. If not treated, it can lead to long term complications such as cardiovascular disease, nerve damage, kidney damage/failure, damage to the blood vessels of the retina which can lead to blindness, cataract, etc.

ALPHA CENTAURI: The nearest star system to the Sun, with a distance of 4.395 light-years (1.35 parsecs, 4.16×10^{13} km). Its light takes more than 4 years to reach the Earth. It is a multiple star in the constellation Centaurus with three components representing the brightest object in the constellation.

ALPHA CLEAVAGE (α -CLEAVAGE): 1. In organic chemistry, especially in mass spectrometry, it is the fission or breaking of the carbon-carbon bond, which originates at an atom which is adjacent to the carbon bearing a specified functional group. 2. In photochemistry, it is the homolytic cleavage of a bond adjacent to a specified group or a bond connecting an atom or a group to an excited chromophore.

ALPHA DECAY (α DECAY): The disintegration of certain radioactive isotopes, causing the spontaneous emission of an alpha (α) particle. An alpha particle is a stable particle formed from two protons and two neutrons and is identical to the nucleus of the helium atom. The general reaction for alpha decay is ${}^Z_X \rightarrow {}^{Z-4}_{A-4}Y + {}^4_2\text{He}$, where X is the parent nuclide, Y is the daughter nuclide, Z is the atomic mass of X and A is the atomic number of X . The atomic number or the number

of protons decreases by two and the mass number decreases by four as a result. Alpha decay commonly occurs in elements such as bismuth with atomic numbers greater than 83. It can occur in some rare-earth elements such as neodymium and lutetium with atomic numbers between 60 and 71.

ALPHA DIVERSITY (ALPHA RICHNESS): The biodiversity within a particular habitat or community usually expressed as the species richness of the area. The alpha diversity of component communities combined with the intercommunity beta diversity gives a measure of regional species diversity or gamma diversity.

ALPHA EFFECT: A positive deviation of an alpha (α)-nucleophile when the atom adjacent to a nucleophilic site bears a lone pair of electrons.

ALPHA ELIMINATION (α -ELIMINATION): 1. A transformation in which the central atom is commonly a carbon. The reverse reaction is called α -addition. 2. A term applied to a reaction by which a group attached to the alpha carbon of an excited chromophore is expelled either as an odd electron species or as an ionic species.

ALPHA HAEMOLYSIS/HAEMOLYSIS (α -HAEMOLYSIS): A partial haemolysis (disintegration of red blood cells, with the release of haemoglobin) on blood agar plates (nutrient culture medium that is enriched with blood and used for the growth of certain strains of bacteria) as indicated by a green discolouration around the bacterial colony. It is caused by hydrogen peroxide produced by the bacterium, oxidising haemoglobin producing the green oxidised derivative, methemoglobin. Examples of α -haemolysis are *Streptococcus pneumoniae* that causes bacterial pneumonia and some other diseases; and *Streptococcus viridans* which causes dental infections some other diseases.

ALPHA HELIX: The most common form of secondary structure of proteins in which the polypeptide chain is coiled into a helix. This is a right-handed coiled or spiral conformation, in which every backbone N-H group donates a hydrogen bond to the backbone C=O group of the amino acid four residues earlier. The α -helix is the most regular and the most predictable from sequence, as well as the most prevalent.

ALPHA NAPHTHOL TEST (MOLISCH'S TEST): A sensitive chemical test for the presence of carbohydrates, named after the Austrian chemist, H. Molisch. The test is based on the dehydration of the carbohydrate by sulfuric acid to produce an aldehyde, which condenses with two molecules of phenol (usually β -naphthol), resulting in a red- or purple-coloured compound, which indicates a positive reaction.

ALPHA PARTICLE: Positively charged, high-energy particle emitted from the nucleus of a decaying radioactive atom, with chemical symbol ${}^4_2\text{He}$. It consists of 2 neutrons and 2 protons and is identical to the helium nucleus without an electron.

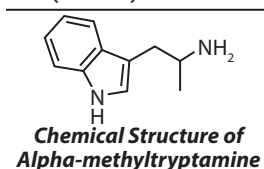
ALPHA RADIATION: A stream or rays of relatively low penetrating power alpha particles emitted by radium and other radioactive substances moving at high speed of up to one tenth of the speed of light ($1/10 \times 299\,792\,458$ m/s) but having small masses. Alpha radiations or rays are the least penetrating of the three radiations (alpha, beta and gamma) and can be stopped or absorbed by a sheet of paper and can be slightly deflected by a strong magnetic or electric field. Although radiation can be harmful, it is also useful. One use of alpha radiation is in smoke detectors. For example, radioactive americium releases alpha radiation, which ionises the air inside a fire detector. Smoke from a fire absorbs alpha radiation, altering the ionisation and triggering an alarm. Alpha rays penetrate only a small distance in dense matter such as living tissue and have been used to cause local destruction of tissue such as in cancer therapy.

ALPHA-ADDITION (α -ADDITION): A chemical reaction resulting in a single reaction product from two or three reacting chemical species. This results in the formation of two new chemical bonds to the same atom in one of the reactant molecular entities.

ALPHA-EXPULSION (α -EXPULSION): A general reaction by which a group attached to the alpha carbon of an excited chromophore is expelled either as an odd electron species or as an anionic species.

ALPHA-IRON (α -IRON; FERRITE; α -FERRITE): Iron with a body-centred cubic structure, which gives steel and cast iron their magnetic properties. It is stable below 910°C (1 183K).

ALPHA-METHYLTRYPTAMINE (AMT): A synthetic derivative of tryptamine with stimulant and hallucinogenic properties. It is used illegally as a club drug.



ALPHA-PARTICLE SCATTERING (RUTHERFORD ALPHA-PARTICLE SCATTERING):

The scattering of alpha particles by thin films of heavy metals, notably gold, a phenomenon explained by Ernest Rutherford in 1911. The patterns produced by such scattering off an extremely thin sheet of gold were early evidence that atoms contain a tiny, positively charged core, what is now called the atomic nucleus. This led to the development of the Rutherford model (planetary model) of the atom and eventually to the Bohr model. Rutherford scattering is also sometimes referred to as **Coulomb scattering** because it relies only upon static electric (Coulomb) forces and the minimal distance between particles is set only by this potential. **Rutherford backscattering** is any technique using high energy particles directed toward a sample, in which the bombarding particles are detected and recorded as function of energy and/or angle. The technique is mostly used for determining depth distributions of elements based on the energy of the backscattered particle. In general, He⁺ or H⁺ particles are used at energies in the order of 100 keV to some MeV.

ALPHABET: In mathematics, the set of symbols from which an n -tuple is drawn to form a word of a code.

ALPHAMAGIC SQUARE: A form of magic square, in which the number of letters in the word for each number, in whatever language is being used, gives rise to another magic square.

ALPHAMETIC (CRYPTARITHMETIC; CRYPT-ARITHMETIC, CRYPTARITHM; WORD ADDITION): A type of mathematical game consisting of a mathematical equation among unknown numbers, in which the letters represent distinct digits form related words or meaningful phrases. The goal is to identify the value of each letter. The game can be extended to puzzles that use non-alphabetic symbols instead of letters.

ALPHANUMERIC (ALPHAMERIC): Describing characters or their codes that represent letters of the alphabet or numerals, particularly in computer applications. In computers designed for English language users, alphanumeric characters are those made up of the combined set of the 26 alphabetic characters, A to Z and the 10 Arabic numerals, 0 to 9. In some usages, the alphanumeric character set may include both upper and lower case letters, punctuation marks and symbols such as @, & and *. For languages other than English (for example, French), alphanumeric characters include letter variations such as é and ç. For some computer usages, such as file naming, alphanumeric characters are strictly limited to the 26 alphabetic characters and 10 numerals; however, for other usages, such as programming, other keyboard symbols are sometimes permitted.

ALPINE BIOME: A major biome of vegetation found in the high mountain regions all around the world. It lies just below the snow line of a mountain with plants generally growing in thin stony soil and are subjected to high light intensity and high wind speeds. Because of the severe climate of the Alpine biome, plants and animals have developed adaptations to those conditions. At high altitudes there is very little CO₂, which plants need to carry on photosynthesis. There are only warm blooded animals in the Alpine biome, although there are insects. Alpine animals have to deal with two types of problems: the cold and too much high UV wavelengths, because there is less atmosphere to filter UV rays from the sun. Alpine animals adapt to the cold by hibernating, migrating to lower, warmer areas or insulating their bodies with layers of fat. **Alpine soil** is mountain soil occurring above the timberline.

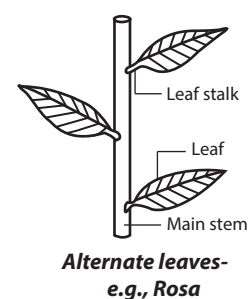
ALT KEY (ALTERNATE KEY): A modifier key on a computer keyboard that is pressed in combination with another key to execute an alternate operation. It is usually located on both sides of the spacebar key. Not all computer keyboards have an Alt key, but it is standard on all PCs, located on either side of the Space bar.

ALTERATIVES: Herbs that gradually restore the proper function of the body and increase health and vitality; herbs which purify the blood. Herbs like these can be used for quite a long time for active cleansing of the blood and purification of vital organs such as the kidneys, liver, spleen and the bowel. Such herbs include corn silk and mahogany.

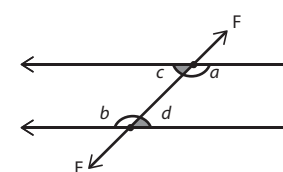
ALTERNANCY SYMMETRY: A topological property of the molecular graphs of alternant hydrocarbons which allows the carbon atoms to be divided into two subsets in such a way that no two atoms of the same subset are adjacent.

ALTERNANT: 1. A conjugated system of π -electrons is termed alternant if its atoms can be divided into two sets so that no atom of one set is directly linked to any other atom of the same set. Alternatively, describing a conjugated molecule whose carbon atoms can be divided into two sets so that members of one set are formally bonded only to members of the other set. An example is naphthalene. 2. In mathematics, a determinant constructed from n functions and n points (not necessarily distinct) by taking the i_j th entry to be the value of the i th function at the j th point or vice versa. 3. Also known as **disjunct**, in logic, it is either of a pair of propositions or formulae related or operated upon by disjunction.

ALTERNATE: 1. A term used to describe two things, which happen or occur or follow one after the other. 2. A term used to describe flowers, buds or leaves that are arranged singly at each node or at different levels on either side of the stem of a plant, as opposed to being in pairs or groups. Examples are Smoke bush (*Cotinus coggygria*), Barberry (*Berberis thunbergii*) and Sweetgum (*Liquidambar styraciflua*). Rose (*Rosa rubiginosa*) also has alternate leaves, which vary from simple to trifoliate, palmate or pinnate.



ALTERNATE ANGLES: Either pair of angles contained between two given lines and a transversal, when a transversal cuts or crosses the two lines. They are non-adjacent and are on the opposite sides of the transversal. Alternate angles are equal if and only if the lines are parallel. For example, in the diagram, angles a and b alternate, which means a is equal to b ($a = b$), so are c and d ($c = d$).



EF is a transversal line.
<c and <d are alternating angles
<a and <b are also alternating angles

Alternate Angles

ALTERNATE HOST: A plant, often weeds, other than the main host, on which a pest or pathogens such as certain rust fungi, must develop to complete their life cycles. When the pathogen has a number of different hosts, it is termed **alternative host**. Alternate hosts, can provide a means for the pathogen to survive when its main host is not available. For example, downy mildew of beet (*Peronospora farinosa*) can overwinter in wild beet. Certain insect pests overwinter as eggs on alternate hosts, for example, blackfly (*Aphis fabae*) migrates to the spindle tree (*Euonymus europaeus*) in the autumn.

ALTERNATE NEWSGROUP (ALT.* HIERARCHY): On Usenet, the "alternative" it is the user-originated newsgroup, containing all newsgroups whose name begins with "alt." that is set up so that anyone can start a newsgroup on any topic without structured discussion and a vote to demonstrate that demand for the newsgroup exists.

ALTERNATING COPOLYMER: A polymer formed of two different monomer molecules that alternate in sequence in the polymer chain.

ALTERNATING CURRENT (AC): Electric current that changes direction periodically. If a graph of the current against time is plotted it is in a wave form, with its value varying periodically around zero. AC can come in a number of forms, as long as the voltage and current are alternating. The most common type of AC is the sine wave. The AC in homes and offices have an oscillating voltage that produces a sine wave. There are three parts to a sine wave: amplitude, frequency and phase. Some other waveforms include the square waves, often used in digital and switching electronics to test their operation; and triangle waves, found in sound synthesis and are useful for testing linear electronics like amplifiers. The frequency of alternating current (AC) is measured in hertz. Typical equations describing an AC current and AC voltage are: $I = I_o \sin \omega t = I_o \sin 2\pi f t$ and $V = V_o \sin \omega t = V_o \sin 2\pi f t$, where $\omega t = 2\pi f t$ and $f = 1/T$, T is the period of the signal in seconds, f is the frequency in Hz, ωt is the angular velocity in radian per second, I and V are the instantaneous values of the current and voltage respectively and I_o and V_o are the peak current and voltage respectively. The frequency of the domestic supply is different in different parts of the world. In Europe and most parts of the world, it is 50 Hz and in the U.S., it is 60 Hz.

ALTERNATING GROUP: Any subgroup of the symmetric or permutation group on a set S consisting of all even permutations of n objects. It is denoted by A_n , has order $n!/2$ and has index 2 in S_n , the symmetric group of degree n . For $n \geq 5$, A_n is the only proper non-trivial normal subgroup of S_n and is itself simple.

ALTERNATING MULTILINEAR FUNCTION: A multilinear form in which all underlying vector spaces are identical and the interchanging two variables merely changes the sign of the expression.

ALTERNATING SERIES TEST (LEIBNIZ'S ALTERNATING SERIES TEST): A method used to prove that an alternating series with terms that decrease in absolute value is a convergent series; and to establish conditional convergence of an alternating series one need only to verify that the terms decrease monotonically to zero in absolute value. In this case the error after adding n terms is always

less than the magnitude of the next. For example, $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$

is convergent, since $|a_n| = \left| \frac{(-1)^n}{2n+1} \right| = \frac{1}{2n+1}$

decreases monotonically to zero as n tends to infinity; its sum is $\pi/4$.

ALTERNATION: 1. In logic, also known as **disjunction** or **logical sum**, it is a proposition that presents two or more alternative terms, with the assertion that only one or at least one is true. It is an OR statement of the form A OR B. It is true if either A OR B is true. Its symbol is $\vee$. 2. In electricity, it is a single fluctuation in the absolute value of an alternating current or voltage from zero to a maximum and back to zero. It is equal to a half cycle.

ALTERNATION OF GENERATIONS (METAGENESIS; HETEROGENESIS): The occurrence in one life cycle of an organism of two or more different forms, usually an alternation of a sexual with an asexual form. The sexual phase, called the gametophyte, produces gametes or sex cells; the asexual phase or sporophyte, produces spores asexually. The two phases or generations, are often morphologically and sometimes chromosomally, distinct. It occurs more commonly in algae, fungi, mosses, ferns and seed plants; many invertebrates, for example, protozoans, jellyfish and flatworms also have an alternation of sexual and asexual generations, but the alternation of haploid and diploid generations is unknown.

ALTERNATION THEOREM: The result, for continuous functions $g_1, \dots, g_n$ on $[a, b]$ that satisfy the Haar condition (as holds for $1, x, \dots, x^n$), that in order for a generalised polynomial P to be the best approximation in Chebyshev norm to a continuous function f it is necessary and sufficient that the error function $r = f - P$ possesses at least $n + 1$ alternations, points at which $r(x_i) = -r(x_{i-1}) = \pm \|r\|_\infty$.

ALTERNATIVE COMPLEMENT PATHWAY: One of three pathways in the complement system of innate immunity capable of initiating the processes that result in C3 activation for the immune system's natural defence against pathogens and invading foreign agents. C3 is activated directly by pathogens, or by foreign materials and damaged tissues. Complement system of innate immunity is independent of antibody activity. It is part of the innate immune system and one of the major means by which the body recognises foreign agents and pathogens that may be harmful to it. C3-C9 are abbreviated versions of **complement components 3 and 9**, which are plasma proteins of the innate immune system. The other two pathways in the complement system of immunity are: the **classical pathway** which is triggered by activation of the C1-complex; and the **lectin pathway** which is activated by the binding of MBL (Mannose-Binding Lectin), to carbohydrates on the surfaces of pathogens.

ALTERNATIVE DEVELOPMENT PATHS: A variety of possible scenarios for societal values and consumption and production patterns in all countries, including, but not limited to, a continuation of today's trends.

ALTERNATIVE ENERGY: Any form of energy that does not come from fossil fuels such as oil, gas, coal and oil shales, as well as from sources such as uranium used in nuclear power generation. Alternative energy is renewable and ecologically safe. Examples include water, solar energy, wind power, tidal power, wave power and geothermal energy.

ALTERNATIVE HYPOTHESIS: The hypothesis that sample observations are influenced by some non-random cause and do not conform with a given null hypothesis; the alternative is only accepted if the value of a test statistics is sufficient, at a chosen significance level, to reject the null. It is denoted by H_1 or H_a .

ALTERNATIVE MEDICINE: Any type of medical practice that does not use synthetic drugs or surgery to treat a disease. The emphasis is on maintaining health with diet and exercise and on dealing with the underlying causes rather than just the symptoms. Examples include homeopathy, traditional medicine, chiropractic and acupuncture.

ALTERNATIVE RESPIRATORY PATHWAY (ALTERNATIVE OXIDASE): A respiratory pathway that oxidises NADH (reduced form of nicotinamide adenine dinucleotide) to NAD^+ (oxidised form of nicotinamide adenine dinucleotide) and generates water, resulting in fewer molecules of adenosine triphosphate (ATP) compared to cytochrome pathway. As a result, it acts as an alternative to the electron transport pathways common to both plants and animals in the normal electron transport chain. The pathway is found in plants, algae, bacteria such as *E. coli* and some protozoa.

ALTERNATIVE SPLICING (DIFFERENTIAL SPLICING): The splicing of the RNA transcript of a gene in different ways by the cellular machinery during RNA processing to create distinct mature messenger RNAs (mRNA) encoding different proteins. This is responsible for the disparity between the relatively small numbers of human genes identified so far.

ALTERNATIVE THEOREM: Any theorem stating that of two systems of equations or inequalities one or the other always has a solution.

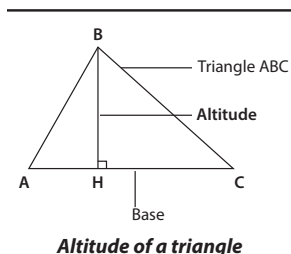
ALTERNATOR: A dynamo that produces an alternating current. It consists of coils that rotate in the magnetic field produced by one or more permanent magnets or electromagnets.

ALTIMETER: A device used to measure the height of the land surface or any object such as an airplane above sea level. There are different types of altimeters, which include barometric altimeters (the counter pointer altimeter and the counter drum pointer altimeter), which use air pressure to determine the altitude; the radio altimeter, which uses radar to determine altitude; and global positioning system (GPS), which uses satellites to determine an object's altitude.

ALTIMETRY: The science of measuring altitude or elevation. Satellite altimetry involves using an instrument, commonly a laser, to measure the distance between the satellite and the ground.

ALTITUDE: 1. The perpendicular distance from the vertex to the base in a geometrical figure. Alternatively, any line segment between a vertex and a side of a polygon that is perpendicular to the side and is the longest such perpendicular through that vertex, especially, any perpendicular from a vertex to the opposite side of a triangle. 2. The height of something above a specific level, especially above sea level or the Earth's surface.

ALTITUDE OF A TRIANGLE: The perpendicular distance from any of a triangle's vertices to the opposite side.



ALTITUDE SICKNESS (ACUTE MOUNTAIN SICKNESS; ALTITUDE ANOXIA): A condition,

characterised by impaired night vision, shortness of breath, fatigue, headache, rapid pulse, loss of appetite, insomnia and nausea. It is caused by lack of oxygen in the blood and tissues due to low atmospheric pressure. This often occurs at high altitudes of above about 5,000 metres above sea level, where there is very little oxygen. In aircrafts on high-altitude flights, it is prevented by using pressurised cabins.

ALTITUDINAL GRADIENT: A gradient of cooler, drier conditions occurring as latitude increases.

ALTOCUMULUS CLOUDS: Clouds composed of flattened globules that may be arranged in groups. They are small, white, puffy clouds composed primarily of water droplets that sometimes slowly drift across the sky and forms at altitudes of between 2000 - 6000 metres above the ground with a vertical velocity of 0.05 ms⁻¹ to 0.1 m s⁻¹. Altocumulus clouds usually form in a layer of moist air, where air currents undulate gently, like waves on the sea. As a wave rises, water vapour condenses and forms the cloud. In the wave troughs, water evaporates. There the cloud is thinner or the sky may be clear, producing the bands of cloud that are sometimes seen.

ALTRICIAL SPECIES: Those mammals or bird species whose newly hatched or newly born are relatively immobile, lack hair or down feather, closed eyes in certain cases, lack certain other important features which can make them independent, etc and must be cared for by adults. Examples are birds such as herons, hawks, woodpeckers and owls and mammals such as marsupials, cats, dogs and humans. On the other hand, precocial animals give birth or hatch young which have open eyes, have hair or down feathers, have large brains and are immediately mobile and somewhat able to flee from or defend themselves against, predators. Examples include ducks and turkeys, whose young are ready to leave the nest in a day or two. For mammals, most ungulates are precocial, being able to walk almost immediately after birth.

ALTRUISM (ALTRUISTIC BEHAVIOUR): A social interaction by an animal, which is not an act of conscious behaviour, which decreases its chances of survival or reproduction, while increasing those of another member of the same species. Vampire bats regularly, for example, regurgitate blood and donate it to other members of their group who have failed to feed that night, so that they do not starve. Similarly, vervet monkeys give alarm calls to warn fellow monkeys of the presence of predators, even though in doing so they attract attention to themselves and increase their personal chance of being attacked. Also, in social insect colonies such as ants, wasps, bees and termites, sterile workers devote their whole lives to caring for the queen, constructing and protecting the nest, foraging for food and tending the larvae.

ALU (ARITHMETIC LOGIC UNIT): In computer science, it is a digital circuit, which is part of a computer processor (CPU) that is used to perform arithmetic and logic operations.

ALU FAMILY: A group of closely related DNA sequences within the genome of humans and other primates, arising from a single ancestral sequence of about 300 base pairs long in humans and about 500,000 to 600,000 copies widely dispersed among all 46 chromosomes copies

in the human genome. Alu insertions have been involved in several inherited human diseases and in various forms of cancer. The name ALU comes from the restriction endonuclease Alu I that cleaves it.

ALULA: A small structure consisting of a set of quill-like feathers composed of a digit bone and two to six feathers located close to the base of the primary feathers at the joint between the hand-wing and arm-wing of birds. It plays a part in increasing or decreasing the bird's lift by affecting the airflow of the wings.

ALUM: A group of double salts, with the chemical formula A₂SO₄·B₂(SO₄)₃·24H₂O, where A is a monovalent metal and B a trivalent metal, occurring naturally in various minerals and can be prepared and purified by crystallisation from their solutions. The most important alums are those of potassium sulfate (potassium alum or potash alum, K₂SO₄ × Al₂(SO₄)₃ × 24H₂O), ammonium sulfate and sodium sulfate. Alternatively, it may be defined as the common name for aluminium potassium sulfate, which is a double salt whose crystals contain two different metal ions or any double sulfate of a monovalent metal or radical such as sodium, potassium or ammonium with a trivalent metal such as aluminium and iron. The commonest is the double sulfate of potassium and aluminium (aluminium potassium sulfate - (KAl(SO₄)₂·12H₂O)₂), which is a white crystalline powder that is readily soluble in water. It is used for paper-sizing agents, flocculating agents in water treatment, mordants in dyeing and in pickles, baking powder, fire extinguishers and medicines.

ALUMINA (CORUNDUM; ALUMINIUM OXIDE): A family of inorganic compounds, with chemical formula Al₂O₃, density 3.95-4.1 g/cm³, melting point 2072 °C (2345 K) and boiling point 2977 °C (3250 K). It is a white powder, soluble in most alkalis or caustic, strong acids but insoluble in water. It is widely distributed in clays, slates and shale. It is used in the production of aluminium metal, as an abrasive due to its hardness and as a refractory material due to its high melting point, as well as for ceramics and electrical insulation.

ALUMINATE: A salt formed when aluminium hydroxide or γ-alumina is dissolved in solutions of strong bases, such as sodium hydroxide. Aluminates exist in solutions containing the aluminate ion, commonly written [Al(OH)₄]⁻. Examples of aluminate compounds include sodium aluminate (Na₂Al₂O₄) and potassium aluminate (K₂Al₂O₄), both being alkali-ion precipitates of the tetrahedral monomer, obtained from an alkaline solution. Aluminate can also refer to the polyatomic (molecule that has more than two atoms) ion, for example, chlorate (ClO₃⁻) and hydroxide (OH⁻).

ALUMINISING (ALUMINIZING): 1. The process of coating a mirror with aluminium. The aluminium is vapourised in a vacuum, causing a film of metal, about 100 nanometres thick, to be deposited on the glass. Layers of other materials, such as silicon dioxide, may subsequently be laid down to give protection. Aluminium offers consistently high reflectance throughout the visible, near-infrared and near-ultraviolet regions of the spectrum. 2. A process used to inhibit corrosion due to oxidation, sulfidation and carburisation in industry by protecting the surface of steels, stainless steels and nickel alloys operating in severe high temperature environments of (500-800 °C). It is a high temperature chemical process whereby aluminium vapours diffuse into the surface of the base metal forming new metallurgical aluminide alloys. Aluminium coating on steel or other materials increases their resistance to oxidation at high temperatures. It may also increase the corrosion-resistance of steel in hydrocarbons and sulfurous atmospheres.

ALUMINIUM: A lightweight soft silvery white metal element that is a member of Group 13 (IIIA) of the Periodic Table. Its chemical symbol is Al, with atomic number 13, melting point 660 °C (933 K), boiling point 2519 °C (2792K), relative density 2.7 and relative atomic mass 26.98. Its specific weight is 2.7 g/cm³, (one-third that of steel). It is the most abundant metal in the earth's crust and the third most abundant element after oxygen and silicon. Its main source is bauxite ore. It is ductile, malleable, easy to handle, has low density, resists corrosion due to the phenomenon of passivation and its alloys can be very strong. It is used widely in construction, transport, making cooking utensil, in aerospace industry, overhead electric cables, etc.

ALUMINIUM BRONZE: Alloy of copper-aluminium that may also contain iron, manganese, nickel or zinc. The composition of aluminium ranges from 5% to 11%.

A

ALUMINIUM CHLORIDE: A white or yellowish deliquescent solid with the chemical formula AlCl_3 . It is the main compound of aluminium and chlorine. It has a low melting and boiling point. Aluminium chloride is hygroscopic and has a very high affinity for water. It fumes in moist air and hisses when mixed with liquid water as the Cl^- ions are displaced with H_2O molecules in the lattice to form the hexahydrate ($\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$). It is used as a catalyst in the cracking of petroleum hydrocarbons. It is also a catalyst in certain other organic reactions, especially the Friedel-Crafts reaction.

ALUMINIUM ETHANOATE (ALUMINIUM ACETATE): A white solid, with the chemical formula $\text{Al}(\text{OOCCH}_3)_3$, which can only be made in the absence of water. For example, ethanoic anhydride and aluminium chloride at 180°C (453 K), decomposes on heating. It is slightly soluble in cold water and decomposes in warm water. It is used in dyeing as a mordant, particularly in combination with aluminium sulfate (red liquor); in the paper and board industry for sizing and hardening; and in tanning. It was previously used as an antiseptic and astringent.

ALUMINIUM HYDROXIDE: A white crystalline compound, with the chemical formula $\text{Al}(\text{OH})_3$, melting point 300°C (573 K) and relative density 2.42–2.52. The compound occurs naturally as the mineral gibbsite (monoclinic). In the laboratory it can be prepared by precipitation from solutions of aluminium salts. It is used as a filtering medium and in ion-exchange and chromatography devices, as an antacid, as an adsorbent, emulsifier, mordant, in the manufacture of paper, ceramics, printing inks, detergents, for waterproofing fabrics and in dentifrices and antiperspirants.

ALUMINIUM ORE: The natural raw material from which aluminium is extracted. The main ore is bauxite.

ALUMINIUM OXIDE (ALUMINA; CORUNDUM): A white crystalline amphoteric compound, which is the principal source of aluminium. It occurs in bauxite and its chemical formula is Al_2O_3 .

ALUMINIUM POTASSIUM SULFATE (POTASH ALUM; ALUMINIUM POTASSIUM SULFATE DODECAHYDRATE): A colourless or white crystalline compound with the chemical formula $\text{Al}_2(\text{SO}_4)_3 \cdot \text{K}_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$, relative density 1.757, molar mass 474 g/mol , melting point $92\text{--}93^\circ\text{C}$ ($365\text{--}366\text{ K}$) and boiling point 200°C (473 K). It occurs naturally as the mineral kalinite salt and can be prepared by recrystallisation from a solution containing equimolar quantities of potassium sulfate and aluminium sulfate. It forms cubic or octahedral crystals that are soluble in cold water, very soluble in hot water but insoluble in ethanol and acetone. Its uses include as a mordant for dyeing and in the tanning and finishing of leather goods (for white leather).

ALUMINIUM SULFATE: A white crystalline compound, with the chemical formula $\text{Al}_2(\text{SO}_4)_3$. It occurs naturally in the rare mineral alunogenite ($\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$) and may be prepared by dissolving aluminium hydroxide or china clays (aluminosilicates) in sulfuric acid. It is used as a foaming agent in fire extinguishers, as a flocculating agent in water treatment and sewage works and as a mordant in dyeing.

ALUMINIUM(III) OXIDE HYDRATE: The International Union of Pure and Applied Chemistry's (IUPAC) name for **hydrated aluminium oxide** also known as **aluminium hydrate**, **aluminium hydroxide** and **hydrated alumina** with the chemical formula $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ or $\text{Al}(\text{OH})_3 \cdot \text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ or $\text{Al}(\text{OH})_3$. It is a white powder, which is insoluble in water but soluble in hydrochloric or sulfuric acid and sodium hydroxide. It is used in the manufacture of ceramic glasses and in paper coating.

ALUMINOSILICATE: Any of colourless chemical compounds, with chemical formula $3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$, consisting of alumina and silica with various metals. They are usually found in minerals and rocks.

ALUNOGENITE: A mineral form of hydrated aluminium sulfate with chemical formula $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$.

ALVAREZ EVENT: A hypothesis that explains that the mass extinction

of the dinosaurs and many other living things was caused by the giant collision or impact of a large asteroid on the Earth during the Cretaceous-Paleogene extinction event.

ALVAREZ, LUIS WALTER (1911–88): An American experimental physicist, who spent 42 years at the University of California, Berkeley (1936 – 1978). After working on radar and the atomic bomb during World War II, he concentrated on particle physics. In 1938, he discovered that some radioactive elements decay, when an orbital electron merges with the atom's nucleus, producing an element with an atomic number smaller by one, a form of beta decay. In 1939, together with Felix Bloch (1905 – 83), they made the first measurement of the magnetic moment of the neutron. In 1959, he built the first large bubble chamber and developed the technique for using it to study charged particles, for which he was awarded the 1958 Nobel Prize for physics.

ALVEOLA (ALVEOLAE, ALVEOLAS): 1. A small depression or pit. 2. Pits arranged in a honey-comb like pattern. 3. One of the pits in the naked receptacle of composite plants. 4. A pore of such fungi as *Polyporus*. 5. The pitted peritheciium in certain fungi.

ALVEOLAR MACROPHAGE: A highly active and aggressive primary phagocyte of the innate immune system, located on the epithelial lining of the lung alveoli, which ingests and destroys any inhaled particles and micro-organisms.

ALVEOLATES: An assemblage of unicellular eukaryotic organisms, that are believed to have common ancestry on certain morphological similarities and molecular systematic. They comprise the ciliates, the apicomplexans and the dinoflagellates.

ALVEOLUS: Singular (Alveoli) 1. A tiny thin-walled air sac found in large numbers in each lung, through which oxygen enters and carbon dioxide leaves the blood. 2. A socket in the jaw bone in which a tooth is rooted.

ALYTIDAE: A family belonging to the order Anura, containing two genera: Alytes (midwife toads) and Discoglossus (painted frogs). They possess thick, disk-shaped tongue resembling a small, round saucer instead of the long, thin tongue of most other species of frogs. They have large eyes usually with vertical pupils that look like top-to-bottom slits or somewhat heart-shaped pupils that are wider at the top than at the bottom.

ALZHEIMER'S DISEASE (SENILE DEMENTIA OF THE ALZHEIMER TYPE, SDAT): A neurological disease which is characterised by progressive loss of memory. It was named after German psychiatrist and neuropathologist, Alois Alzheimer in 1906, who first described it. Early symptom is inability to acquire new memories, such as difficulty in recalling recently observed facts. Generally, it is diagnosed in people over 65 years of age, although it can occur much earlier. No treatments to delay or halt the progression of the disease are as yet available.

AM 0 SUNLIGHT (EXTRATERRESTRIAL IRRADIANCE): Solar irradiance in space just above the atmosphere of the Earth on a plane perpendicular to the direction of the Sun. In other words, it is the spectrum outside the atmosphere. AM is the acronym for **air mass**, which is the optical path length through the Earth's atmosphere for light from a celestial source. Air mass normally indicates relative air mass, the path length relative to that at the zenith at sea level. The value of the air mass is often given by appending its value to AM, so that AM1 indicates an air mass of 1, AM2 indicates an air mass of 2, etc. On average the extraterrestrial irradiance is $1367\text{ Watts/metre}^2$ (W/m^2). This value varies by $\pm 3\%$ as the Earth orbits the Sun. The Earth's closest approach to the Sun occurs around January 4th and it is furthest from the Sun around July 5th. Solar cells used for space power applications, such as those on communications satellites are generally characterised by AM0.

AM 1 SUNLIGHT (TERRESTRIAL GLOBAL IRRADIANCE): The solar irradiance at sea level, traversing the atmosphere, when the direction of the Sun is perpendicular to the surface of the Earth. In other words, it is the spectrum after travelling through the atmosphere to sea level with the Sun directly overhead. It is a useful range for estimating performance of solar cells in equatorial and tropical regions.

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AMACRINE CELL: A relay neuron or interneuron in the retina that synapses with the rod and cone at the affector end and effector end, respectively to transmit nerve impulses of the eye from the rods to the cones.

AMAGAT: A dimensionless measure of density in terms of density at standard temperature and pressure (STP).

AMALGAM: Alloy of mercury and one or more other metals. Most metals will combine with mercury to form amalgam, except iron and platinum. Amalgams of tin and silver, with minor amounts of copper and zinc, are used in dentistry to fill teeth. Sodium amalgam is used in manufacturing chlorine and sodium hydroxide by electrolysis of brine.

AMALGAMATION: 1. The process of combining mercury and one or more metals to form an amalgam. 2. In metallurgy, it is the process of extracting precious metals such as gold and silver from their ores by treatment with mercury.

AMALGAM LAMP: Intense source of ultraviolet radiation produced by an electrical discharge in a lamp with the inner side covered by an amalgam of mercury with another element such as indium or gallium to control the vapour pressure of the mercury. These lamps have 2-3 times the UV output for the same wavelength as the standard low-pressure mercury lamps.

AMALGAMATE: To alloy a metal with mercury or be alloyed with mercury.

AMARANTH: 1. Amaranth family or Amaranthaceae is a family of flowering plants that contains about 165 genera and 2040 species, which are annual or perennial herbs or subshrubs and a few vines or trees. It is distributed nearly worldwide. The leaves are simple and hairy, usually arranged alternately along the stems. The stems, roots, leaves, or flowers of many species are red in colour. The flowers are small, often borne in dense spikes, usually with bracts and bracteoles and can be bisexual or unisexual. The fruits are achenes, capsules or berries. Some species such as garden beets, chard, sugar beets, and mangel-wurzel; lamb's quarters (*Chenopodium album*), and spinach (*Spinacia oleracea*) are cultivated for food. Cultivated ornamentals in the family include globe amaranth (*Gomphrena*) and cockscomb (*Celosia*). One species, *alteranthera repens* (commonly known as khaki burweed or paper thorns), has medicinal properties; the entire plant is used for the treatment of neurosis, constipation and piles. 2. A synthetic red food dye.

AMATEUR RADIO (HAM RADIO): A popular hobby and service that brings people, electronics and communication together. People use ham radio to talk across town, around the world or even into space, all without the Internet or cell phones.

AMATOL: An explosive consisting of ammonium nitrate and TNT (trinitrotoluene).

AMAZON: 1. The greatest river in South America and the largest drainage system in the world in terms of the volume of its flow and the area of its basin. The total length of the river is about 6,400 km, making it the longest river in the world after the Nile River, but that is contentious as some people regard it as longer than the Nile River. The Amazon drainage basin covers an area of about 7,500,000 km². 2. **Amazon.com** is a United States based online shopping site and the world's largest and a prominent cloud services provider with over 200 million unique visitors (across all platforms) in December 2017.

AMBARELLA (HOG PLUM; GOLDEN APPLE; *Spondias dulcis*): A fast-growing tropical deciduous plant in the family Anacardiaceae. It is a fast growing tree that can reach between 12-18 metres in height. It has pinnately compound leaves which are obovate-oblong or elliptical and glossy. The small white flowers are borne in large bunches with male, female and perfect flowers within each cluster.

The long-stalked fruits are produced in bunches of a dozen or more. It is used as a food source, especially the leaves which are used as salad and seasoning for foods.

AMBER: A yellow or reddish-brown fossil resin, which has the property of acquiring an electrical charge when rubbed. The resin was exuded by certain extinct trees and other plants and often contains preserved insects, flowers or leaves that were trapped by its sticky surface before the resin hardened. Amber is used for jewellery and ornaments.

AMBIDENT: A term most commonly applied to conjugated nucleophiles, whose molecular entities each possesses two alternative and strongly interacting distinguishable reactive centres, to either of which a bond may be made in a reaction. The centres must be connected in such a way that reaction at either site stops or greatly retards subsequent attack at the second site. An example is the enolate ion in which electrophilic attack can occur at either the oxygen atom or the beta-carbon atom.

AMBIDENTATE: It describes or refers to a ligand that can coordinate at two different sites. For example, the NO₂ molecule can coordinate through the N atom (the nitro ligand) or through an O atom (the nitrido ligand). Complexes that differ only in the way the ligand coordinates display linkage isomerism.

AMBIENT: A term describing environmental conditions, such as pressure and temperature, present in the immediate environment of a given location. For example, ambient temperature is the temperature of the surrounding medium, such as gas or liquid, which comes into contact with the apparatus.

AMBIGUOUS CASE: If the two sides of a triangle are known and an acute angle is not the angle between the two sides.

AMBIPAROUS BUD: A bud containing young vegetative leaves and young flowers.

AMBO-: A prefix used to denote that a substance is present as a mixture of racemic diastereoisomers in unspecified proportions. For example, if L-alanine is reacted with DL-leucine, the resulting dipeptide can be described as L-alanyl-leucine, to indicate the mixture.

AMBULACRA: Any of five radial bands on the ventral surface of echinoderms, such as the starfish and sea urchin, where the tube feet protrude as well as withdraw.

AMDAHL'S LAW: In computer programming, it states that in a program with parallel processing, relatively few instructions that have to be performed in sequence will have a limiting factor on program speedup such that adding more processors may not make the program run faster.

AMELOBLAST: Cells that are manifested only during tooth development and secrete enamel forming the hard tooth covering of the crown.

AMENORRHOEA: The absence of menstrual period in females that is not caused by childhood or immaturity, pregnancy, menopause and breastfeeding. Amenorrhoea may be caused by problems with the reproductive system such as hormonal (oestrogen and progesterone) imbalances. Other causes include weight loss, weight gain, emotional stress, excessive exercises, etc.

AMENSALISM: A symbiotic relationship between organisms in which a weaker organism is inhibited or damaged by a stronger organism, whereas the latter is neither inhibited nor stimulated or an interaction between populations of two species that harm one but not the other. For example, penicillium secretes penicillin, a chemical that kills bacteria. Also, when sheep or cattle make trails in grass that they trample on, they kill the grass. Another example is the black walnut tree (*Juglans nigra*), which secretes juglone, a chemical that harms or kills some species of neighbouring plants, from its roots. This interaction may still increase the fitness of the non-harmed organism though, by removing competition and allowing it access to greater scarce resources.

AMENT (CATKIN): An inflorescence made up of a dense spike or raceme of apetalous, unisexual flowers. Examples are Betulaceae and Salicaceae.

AMENTACEOUS: In botany, it refers to a catkin-like structure or bearing.

A

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI): US national standards body founded in 1918 concerned with the development of technology standards. It enhances the global competitiveness of U.S. business and the American quality of life by promoting and facilitating voluntary consensus standards and conformity assessment systems to promote their integrity.

AMERICAN STANDARD CODE FOR INFORMATION INTERCHANGE (ASCII): A coding system used in computing, in which numbers are assigned to letters, digits and punctuations.

AMERICIUM: A silvery, shiny radioactive metallic transuranic element belonging to the actinoids, with the chemical symbol Am and atomic number 95, melting point 1176 °C (1449 K), boiling point 2011 °C (2284 K), density 12 g cm⁻³ and the mass number of the most stable isotope is 243. It was discovered in 1944. It is obtained from the beta decay of plutonium and it is a source of alpha particle used in research. It is also commonly used in smoke alarms.

AMES TEST (SALMONELLA MUTAGENESIS): A test to determine the effects of a chemical on the rate of mutation in bacterial cells and its likelihood of causing cancer in other organisms. It is widely used in screening chemicals, drugs and cosmetics for possible carcinogenic activity.

AMETABOLOUS: It refers to insects that exhibit a life cycle of incomplete metamorphosis or shows no metamorphosis. Such insects have no distinct structural differences between their young (nymph) and the adult (imago) forms. Examples are grasshoppers, cockroaches, cotton strainers and termites.

AMETHYST: A variety of quartz (SiO₂), which is coloured violet by the presence of small quantities of impurities such as manganese or iron. It is used as a semi-precious stone.

AMICABLE NUMBERS: Two different numbers so related that the sum of the proper divisors of each is equal to the other. For example, the pair of numbers 220 and 284 are said to be amicable numbers in the sense that the sum of the proper positive divisors of each, sum to the other. For

$220 \Rightarrow 1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 284$; and for $284 \Rightarrow 1 + 2 + 4 + 71 + 142 = 220$.

AMIDES: 1. Any organic chemical derived from carboxylic acids by the replacement of the hydroxyl group (-OH) with an amino group (-NH₂). The O=C—N group in an amide forms the amide group. Chalcogen replacement analogues are called thio-, seleno- or telluro-amides. Compounds having one, two or three alkyl groups on a given nitrogen atom may be designated as primary, secondary and tertiary amides, respectively. The three simplest amides are methanamide (carbamide or formamide, HCONH₂), ethanamide (acetamide, CH₃CONH₂) and propanamide (CH₃CH₂CONH₂), which are known as primary amides. Referring to Fig III, R¹ = H or alkyl (H(CH₂)_n group, R² = R³ = H gives a primary amide, for example, methanamide (HCONH₂), whose structure is shown in Fig IV. If R² = H and R³ = alkyl, it gives secondary amides, for example, N-methylethanamide, whose

structure is shown in Fig V. If R² and R³ = alkyl, it gives tertiary amide, for example, N-methyl-N-ethylethanamide, whose structure is shown Fig VI. Primary amides with the general formula RCONH₂ are named by replacing the -oic acid or -ic acid ending with -amide. Secondary amides (RCONHR) and tertiary amides (RCONR₂) are named by using an upper case N to designate that the alkyl group(s) is/are on the nitrogen atom. Alkyl groups attached to the nitrogen are named as substituents. The letter N is used to indicate they are attached to the nitrogen. 2. Metal derivatives of ammonia and amines, in which a cation replaces a hydrogen atom on nitrogen. Such compounds are also called azanides, for example, lithium diisopropylamide.

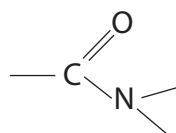


Fig. I:
General Structure
of Amide Group;
the O=C—N group
forms the amide group

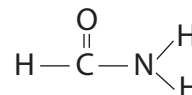


Fig. IV: Structure of
Methanamide
(a Primary Amide)

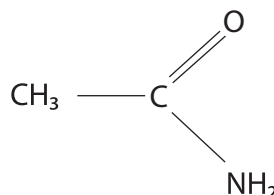


Fig. II: Structure of
Ethanamide, an example
of a Primary Amide

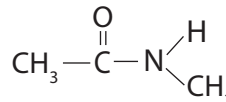
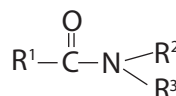


Fig. V: Structure of
N-methylethanamide



R² R³ = alkyl gives
tertiary amide

Fig. III: Structure of
Amides

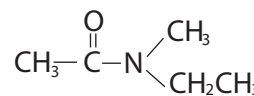


Fig. VI: Structure of
N-methyl-N-ethylethanamide

AMIDINES: Derivatives of oxoacids R_nE(=O)OH in which the hydroxy group is replaced by an amino group and the oxo group is replaced by =NR. Amidines include carboxamidines, sulfinamidines and phosphinamidines, R₂P(=NR)NR₂. In organic chemistry an unspecified amidine is commonly a carboxamidine. **AMIDRAZONES (AMIDE HYDRAZONE; HYDRAZIDE IMIDE):** Compounds having the structure RC(=NH)NHNH₂ or RC(NH₂)=NNH₂ derived from carboxylic acids. Also included are N-hydrocarbyl derivatives.

AMINALS (GERMINAL DIAMINES): Compounds having two amino groups attached to the same carbon, R₂C(NR₂)₂.

AMINASE: One of a group of enzymes that can catalyse the hydrolysis of amines.

AMINATION: The process in which an amine group (NH₂) is introduced into an organic molecule. Enzymes which catalyse the reaction are termed **aminases**.

AMINE: Any of a class of organic chemical compounds in which one or more of the hydrogen atoms of ammonia have been replaced by other groups of atoms such as an alkyl or aryl group. The four categories of amine include the following: (i) **Primary amines** with the general structural formula RNH_2 have two hydrogen atoms and one alkyl or aryl group attached to the nitrogen to form the amine or amino group $-\text{NH}_2$. Examples of primary aliphatic amines include methylamine (aminomethane), ethylamine (aminoethane), propylamine, 2-aminopropane and butylamine. In primary aromatic amines, the amino or amine group is directly attached to the aromatic benzene ring. Examples include phenylamine or aniline, 1,3-diaminobenzene and 2-methylphenylamine. (ii) **Secondary amines**, with the general structural formula R_2NH have one hydrogen atom and two alkyl or aryl groups attached to the nitrogen (N) or no hydrogen if one of the substituent bonds is double. Examples of secondary aliphatic amines include dimethylamine, diethylamine, dipropylamine, ethylpropylamine and methylethanolamine; examples of secondary aromatic amines include diphenylamine, N-methylphenylamine and N-ethylphenylamine. (iii) **Tertiary amines** with the general structural formula R_3N have all three hydrogen atoms replaced by organic substituents. Examples of aliphatic tertiary amines include trimethylamine, ethyldimethylamine and diethylmethylamine. Examples of aromatic tertiary amines include N,N-dimethylphenylamine, N,N-diethylphenylamine and triphenylamine. (iv) **Cyclic amines** are either secondary or tertiary amines. Examples include the 3-member ring aziridine and the six-membered ring piperidine. N-methylpiperidine and N-phenylpiperidine are examples of cyclic tertiary amines. (v) **Quaternary amine** results when a fourth hydrogen is added and has a positive charge on the nitrogen atom. Examples include tetramethylammonium chloride and tetrapropylammonium chloride.

AMINE OXIDES: Compounds derived from tertiary amines by the attachment of one oxygen atom to the nitrogen atom $\text{R}_3\text{N}^+-\text{O}^-$. It also includes the analogous derivatives of primary and secondary amines. Examples are pyridine N-oxide, a water-soluble crystalline solid and N-methylmorpholine N-oxide, which is an oxidant. Amine oxides are surfactants commonly used in consumer products such as shampoos, conditioners, detergents and hard surface cleaners.

AMINIUMYL RADICAL IONS: Radical cations, R_3N^+ derived from trialkylammonium ions, R_3NH^+ by removal of a hydrogen atom. Aminiumpy radical ions are, except for H_3N^+ , synonymous with the ammoniumpy radical ions.

AMINO: A term used to describe a chemical compound containing the $-\text{NH}_2$ group of atoms, combined with a non-acidic radical.

AMINO ACID: A class of organic compounds with the general formula $\text{R}-\text{CH}(\text{NH}_2)\text{COOH}$, in which a carbon atom has bonds to an amino group ($-\text{NH}_2$), a carboxyl group ($-\text{COOH}$), a hydrogen atom ($-\text{H}$) and an organic side group ($-\text{R}$). They are, therefore, both carboxylic acids and amines. The physical and chemical properties unique to each result from the properties of the R group. Amino acids are the building blocks

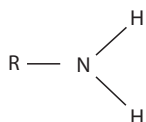


Fig I:

Chemical Structure of Primary Amine

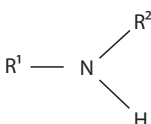


Fig II:

Chemical Structure of Secondary Amine

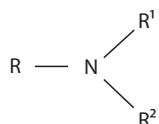
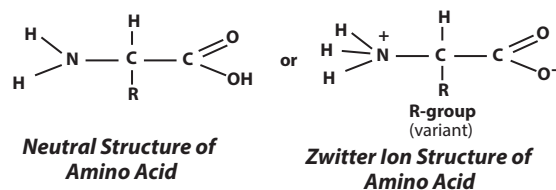


Fig III:

Chemical Structure of Tertiary Amine

for proteins. There are over 100 natural amino acids, but only 20 make up the proteins of all living organisms. Humans can synthesise 10 of them but the other 10 comprising the essential amino acids of arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine, must be consumed in the diet. Amino acids share a common structure except for one chemical group (R, side chain) attached to the central carbon atom, as shown in the diagram.



AMINO ACID ACTIVATION: The first stage of synthesis of proteins, in which the amino acid is attached to transfer RNA (tRNA) using ATP (Adenosine triphosphate) for energy. This is done by a process called aminoacylation, or a process in which an enzyme called aminoacyl-tRNA synthetase adds the correct amino acid to its tRNA.

AMINO ACID RACEMISATION/RACEMIZATION (AAR): A dating technique used in archaeology which is based on the relative amounts of optical isomers of an amino acid in a sample. In most organisms, the L-isomer of the amino acid is the one produced by metabolism. When the organism dies, this isomer slowly converts into the D-form and eventually equilibrium is reached, in which the forms are present in equal amounts. A particular application in forensic science involves measuring the D/L ratio of aspartic acid in the dentine of teeth. Once a tooth has fully formed, the dentine is isolated by the enamel and then racemisation takes place in the living subject at a fairly constant temperature and moisture level.

AMINO ACID SEQUENCE (PEPTIDE SEQUENCE): The order in which amino acid residues, connected by peptide bonds, lie in the chain in peptides and proteins. The sequence is generally reported from the N-terminal end containing free amino group to the C-terminal end containing free carboxyl group. Peptide sequence is often called **protein sequence** if it represents the primary structure of a protein.

AMINO GROUP: A functional group with the general formula $-\text{NRR}'$, where R and R' may be hydrogen atoms or organic radicals. The commonest form is $-\text{NH}_2$. Compounds containing amino groups include amines and amino acids.

AMINO SUGAR: A monosaccharide or simple sugar in which a non-glycosidic hydroxyl group is replaced by an amino or substituted amino group and are components of glycoproteins. The most commonly occurring amino sugars are glucosamine and galactosamine.

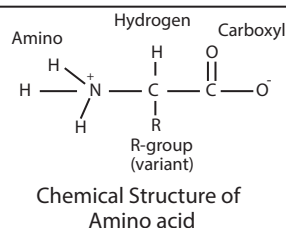
AMINOACYL/ACCEPTOR SITE (A SITE): The site on the ribosome which is the binding site for charged t-RNA molecules during protein synthesis.

AMINOACYL-TRNA SYNTHETASES: A family of enzymes, at least one for each amino acid that play a major role in translation during protein synthesis, by catalysing the attachment of an amino acid to its specific tRNA molecule. The amino acid is esterified through its carboxyl group to the 3'-hydroxyl group of the 3'-terminal adenosine of a molecule of transfer RNA.

AMINOGLYCOSIDE: Any group of antibiotics derived from species of Streptomyces or Micromonosporum that are introduced into the body by injection and effective against gram-negative aerobes and some anaerobic bacilli. Examples are streptomycin, neomycin and gentamicin.

AMINOPEPTIDASE: Any enzyme produced by the intestinal mucosa that cleaves amino acids from the N-terminus of peptides or polypeptides.

AMINOPHENOL: Any of various organic compounds with chemical formula $\text{H}_2\text{NC}_6\text{H}_4\text{OH}$, which are used as reducing agents, especially as photographic developers and for making dyes. Examples include



amidol (the dihydrochloride of 2,4-diaminophenol), metol (the hemisulfate of 4-methylaminophenol) and rhodinol (4-methylaminophenol).

AMINOPLASTIC RESIN: Synthetic resin derived from the reaction of urea, thiourea or melamine with an aldehyde, particularly formaldehyde. They are used as adhesive and as coating for paper and textiles.

AMINOTRANSFERASE (AMINOPHERASE; TRANSAMINASE): An enzyme that catalyses the transfer of an amino group from an alpha-amino acid to an alpha-ketoacid, as in the transfer of glutamic acid to oxaloacetic acid to form aspartic acid and alpha-ketoglutarate.

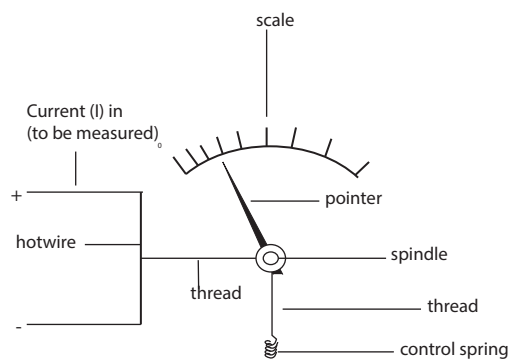
AMINOXIDES: The anion $\text{H}_2\text{N}-\text{O}^-$, aminoxide and its N hydrocarbyl derivatives ($\text{R}_2\text{N}-\text{O}^-$) formally derived from hydroxylamines ($\text{R}_2\text{N}-\text{OH}$), by removing a hydron from the hydroxy group, for example, $(\text{CH}_3)_2\text{N}-\text{O}^-$ dimethylaminoxide.

AMINOXYL RADICALS: The uncharged form of an aminoxide derived from hydroxylamines by removal of a proton. Its chemical structure is represented as $\text{R}_2\text{NO}^\cdot \rightleftharpoons \text{R}_2\text{N}^+-\text{O}^-$.

AMINYL RADICALS: The nitrogen-centred radical $\text{H}_2\text{N}^\cdot$, formally derived by the removal of a hydrogen atom from ammonia and its hydrocarbyl derivatives $\text{R}_2\text{N}^\cdot$.

AMITOSIS (DIRECT NUCLEAR DIVISION): Cell division by simple division of the nucleus and cytoplasm, without the nuclear changes that usually occur during cell reproduction.

AMMETER: An instrument used to measure the current flowing through an electric circuit. The main types include moving coil ammeter, moving iron ammeter and the thermo ammeter. Ammeters, as well as voltmeters and ohmmeters, are designed with the use of a sensitive current detector such as a galvanometer. An ammeter is always connected in series in a circuit. When a resistor with a low internal resistance is connected across (parallel) a galvanometer, it can be used as a current measuring instrument. The low internal resistance resistor serves as a by pass (shunt) and allows most of the current to pass through so that the desired current flows through the meter at its full-scale deflection.



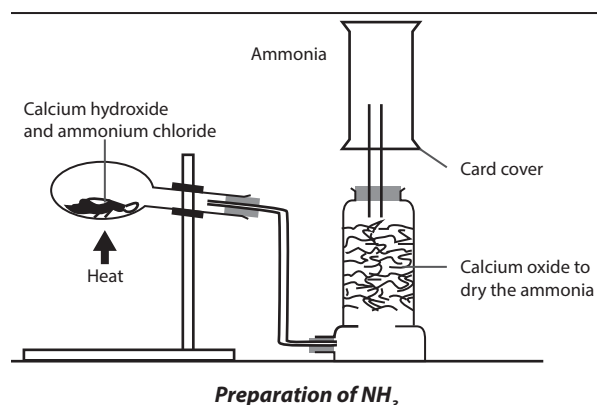
Hot-wire ammeter

AMMINE: A ligand of ammonia molecule (NH_3), used to prepare complex species by treating aqueous solutions of metal salts with ammonia or, in some instances, by the action of dry gaseous or liquid ammonia on the anhydrous salt. An example of an ammine is the tetraamminecopper(II) ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$.

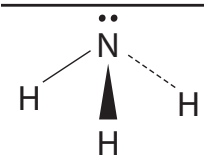
AMMINE COMPLEXES: Complex species that contain ammonia molecules bonded to metal ions. Examples of ammine complexes are $\text{Cr}(\text{III})$, $\text{Co}(\text{III})$, $\text{Ni}(\text{II})$, $\text{Cu}(\text{II})$ as well as several platinum group metals.

AMMONIA (AZANE; HYDROGEN NITRIDE; TRIHYDROGEN NITRIDE; NITRO-SIL): A colourless, pungent smelling gas, with the chemical formula NH_3 , melting point -77.73°C (195 K) and boiling point -33.34°C (240 K), which is lighter than air and forms a minute proportion of the atmosphere. It is very soluble in water, giving an alkaline solution. It turns moist red litmus paper blue and forms white

fume with concentrated hydrochloric acid. It is found in volcanic gases and as a product of decomposition of animal and vegetable matter. It may be produced from coal or by the Haber-Bosch process. It is used to make fertilisers and nitric acid.



AMMONIA CLOCK: A form of atomic clock in which the frequency of a quartz oscillator is controlled by the vibrations of excited ammonia molecules. The ammonia molecule (NH_3) consists of a trigonal pyramid with a nitrogen atom at the apex and one hydrogen atom at each corner of the triangular base. When the molecule is excited once every 20.9 microseconds the nitrogen atom passes through the base and forms a pyramid at the other side, a process known as inversion. After 20.9 microseconds it inverts to its original position. This inversion has a frequency of 23 870 hertz and ammonia gas will only absorb excitation energy at exactly this frequency. By using a crystal oscillator to feed energy to the gas and a suitable feedback mechanism, the oscillator can be locked to exactly this frequency.



Chemical structure of Ammonia Molecule

AMMONIA OXIDATION TEST: A test which is conducted to evaluate ammonia oxidation rate for nitrifiers.

AMMONIACAL: A term which describes a solution in which the solvent is aqueous ammonia.

AMMONIFICATION: 1. Treatment with ammonia or an ammonium compound. 2. A process that takes place in the soil in which bacteria and fungal decomposers convert proteins from dead organisms and urea from excrement into ammonia.

AMMONIFYING BACTERIA: Bacteria or fungi that break down protein and other nitrogenous substances to produce ammonia.

AMMONITE: 1. A mixture of dried animal wastes, used as a fertiliser. 2. An explosive consisting of ammonium nitrate and trinitrotoulene (TNT). 3. Extinct cephalopod mollusc of the order Ammonoidea. They are related to the modern nautilus.

AMMONIUM: The univalent ion, NH_4^+ or positively charged polyatomic group with the chemical formula NH_4 . It is formed by the reaction of ammonia (NH_3) with an acid.

AMMONIUM CARBONATE: A white crystalline solid, with the chemical formula $(\text{NH}_4)_2\text{CO}_3$, density 1.50 g/cm^3 , melting point 58°C (331 K), decomposes at a boiling point and readily sublimates at room temperature and breaks down slowly into its constituent gases: ammonia, carbon dioxide and water. It is manufactured by heating a mixture of ammonium chloride and calcium carbonate and recovering the product as a sublimed solid. It is used in dyeing, wool preparation and in baking powder.

AMMONIUM CHLORIDE (SAL AMMONIAC): A volatile salt with the chemical formula NH_4Cl that forms white crystals around volcanic craters. It is prepared synthetically for use as electrolytes in dry-cell batteries, as flux in soldering, tinning and galvanising and as fertilisers and dyes.

AMMONIUM FIXATION: The adsorption or absorption of ammonium ions (NH_4^+) by the inter-layer sites of the clay minerals of the soil in such a way that they are neither water soluble nor readily exchangeable.

AMMONIUM HYDROGEN CARBONATE (AMMONIUM BICARBONATE; BICARBONATE OF AMMONIA; HARTSHORN; POWDERED BAKING AMMONIA): A white crystalline compound, with the chemical formula NH_4HCO_3 , formed naturally as a decay product of nitrogenous matter and is made commercially by various methods including the action of carbon dioxide and steam on a solution of ammonium carbonate, heating commercial ammonium carbonate and the interaction of ammonia, carbon dioxide and water vapour. It is used in baking powders and medicines.

AMMONIUM HYDROXIDE (AMMONIA SOLUTION): An alkali, with the chemical formula NH_4OH which is made by dissolving ammonia in water. The solution contains hydrates of ammonia. It is used in making soaps and fertilisers. Although its name suggests a salt with composition $[\text{NH}_4^+][\text{OH}^-]$, it is not actually possible to isolate samples of NH_4OH .

AMMONIUM ION: Ion NH_4^+ formed when ammonia accepts a proton from an acid. The chemical properties of ammonium salts are frequently very similar to those of equivalent alkali-metal salts. When heated with an alkali it produces ammonia gas. Aqueous solution is acidic.

AMMONIUM NITRATE: A colourless, crystalline solid with the chemical formula NH_4NO_3 , density 1.725 g/cm^3 , melting point 169.6°C (442.75 K), which decomposes at 210°C (483 K) and is prepared by the neutralisation of nitric acid with ammonia. It is used in fertilisers, herbicides, insecticides and explosives.

AMMONIUM PHOSPHATE (TRIAMMONIUM PHOSPHATE): A colourless, crystalline compound with chemical formula $(\text{NH}_4)_3\text{PO}_4$, which is obtained from mixing concentrated solutions of ammonia and phosphoric acid or the addition of excess of ammonia to the acid phosphate $(\text{NH}_4)_2(\text{HPO}_4)$. It is used as fertiliser and fire retardant.

AMMONIUM SULFATE: A white crystalline compound, with the chemical formula $(\text{NH}_4)_2\text{SO}_4$, which is used as fertiliser and for water purification.

AMMONIUM THIOCYANATE: A colourless, water-soluble crystalline compound with chemical formula NH_4SCN , density 1.305 g/cm^3 , melting point 149.5°C (423 K), which decomposes at 170°C (443 K). It is made from ammonia and carbon sulfide in ethanol. On heating, it turns into its isomer thiourea, $\text{SC}(\text{NH}_2)_2$. Its solutions give a characteristic blood-red colour with iron (III) compounds and so are employed as a test for ferric iron. It is used as a rapid fixative in photography and as an ingredient in making explosives.

AMMONOTELIC: A term which describes animals, especially aquatic ones, that excrete nitrogenous waste in the form of ammonia.

AMNESIA: The complete or partial loss or impairment of memory, such as facts, information and experiences, but not identity. It is usually caused by disease, brain injury, use of some drugs, shock or as a symptom of psychological trauma. Symptoms include short-term memory or inability to retain new information, while long-term memories from before the event are not affected (**anterograde amnesia**); and inability to recall past events and information before onset of amnesia, called **retrograde amnesia**.

AMNIOCENTESIS: A method of prenatal testing in the early second trimester of pregnancy in which amniotic fluid is withdrawn from the uterus through a needle and analysed to detect biochemical or chromosomal disorders of an unborn baby. Some of the disorders tested for include Down syndrome and hereditary genetic disorders such as Tay-Sachs disease, sickle cell disease, haemophilia, muscular dystrophy and cystic fibrosis.

AMNION: The innermost of the three membranes in mammals which form a sac that enclose the foetus within the uterus. It contains the amniotic fluid which helps to cushion the embryo.

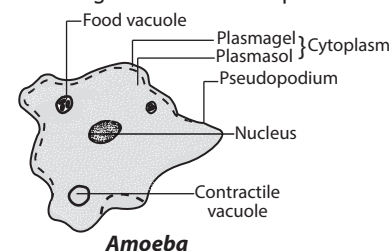
AMNIOTE: Higher vertebrates comprising reptiles, birds and mammals whose embryos are totally enclosed in a fluid filled sac called the amnion.

AMNIOTIC EGG: An egg with liquid-filled sac in compartments with a food sac and a waste sac in which the embryo develops.

AMNIOTIC FLUID: The liquid in the amnion that surrounds and protects the embryo in higher vertebrates.

AMNIOTIC SAC: The two-layered (the inner amnion and the outer chorion) membranous bag that surrounds the foetus or embryo in the uterus. It is filled with watery amniotic fluid in which the embryo or foetus is suspended as pregnancy advances and normally ruptures in early labour to release the amniotic fluid that lubricates the birth canal.

AMOEBA: A microscopic protist or single-celled microscopic animals found in ponds, ditches, moist soil or as parasites in digestive tract of humans and animals. Its body consists of a granular and colourless cytoplasm that changes shape constantly. It engulfs food and moves about by extending finger-like projections of protoplasm. Some species are harmful parasites.



AMOEBIASIS: An infection or disease affecting the bowel, which is caused by the amoeba *Entamoeba histolytica*.

AMOEBIIC: Of or caused by amoeba, for example, amoebic dysentery.

AMOEBOCYTE: An animal cell, such as a leukocyte or white blood cell whose location is not fixed and is, therefore, able to wander through the body tissues, engulfing particles. Amoebocytes are named after their resemblance, especially in their movement, of amoeba.

AMOEBOID: Having no definite shape to the cell and able to alter shape. **Amoeboid movement** is a type of motility by amoeba and other cells in multicellular animals including leukocytes (white blood cells). It is a crawling-like type of movement accomplished by protrusion of cytoplasm of the cell involving the formation of pseudopodia.

AMOEBOZOANS: An assemblage of eukaryotic protists, based largely on molecular systematic that includes the plasmodial or true slime moulds, the dictyostelid cellular slime moulds, the lobose amoebas and the archamoebas, such as the pelobionts and many other features of eukaryotic cells, apart from a membrane-bounded nucleus.

AMOLED (ACTIVE-MATRIX ORGANIC LIGHT-EMITTING DIODE): A type of organic light-emitting diode (OLED) used in screen displays consisting of a thin film transistor of an organic substance, often thin films of polymer sandwiched between two conductors that emits light when an electric current is applied. AMOLED has a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes. The screen is lighter, thinner, and durable. AMOLED screen technology has been used in smartphones, television screens, computer monitors, etc. Another type of organic light-emitting diode (OLED) is the passive-matrix organic light-emitting diode (PMOLED) that does not contain a storage capacitor and so the pixels in each line are actually off most of the time and needs more power to make them brighter.

AMORPHOUS: 1. Without any clear shape, form or structure. 2. Not obviously belonging to any category or type. 3. A term describing a solid that is not crystalline, i.e. one that has no long-range order in its lattice. Many powders that are described as amorphous in fact are composed of microscopic crystals, as can be demonstrated by X-ray diffraction. Glasses are examples of true amorphous solids.

AMORPHOUS CARBON: A carbon material without long-range crystalline order. Short-range order exists, but with deviations of the interatomic distances and/or interbonding angles with respect to

A

the graphite lattice as well as to the diamond lattice. In mineralogy, amorphous carbon is the name used for coal, soot and other impure forms of the element or carbon that are neither graphite nor diamond. In a crystallographic, however, these materials are not truly amorphous, but are polycrystalline or nanocrystalline materials of graphite or diamond within an amorphous carbon matrix.

AMORPHOUS MINERAL: 1. A mineral that lacks a definite crystalline structure. 2. A mineral that has a definite crystalline structure, but appears amorphous because of the small crystallite size.

AMORPHOUS SEMICONDUCTOR: A non-crystalline semiconductor material that has no long-range order. It can be deposited in thin films at low temperatures onto a variety of substrates. A layer of amorphous semiconductor, such as copper indium diselenide, cadmium telluride, gallium arsenide or amorphous silicon, a few microns or less in thickness, is used to make solar cells. Although easier and cheaper to make than a crystalline semiconductor, it is less efficient and slowly degrades over time.

AMOUNT CONCENTRATION (AMOUNT OF SUBSTANCE CONCENTRATION; MOLARITY; SUBSTANCE CONCENTRATION): Amount of a constituent divided by the volume of the mixture. For entities B it is often denoted by $[B]$. In other words, it is the number of moles contained or dissolved in one cubic decimetre of a solution. The common unit is mole per cubic decimetre (mol dm^{-3}) or mole per litre (mol L^{-1}). Its symbol is c , sometimes denoted by M .

AMOUNT FRACTION (MOLE FRACTION): The amount of a constituent divided by the total amount of all constituents in the mixture. Amount fraction is equal to the number of entities of one constituent divided by the total number of entities in the mixture.

AMOUNT OF SUBSTANCE: The amount of chemical or elementary entities or pure substance (atoms, molecules, ions or electrons) containing the same number of chemical units as there are atoms in 12g of carbon-12. The SI unit of amount of substance is the mole. Its symbol is n .

AMPERE (AMP): The SI unit of electric current, with the symbol A. One ampere, is defined as the constant current that, if maintained in two straight parallel conductors of infinite length and negligible circular cross-section and placed 1 metre apart in vacuum, would produce between these conductors a force of 2×10^{-7} newton per metre. One ampere is equal to 6.28×10^{18} electrons flowing around a circuit each second. A current of about 0.25 ampere lights a 60 watt bulb in a table lamp. It measures the rate of flow of a charge. It is named after the French physicist André Marie Ampère (1775-1836).

AMPERE BALANCE (CURRENT BALANCE; KELVIN BALANCE): An electromechanical device used to measure current absolutely, on the basis of definition of the ampere.

AMPERE HOUR: A practical unit of electric charge equal to the charge flowing for one hour through a conductor with a current of one ampere. It is equal to 3600 coulombs.

AMPERE TURN: The SI unit of the magneto motive force produced when a current of one ampere flows through one turn of a magnetising coil.

AMPÈRE, ANDRÉ-MARIE (1775-1836): French physicist who is best known for analysing and putting electromagnetism (electrodynamics) on a mathematical basis. The ampere was named after him.

AMPERE'S LAW (AMPERE-LAPLACE LAW): A law used to determine the magnetic flux density B and the distance r from a conductor of length l , carrying a current I . It is of the form $dB = (\mu_0 I \sin \theta dl) / \pi r^2$ where dB is the infinitesimal element of the magnitude of the magnetic flux density at a distance r at point P from the element length dl of a conductor, μ_0 is the magnetic permeability of free space, l is the length of the line joining the element of the conductor and P .

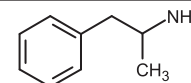
AMPERE'S RULE: A rule developed by the French physicist André Ampère which connects the direction of an electric current and its associated magnetic currents. It states that if a person were travelling along a current carrying wire in the direction of conventional current flow (from the positive to the negative terminal of a battery) and carrying a magnetic compass, then the north pole of the compass needle would be deflected to the left-hand side.

AMPEROMETRIC DETECTION METHOD: A detection method in which the current is proportional to the concentration of the species generating the current. **Amperometry** is the chemical analysis by such methods or techniques that may involve the measurement of electric current.

AMPEROMETRIC TITRATION: A method of determining the chemical composition of a solution by measuring the current passing through a cell containing the solution. The potential is held constant during titration for both the indicator and reference electrodes with changes in the current being measured. The current flowing through the cell is measured as a function of the substance being titrated.

AMPERSAND: A common symbol, '&' for 'and'. It also has several uses in the computer software field, for example, in programming languages, including Visual Basic (to combine variables and literal text). It was incorporated into ASCII (American Standard Code for Information Interchange), the de facto standard for the character encoding used by computers and communications equipment to represent text.

AMPHETAMINE: A colourless, volatile compound, 2-amino-3-phenyl propane or a derivative of this compound, with chemical formula $C_9H_{13}N$, that stimulates the central nervous system by causing the release of the transmitters noradrenaline and dopamine from nerve endings. It inhibits sleep, suppresses appetite and has variable effects on mood. Its prolonged use can lead to addiction.



Chemical Structure of Amphetamine

AMPHIBIA: A Greek word which means "double life" and used to describe: (i) A group of animals including frogs, toads, salamanders and newts that live in both water and on land but which returns to water to breed or a member of a class of vertebrates which generally spend their larval (tadpole) stage in fresh water and undergo metamorphosis, transferring to land at maturity and usually returning to water to breed. (ii) Aircraft that can take off from or land on either land or water or a vehicle that can move over land or water.

AMPHIBOLE: A large group of rock-forming metasilicate minerals present in many igneous and metamorphic rocks. They have a structure of silicate tetrahedra linked to form double endless chains, in contrast to the single chains of the pyroxenes, to which they are closely related.

AMPHIBOLIC PATHWAY: A biochemical or metabolic pathway that employs both catabolism and anabolism. An example of this process is the Krebs's cycle in which there is a breakdown (catabolism) of a four-carbon compound that is added to a two-carbon compound to build up anabolism, a six-carbon compound.

AMPHICRIBAL: A term used to describe a concentric vascular bundle that has the phloem surrounding the xylem.

AMPHIDIPOID: A term which describes an organism, cell or nucleus that contains diploid sets of chromosomes originating from two different species or having a diploid set of chromosomes derived from each parent.

AMPHIGASTRIA: The leaves that form in a row on the undersurface of the stem of a leafy liverwort, scale mosses and other *Hepaticæ*. They are smaller than the leaves on the upper surface and are often only seen just below the apex, having been shed lower down.

AMPHIMIXIS: True sexual reproduction, involving the fusion of male and female gametes leading to the formation of a zygote.

AMPHIOXUS (LANCELET): A Greek word meaning "pointed at both ends" as found in the lancelets of the cephalochordates, which are very small ell-like organisms buried in soil.

AMPHIOXUS EGG: A type of egg found in holoblastic cleavage in which there is little amount of yolk. Holoblastic type of cleavage is commonly seen in eggs containing moderate to sparse amount of yolk, as seen, for example, in amphibians, mammals, echinoderms, annelids, flatworms and nematodes.

AMPHIPHATHIC: 1. A term used to describe a molecule, such as a detergent, which is amphiphilic or containing both hydrophobic and hydrophilic groups. This assumes that one of the two phases is aqueous and the other non-aqueous. If both are non-aqueous such as oil or air, molecules containing organophilic and organophobic groups may be amphipathic and surface active. 2. It often refers to the surface(s) on a protein, particularly an alpha helix, where one surface of the alpha helix has hydrophilic amino acids and the opposite face has hydrophobic or lipophilic amino acids.

AMPHIPHILIC: A term which describes a molecule which contains both a hydrophilic and hydrophobic groups such as soaps and detergents. The existence of distinct polar (hydrophilic) and nonpolar (hydrophobic) regions in the molecule promotes the formation of *micelles* in dilute aqueous solution.

AMPHIPHLOIC: Pertaining to the central vascular cylinder of stems having phloem on both sides of the xylem, for example, as in a solenostele.

AMPHIPHLOIC SIPHONOSTELE (SOLENOTELE): A monostele type of siphonostele, found in certain ferns, such as *Marsilea* and *Adiantum*, which appears in cross-section as 1 ring of phloem around the outside of the xylem and another around the inside of the xylem ring, but outside the pith.

AMPHIPOD: An order of crustaceans (Order Malacostraca) including marine and freshwater species. They are characterised by a compressed body flattened and the absence of a carapace. An example is the beach flea. Some amphipods are nonpigmented cave-dwellers.

AMPHIPROTIC: Self-ionising solvent, which possesses both characteristics of Brønsted acids and bases. Examples are water (H_2O) and methanol or wood alcohol (CH_3OH), in contrast to aprotic solvent.

AMPHISBAENIAN: A group of legless, worm-like, long reptiles characterised by short tails and ring shaped scales that are well adapted to burrowing.

AMPHISTOMATIC: In botany, it refers to a leaf that has stomata on both upper and lower surfaces.

AMPHITHECIUM: The outer layer of the sporophyte of mosses and liverworts that develops into the outer parts of the spore-bearing capsule. In *Sphagnum* and *Anthoceros*, the sporogenous tissue develops from the amphithecium, whereas in other bryophytes it develops from the endothecium.

AMPHITRICHIOUS FLAGELLUM: Having one flagellum located at each end of the bacterial cell, enabling the bacteria to move forward or reverse directly.

AMPHITROPOUS OVULE: A half inverted and straight ovule, which has turned back 90 degrees on its stalk, with the hilum being lateral.

AMPHIVASAL: Describing a concentric vascular bundle in which the xylem surrounds the phloem.

AMPHOLYTE: A substance that conducts electric current as a result of dissociation into positively and negatively charged particles. The most familiar ampholytes are acids, bases and salts, which ionise when dissolved in such solvents as water or alcohol.

AMPHOTERIC: A term used to describe the ability of some chemical compounds or elements to behave either as an acid or a base depending on their environment. For example, aluminium and zinc react with both acids and alkalis. Another example is water in Brownstend-Lowry sense.

AMPHOTERIC OXIDE: A substance such as aluminium oxide (Al_2O_3) and zinc oxide (ZnO), which can behave both as a basic and acidic

oxide. For example, when aluminium oxide (Al_2O_3) reacts with hydrogen chloride (HCl), it acts as a base to form the salt aluminium chloride ($AlCl_3$). When reacted with sodium hydroxide ($NaOH$), it acts as an acid to form sodium aluminate ($NaAlO_2$).

AMPLEXICAUL: Stipules of a sessile leaf such as the upper leaves of henbit (*Lamium amplexi-caule*) clasping the stem but not entirely encircling it.



Amplexicaul leaf

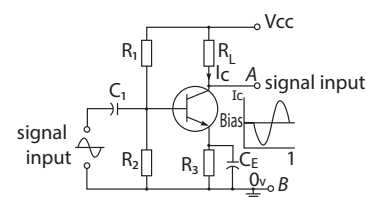
AMPLEXUS (PSEUDOCOPULATION): A type of mating in anuran amphibians (frogs and toads) which does not involve the insertion of an intromittent organ (penis) by the male into a female and the eggs are fertilised outside the female's body. The male grasps a female with his front legs as part of the mating process and the eggs are fertilised as they are released from the female's body.

AMPLIFICATION: 1. In biology, it is usually a large replication of genetic material. 2. In electricity, it is the increase of the amplitude or strength of an electrical signal.

AMPLIFICATION REACTION: A reaction which replaces the conventional reaction used in a particular determination so that a more favourable measurement can be made. The sequence can be repeated to provide a further favourable increase in measurement.

AMPLIFIER: In electronics, it is a device that magnifies or increases the strength of a signal, such as a radio signal or sound. A sine wave (voltage) applied to the input terminal produces a much larger sine wave signal across the output signals. The output signal is inverted compared with the input signal. When the input voltage (signal) is at a peak, the output voltage (signal) is at a trough and vice versa.

Amplifiers are classified by the size of the signal (large or small), its physical configuration and how it processes the input signal (how the input signal and current flowing in the load relates). All amplifiers provide a certain amount of gain or amplification factor. It is the relationship that exists between the signal measured at the output with the signal measured at the input and is usually defined as the mean ratio of the signal output of a system to the signal input of the same system. It is usually expressed by the logarithmic decibel (dB) units. The three different kinds of amplifier gains are voltage gain, current gain and power gain. A gain may be linear or non-linear. In a linear gain, the output signal strength varies in direct proportion to the input signal strength and delivers more power to a load. In a nonlinear amplification, a change in input does not produce a proportional change in output. The four basic types of amplifiers include the following: (i) voltage amplifier, which converts a small input voltage into a much larger output voltage, hence an increased power. It is used to increase signals, for example, to boost an audio signal to speakers. (ii) current amplifier, which converts a small input current into a much larger output current; (iii) transconductance amplifier or operational transconductance amplifier, which converts an input voltage to an output current. Its applications include as variable frequency oscillators, variable gain amplifiers and current-controlled filters. (iv) Transimpedance or transimpedance amplifier converts an input current to an output voltage. It is used to convert low-level photodiode currents to usable voltage signals. In the diagram, R_1 and R_2 are potential dividers that provide the necessary R_1 base-bias; R_L is the load that produces the output across AB; C_1 is the capacitor that isolates the d.c. component in the input signal from the current; C_2 is the large capacitor that prevents undesirable feedback of the amplified signal to the base-emitter circuit; and R_3 is the emitter resistance that stabilises the circuit for excessive temperature rise. As the collector current rises, the potential

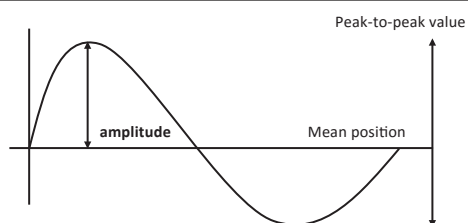


Single stage amplifier

difference between the emitter and base lowers and the collector current is automatically lowered.

A

AMPLITUDE: 1. The farthest distance that a vibrating or oscillating system such as a pendulum travels from a mean or equilibrium point; or the maximum displacement of a wave, for example, a sine wave from its mean position during a cycle. Since the sine function varies from +1 to -1, the amplitude is 1. A sine wave is given by the formula $a\sin(\omega t)$, where a is the amplitude. The energy associated with either a simple harmonic oscillation or a waveform is proportional to the square of the amplitude. With sound waves, the greater the amplitude, the louder is the sound. 2. In mathematics, it is also known as **argument**, **anomaly** or **azimuth**, it is the angle that describes the direction of a complex number when it is plotted in a complex plane using polar coordinates. It is measured in radians as an angle in standard position.



Amplitude of a sinusoid or oscillating system

AMPLITUDE MODULATION (AM): The process of varying or modulating the amplitude of a high frequency carrier wave according to that of a low-frequency analogue audio signal for transmission in order to make the audio-frequency radio waves travel longer distances without change in the quality of the input information. In this type of transmission the maximum displacement of the carrier waves is increased or decreased as the signal increases or decreases.

AMPULLA: A slight swelling at the base of each semicircular canal in the inner ear, which detects the movement of the head. The sense of balance largely depends on sensitive hairs within the ampulla responding to movements of fluid within the canal by cytoplasmic streaming.

AMR (ADAPTIVE MULTI-RATE): A codec based on an audio compression format that is used for speech coding. It is a preferred codec for the Global System for Mobile Communications (GSM) standard and used by most third-generation (3G) cellular systems. AMR codec provides speech rates ranging from 4.75 to 12.2 kbits/s. Its advantages are providing superior speech quality, better power control, increased resistance to interference and errors, etc.

AMU UNIT (ATOMIC MASS UNIT): A unit used to indicate mass on an atomic or molecular scale. Its value is precisely one-twelfth the mass of an atom of carbon-12, which has six protons and six neutrons in its nucleus.

AMYGDALA: An almond-shaped mass of nerve cells located deep in the cerebral hemispheres of the brain and forms part of the limbic system. The amygdala is concerned with decoding emotions such as pleasure and fear, particularly those that are threatening to the organism. In humans, it is the most sexually-dimorphic brain structure and shrinks by more than 30% in males upon castration. Consequently, many sensory inputs converge in the amygdala to inform it of potential dangers in its environment. This sensory information comes to the amygdala either directly from the sensory thalamus or from the various sensory cortices. The amygdala also receives numerous connections from the hippocampus. Since the hippocampus is involved in storing and retrieving explicit memories, its connections to the amygdala may be the origin of strong emotions triggered by particular memories.

AMYGDALIN: A bitter crystalline compound with molecular formula $C_{20}H_{27}O_{11}N$, which is a glucoside found in bitter almonds, cherry kernels, apricots and other fruits and vegetable products.

AMYL ALCOHOL (PENTYL ALCOHOL): Any of eight isomers of the alcohol composition with the chemical formula $C_5H_{11}OH$ but different structures. It is synthesised from hydrocarbons or extracted from fusel oil. Mixtures of these compounds are used as solvents for resins and oily materials and in the manufacture of other chemicals, especially amyl acetate, a solvent for nitrocellulose lacquers. Commercial amyl alcohol, a mixture of these isomers, has a sharp odour and is used as a solvent.

AMYL COMPOUND: An organic compound containing the amyl group C_5H_{11} , which has eight isomers. Amyl compounds include: (i) **amyl alcohols**, synthesised from hydrocarbons or extracted from fusel oil and used as a solvent; (ii) **isoamyl nitrite** ($C_5H_{11}ONO$), a vasodilator used to dilate the blood vessels of patients with some forms of heart disease such as angina; (iii) **amyl acetate** also known as **banana oil** with a pleasant, fruity smell and used as a solvent for nitrocellulose and as a flavouring for candy.

AMYL GROUP (PENTYL GROUP): Any of the isomeric arrangement of the monovalent alkyl group, with the chemical formula C_5H_{11} .

AMYL NITRITE: A pale brown volatile liquid organic compound, with chemical formula $C_5H_{11}ONO$, which is normally used in medicine to dilate the blood vessels of patients with some forms of heart disease such as angina.

AMYLASE (DIASTASE): Any enzyme which catalyses the hydrolysis of starch to dextrin and dextrin to maltose. It is common in plants and animals. The α -amylases occur in animals and include pancreatic and salivary amylase; the β -amylases occur in higher plants.

AMYLIN: The insoluble covering (cellulose) of starch grains.

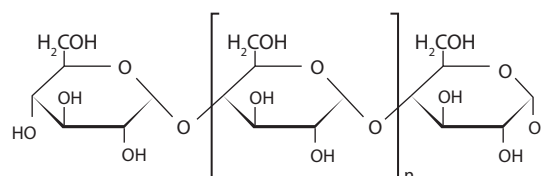
AMYLOID: A hard waxy deposit consisting of protein and polysaccharides that result from the degeneration of tissue.

AMYOLECTIN: A highly branched polysaccharide component of starch stored in plants. It gives a red-violet iodine-potassium iodide solution.

AMYOPLAST: An organelle in plants that stores starch and often found in non-photosynthetic tissues. Examples are root tubers (e.g. potato) and fruits.

AMYOLOPSIN: A pancreatic enzyme, α -amylase, that converts insoluble starch into soluble maltose.

AMYOLOSE: A straight-chain polysaccharide component of starch stored in plants, which is a constituent of most starch. It is a straight chain with α (1 $\rightarrow$ 4) linkages. In water, amylose reacts with iodine to give a characteristic blue colour.



Chemical Structure of Amylose

AMYLUM (STARCH): The white, granular, polysaccharide with the general chemical formula $(C_6H_{10}O_5)_n$, where n may range from 100 to several thousands. It is manufactured in the green leaves of all green plants from excess glucose produced during photosynthesis and serves the plant as a reserve food supply. It is used by humans and animals to produce energy, thus forming the most important component in their diet. The starch is broken down into its constituent sugar molecules, which then supply energy to the tissues. In the dark the starch thus formed is rapidly broken down to sucrose and transported to other organs. It is contained in large amounts in plants such as the roots of cassava (*Manihot esculenta*), which yield tapioca, the rhizomes of arrowroot (*Maranta arundinacea*), the stem pith of the some sago palms such as species of *Metroxylon* and *Arenga*, the tubers of potato and the grains of various cereals, especially maize, wheat, rice and sorghum.

Starch is processed to produce many of the sugars in commercial foods. It can be used as a thickening, stiffening or gluing agent. It is also used as adhesive in the papermaking industries.

ANABATIC WIND: Warm wind that blows up a slope of a hill or mountain. It is generated by the effects of local heating as a result of the temperature difference that occurs when warm air comes in contact with the ground.

ANABIOSIS: A temporary state of reduced metabolism in organisms with metabolic activity absent or undetectable, especially one in which certain aquatic invertebrates are able to survive long periods of drought.

ANABOLIC STEROID: A biochemical such as androgens which occurs naturally in the body or man-made male sex hormones that stimulate muscle growth. Bodybuilders and athletes often use anabolic steroids to build muscles and improve athletic performance, although it is illegal. It is medically used to treat delayed puberty and loss of muscle by disease. When used, it can produce many side effects such as acne, aggressive behaviour, kidney problems, cancers and increased blood pressure.

ANABOLISM: A biochemical reaction that results in the build up of new complex compounds from the combination of simple precursors usually preceded by catabolism (breakdown) of existing complex compounds. It requires energy in the form of adenosine triphosphate (ATP) which is derived from the breakdown process.

ANABRANCH: A diverging branch of a stream or river that loses itself in sandy soil or rejoins the main flow downstream.

ANACONDA: The largest snake (*Eunectes murinus*) in the world that belongs to the genus *Eunectes* and family Boidae (boas and pythons), which may reach up to 9 metres long. It is a native of tropical South America. It has recurved sharp teeth in both the lower and upper jaws. It feeds mostly on mammals or birds which it kills by coiling around and constricting them to death.

ANADROMOUS: A term for the migration of certain fish, such as salmon, that spend most of their lives in oceanic water before moving to breed in the upper reaches of rivers and streams.

ANAEMIA: A condition caused by deficiency in haemoglobin or the oxygen carrying component of red blood cells or depletion of red blood cells. Symptoms include tiredness, paleness of skin, breathlessness, palpitations and poor resistance to infection.

ANAEROBE: Organisms that can obtain energy from food without using oxygen or an organism that does not require oxygen. They include many bacteria, yeasts and internal parasites. Many anaerobes are sensitive to free oxygen and some cannot survive if oxygen is present. **Obligate anaerobes** grow only in the absence of oxygen. **Facultative anaerobes** can grow either in the presence or in the absence of molecular oxygen.

ANAEROBIC DIGESTER (BIODIGESTER): A device that uses bacteria to break down organic matter such as agricultural waste, waste food, compost, manure or sewage waste in the absence of oxygen to produce biogas (mostly methane (CH₄) and carbon dioxide (CO₂) and other trace gases) in a process called anaerobic fermentation. It includes an oxygen-free closed tank with a decomposition space for the organic materials and a valve system that can be opened to release the methane gas and control the pressure within the tank. The organic waste decomposes in the bottom of the device as bacteria break down the waste material. When fully decomposed, the waste is used as compost and fertiliser. Methane gas, carbon dioxide and other gases rise to the surface and can be collected from the valve to be used as a fuel source in many applications. For example, domestically, it can be used for heating and lighting. The gas can also be used in devices such as boilers to produce steam. It can also be refined into bio-methane for use in motor vehicles, for generation of electricity and to produce heat. Often this requires that it is purified to remove carbon dioxide and the water content. This is a renewable energy that is commonly used.

ANAEROBIC DIGESTION (METHANIZATION): It is a series of biological processes in which diverse microorganisms break down

bio-degradable materials in the absence of oxygen, using the process of fermentation to break down organic matter from animals, plants or sewage to produce biogas, with the process taking place within a centralized system in a unit called an anaerobic digester, biogas reactor or biodigester. An anaerobic digester is a unit designed for processing organics through the process of anaerobic decomposition. They take manure, sludge and other organic waste materials and produce methane (natural gas) fuel. From the foregoing, it is clear that the process of anaerobic digestion is used for industrial or domestic purposes to manage waste or produce fuel.

ANAEROBIC RESPIRATION: A form of respiration that does not require the presence of oxygen for the release of energy from food molecules such as glucose. This type of respiration has only 5% of the efficiency of aerobic respiration. Both brewing and baking industries use this process.

ANAESTHETIC: A substance that produces a loss of feeling or sensation. General anaesthetic produces loss of consciousness. Local or regional anaesthetic renders a specific area insensible to pain. A nerve disorder may also cause anaesthesia.

ANAGENESIS: The degree or rate of evolutionary divergence or the evolution of species involving an entire population with no branching or splitting of the population. It can also be defined as any evolutionary process that results in biological improvement.

ANAL: Relating to or situated near the anus. **Anal pore** is a tiny opening through which certain protozoans and other unicellular organisms eliminate waste.

ANALAR REAGENT: A chemical reagent of high purity with known contaminants for use in chemical analyses.

ANALGESIC: Drugs such as aspirin, paracetamol and morphine that relieve pain for short- or long-term. It does not cause loss of sensation or of consciousness as does anaesthetics.

ANALOGOUS STRUCTURES: Body parts that are similar in functions to a structure or organ in another organism but have evolved in different ways. For example, the wings of insects and birds have the same purpose of giving ability to fly but have different origins.

ANALOGUE: 1. A quantity which changes continuously. By contrast a digital quantity varies in series of distinct steps. 2. A term which describes a signal that is wavy in form rather than being pulsed (digital). At any moment it can have any value in the range allowed, rather than only a few. 3. A way of recording values by means of a continuous wave or motion. Examples are given by the hands of a clock and the typical speedometer in a motor vehicle. The pendulum clock is an analogue clock for measuring time.

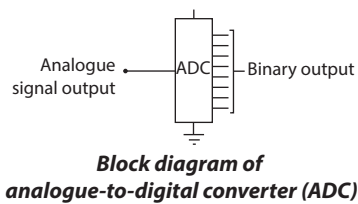
ANALOGUE COMPUTER: A computing device that performs calculations through the interaction of continuously varying physical quantities such as voltages. An analogue computer is said to operate in real-time and can therefore be used to monitor other events as they happen. Analogue computers are time-consuming to set up and operate. Most work once done on analogue computers is now carried out on digital computers, which are simpler and quicker to use.

ANALOGUE DEVICE: A mechanical or electrical device that operates with variables represented by continuously measured quantities such as pressures, resistances, rotations, temperatures and voltages. They are used in monitoring systems, simulation processes, etc. Some analogue devices such as the slide rule for computation have been largely replaced by digital systems such as the electronic calculator.

ANALOGUE SIGNAL: Current or voltage that conveys or stores information and varies continuously in the same way as the information it represents. It is a signal whose amplitude can take any value between reasonable limits and which can vary with time. Analogue signals are prone to interference and distortion. An example of an analogue signal is sound wave.

ANALOGUE TELEPHONE ADAPTER: A device that can be used to connect a standard telephone to a broadband cable or DSL network to enable calls over the Internet.

ANALOGUE-TO-DIGITAL CONVERTER (ADC): A type of integrated circuit that converts continuously varying analogue signals from instruments and sensors that monitor conditions, such as sound, movement and temperature into binary code or digital signal. The basic conversion steps include the following: An analogue input signal is converted from some continuously varying physical value such as pressure in air or frequency or wavelength of light, into a sequence of digital values, called samples at regular intervals. The amplitude at each interval is quantised or given a value and the values are mapped into a series of binary digits or bits. Coding is the process of converting the quantised signals into a digital representation, which is transmitted as a digital signal to the receiver, where it is decoded. Once the digital signals are decoded and processed, they are converted back to an analogue signal and transmitted to a device such as a speaker, which requires an analogue signal. ADCs are found in digital desk phones and cell phones, digital cameras, camcorders or scanners.



ANALOGUE VIDEO: A video signal that represents an infinite number of images (video) through the use of a rapidly changing signal, which indicates the luminance and colour information of red, green and blue intensities, called the analog signal, within the video signal.

ANALOGY: 1. Equivalence in biological function between body parts or organs that have appeared independently in different plants and animals. 2. A general similarity between two problems or methods, used to indicate the results of one problem from the known results of the other.

ANALYSER: 1. A device, used in the polarisation of light, that is placed in the eyepiece of a polarimeter to observe plane-polarised light. The analyser, which may be a Nicol prism or Polaroid, can be oriented in different directions to investigate in which plane an incoming wave is polarised or if the light is plane polarised. If the analyser is horizontal when extinction of light takes place, the polarisation of light must have been in the vertical plane. The intensity of a beam of light transmitted through an analyser is proportional to $\cos^2\theta$, where θ is the angle between the plane of polarisation and the plane of the analyser. Extinction is said to be produced by 'crossing' the polariser and analyser. 2. The left cerebral hemisphere of the brain that, in sport, is concerned with learning new skills, correcting flaws in technique and developing strategy.

ANALYSIS: 1. The process of breaking down a complex topic or substance into its constituent parts to learn what they do and how they relate to one another. This is used commonly by scientists in studies to understand complex things. For example, chemists may use **qualitative analysis** to identify the essential components of a particular chemical compound; use **quantitative analysis** to identify the proportions of components in a substance, or to break down chemical processes and examine the chemical reactions between elements of matter. In biological science, for example, blood may be examined in a laboratory to analysis its composition. **Data Analysis** is the process of collecting and organising data so that statistical and logical techniques can be used to answer questions, solve problems, and derive important information. 2. A branch of mathematics based on convergence and limits of continuous functions. It is derived out of calculus, with fields such as classical and complex analysis and more abstract functional analysis and the theory of linear operators. Examples are series, logarithmic functions, trigonometrical functions and calculus.

ANALYSIS OF BINARY DATA: The study of how probability of success depends on explanatory variables and grouping of materials. The analysis of binary data also involves goodness-of-fit tests of

simple binary variables to a theoretical distribution, as well as the study of 2 by 2 contingency tables and their subsequent analysis.

ANALYSIS OF RESIDUALS: It is used to test the validity of statistical model and to control the assumptions made on the error term. It may also be used for outliers detection.

ANALYSIS OF VARIANCE: A technique that consists of separating the total variation of data set into logical components associated with specific sources of variation in order to compare the mean of several populations. This analysis also helps to test certain hypotheses concerning the parameters of the model or to estimate components of the variance. The sources of variation are globally summarised in a component called **error variance**, sometimes also called **within treatment mean square** and another component that is termed **"effect" or treatment**, sometimes called **between-treatment mean square**.

ANALYST: 1. A person who specialises in analysis or a person who analyses a complex process or operation in order to improve its efficiency, especially by applying mathematical or scientific means. 2. In computer science, a **systems analyst** is an information technology specialist responsible for the analyses, designs, modification and implementation of information systems to ensure compatibility and user effectiveness.

ANALYTIC: 1. In complex function, also known as **regular** or **holomorphic**, it is a function that is single-valued and differentiable at every point of its domain and in consequence possessing derivatives of all orders and agreeing with its Taylor series locally. The function may not be differentiable everywhere, it is only analytic at points where the function is differentiable. For example, $\exp z$ is analytic on the complex plane and $\log z$ is analytic on $\mathbb{C} \setminus (-\infty, 0)$. 2. Also known as **real analytic**, of a real function, possessing derivatives of all orders and agreeing with its Taylor series locally. 3. In logic, of a proposition, true by virtue of the meanings of the words alone without reference to the facts; alternatively, true or false by virtue of meaning alone.

ANALYTIC CONTINUATION: 1. The construction of an analytic function whose restriction to a given domain is a given analytic function. 2. A procedure to extend the domain of a given analytic function. For example, $\sin z$ on the complex plane is the analytic continuation of $\sin z$ on the real line, since there is no other analytic function on $\mathbb{C}$ of which the restriction to the reals is the sine function.

ANALYTIC/ANALYTICAL DATABASE: A type of read only system that stores, manages and consumes big data. An analytical database is specifically designed to support business intelligence solutions and analytical applications.

ANALYTIC FUNCTION (HOLOMORPHIC FUNCTION; REGULAR FUNCTION): A complex function that is single-valued and differentiable at every point of its domain and in consequence possessing derivatives of all orders and agreeing with its Taylor series locally. The function may not be differentiable everywhere, it is only analytic at points where the function is differentiable. For example, $\exp z$ is holomorphic on the complex plain and $\log z$ is holomorphic on $\mathbb{C} \setminus (-\infty, 0)$.

ANALYTIC GEOMETRY (ANALYTICAL GEOMETRY; CARTESIAN GEOMETRY; COORDINATE GEOMETRY): The branch of mathematics that studies geometrical relations using algebra with coordinates as references. An example of coordinate geometry is plotting points, lines and curves on an x - and y -axis.

ANALYTIC PROOF (ANALYTIC METHOD): Using the coordinate system and algebraic reasoning to prove the validity of a geometric statement. Some of the methods used include axioms or postulates; definitions; assumptions; etc.

ANALYTIC SET (SOUSLIN SET): In mathematics, the continuous image of a Polish space. Analytic subsets of Polish spaces are closed under countable unions and intersections, continuous images and inverse images.

ANALYTIC STRUCTURE: A covering of a Euclidean topological space by open sets, such that, whenever two members of the cover overlap, the coordinate transformations in both directions are analytic in their intersection.

ANALYTICAL CHEMISTRY: A branch of chemistry that deals with the determination of the nature and chemical composition of substances. Qualitative analysis determines the identities of the substances in a given sample, whilst quantitative analysis determines how much of a particular substance is present.

ANALYTICAL INSTRUMENT: A device or a combination of devices used to carry out single or multiple analytical procedures. The analytical process is all or part of the analytical procedure that encompasses all steps from the introduction of the sample or the test portion to the production of the result. An example of analytical instrument is the gas chromatography mass spectrometer, used in analysing fluid samples.

ANALYTICAL INTERCOMPARISON: A procedure which gives insight to the accuracy of results of analytical procedures by comparing the results obtained in the analyses of identical samples at different laboratories and preferably with different analytical methods.

ANALYTICAL PORTION (TEST PORTION): The amount or volume of the test sample taken for analysis, usually of known weight or volume.

ANALYTICAL PYROLYSIS: The characterisation, in an inert atmosphere, of a material or a chemical process by a chemical degradation reaction(s) induced by thermal energy.

ANALYTICAL QUALITY CONTROL (AQC): Procedures which ensure that the results of laboratory analysis are consistent, comparable, accurate and within specified limits of precision.

ANALYTICAL RADIOCHEMISTRY (RADIO ANALYTICAL CHEMISTRY): The part of analytical chemistry in which the application of radioactivity is an essential step in the analytical procedures.

ANALYTICAL SOLUTION: 1. An exact solution to a problem that can be calculated symbolically by manipulating equations, unlike a numerical solution. 2. Also known as **test solution**, it is the solution prepared from the test portion for the analytical procedure. The proportions of test portion and solvent are normally known.

ANALYTICAL UNIT (ANALYSER): 1. An assembly of subunits comprising suitable apparatus permitting the introduction and removal of the gas, liquid or solid to be analysed and/or calibration of materials. It is also a measuring cell or other apparatus which, from the physical or chemical properties of the components of the material to be analysed, gives signals allowing their identification and/or measurement. They may be signal processing devices (amplification, recording) or, if need be, data processing devices. 2. In statistics, analytical units represent real or artificially constructed units, for which statistics are compiled.

ANAMNIOTE: Referring to fishes and amphibians, whose embryo lacks an amnion. The embryos and larvae therefore develop in water.

ANAMORPH: A stage of fungal reproduction, where cells are asexually formed by the process of mitosis.

ANAMORPHOSIS: The process of distorting the perspective of an image to such an extent that its normal appearance can only be viewed from a special angle or with a special instrument.

ANANAS: A genus of tropical American plants in the family Bromeliaceae. They have sword-shaped leaves and a fleshy compound fruit composed of the fruits of several flowers. Plants in the species have tough leaves that are lanceolate with serrated or spiny margin and grow in large rosettes, arising from the crown basally. From the heart of the rosette arises flowers, each having its own sepals. These flowers grow into a compact head on a short, robust stalk. The sepals become fleshy and juicy, developing into a pseudocarp fruit with a rosette of leaves as a crown. The most commercially important species in the genus is *Ananas comosus*, commonly known as pineapple. It is a tropical plant with a short thick stem and long spiny succulent

leaves. It is cultivated for its large edible multiple fruit, also called pineapple. The fruit has a yellowish juicy flesh. It is typically covered with floral bracts and topped with short spiny leaves. A determinate inflorescence forms 15-20 months after planting. After fertilisation, the many lavender flowers fuse and become fleshy to form the fruit, which ripens in 5-6 months. The plant does not contain any viable seeds and the crowns from the top of the fruit, the slips (tiny plantlets that grow at the base of the fruit on the fruit stalk) as well as the suckers (little plantlets that grow between the leaves of the mature pineapple) can be used for propagation. Plants propagated by the slips and suckers mature more quickly than those propagated by the crowns. Matured or ripe fruits can be peeled and eaten raw and can be processed into a syrupy juice. It can be used to make wine and the leaves are used to produce. Medicinally, the plant has been used traditionally to induce menstruation and abortion, kill parasitic amoebas and expel worms.

ANANDROUS: A flower lacking an androecium or without stamen as in pistillate flowers.

ANANTHOUS: A plant without flowers.

ANAPHASE: The third stage of cell division, mitosis and meiosis, in which the chromosomes move to opposite ends of the nuclear spindle. In the first stage of anaphase, homologous chromosomes separate and move to opposite ends and in the second stage the chromatids of each chromosome move apart to the opposite ends of the spindle as in mitosis. Other stages in cell division are the prophase (first stage), the metaphase (second stage) and the telephase (final stage).

ANAPHYLAXIS: A hypersensitivity reaction to the ingestion or injection of a substance such as a protein or drug resulting from prior contact with a substance. It may also be defined as an abnormal immune response that occurs when an individual, who was previously exposed to a particular antigen is re-exposed to the same antigen. Anaphylaxis may be caused by an insect bite, injection of a drug such as penicillin or an excessive reaction to an allergen such as pollen, dust or animal hair or certain foods. It is caused by the release of histamine and similar substances, which may produce a localised or a more generalised and severe one with difficulty in breathing, pallor or drop in blood pressure, unconsciousness and possibly heart failure and death.

ANAPLASMOSIS: An infectious disease caused by rickettsial bacterium, which invades the red blood cells of affected animals, causing anaemia and, sometimes, death. It affects cattle, buffalo, sheep, goat, deer, etc.

ANAPLEROTIC: A term that describes a metabolic pathway that replenishes an intermediate in one of the main metabolic pathways.

ANALYSIS: The process of breaking down a topic or substance (into simpler elements) to study its nature, or to determine its essential features and their relations in order to gain a better understanding of it. An example is examining a specimen of urine in a laboratory to discover all of its components.

ANAPSID: An extinct subclass of reptiles excluding that have no opening in the temporal region of the skull. The only living anapsid is the turtle.

ANASTIGMATIC LENS: A compound lens corrected for astigmatism (a defect in a lens or mirror that prevents light rays from meeting at a single point, producing an imperfect image) and curvature of field.

ANASTOMOSING: Rejoining after branching and forming a complex network as in the venation of Dicotyledonous leaves.

ANASTOMOSIS: 1. A cross-connection of veins in a leaf forming a network. 2. A type of multiple interconnected, co-existing channel belts on alluvial plains that most often seems to form under relatively low-energetic conditions near a local base level. Examples are the Tonlé Sap River in Central Cambodia and the Upper Columbia River in British Columbia, Canada. 3. A union or joining together of blood vessels or other severed tubular hollow organs to form a continuous channel, as between two parts of the intestine. 4. The connection of separate parts of a river and its branches to form a network.

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ANATOMICAL POSITION: A standard position of the body with the anterior view of the human body shown standing erect, facing directly forward, feet pointed forward and slightly apart and arms hanging down at the sides with palms facing forward. The various parts of the body are then described in relation to certain imaginary planes.

ANATOMY: The study of the structure of the body of plant or animal and its component parts or the physical structure, especially the internal structure, of an animal, plant or other organism or of any of its parts.

ANATROPOUS OVULE: In botany, it is a form of ovule orientation in the ovary in which the funiculus has lengthened and the ovule is completely inverted 180° so that the micropyle is facing downward and lying near the base of the funiculus.

ANCESTOR: A distant or early form of an organism from which later forms of life originate. An example in humans is great great-grandfather.

ANCESTRAL (ANCESTRAL RELATION): 1. In logic, a relation, derived from a given relation, that holds between two elements of its field whenever there is a chain of instances of the given relation leading from the first element to the second. Thus the ancestral of 'parent of' is 'ancestor of', since x is an ancestor of y if and only if there is a sequence of individuals each of whom is the parent of the next and of whom the first is x and the last y . 2. The set of elements that have the ancestral of a given relation to a given element of its domain. Thus the ancestral of 5 under the successor relation on the positive integers is the class $\{1, 2, 3, 4\}$. In a tree, a node A is a member of the ancestral of a node B if and only if there is a path from the root to B that includes A .

ANCESTRAL TRAIT: A character shared by a group of organisms because they originate from a common descent and related through DNA.

ANCHOR RING (TORUS): A solid generated by rotating a circle about an external line in its plane, also called an anchor ring. It has the shape of the inner tube of a tyre. If r is the radius of the rotating circle and R the distance between the centre of the circle and the axis of rotation, the volume of the torus is $2\pi^2 Rr^2$ and the surface area is $4\pi^2 Rr$.

ANCILLARY BENEFITS: The side effects of policies aimed exclusively at climate change mitigation. Such policies have an impact not only on greenhouse gas emissions, but also on resource use efficiency, like reduction in emissions of local and regional air pollutants associated with fossil-fuel use, and on issues such as transportation, agriculture, land-use practices, employment, and fuel security.

ANCILLARY SERVICES: 1. In health care, services such as radiology, pharmacy, and physical therapy that are provided to patients. They do not include room, board, medical and nursing services. 2. In electrical power grid network, things that support the transmission of electricity from its generation site to the customer to ensure voltage/frequency stability and continuous power supply.

ANCIPITAL: A structure or organ which is two-edged. An example is the winged stem of *Sisyrinchium*.

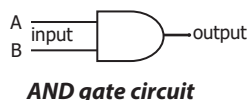
AND: The word "AND" or " $\wedge$ " is a connective word used in logic. The sentence " p AND q " is true only if both sentence p as well as sentence q are true. The operation of AND is illustrated by the truth table as shown. When used in a truth table, AND is often represented by the symbol $\wedge$. An AND sentence is also called a conjunction.

AND GATE (AND CIRCUIT; AND ELEMENT)

A digital logic gate that simulates the function of the logical operator AND, emitting signals only if all input wires receive coincident signals. For example, a HIGH output (1) results only if both the inputs to the AND gate are HIGH (1). If neither or only one input to the AND gate is HIGH, a LOW output results.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Truth Table



ANDALUSITE (ALUMINIUM SILICATE): A white to pinkish mineral, with the chemical formula, Al_2SiO_5 , which crystallises as square or rhombus-based prisms. It is composed of aluminium silicate and it is used as gems.

ANDERSON, CARL DAVID (1905 – 1991): An American physicist, who became a professor at the California Institute of Technology where he worked mainly in particle physics. In 1932 he discovered the positron in cosmic radiation, for which he was awarded the 1936 Nobel Prize in Physics. He also discovered the mu-meson (muon) in 1936.

ANDESITE: A volcanic igneous rock, which is intermediate in silica content between rhyolite and basalt. It is characterised by a large quantity of feldspar minerals, giving it a light colour.

ANDON: A visual control that indicates quality or process problem, in a manufacturing line or work process, often accompanied by audible alarms to alert management, supervisors, and other workers when there is a change in status.

ANDREEV REFLECTION (AR): A process named after the Russian physicist, Alexander F. Andreev, who first described it in 1964. It is a type of particle scattering which occurs at the interface between a normal conductor of electricity and superconductor. A charge (normal current) is transferred from normal conductor to the super conductor, avoiding the forbidden single-particle transmission within the superconducting energy gap by a dual transfer process across the interface.

ANDREW FILE SYSTEM (AFS): A location independent file system that uses the client/server model, where all the data is stored on file server machines. File are transferred to client machines as necessary and cached locally.

ANDREWS TITRATION: A titration used to estimate the amount of reducing agents. The reducing agent being estimated is dissolved in concentrated hydrochloric acid and titrated with a solution of potassium iodate. A drop of tetrachloromethane is added to the solution. The end point of the titration is reached when the iodine colour disappears from this layer. This is due to the reducing agent being oxidised and the iodate being reduced to ICl .

ANDROCLINIUM (CLINANDRIUM): A portion of an orchid column in which the anther is concealed in a flower.

ANDRODIOECIOUS: A term used to describe plant species in which male and hermaphrodite flowers are borne on separate plants. An example is *Caltha* (marsh marigolds).

ANDROECIUM: The male reproductive part of a flower comprising a number of stamens.

ANDROGEN: Any of a group of hormones that mainly influence the development of the male reproductive system, of which testosterone is the most important. They are mainly induced in the production of male secondary sexual characters such as facial hair in human.

ANDROGENESIS: 1. A male parthenogenesis (a process in which an unfertilised egg develops into a new individual) in which the embryo contains only paternal chromosomes because the male nucleus has dominated the female nucleus during fertilisation. 2. The development of secondary male characteristics such as deepening of voice and more body hair by using androgen hormones.

ANDROGENETIC ALOPECIA (ANDROGENIC ALOPECIA; BALDNESS): The loss of hair from the upper scalp, common in older men, but which can occur as from early twenties and progresses at different rates in different men. In men it is also called **male pattern hair loss**. It also affects women. Its onset and extent are influenced by genetic makeup and the level of male sex hormones. It is sex-linked; the gene is located on the sex chromosome.

ANDROGYNOPHORE: A single stalk supporting the androecium and gynoecium in some flowers.

ANDROGYNOUS: Plants having both stamen and pistil in the same flower or an inflorescence having both stamen (male) and pistil (female) parts. They are, therefore, hermaphrodites (bisexual flowers) or have both male and female characteristics and also

known as perfect flowers. The staminate flowers are borne above the pistillate. All the male reproductive organs of a flower including stamens are called **gynoecium**; and the female counterparts are known as **androecium**. The flower is male, if just one androecium is present; if only one gynoecium is developed, then it is a female flower. Examples of androgynous plants include lily, rose and most plants with large showy flowers.

ANDROHERMAPHRODITIC: Inflorescence with staminate flowers inside or above and hermaphroditic outside or below.

ANDROID: 1. Possessing human features. 2. Also known as humanoid, it is an automaton that resembles a human, created from biological materials. 3. In telecommunications, it is a Linux based operating system originally developed by Google and released in November 5, 2007 and used for smartphones such as the Motorola Droid, the Samsung Galaxy, LG Electronics G2 and Google's own Nexus One; tablets and laptops. Android is open source and developers can modify and customise the OS for each phone. Android is a Linux OS and its applications are programmed in Java. Users download applications from Google's Play Store (formerly Android Market), the Amazon Appstore and other online sources.

ANDROMEDA: 1. Major constellation of the northern hemisphere which is visible in autumn. Its main feature is the Andromeda galaxy. 2. Any of several shrubs of the genera *Pieris* or *Andromeda* or their relatives.

ANDROMEDA GALAXY: A galaxy in the constellation Andromeda, with a diameter of about 200,000 light-years and 2.2 million light years away from Earth. It is the most distant object visible to the naked eye, appearing as a milky blur.

ANDROMONOECIOUS: Plant species in which male and hermaphrodite flowers are borne on the same plant. An example is *Aesculus hippocastanum* (horse chestnut).

ANDROPHORE: An extension of the receptacle between the petals and the stamens on which the androecium and gynoecium are borne. It is seen in many members of the *Capparaceae*.

ANDROSPORE (ANDROGONIDIUM): A specialised zoospore of the algal family *Oedogoniales* that develops into a small male plant that produces true spermatozooids.

ANCHOIC: Having a low degree of reverberation with little or no reflection of sound. **Anechoic chamber** also known as **dead room** or **free-field room** is a test room in which all surfaces are lined with a sound-absorbing material to reduce reflections of sound to a minimum.

ANEMOMETER: An instrument for measuring the speed of the wind or any other flowing fluid. Examples are the simple vane anemometer and the hot wire anemometer. To determine the velocity, an anemometer detects change in some physical property of the fluid or the effect of the fluid on a mechanical device inserted into the flow.

ANEMOPHILY (WIND POLLINATION): A type of pollination in which the pollen grain is carried by the wind. Anemophilous plants may be either gymnosperms (non-flowering) or angiosperms (flower-producing). Examples of plants pollinated by wind are grasses, conifers, sweet chestnuts and members of the hickory family. Anemophilous flowers have pollens, which are light and non-sticky so that they can be transported by air currents. They possess well exposed stamen so that the pollens are well exposed to wind currents. They have large and feathery stigma to easily trap air borne pollen grains.

ANENCEPHALY: A neural tube defect that occurs during embryonic development (usually between the 23rd and 26th days of pregnancy) resulting in the absence of a major portion of the brain, skull and scalp. It occurs when the cephalic (head) end of the neural tube fails to close. Babies born with such defects are usually blind, deaf, unconscious and live for only a few hours. It can be diagnosed prenatally through screening of the mother's blood for the level of alpha-fetoprotein (AFP) and an ultrasound examination.

ANERGIZED CULTURES (HABITUATION): The process of becoming accustomed to a new environment. The term is applied specifically to the ability of certain long-established callus cultures to synthesise auxin and hence become independent of exogenous supplies.

ANEROID BAROMETER: A device, devoid of a liquid, which is used for measuring atmospheric pressure. Variations of atmospheric pressure are indicated by the relative bulges of a thin elastic metal disk covering a partially evacuated chamber.

ANEUPLOID: A description of a nucleus, cell or organism in which one or more chromosomes have been added to or deleted from the complete set, so that the total number of chromosomes is not an exact multiple of the haploid number.

ANEURYSM: A widening or ballooning of a portion of an artery due to weakness in the wall of the blood vessel. An aneurysm that grows and becomes large enough can burst, causing dangerous, often fatal, bleeding inside the body.

ANEUSPORY: The production of an unusual number (usually two, instead of the normal four) of a spore mother cell, through a modification of the meiotic process. It is a form of apomixis, found in the formation of megaspores in dandelions (*Taraxacum*).

ANFO: Abbreviation of Ammonium Nitrate-Fuel Oil, which is used extensively as a blasting agent in mining and quarrying. The proportions are approximately 94% ammonium nitrate and 6% fuel oil. The oil used is most often No. 2 fuel oil or diesel fuel, but sometimes kerosene, coal dust or even molasses are used.

ANGER CAMERA (GAMMA CAMERA; SCINTILLATION CAMERA): A camera in which a large diameter scintillator is coupled to an array of photomultiplier tubes by fibre optics. It is used to image gamma radiation emitting radioisotopes, a technique known as scintigraphy. Scintigraphy includes early drug development and nuclear medical imaging to view and analyse images of the human body or the distribution of medically injected, inhaled or ingested radionuclides emitting gamma rays.

ANGINA (ANGINA PECTORIS): A severe pain occurring in the chest due to inadequate supply of oxygen through the blood to the heart muscle because of a coronary artery disease such as narrowing of a coronary artery. This is caused by the deposition of fatty materials, cholesterol and other substances in and on the artery walls resulting in the hardening and narrowing of the artery walls and the formation of plaque, which can restrict blood flow to the tissues and organs. If the plaque is not ruptured, it can cause coronary artery disease resulting in angina. Its symptoms include squeezing, pressure, heaviness, tightness or pain in the chest, faintness and difficulty in breathing. It can be treated by drugs or bypass surgery.

ANGIOGRAPHY: An X-ray imaging procedure that enables blood vessels, including major arteries and veins. Water-soluble ionic or nonionic X-ray contrast media is injected into the blood stream of arteries or veins to investigate coronary artery disease or any disruption of blood supply to the brain. The blood vessels being examined appear in clear silhouette on an X-ray screen.

ANGIOPLASTY: A surgical procedure for repairing damaged or diseased blood vessels by inserting a balloon-tipped catheter into arteries. It is used to widen narrowed or obstructed arteries caused by fatty substances blocking the artery to restore blood flow to the heart muscle during a heart attack.

ANGIOSPERM: Flowering plants, which constitute the most flowering plants, in which the ovules are enclosed within an ovary, which mature into fruits. They include herbaceous plants, shrubs, grasses and most trees. They are divided into monocotyledons which possess one seed leaf in the embryo and dicotyledons with two seed leaves in the embryo. Examples include rhododendron, apple, sweetbay magnolia, oaks, maples and dogwoods. **Gymnosperm** refers to woody vascular plants in which the ovules are carried naked on the scales of a cone. That is their seeds are not enclosed in an ovule. They usually have

needles that stay green throughout the year. Examples are pines, cedars, spruces and firs.

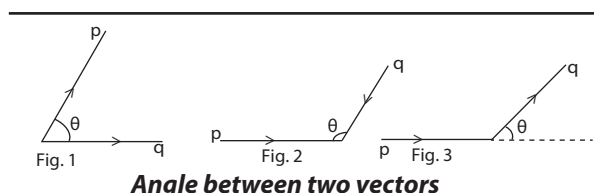
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ANGIOTENSIN: Either of three related polypeptide hormones that raise blood pressure through constriction of vascular smooth muscles and stimulates the release of aldosterone from the adrenal glands. The first of these, angiotensin I, is itself inactive and is converted to the second, active form, angiotensin II, by the action of a converting enzyme. Angiotensin II causes constriction of the small blood vessels in tissues, resulting in an increase in blood pressure. Angiotensin III, may be as active on the adrenal gland as angiotensin II and may also cause constriction of the small blood vessels.

ANGLE: A measure of the turn or rotation between two lines or planes that meet; it is the inclination of one line relative to another. It can also be defined as a pair of rays that share a common end point but do not lie on the same line. The magnitude of an angle can be given in one of two units. The **degree**, with the symbol $^{\circ}$ is one of the units often used and it is defined as one complete rotation divided by 360° . The other unit is the **radian**, a standard unit of angular measure, used in many areas of

mathematics. Its symbol is the rad. Radian can be defined as a unit of measurement of angle equal to 57.3° , equivalent to the angle subtended at the centre of a circle by an arc equal in length to the radius. In other words, radian is a unit of measurement of angle that is equivalent to the angle subtended at the centre of a circle by an arc equal in length to the radius. One radian is equal to 57.2958 degrees. In the diagram, the angle θ in radians = arc length divided by radius of the circle: $\theta = AB/OA$. An angle's measurement in radians is numerically equal to the length of a corresponding arc of a unit circle. Another unit, the **steradian**, with the symbol sr or squared radian is the SI unit of solid angle. It is used in three-dimensional space and functions. A right angle has exactly 90° (or $\pi/2$ radians), an acute angle is less than 90° , an **obtuse angle** measures more than 90° but less than 180° and a **straight angle** has exactly 180° . Any angle that is at most 180° is called a **convex angle**. Any angle that is at most 180° is called **convex angle**. An angle between 180° and 360° is called a **reflex angle**. An angle of exactly 360° is a complete turn or one revolution. **Complementary angles** add to give 90° , **supplementary angles** make a total of 180° . In spherical geometry, a **spherical angle** is that formed by intersecting arcs of two great circles. Its magnitude is equal to that of the angle between the two planes of the great circles.

ANGLE BETWEEN TWO VECTORS: Angle between two vectors p and q . If p and q are two non-zero vectors and θ is the angle between them, then the angle between is defined by $\theta = \cos^{-1} \frac{p \cdot q}{|p||q|}$. This is the dot product or scalar product.



Angle between two vectors

ANGLE BRACKETS: 1. In mathematics, the pair of symbols $\langle$ and $\rangle$, used together to indicate quantities to be treated as a unit. The pair also indicate that the terms lying between them form a sequence or ordered n -tuple or to denote the inner product of two vectors; as well as to denote the subgroup or ideal generated by a subset of a given group or ring. 2. In computer technology they are commonly

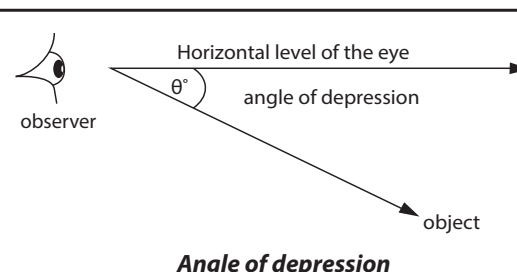
used to enclose a code and some other things, for example, HTML tags and PageMaker tags.

ANGLE MODULATION (SINUSOIDAL ANGULAR MODULATION):

The variation in the angle or phase of a sine-wave carrier to transmit data, instead of varying the amplitude. These variations are controlled by both the frequency and the amplitude of the modulating wave. When used in an analogue manner, it is referred to as either frequency modulation (FM) or phase modulation (PM), depending upon whether the instantaneous frequency or instantaneous phase varies with the modulation. When used in a digital modulation format, it is referred to as either frequency-shift keying (FSK) or phase-shift keying (PSK). Both FSK and PSK can be used with either a binary or an M -ary alphabet, where M is an integer greater than 2.

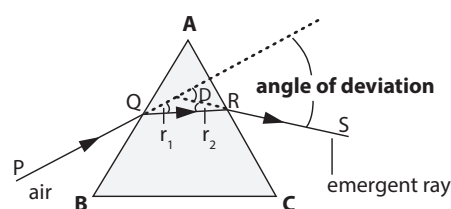
ANGLE OF DECLINATION: Angle at a particular point on the Earth's surface between the direction of the true or Geographic North Pole and the Magnetic North Pole.

ANGLE OF DEPRESSION: The angle through which the eye must be lowered to observe an object below the horizontal level of the eye of the observer or on the ground.



Angle of depression

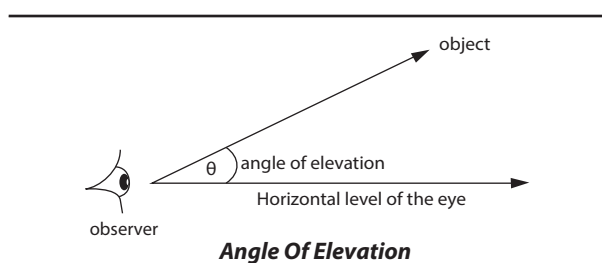
ANGLE OF DEVIATION: The angle through which the emergent ray deviates from the direction of incident ray. When a ray of light passes through a prism it undergoes two refractions. Referring to the diagram, if PQ is a monochromatic light incidenting on side AB at Q , it is refracted and travels along QR and again gets refracted and emerges out along RS . The sum of deviations at Q and R is the total deviation, D . Minimum deviation D_M is obtained, when the deviation at Q (r_1) equals the deviation at R (r_2), that is at minimum deviation, ($r_1 = r_2 = r$) $\Rightarrow 2r = D_M$; therefore $r = D_M/2$.



Angle of deviation in a prism known as angle of minimum deviation

ANGLE OF DIP (DIP ANGLE; MAGNETIC DIP; MAGNETIC INCLINATION): The angle made with the horizontal by the Earth's magnetic field lines at any specific location or by a compass needle when the compass is held in a vertical orientation. It varies at different points on the Earth's surface, for example, it is 0° at the magnetic equator and 90° at each of the magnetic poles. If the value is positive at the point of measurement, the magnetic field of the Earth is pointing downward, into the Earth; and if negative, it is pointing upward.

ANGLE OF ELEVATION: The angle between the horizontal line and the line of sight to an object at a higher level. In other words, it is the angle through which the eye must be raised to observe an object above the horizontal level of the eye of the observer.



ANGLE OF FRICTION: The angle of a plane to the horizontal when a body placed on the plane will just start to slide. It is given by " λ ", which is the angle $\tan \lambda = \mu_s$, where μ_s is the coefficient of static friction.

ANGLE OF INCIDENCE: The angle made by a light ray or wave falling on a surface and the line perpendicular to that surface. In other words, it is the angle between a ray falling on a surface and the perpendicular drawn to the surface at the point at which the ray strikes the surface. The ray can be formed by any wave, such as optical, acoustic, microwave or X-ray.

ANGLE OF OPTICAL ROTATION: Angle through which plane polarised light is rotated clockwise, as seen when facing the light source, in passing through an optically active medium.

ANGLE OF PROJECTION: The angle formed between the direction of a projected particle and the horizontal.

ANGLE OF REFLECTION: The angle between a ray of light reflected from a surface and the perpendicular to that surface.

ANGLE OF REFRACTION: The angle between a ray that is refracted at a surface between two different media and the perpendicular (normal) to the surface at the point of refraction. When a ray of light passes from a less denser medium such as air into a denser medium such as glass, it is bent towards the normal so the angle of refraction is less than the angle of incidence.

ANGLE STRAIN (BAEYER STRAIN): The tension that develops in a molecule when bond angles deviate from the normal value. For example, in cyclopropane, C_3H_6 , the angles between the bonds of the ring differ from the normal tetrahedral angle of $109^\circ 28'$, which is found in cyclohexane, leading to angle strain in the molecule. The presence of angle strain in a molecule indicates that in a specific chemical conformation there exist bond angles that deviate from the ideal bond angles required to achieve maximum bond strength.

ANGLE-RESOLVED PHOTOELECTRON SPECTROSCOPY (ARPES): A technique for studying the composition and structure of solid surfaces by measuring both the kinetic energy and angular distribution of photoelectrons ejected from a surface by electromagnetic radiation.

ANGLESITE: A mineral form of lead (II) sulfate, with the chemical formula $PbSO_4$, which is usually formed by the oxidation of galena. Due to the fact that it does not occur in large enough quantity, it is of no particular commercial value.

ANGORA: Wool from wool-producing rabbits. Wool from Angora goat is called mohair.

ANGSTROM: A unit of length named after the Swedish scientist Anders Jonas Angstrom; formerly used to measure wavelengths and intermolecular distances but now replaced by the nanometre. Its symbol is Å. One Å is equal to 10^{-10} metres or one-ten millionth of a millimetre. It is used for atomic measurements and the wavelengths of electromagnetic radiation.

ANGSTROM, ANDERS JONAS (1814 – 1874): Swedish physicist and astronomer, who worked mainly with emission spectra and demonstrating the presence of hydrogen in the sun. He also worked out the wavelengths of Fraunhofer lines. The spectra wavelength angstrom was named after him.

ANGULAR: 1. Of or pertaining to measurements in terms of angles.
2. In physics, of or pertaining to physical properties or quantities related to a revolving body that are measured in reference to its axis of revolution.

ANGULAR ACCELERATION: The rate of change of angular velocity with respect to time. In SI units, it is measured in radians per second squared (rad/s^2). Its symbol is α (alpha). The average angular acceleration of an object whose angular velocity changes from ω_1 to ω_2 over time t is $(\omega_2 - \omega_1)/t$. Instantaneous acceleration is the value approached as t becomes smaller.

ANGULAR COBBLY: Irregular, angular rock mineral fragment of 7.5 to 25 cm diameter that are not regarded as part of the soil mass.

ANGULAR COORDINATE DIAGRAM (CORRELATION DIAGRAM; WALSH DIAGRAM): A diagram that relates the energy levels of separate atoms to the energy levels of diatomic molecules and united atoms, the two limiting states being correlate with each other. It explains why molecules are more stable in certain spatial configurations. It is used for making predictions about the geometries of small molecules.

ANGULAR DIAMETER (ANGULAR SIZE; APPARENT DIAMETER): The angle that the actual diameter of an object makes in the sky. The Moon, with an actual diameter of 3,476 kilometres, has an angular diameter of $29' 21''$ to $33' 30''$, depending on its distance from Earth.

ANGULAR DISPERSION: A measure of the angular separation of light rays of different wavelength or colour traversing a prism or diffraction grating. It is equal to the rate of change of the angle of deviation with respect to the change in wavelength.

ANGULAR DISPLACEMENT: A measure of the angle in radians of the rotation of an object about an axis. Angular displacement has both magnitude and direction. Conventionally, clockwise movements are described as positive (+) and anticlockwise movements as negative (-).

ANGULAR DISTANCE (ANGULAR SEPARATION; APPARENT DISTANCE; APPARENT SEPARATION): The angle between two objects as perceived by an observer from a location different from either of these objects.

ANGULAR DISTRIBUTION: The distribution in angle, relative to an experimentally specified direction, of the intensity of photons or particles resulting from a nuclear or extranuclear process.

ANGULAR FREQUENCY (PULSATANCE; ANGULAR SPEED; RADIAL FREQUENCY; CIRCULAR FREQUENCY; ORBITAL FREQUENCY; RADIAN FREQUENCY): A quantity proportional to the frequency of a periodic phenomenon but having the dimensions of angular velocity.

ANGULAR MAGNIFICATION (MAGNIFYING POWER): The ratio of the angle subtended by the image at the eye to the angle subtended by the object at the eye. It is represented by angular magnification, $M_\theta = \frac{\theta_i}{\theta_o}$, where θ_o is the angular size of an object and θ_i is the angular size of the image after passing through an optical system.

ANGULAR MASS (MASS MOMENT OF INERTIA; MOMENT OF INERTIA; ROTATIONAL INERTIA): A measure of an object's resistance to changes to its rotation, which is the sum of all the products formed by multiplying the magnitude of each element of mass by the square of its distance from the axes. Its symbol is I . The moment of inertia of an object about a given axis describes how difficult it is to change its angular motion about that axis. If a body

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consists of particles of mass m , a perpendicular distance r from an axis of rotation, the moment of inertia I of the body about that axis of rotation is given by $I = \sum mr^2$. The basic equation is $T = I\alpha$, where T is the torque causing angular acceleration α about the specified axis. The moment of inertia of a body is an important quantity because it is the analogue for rotational motion of mass for linear motion.

ANGULAR MEASURE: Measurements which are done in degrees and radians, with degrees normally used in basic measurements; while radians are used in advanced work.

ANGULAR MOMENTUM: The momentum that a body has as a result of its rotation about an axis or the momentum of a body, calculated as the product of its mass and its angular velocity or the product of the angular velocity of a body and its moment of inertia about the axis of rotation. Its symbol is L . The angular momentum, of an orbiting body of mass, m , travelling at a velocity, V , along a circular orbit of radius, r , is given by $L = mvr$. Angular momentum is conserved; in that if the radius of the orbit is increased, the other values such as velocity will also increase so as to compensate the effect and maintain, the value of angular momentum.

ANGULAR OVERLAP MODEL (AOM): A method used to describe transition metal–ligand interactions and main group stereochemistry, whose basic assumption is that the strength of a bond formed using atomic orbitals on two atoms is related to the magnitude of overlap of the two orbitals.

ANGULAR RESOLUTION: A quantitative measure of the ability of any image-forming device such as an optical instrument or radio telescope, a microscope, a camera or an eye, to distinguish small details of an object. It is a major determinant of image resolution.

ANGULAR SEPARATION: The angle, measured from the observer, between two celestial objects, such as the components of a binary star system.

ANGULAR SIZE: The angle subtended by the observer by a diameter of a distant spherical body which is perpendicular to the line between the observer and the centre of the body. The angle may be measured in units of radians or in units of degrees, arc minutes (1 degree = 60 arc minutes) and arc seconds (one arc minute = 60 arc seconds). For example, the angular size of the moon in the sky is 30 arc minutes. The human eye cannot resolve details smaller than a few arc minutes.

ANGULAR SPEED: It is the magnitude of the angular velocity or the speed element of angular velocity. The average angular speed (ω) is obtained by dividing the angular distance through which the body rotates by the time taken, expressed mathematically as $\omega = \theta/t$, where θ is the angular distance and t the time taken in seconds.

ANGULAR VELOCITY: A vector quantity, which is the rate of angular displacement with respect to time. It specifies the angular speed or the rotational speed of an object and the axis about which the object rotates. The SI unit of angular velocity is radians per second (rad s^{-1}). It may also be measured in degrees, radians or revolutions per unit time. Its symbol is ω (omega).

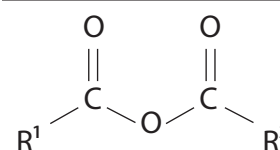
ANGUSTISEPTATE: A fruit flattened at right angles to the septum which crosses the narrowest diameter.

ANHARMONIC OSCILLATOR: An oscillating system that is not oscillating in simple harmonic motion or the deviation of a system from being a harmonic oscillator. In this case, the system can be approximated to a harmonic oscillator and the anharmonicity can be calculated using perturbation theory. If the anharmonicity is large then other numerical techniques have to be used. Assuming

that ω is the fundamental frequency of the oscillator, then as a first approximation, the frequency shift $\Delta\omega = \omega - \omega_0$ is proportional to the square of the oscillation amplitude A . That is, $\Delta\omega \propto A^2$. In a system of oscillators with natural frequencies $\omega_a, \omega_b, \dots$. Anharmonicity results in additional oscillations with frequencies $\omega_a \pm \omega_b$. Anharmonicity also modifies the profile of the resonance curve, leading to interesting phenomena such as the foldover effect and superharmonic resonance. A harmonic oscillator obeys Hooke's Law and is an idealised expression that assumes that a system displaced from equilibrium responds with a restoring force whose magnitude is proportional to the displacement. Anharmonic oscillators exhibit chaotic or nonlinear behaviour. Some examples of anharmonic oscillators are the large-angle pendulum, which exhibits chaotic behaviour as a result of its anharmonicity; nonequilibrium semiconductors that possess a large hot carrier population, which exhibit nonlinear behaviours of various types related to the effective mass of the carriers; and ionospheric plasmas, which also exhibit nonlinear behaviour based on the anharmonicity of the plasma.

ANHARMONICITY: The extent to which the oscillation of an oscillator differs from simple harmonic motion. In molecular vibrations the harmonicity is very small near the equilibrium position, becoming large as the vibration moves away from the equilibrium position and is very large as dissociation is approached.

ANHYDRIDES: 1. Compounds obtained by the removal of water from another compound usually an acid. 2. Substances, which react with water to produce acids. For example, SO_3 is an anhydride of H_2SO_4 .



Chemical Structure of Anhydrides

ANHYDRITE (CALCIUM

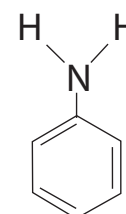
SULFATE): A naturally occurring anhydrous calcium sulfate (CaSO_4), which differs chemically from gypsum by having no water of crystallisation. It changes into gypsum under humid conditions. It is used in plasters and cement, as a drying agent.

ANHYDROBIOSIS: A strategy used by some organisms to survive extreme dry spells. It is a form of cryptobiosis, which is a state of suspended animation which certain organisms enter in response to desiccation without disappearing altogether. During anhydrobiosis organisms such as bacteria, yeasts, plants and many invertebrates such as nematodes and bdelloid rotifers slowly lose body water by evaporation and contract into a smaller undetectable shape in response to dry temperatures. When water becomes available, they revert to active life form.

ANHYDROUS: Having or containing no water. The term usually refers to salts with no water of crystallisation. It is also used for liquids, which are perfectly dry. Examples are anhydrous ether or ether from which all traces of water has been removed and white anhydrous copper(II) tetraoxosulfate(VI).

ANILINE (AMINOBENZENE; PHENYLAMINE):

An aromatic amine, with the chemical formula $\text{C}_6\text{H}_5\text{NH}_2$, density 1.0217 g/cm^3 , molar mass 93.13 g/mol , melting point -6.3°C (267 K) and boiling point 184.13°C (457 K), which is a pale-brown oily liquid obtained chemically from coal tar. On exposure to sunlight the compound turns brown. It is used in making dyes, pharmaceuticals, rubber products, etc.



Chemical Structure of Aniline

ANILINE STAIN: Any of a group of dyes derived from aniline that is used to stain biological material. Aniline sulfate and aniline hydrochloride both stain lignin yellow. Aniline blue is often used as a counterstain with safranin.

ANIMAL: Any multi-cellular living organism of the kingdom Animalia, which is unable to manufacture its own food and typically having specialised sense organs and nervous system and able to respond rapidly to stimuli. Animals develop from embryos formed by the fusion of haploid eggs and sperm. Animals are divided into two groups, known as vertebrates and invertebrates. Vertebrates have a backbone or spinal column, made of interlocking units called vertebrae as in a dog or cow. Invertebrates lack backbones as in snails, lobsters, worms, crabs scorpions, etc. The size of animals ranges from only 0.05 millimetres long in microscopic organisms to 30 metres long in whales. Animals are the only living things that have evolved nervous systems and sense organs that monitor their surroundings. Animals differ from plants in having cells without cellulose walls, do not undergo photosynthesis and therefore cannot produce their own food and obtain energy by feeding on other organisms and organic matter, which they digest to obtain energy. They also require more complex food materials such as proteins. They are organised to a greater degree of complexity and have the capacity for spontaneous movement and rapid motor responses to stimulation and also show flexible patterns of behaviour that can be shaped by past experience.

ANIMAL BEHAVIOUR: 1. The activities that constitute an animal's response to its external environment. 2. The scientific study of the behaviour of animals either by comparative psychologists or by ethologists. It includes investigating the relationship between animals and their physical environment as well as to other organisms, how animals find and defend resources, avoid predators, choose mates and reproduce and care for their young.

ANIMAL CELL: Eukaryotic cell enclosed by a plasma membrane and containing a membrane-bound nucleus and organelles. The nucleus is a membrane-bounded sac containing the genetic material of the cell. Unlike the eukaryotic cells of plants and fungi, animal cells do not have a cell wall. Organelles perform so many functions, which include producing hormones and enzymes and providing energy for animal cells. Some structures and organelles found in typical animal cells include nucleus, mitochondria, Golgi apparatus, endoplasmic reticulum, ribosomes, lysosomes, peroxisomes, cytoskeleton, cilia and flagella and peroxisomes. Most animal cells are between 1 and 100 micrometres and are visible only with the aid of a microscope.

ANIMAL DROPPING (ANIMAL WASTE): Faecal matter of animals commonly used to fertilise soil.

ANIMAL HUSBANDRY: The care and management of farm animals or the branch of farming concerned with the keeping of animals for products such as milk, meat, hide, skin, etc. Examples of farm animals are cattle, goats and pigs. Cattle give meat, milk and hide; goats give milk, meat and skin; and pigs give meat (pork) and lard. The external features of a cow, a goat and a pig are shown in the following diagrams:

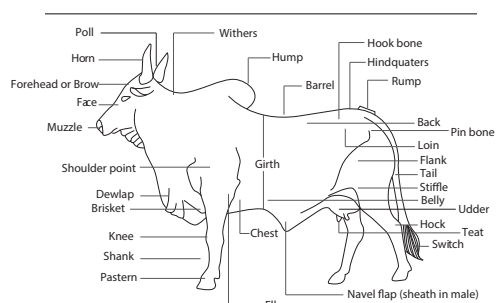


Fig. I: The external features of a cow

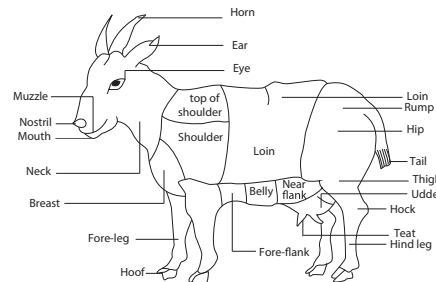


Fig. II: The external features of a goat

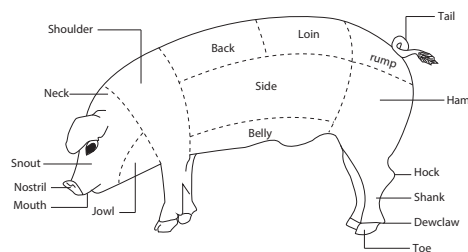


Fig. III: The external features of a pig

Examples of animals used in animal husbandry

ANIMAL MODEL: A laboratory living animal used for medical research into human diseases because it has specific characteristics that resemble a human disease. Scientists can create animal models, usually laboratory mice, by transferring new genes into them.

ANIMAL POLE: That part of an egg that is opposite to the vegetal pole and contains the nucleus, most of the cytoplasm and the least yolk. It is usually the most active part of the protoplasm. During meiosis of the oocyte the polar bodies are expelled at the animal pole.

ANIMAL POPULATION: All of the animals of the same species that live in the same place at the same time and capable of interbreeding or reproduction. An example is elephant population.

ANIMAL SCIENCE (ANIMAL BIOLOGY; ZOOLOGY): The use of scientific principles and practices to study animal growth, care and management such as breeding, production, nutrition, animal agribusiness, behaviour and well being and biotechnology. The field of study may include anatomy, genetics, microbiology, animal behaviour and management, nutrition, physiology and reproduction.

ANIMAL WELFARE: The well being of farm animals. An animal is in a good state of welfare if it is healthy, comfortable, well nourished, safe, able to express innate behaviour, and if it is not suffering from unpleasant states such as pain, fear, and distress. Good animal welfare requires disease prevention and veterinary treatment, appropriate shelter, management, nutrition, humane handling and humane slaughter/killing.

ANIMALIA: The kingdom to which all animals belong.

ANIMATE: 1. In biology, it refers to living or having life or relating to animal life as distinct from plant life. 2. To give inanimate characters

the appearance of movement on a screen. Called **animation** or **computer animation**, it is a visual digital display technology that simulates lifeless objects to life on a screen or monitor. An example of animation is a cartoon shown on a television.

ANIMATED GIF (ANIMATED GRAPHICS INTERCHANGE FORMAT): A graphic image format often used on World Wide Web (www) pages that provides moving icons and banners.

ANIME: A type of Japanese comic book and video cartoon animation in which the main characters have large doe-like eyes using characters and images that are drawn rather than real.

ANIMISM: The belief that animate or inanimate objects including the universe possess souls or spirits. It is a fundamental system of belief in certain religions, particularly prior to the development of organised religion.

ANIOKOS: A form of programmed cell death induced in certain cell types when they become detached from their extracellular matrix or from a surface.

ANION: A monoatomic or polyatomic species having one or more elementary charges of the electron. Anions can be formed from nonmetals by reduction or from neutral acids or polar compounds by ionisation. Anionic species include Cl^- , SO_4^{2-} and CH_3COO^- . Highly coloured intermediates in organic reactions are often radical anions (anions containing an unpaired electron). Salts are made up of anions and cations. In an electrolytic cell anions are attracted to the anode or positive electrode.

ANION EXCHANGE: The process of exchanging anions between a solution and an anion exchanger. **Anion exchanger** is an ion exchanger with anions as counter-ions. The term anion-exchange resin may be used in the case of solid organic polymers. The base form of an anion exchanger is the ionic form in which the counterions are hydroxide groups (OH^- form) or the ionogenic groups form an uncharged base such as ^-NH . **Anion exchange capacity** is the total amount of exchangeable anions that a soil can adsorb. It is expressed as milliequivalents per 100 g of soil or other adsorbing material such as clay.

ANIONIC POLYMER: Polymer composed of negatively charged macromolecules and an equivalent amount of counteranions. If a substantial fraction of constitutional units carries negative charges, then an anionic polymer is a polyelectrolyte. Anionic polymer is not the same as a polymer prepared by anionic polymerisation.

ANIONOTROPIC REARRANGEMENT (ANIONOTROPY): An organic reaction where a migrating group moves with its electron pair from one atom to another.

ANISE: A plant, *Pimpinella anisum*, of the family *Umbelliferae* which is native to the Mediterranean but cultivated widely. The seed has a strong aromatic flavour and oil obtained by distilling the seed is used in medicine and for flavouring.

ANISOCARPOUS: In botany, it refers to a flower whose number of carpels is different from the number of stamens, petals and sepals. Examples of plants with anisocarpaceous flower include members of the olive family (*Oleaceae*), Milk-weeds (*Asclepiadaceae*) family and *Gentianaceae*.

ANISOCOTYLOUS: In botany, it refers to unequal cotyledons such as those having different size and/or shape.

ANISOGAMY (HETEROGAMY): Sexual reproduction involving the fusion of two dissimilar gametes differing either in size only or in size and form. The smaller gamete is considered to be the male (sperm cell), whereas the larger gamete is regarded as the female (egg cell). There are several types of anisogamy. Both gametes may be flagellated and thus motile. Alternatively, neither of the gametes may be flagellated. This situation occurs for example in some algae and plants. In the red alga *Polysiphonia*, large non-motile egg cells are fertilised by small, non-motile spermatia. In flowering plants, the gametes are non-motile cells within gametophytes. The form of heterogamy that occurs in animals is **oogamy**.

ANISOLATERAL: Having unequal sides.

ANISOMEROUS: A flower with different number of parts usually less than the other floral whorls. A typical example is with flowers with five sepals and petals, but with only two stamens.

ANISOPETALOUS: In botany, it refers to petals that are unequal.

ANISOPHYLLOUS: In botany, it refers to unequal leaves such as in shapes and sizes. Examples of some plants with such leaves are some conifers and many aquatic plants.

ANISOSTYLOUS: Having unequal styles.

ANISOTROPIC: 1. Denoting a medium in which certain physical properties are different in different directions. Wood, for instance, is an anisotropic material, because its strength along the grain differs from that perpendicular to the grain. It is the property of molecules and materials to exhibit variations in physical properties along different molecular axes of the substance. Single crystals that are not cubic are anisotropic with respect to some physical properties, such as the transmission of electromagnetic radiation. 2. In botany, it refers to a plant that responds unequally to an external stimulus in different parts.

ANISTERUM: A serum that contains antibodies, either to fight particular antigens of known specificity or a broad mixture of antibodies generally used to give short term passive immunity.

ANKLE: The hinge joint between the foot and the leg. The uppermost bone of the foot, called the talus or the ankle-bone, fits between the two bony protuberances formed by the lower ends of the tibia or the shin bone and fibula.

ANKYLOSIS: The abnormal fusion or stiffening of a joint, restricting movement. It may be caused by injury, surgery, rheumatoid arthritis and tuberculosis.

ANNATTO (ACHIOTE; *Bixa orellana*): A flowering shrub or small tree and the only member of the family *Bixaceae*, native to the tropical regions of America. It grows to about 5-10 metres tall. The leaves are ovate with a round, heart-shaped base and a pointed tip. The flowers are pink, white or pinkish-white. The pod-like fruit is heart-shaped containing about fifty small reddish-orange seeds. The seeds are enclosed in a reddish aril, which is the source of the annatto dye, containing the soluble reddish-orange carotenoid pigment known as bixin. It is used as a colourant and condiment for traditional dishes; and in production, it is used to add colour to some drinks and foods such as butter, cheese, popcorn, drinks and breads. Medicinally, the plant is used to treat desentery and fever.

ANNEALING: A process of heating a material, usually glass or metal for a given time at a given temperature; followed by slow cooling, to increase ductility and strength.

ANNELID (SEGMENTED WORM): An invertebrate organism, belonging to the phylum *Annelida*, with a flat body that is divided into segments. They include the earthworms, leeches and marine worms such as lugworms. They are triploblastic, bilaterally symmetrical and eucoelomate. Besides, they have a body wall with both longitudinal and circular muscle layers which, along with segmentation allows them to be quite mobile. They also have a complete digestive tract, a nervous system which shows some degree of cephalisation, a closed circulatory system and an excretory system which is based on nephridia. The three major classes within the phylum *Annelida* are the Class *Polychaeta*, which are mostly marine worms, such as *Nereis* (the clamworm), the Class *Hirudinea* which are the leeches (predominantly freshwater), such as *Hirudo* and the Class *Oligochaeta*, which are mostly freshwater and terrestrial worms, such as *Lumbricus* (the earthworms). They aerate soil so that roots and water can move freely and efficiently. They also convert organic debris in oceans into carbon dioxide which is used by marine planktons for photosynthesis. Earthworms serve as food for birds. Sometimes large earthworms are used for dissection in academia, to teach the basic internal organs. Leeches are also annelids and help with healing wounds. Large earthworms or night crawlers (*Lumbricus terrestris*), are cultivated and sold as bait for freshwater fishes and as humus builders in gardens. The abundance of the sludge worm *Tubifex*, near sewer outlets serve as indicators of water pollution.

ANNIHILATION: 1. In radiochemistry, it is an interaction between a particle and its antiparticle in which they both disappear. 2. In physics, it is the process in which the entire mass of two colliding particles, one of matter and one of antimatter, is converted into radiant energy (according to Einstein's famous formula $E = mc^2$) in the form of gamma rays or mesons. It occurs within when a particle meets in anti-particle. For example, in the case of a collision between an electron and a positron (anti-electron), which have a combined rest mass of 1.02 MeV, the result is two gamma rays each with a characteristic energy of 0.511 MeV. When nucleons annihilate each other the result is mesons. Electron-positron annihilation is made use of in PET scanning, where the source of each pair of gamma photons can be traced allowing an image to be constructed that shows where a positron-emitting tracer is located in a patient's body.

ANNIHILATOR: In mathematics, it has the following definitions: (i) The collection of all functions of some prescribed type that takes value zero on every member of a given set. (ii) The linear subspace of all continuous linear functionals whose value is zero on every member of a given set in a normed space. (iii) The orthogonal complement of a set in Hilbert space. (iv) The set of elements of a product with every member of a given subset of a module over the ring is the zero element of the ring; this set is an ideal of the ring and is denoted $\text{Ann } X$, where X is the given subset.

ANNONA: A genus of more than 160 species of evergreen or semi-deciduous flowering shrubs or small trees in the pawpaw/sugar apple family, Annonaceae in the major group *Angiosperms* (**Flowering plants**). Annona species generally have trunks with thin bark and broad, shallow depressions or fissures which join together and are scaly giving rise to slender, stiff, cylindrical and tapering shoots with raised pores and naked buds. The leaf blades can be leathery or thin and rather soft or pliable, bald or hairy. The flowering stalks rise from axils or occasionally from axillary buds on main stems or older stems or as solitary flowers or small bundle of flowers. The fruit is fleshy, ovate to spherical, consisting of many individual small fruits or syncarps, with one syncarp and seed per pistil. The seeds are bean-like with tough coats; the seed kernels are toxic. The fruit pulp of some of the species is yellow, sweet and similar to pineapple and used primarily as food. Some of the species are used in traditional medicines for the treatment of a variety of diseases including worms parasitic on the intestines or skin (notably guinea worms), diarrhoea, fever, abdominal pain, leprosy, gastroenteritis, lung infections, toothaches, dizziness, indigestion, chest pains and venereal diseases and snakebites. The bark of some species can also be used to produce yellow-brown dye and insecticide. Some of the species include *Annona senegalensis*, commonly known as African custard apple, wild custard apple or wild soursoap; *Annona muricata* with the common name soursoap; *Annona scleroderma* with the common name wild red custard apple; *Annona amambayensis*; and *Annona glabra* commonly known as pond apple, alligator apple or monkey apple.

ANNOTATION: 1. Addition of explanatory or critical comments to a text or an explanatory or critical comment that has been added to a text. 2. In bioinformatics, descriptive notes that are attached to a database entries such as chromosomal loci or nucleic acid or amino acid sequences detailing pertinent features such as transcripts, markers, structure, function, disease associations and citations to relevant journal articles.

ANNOYWARE (NAGWARE): In computing, it is any software that is free for a trial period and afterwards periodically reminds (usually through pop-up windows or alerts) the user to register or purchase the product.

ANNUAL NETTLE (DWARF NETTLE; SMALL NETTLE; *Urtica urens*): An annual stinging temperate weed, *Urtica urens* of the genus *Urtica*. It has medicinal and therapeutic uses and important in the diet of many farmland birds.

ANNUAL PERCENTAGE RATE (APR; INTEREST RATE): Annual rate at which interest is charged on a loan agreement. This is always an important consideration when borrowing money. There are several classifications of APRs. For example, a GH¢1000 loan repaid fully after one year with GH¢80 interest plus a GH¢ 10 service fee, has a total finance charge of GH¢90 and so has an APR of 9%. The **nominal APR** is the interest rate that is stated on a loan. The **effective APR** includes fees that have been added to the balance. The effective APR on a credit card or loan might be higher than the nominal APR. A **fixed APR** generally remains the same throughout the life of the loan. However, in the case of credit cards a fixed APR can change if the card issuer notifies you 45 days in advance. A **variable APR** can change without notice and is based on another interest rate, like the prime rate. **Credit Card APRs** could have several APRs for each type of balance: one for purchases, one for balance transfers and one for cash advances. The APR for cash advances and balance transfers tends to be higher than the APR for purchases.

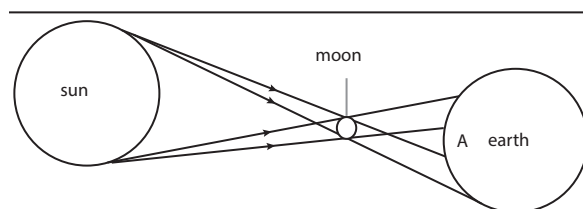
ANNUAL PLANT: A plant such as tomato, sunflower and marigold that completes its life cycle within one year or less than a year, during which it germinates, grows to maturity, bears flowers, produces seeds and then dies. Plants that live more than 1 year may be either biennial or perennial.

ANNUAL RHYTHM (CIRCANNUAL RHYTHM): Periodic biological fluctuation in an organism that corresponds to and is in response to, periodic environmental change. Examples of such change include cyclical variations in the relative position of the Earth to the Sun and to the Moon and in the immediate effects of such variations, for example, day alternating with night, high tide alternating with low tide.

ANNUAL RING (GROWTH RING): One of a set of concentric rings visible on the wood of a cut tree trunk. Each ring represents a period of growth and may be used to determine the age of the tree.

ANNULAR: Shaped like a ring. The annular markings on the transverse section of the stem of a tree indicate its age.

ANNULAR ECLIPSE: Eclipse of the Sun by the Moon when a ring of sunlight can be seen around the Moon or solar eclipse in which the Moon does not completely obscure the Sun and a thin ring of sunlight remains visible. Eclipse of the Sun by the Moon occurs when the Moon is at its farthest point from the Earth.



Annular eclipse of the Sun

ANNULAR PARALLAX: The parallax (the angle between two imaginary lines from two different observation points meeting at an astronomical object, used for measuring a star's distance from Earth) of a celestial object resulting from the movement of the earth in its orbit during a year. It reflects the orbit of the earth around the sun. The motion is a circle at the ecliptic poles and a line on the ecliptic.

An object at a distance of 1 parsec undergoes an annual parallax with a semi-major axis of 1 arcsec. Its symbol is π .

A

ANNULAR THICKENING: A type of thickening of wall of a xylem vessel or tracheid, in form of rings at intervals along its length, which occurs in tree and particularly the xylem tissue which transports water. The thickening of the cells provides stability within the structure of the plant.

ANNULATION (ANNELEMENT): A transformation involving fusion of a new ring to a molecule via two new bonds.

ANNULENES: Organic hydrocarbons that have molecules containing simple single rings of carbon atoms linked by alternating single and double bonds. Such compounds have even numbers of carbon atoms. Its general formula is C_nH_n (n is an even number) or C_nH_{n+1} (n is an odd number). In systematic nomenclature an annulene with seven or more carbon atoms may be named $[n]$ annulene, where n is the number of carbon atoms, for example, $[9]$ annulene for cyclonona-1,3,5,7-tetraene.

ANNULENYLIDENES: Carbenes, derived by formal insertion of a divalent carbon atom into an even-membered annulene, for example, cycloheptatrienyliene.

ANNULISED PERCENTAGE RATE (APR): The actual cost of borrowing money from a lender.

ANNULUS: 1. A ring-shaped geometric plane figure formed between two concentric circles of different radii, R and r . Its area is $\pi(R^2 - r^2)$, where $R > r$. 2. The ring of differentially thickened cells that encircles the sporangium of certain fern, which aids in spore dispersal. 3. Also called **velum**, it is the remnants of the ruptured partial veil, found around the stipes of certain mushrooms or toadstool.

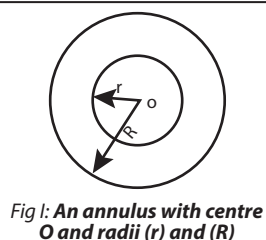


Fig I: An annulus with centre O and radii (r) and (R)



Fig II: Annulus (velum)

ANODE: In an electrolytic cell, it is the positive electrode or terminal towards which negative particles (anions and electrons) move. In other words, it is the electrode at which oxidation takes place. In electrochemical cell, it has negative polarity. In an electronic vacuum tube the anode attracts electrons from the cathode.

ANODISING (ANODIZING): A process of electrically depositing an oxide film from aqueous solution onto the surface of a metal, usually aluminium, which serves as the anode in an electrolytic cell. The process increases the resistance to corrosion (passivation), electrical insulation, improving paint adhesion, etc., of the metal.

ANODYNES: A group of analgesic or pain-relieving herbs that reduce the intensity of pain and relax the body by reducing the sensitivity of the brain or nervous system. Anodynes are applied topically or externally. Examples of such herbs are dandelion, wild lettuce, deadly nightshade and thorn apple.

ANOESTRUS: A situation where a female does not come on heat at the usual time or nonoccurrence of oestrus so that the female is not sexually receptive at any time. It is common both in the maiden gilt and in the sow at post-weaning time.

ANOIKIS: A form of programmed cell death (apoptosis) induced in certain cell types when they become detached from their extracellular

matrix or from a surface. Epithelial, endothelial and many other tissue cells depend on continual interchange of signals with neighbouring cells and their immediate environment in order to survive.

ANOMALOUS DISPERSION: The sudden change in the refractive index of a material for wavelengths in the vicinity of absorption bands in the spectrum of the material.

ANOMALOUS EXPANSION OF WATER: The unusual characteristic of expansion of water as it is cooled from 4°C (277 K) to 0°C (273 K), contrary to the fact that most substances contract when they are cooled. Water therefore has a greater density at 4°C than ice at 0°C , enabling ice to float in water. The water at the bottom of a pond in winter is warmer than that at the surface; making ponds and lakes freeze slowly from the surface downwards and usually remaining liquid near the bottom, where aquatic life is more likely to survive.

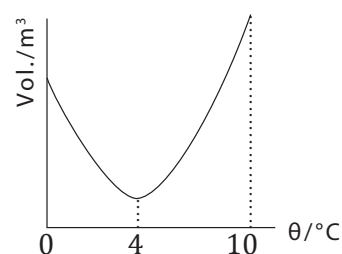


Fig I: Anomalous expansion of water ($\text{Vol./m}^3 - \theta/^\circ\text{C}$)

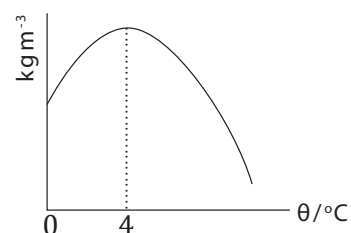


Fig II: Anomalous expansion of water ($\text{Density/kgm}^{-3} - \theta/^\circ\text{C}$)

ANOMALOUS MAGNETIC MOMENT: The difference between the observed gyromagnetic ratio of the electron and the value of exactly two predicted by Dirac's theory of the electron. The discrepancy is resolved using quantum electrodynamics.

ANOMALY: 1. Deviation or departure from the normal order, form or rule. 2. The angle between a planet's position, the Sun and the point in the planet's orbit when it is closest to the Sun.

ANOMERS: The sugar isomers that differ in configuration about the anomeric carbon, i.e., the C1 atom of an aldose or the C2 atom of a ketose. The α - and β -forms of glucose are examples of anomers.

ANOMOCYCTIC STOMATE: A stomata lacking differentiated subsidiary cells.

ANONYMOUS FILE TRANSFER PROTOCOL (FTP): In computer science, it is a method for downloading files from a website without the need for identification. When you access an anonymous FTP site, you just enter the

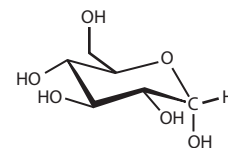


Fig. I: α -form of glucose

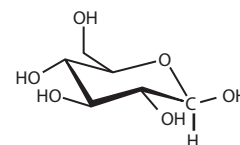


Fig. II: β -form of glucose

Examples of anomers

word anonymous or ftp without being prompted for your username and password. When prompted by the host for a username, you can use an e-mail address or simply the word "guest".

ANOPHELES: A mosquito belonging to the genus *Anopheles*, that includes some that can carry and transmit malaria parasite to humans.

ANOPLURA (SUCTION LOUSE): An order of a group of wingless insects that are parasitic and feed on blood of their host with typical example as the body louse.

ANOREXIA: Eating disorder characterised by low body weight. The sufferer has an obsessive fear of gaining weight.

ANORTHITE: A mineral occurring in igneous rocks, consisting of calcium aluminium silicate with the chemical formula $\text{CaAl}_2\text{Si}_2\text{O}_8$, Mohs scale hardness 6, specific gravity 2.76 and melting point 1550 °C (1823 K). It occurs in the form of vitreous white or grey crystals. Anorthite is one of the end-members of the feldspar group.

ANORTHOSITE: A coarse-grained basic igneous rock composed mainly of plagioclase feldspar (95%), with small amounts of pyroxene (4%), olivine and iron oxides. Usually light in colour, it may have bands of darker minerals. Anorthosite forms deep underground and in dikes and intrusions. It makes up about 60% of the Earth's crust. Anorthosite was found in all the rocks returned from the Moon. All anorthosites found on Earth consist of coarse crystals, but some samples of the rock taken from the Moon are finely crystalline.

ANOXIA: A severe form of hypoxia characterised by an absence of oxygen supply to an organ or a tissue. Symptoms include altered thought processes, dizziness, breathing difficulties or confusion. If it is not immediately reversed, the condition can lead to collapse or death. If the condition does not involve total oxygen deprivation, it is called **hypoxia**. **Anoxic** means lacking or not involving or requiring oxygen. An example is a culture of anaerobic microorganisms.

ANOXIC: Referring to a condition of an organism with extreme absence of oxygen in the blood, tissue organs or the entire body.

ANOXIC REACTOR: A biological treatment plant designed to remove oxygen and to recycle waste by anaerobic reactions.

ANOXYGENIC PHOTOSYNTHESIS: A type of photosynthesis seen in green and purple bacteria in which oxygen is not produced, but the captured light is converted to ATP (Adenosine triphosphate). Water is therefore not used as an electron donor.

ANSA COMPOUNDS: Benzene derivatives having *para* positions or *meta* connected by a bridge 10 to 12 atoms long. Also, any arene bridged by a chain constrained to lie over one of the two faces of the arene.

ANSERIFORMES: An order of birds which are specialised for aquatic life, most of them found in all parts of the world, except for the Antarctic. The order consists of about 180 living species in three families: Anhimidae, referred to as the screamers; Anseranatidae, the magpie goose; and Anatidae, the largest family. Examples of birds in the order are the ducks, geese and swans, known collectively as waterfowl. Anseriformes generally have long necks and web between three forward-pointing toes and flattened lower bill with cone-shaped upper one, used to feed on various diets. Some are vegetarians and a few filter-feeders. Some eat molluscs and others depend on fish. They lay numerous eggs in nests built on the ground. Some species, such as waterfowl are used as food, and are reared for their eggs, and meat.

ANSI (AMERICAN NATIONAL STANDARDS INSTITUTE): US national standards body founded in 1918 concerned with the development of technology standards. It enhances the global competitiveness of U.S. business and the American quality of life by promoting and facilitating voluntary consensus standards and conformity assessment systems to promote their integrity.

ANT: A social insect belonging to the family *Formicidae* and to the order *Hymenoptera*. More than 12,500 out of an estimated total of

22,000 species have been classified. The body of an ant is divided into three sections: the head, the thorax and the gaster (the narrow waist is actually within the abdomen, so the part of the abdomen behind the waist is called the gaster). They are easily identified by their elbowed antennae and a distinctive node-like structure that forms a slender waist. Ants thrive in most ecosystems and may form 15–25% of the terrestrial animal biomass. They form colonies that range in size from a few individuals living in small natural cavities to highly organised colonies which may occupy large territories and consist of millions of individuals. These larger colonies consist mostly of fertile males called **drones**, one or more fertile females called **queens** and sterile wingless females forming castes of "workers", "soldiers" or other specialised groups. Some ants are used as food by some people, others for medication and rituals. Some species play the role of biological pest control agents. However, their ability to exploit resources brings ants into conflict with humans, as they can damage crops and invade buildings. Their long co-evolution with other species has led to mimetic, commensal, parasitic and mutualistic relationships.

ANTACIDS: Substances such as magnesium hydroxide, sodium bicarbonate or aluminium hydroxide used to increase the pH of stomach juices and, therefore, relieves indigestion.

ANTAGONISM: 1. The interaction of two substances such as drugs, hormones or enzymes having opposing effects in a system in such a way that the action of one (the **antagonist**) partially or completely inhibits the effect of the other (the **agonist**). For example, one group of anti-cancer drugs acts by antagonising the effects of certain enzymes controlling the activities of the cancer cells. 2. An interaction between two muscles in which contraction of one, the antagonist, prevents that of the other, the agonist. 3. An interaction between two organisms such as moulds and bacteria, in which the growth of one is inhibited by the other. 4. In ecology, it is an association between two organisms occupying the same ecological niche in which there is rivalry or feud between them ending in death, injury, or inhibition of one of them.

ANTAGONIST: A drug that inhibits the effect of an agonist in such a way that the combined biological effect of the two substances becomes smaller than the sum of their individual effect.

ANTAGONISTIC MUSCLES (ANTAGONISTIC PAIR): A pair of muscles voluntarily attached to movable joints in pairs allowing coordinated movement of the skeletal joints. For example, the extension of the arm requires one set of muscles to relax while another set contracts. The individual component of antagonistic pairs can be classified into extensors and flexors. One muscle, the flexor muscle bends the joint, while the other, the extensor muscle straightens it. Since the muscles work against each other they are called an antagonistic pair.

ANTARCTIC CIRCLE: An imaginary line that encircles the South Pole at latitude 66°32' S, encompassing the continent of Antarctica and the Antarctic Ocean.

ANTARCTICA: A great mass of land around the South Pole with an area of 14,300,000 square kilometres and covered mostly by ice (up to an average of 2,450 metres thick) and snow, consisting of plateau (1800–3000 metres) and mountain peaks rising to 4500 metres high. It is a continent (the fifth largest), with temperatures below freezing point year round. It is uninhabited by humans, except for research stations. Some of the wildlife that live there, at least for part of the year includes seals, penguin, albatrosses, whales, etc. There are only two species of flowering plant, Antarctic hair grass and Antarctic pearlwort, but is inhabited by a variety of mosses, liverworts, lichens and macrofungi.

ANTECEDENT: In a conditional statement, it is the clause that expresses the condition. For example, if 5n is a multiple of 5, then it is divisible by 5. The statement "5n is a multiple of 5" is the antecedent.

ANTENATAL (PRENATAL): Occurring or existing before birth.

ANTENATAL CARE: Medical examination of a pregnant woman.

A

ANTENNA: 1. A long whip-like jointed, mobile and paired appendage on the head of many arthropods. They are used for smelling, tasting, touching, etc. 2. Another name for the aerial in telecommunications which is used in sending and receiving radio signals. The design depends on the wavelength and amplitude of the signals. A **transmitting antenna** is a combination of conductors which converts alternating current electrical energy into radio waves.

ANTENNA DIRECTIVITY: The ratio of the maximum sensitivity of the power radiated per unit solid angle, to the average power radiated per unit solid angle. It varies directly as the effective aperture of the antenna and inversely as the square of the wavelength.

ANTENNA TEMPERATURE (T_A ; ANTENNA NOISE TEMPERATURE): The quantity that describes how much noise an antenna produces in a specific environment. It is a practical unit for the power output per unit frequency from a receiving antenna, which is the power per unit bandwidth (p) of the signal received by an antenna after losses in the detector system. Antenna temperature is not the physical temperature of the antenna as measured by a thermometer.

ANTENNAL GLAND (GREEN GLAND): Either of a pair of organs found in the third segment of adult crustaceans, which opens to the exterior at the base of the second antenna and functions as osmoregulatory organs.

ANTEPETALOUS (ANTIPETALOUS): A structure or organ directly in front of or opposite the sepals.

ANTERIOR: The front part of an organism, usually the part that goes forward first when the animal is moving.

ANTESEPALOUS (ANTISEPALOUS): A structure or organ directly in front of or opposite the petals.

ANTHELMINTICS (VERMIFUGE): A group of anti-parasitic drugs or herbs used to expel or kill parasitic worms such as flukes, tapeworms, nematodes, flatworms, round worms, and other internal parasites from the body. Examples of such herbs include pawpaw (seeds, roots, leaves, etc.), aloe vera, neem, garlic, parsley and clove. Anthelmintic herbs are safe to use and have few or no side effects.

ANTHER: The part of the stamen of a flower containing pollen grains. It is usually borne on a slender stalk or filament and has two lobes each containing two chambers or pollen sacs within which the pollen grains are found.

ANTHER CULTURE: A culture method in which anthers are made to divide and grow into either callus or embryonic tissue in sterile nutrient medium. If anthers are taken from a plant at a certain stage of development and cultured under appropriate conditions then embryoids and subsequently haploid plantlets may be induced to form from the pollen grains. In some species it is possible to culture isolated pollen grains.

ANTHER SAC: A pollen-bearing chamber of the anther.

ANTHERIDIAL CELL: The cell that remains after one or more vegetative cells are cut off from the microspore and from which the antheridium (sperm-producing organ occurring in seedless plants, fungi and algae) develops. In seed plants, it is the generative cell in the pollen grain, which divides to provide the two sperm cells in the pollen tube.

ANTHERIDIOPHORE: A specialised stem-like structure consisting of a stalk and cap that bears the antheridia in some mosses and liverworts. The antheridia are borne in pits on the upper surface of the cap.

ANTHERIDIUM: The male organ of seedless plants such as ferns and mosses that bears the male gamete (sperm cell).

ANTHEROCYTE: A cell that differentiates into an antherozoid without further cell division. The antherozoid is usually released from the antherocyte after it has been discharged from the antheridium.

ANTHEROZOID: Motile male gametes produced by a plant such as algae, bryophytes, pteridophytes and some gymnosperms. Antherozoids are formed in an antheridium and after being released, swim by means of flagella(e) to a female gamete.

ANTHESIS: The period of time that it takes for the flower to fully open or when the anthers emerge; or the time when a flower becomes sexually functional. It occurs at the time between the opening of a flower and the formation of the fruit; or between the opening of the bud and the onset of withering.

ANTHOCARP: A false fruit with some portion of the flower besides the pericarp persisting, as in apples and pears (pome) with the fleshy perianth tube surrounding the pericarp.

ANTHOCEROPHYTA (ANTHOCERATOPHYTA): A phylum comprising about 100 species of simple nonvascular plants, including the hornworts found worldwide in temperate and tropical regions on tree trunks, river banks and other damp areas.

ANTHOCEROS: A genus of hornworts in the family Anthocerotaceae that is widely distributed throughout the world in tropical and warm temperate regions grow on damp soils or on rocks. It is distinguished by having spores that are dark-brown to black. Anthoceros is much larger and complex than those of riccia and marchantia found in moist clayey soils along hills and ditches. Anthoceros exhibits alternation of generation, with the gametophyte generation being dominant, with the gametophyte being typically flattened.

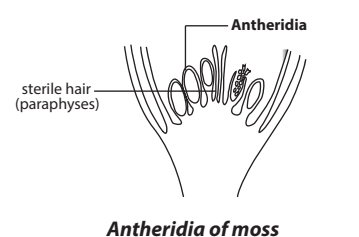
ANTHOCHLOR PIGMENTS: A group of yellow flavonoid flower pigments containing the chalcones and aurones. They turn red on exposure to ammonia. This reaction distinguishes them from the yellow carotenoid pigments, which do not react in this way.

ANTHOCYANESCENCE: The appearance of reddish-purple pigment in tissues that are normally green. It is a symptom of disease, which occurs in response to drought or other environmental extremes, nutritional deficiencies or attacks by parasitic insects and plant pathogens such as the fungus *Taphrina deformans*.

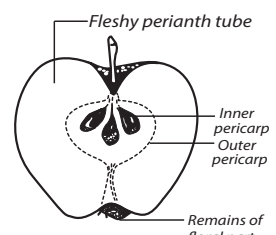
ANTHOCYANIN: A group of water-soluble flavonoid pigments occurring in the cell vacuoles of leaves, stems, roots, flowers and fruits. They are responsible for most of the scarlet, purple, mauve and blue colouring according to pH in flowers and fruits.

ANTHOPHORE: A long stipe or stalk that bears the corolla, stamens and carpel above the receptacle and the calyx.

ANTHOPHYLL CYCLE: An energy efficiency cycle in which xanthophylls are converted from violaxanthin to zeaxanthin to protect chloroplast from intense light and zeaxanthin reconverted



Antheridia of moss



Anthocarp of fruit

to violaxanthin in dim light so that all incident rays are trapped for synthesis of food in plants during photosynthesis.

ANTHOPHYTA (ANGIOSPERMOPHYTA; MAGNOLIOPHYTA):

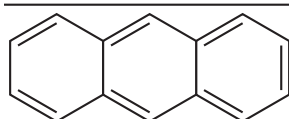
A phylum comprising flowering plants (angiosperms), comprising two classes within the group; the Monocotyledoneae and the Dicotyledoneae. The gametes are produced within flowers and the ovules are enclosed in a carpel. It represents the largest group of plants on Earth.

ANTHOTAXY (INFLORESCENCE): The arrangement of flowers on the flower axis or peduncle.

ANTHOXANTHIN: An accessory photosynthetic pigment which is water-soluble. It is yellow, red or orange, normally consisting of a glucose molecule attached to a flavone or xanthone molecule.

ANTHOZOA: Cnidarians that exist only as polyp forms. The enteron is divided by large radial partitions called mesenteries. They are solitary or colonial animals. The mesoglea contains many amoeboid cells. All species under this class are marine. Examples include sea anemone, coral, sea fan and sea pansy.

ANTHRACENE: A white, glittering, crystalline, aromatic hydrocarbon, with the chemical formula $C_{14}H_{10}$, melting point 215.8°C (488.95 K) and boiling point 341.4°C (614.55 K). It consists of three fused benzene rings and shows a faint blue fluorescence when pure. Anthracene is obtained from the fractional distillation of coal-tar. It is mainly used in the manufacture of dyes.



Chemical Structure of Anthracene

ANTHRACITE: A very hard form of coal which contains more fixed carbon than any other form of coal and the lowest amount of volatile material. It burns with little smoke or flame.

ANTHRACNOSE: A fungal disease characterised by lesions, necrosis and hypoplasia that infects a variety of plants from trees to grasses. Anthracnose diseases are generally caused by certain fungi (usually *Colletotrichum* or *Gloeosporium*) producing spores in tiny, sunken, saucer-shaped fruiting bodies (*acervuli*).

ANTHRAQUINONE: A yellowish crystalline diketone, with the chemical formula $C_{14}H_8(O)_2$, density 1.308 g/cm^3 , melting point 286°C (559 K) and boiling point 379.8°C (652.95 K). It is used in the manufacture of dyes.

ANTHRAX: Infectious, often fatal disease of mainly herbivores such as sheep and cattle caused by the bacterium, *Bacillus anthracis* that can be transmitted to humans.

ANTHRO- (ANTHROPO-): A prefix meaning human or human-like. An example is anthropology.

ANTHROPIC PRINCIPLE: The idea that the universe is the way it is because if it were different humans would not be here to inhabit it. The principle arises from the observation that if the laws of science were even slightly different it would have been impossible for intelligent life to evolve.

ANTHROPOGENIC: Resulting from or influenced by human activities.

Anthropogenic greenhouse gas emissions are greenhouse emissions associated with human activities that result in a net increase in emissions. The sources include greenhouse gases; burning of fossil fuels for energy; livestock in the form of methane released by ruminants such as cows; deforestation; fertilisation; transport-related emissions, mostly in the form of petroleum-based fuels; and aerosols.

ANTHROPOID: Any mammal of the primate suborder Anthroidea. Monkeys, apes (gorilla, chimpanzee and orangutan) and humans are the living anthropoids. Anthropoids have human-like faces, big brains, a relatively flat face, dry nose, small immobile ears and forward-facing eyes.

ANTHROPOID APES: The tailless apes of the family Pongidae of the subfamily Ponginae (apes) comprising the gorillas, chimpanzees, orangutans and the Hylobatidae (gibbons) family. They resemble

most closely human beings (genus *Homo*) and sharing with them a relatively recent common evolutionary ancestor. They have a higher intelligence than the other primates and greater manual dexterity. Humans are also currently classified in the family Pongidae of the subfamily Homininae (humans).

ANTHROPOLOGY: The scientific study of the origin, behaviour and the physical, social and cultural development of humans, which received prominence, following the 19th century evolutionary theories to investigate the human species.

ANTHROPOMETRY: The science dealing with the measurement of human body, particularly stature, body weight, cranial capacity and length of limbs, in samplings of the populations of living people as well as the remains of buried and fossilised humans.

ANTHROPOMORPHIC: Resembling a human being in external appearance.

ANTHROPOMORPHISM: In animal behaviour, it is the attribution of human characteristics to non-human animals.

ANTI-: 1. A prefix for the inverse of a function. For example, antitrigonometric is the inverse of the trigonometric function. 2. In the representation of stereochemical relationships 'anti' means 'on opposite sides' of a reference plane, in contrast to 'syn' which means 'on the same side', as in the following examples: (i) Two substituents attached to atoms joined by a single bond are anti if the torsion angle (dihedral angle) between the bonds to the substituents is greater than 90° or syn if it is less than 90° . (ii) In the older literature the terms anti and syn were used to designate stereoisomers of oximes and related compounds. That usage was superseded by the terms 'trans' and 'cis' or *E* and *Z*, respectively. (iii) When the terms are used in the context of chemical reactions or transformations, they designate the relative orientation of substituents in the substrate or product.

ANTI-ALIASING (AA): An Adobe Photoshop software process of making the jagged edges of graphic objects or fonts smoother by adding additional pixels in-between the edges of an object and its background. It is useful when cutting, copying and pasting selections to create composite images.

ANTI-ALIASING FILTER: A lowpass filter used before analogue-to-digital conversion to limit the bandwidth of a signal being input to a sampled system, such as a digital audio system, so that it is less than one half the system. This ensures that signal components above half the current sampling rate called the Nyquist frequency are removed, thereby eliminating their sampled replicas or aliases in the baseband, in order to properly represent that sound wave.

ANTI-g SUIT (g-SUIT): A tight-fitting suit worn by aviators and astronauts who are subject to high levels of acceleration force (g). It is designed to retard the flow of blood to the lower body in reaction to acceleration or deceleration.

ANTI-MONEY LAUNDERING (AML) SOFTWARE: A type of computer program used by financial institutions to analyse customer data and detect suspicious transactions.

ANTI-SPYWARE SOFTWARE: A program designed to prevent and detect unwanted spyware program installations and to remove those programs if installed.

ANTI-STOKES RADIATION: Electromagnetic radiation occurring in the Raman scattering which is of much lower intensity than the Rayleigh scattering in which the frequency of the radiation is higher than that of the incident light, i.e., displaced towards shorter wavelengths.

ANTI-TOXIN: A substance that acts against a poisonous substance and prevents it from having a harmful effect.

ANTI VIBRATION MOUNT: 1. A structure that locks a hard drive or other or other 3.5-inch device in one place so that the read/write heads and other mechanisms are protected from movement. 2. In the automotive industry, it is a flexible support for an engine that reduces the noise and vibration that passes to the vehicle chassis, most commonly achieved by bonding rubber to metal.

ANTIAROMATIC COMPOUNDS: Compounds that contain $4n$ ($n \neq 0$) π -electrons in a cyclic planar or nearly planar, system of alternating single and double bonds, for example, cyclobuta-1,3-diene.

A

ANTIAROMATICITY: Those cyclic molecules with a conjugated π electron system. The molecule must have $4n$ π electrons, where n is any integer. Antiaromatic compounds are unstable and highly reactive than that of similar nonaromatic species. An example of antiaromatic compound is cyclobuta-1,3-diene. Antiaromatic ones are prone to reactions causing changes in their structural type and display tendency to alternation of bond lengths and fluxional behaviour both in solution and in the solid. In antiaromatic molecules, an external magnetic field induces a paramagnetic electron current.

ANTIATOM: An atom in which all the particles of an ordinary atom are replaced by their antiparticles: electrons by positrons, protons by antiprotons and neutrons by antineutrons. The charges and magnetic moments are the opposite of that of an ordinary atom. An antiatom cannot coexist with an ordinary atom since the two would annihilate each other with the production of energy in the form of high-energy photons.

ANTIBARYON: The antiparticle of a baryon having a baryon number of $B = -1$.

ANTIBIOSIS: The antagonistic association between an organism and the metabolic substances produced by another (example Amensalism). **Amensalism** is any association between two organisms in which one is harmed by the activities of the other. It includes parasitism and also the various ways in which plants protect themselves from pests and predators.

ANTIBIOTIC: A substance produced by certain living cells especially bacteria, yeasts and moulds or an equivalent synthetic substance, which at low concentrations is capable of slowing down the growth of or destroying, bacteria and other microorganisms which cause bacterial disease. Antibiotics cure bacterial infections, but not viral infections such as colds or flu and bronchitis. Examples of antibiotics are penicillin produced by *Penicillium notatum* and streptomycin by *Streptomyces griseus*.

ANTIBIOTIC DISCS: Sterile paper discs containing antibiotic which are used by placing on a culture medium to identify which antibiotic will be effective against a particular bacterium.

ANTIBIOTIC RESISTANCE: The ability of certain micro-organisms to overcome the action of antibiotics. This results from mutations, producing new strains, which are no longer susceptible to the antibiotic, etc.

ANTIBODY: A specialised blood protein (immunoglobulin) secreted by B-lymphocytes (a type of white blood cell in the immune system of an organism). Antibody is characterised by its specific binding to a site of that molecule. It is produced in response to exposure to a foreign molecule (antigen) and circulates in the bloodstream to attack an antigen and render it harmless. Specific antibodies are synthesised in response to the presence of a particular antigen, which may be foreign or potentially dangerous substances such as bacterial toxins, viruses and allergens.

ANTIBODY-DEPENDENT CELL-MEDIATED CYTOTOXICITY, ADCC (ANTIBODY-DEPENDENT CELLULAR CYTOTOXICITY): A type of immune reaction in which Fc receptor-bearing effector cells coat target cells or microbes with antibodies to lyse or destroy them. Fc receptor-bearing effector cells are proteins found on the surface of certain cells, for example, B lymphocytes, macrophages, neutrophils, human platelets, etc., which are involved in destroying invading pathogens or microbes.

ANTIBODY-MEDIATED IMMUNITY (HUMORAL IMMUNITY): Immune reaction that primarily protects against invading viruses and bacteria through antibodies produced by plasma cells.

ANTIBONDING MOLECULAR ORBITAL: An atomic or molecular orbital whose occupation by electrons decreases the total chemical bonding of a molecule. The energy level of antibonding molecular orbital increases as atoms are brought closer together.

ANTIBONDING ORBITAL: A molecular orbital higher in energy than any of the atomic orbitals from which it is derived. It causes a molecule or ion to be unstable when electrons are present and tends to push the atoms apart. Its symbol is a star superscript (*).

ANTICHAIN: A set of pairwise incompatible elements of a partially ordered set (poset).

ANTICHOLINERGIC: Any drug such as atropine, used to treat stomach ulcers and gastritis that blocks the passage of certain nerve impulse in the central nervous system by inhibiting the production of acetylcholine, a neurotransmitter.

ANTICHOLINESTERASE: Any substance that inhibits or inactivates the enzyme cholinesterase, which is responsible for the breakdown of the neurotransmitter acetylcholine at nerve synapses.

ANTICLASTIC: A surface is said to be anticlastic at a point C if it is a turning point which has opposite signs of a curvature in two perpendicular directions at the point, that is one being a maximum and one being a minimum, giving a saddle-point. In other words, of a surface, having opposite curvatures at a given point; specifically, if it is curved convexly along a longitudinal plane section and concavely along the perpendicular section.

ANTICLINAL: 1. At right angle to the surface of an organ or part or inclining in opposite directions from a central axis. For example, cell divisions that take place in a plane that is perpendicular to the surface of the next surface of the organ are called anticlinal. 2. In geology, it refers to an incline downward on both sides from a median line or axis, such as a fold of rock strata; or pertaining to such a fold.

ANTICLINE: A fold on the rocks of the Earth's crust in which the layers or beds bulge downwards to form an arch.

ANTICLOCKWISE (COUNTER CLOCKWISE): Movement in the opposite direction to the movements of the hands of the clock.

ANTICOAGULANT: A substance that suppresses the formation of blood clots. A common one is heparin, produced by the liver and lungs. Anticoagulants are given to prevent thrombosis and used in drawing and storing blood.

ANTICODING STRAND (NONCODING STRAND; TEMPLATE STRAND): The strand in duplex DNA that by convention contains the sequence of bases that is complementary to that of the messenger RNA transcribed from the DNA.

ANTICODON: A sequence of three bases on the transfer RNA that pairs with the bases in the corresponding codon on the messenger RNA. A given molecule of transfer RNA will possess a specific anticodon that only complexes with one particular amino acid. It is this absolute correspondence between an amino acid, an anticodon and a codon, that enables the type and sequence of amino acids in a protein to be determined precisely.

ANTICODON TRIPLET: A triplet of nucleotides in transfer RNA (tRNA) that is complementary to the codon in messenger RNA (mRNA) by determining which amino acid should bind to the synthesising protein.

ANTICOINCIDENCE DETECTOR (ACD): A system on a gamma-ray observatory that triggers when it detects an incoming charged particle (cosmic ray) so that a telescope will not mistake it for a gamma ray.

ANTICONVULSANT: Any drug, such as carbamazepine, ethosuximide (Zarontin), clonazepam (Klonopin), diazepam (Valium), lorazepam that is used to prevent or reduce the severity of epileptic seizures.

ANTICYCLONE: A large-scale circulation of winds that comes from a high-pressure centre with air pressure higher than that of its surroundings, blowing clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere. Anticyclones have wind speeds of about 30 to 50 kilometres per hour, and usually have a diameter of 2400 to 4000 kilometres. Anticyclone typically brings fair weather.

ANTIDEPRESSANT: A drug such as tricyclic compounds and monoamine oxidase inhibitors used in the treatment of depression. Depression can occur if neurotransmitters in the brain do not function effectively.

ANTIDERIVATIVE (INDEFINITE INTEGRAL): The antiderivative of a function $f(x)$ is a function $F(x)$ whose derivative equals $f(x)$. The process of solving antiderivatives is called **antidifferentiation** or **indefinite integration** and its opposite operation is called **differentiation**, which is the process of finding a derivative. For example, the function $F(x) = x^5/5$ is an antiderivative of $f(x) = x^4$. Since the derivative of a constant is zero, x^4 has an infinite number of antiderivatives such as $(x^5/5) - 10$, $(x^5/5) + 2$, etc. Thus, all the antiderivatives of x^4 can be obtained by changing the value of C in $F(x) = (x^5/5) + C$; where C is an arbitrary constant known as the constant of integration. Basically, the graphs of antiderivatives of a given function are vertical translations of each other, with each graph's location depending on the value of C . Antiderivatives are related to definite integrals through the fundamental theorem of calculus and they are important because they can be used to compute definite integrals.

ANTIDEUTERIUM: A highly volatile antimatter hydrogen isotope, which contains one antiproton and one antineutron in its nucleus and one positron orbiting the nucleus.

ANTIDIFFERENTIATION (INTEGRATION): The reverse process of differentiation or a process of solving for antiderivatives. In other words, it is the process of finding an antiderivative for a given function; especially when deriving an indefinite integral or when evaluating a definite integral using the fundamental theorem of calculus rather than as an infinite sum of infinitesimal elements. The symbol $\int f(x)dx$ indicates the operation of antidifferentiation on $f(x)$, which is the function being antidifferentiated or integrated and known as integrand. The antiderivative of a function $f(x)$ is another function $F(x)$, whose derivative is $\int f(x)dx = F(x) + C$. The generalised form of the antiderivative of $f(x)$ with respect to x is given as $\int f(x)dx = F(x) + C$, where C is called the constant of integration or arbitrary constant. This formula works because the derivative of a constant is zero. For example, x^2 is the antiderivative of $2x$, because $2x$ is the derivative of x^2 .

ANTIDISINTERMEDIATION: In commerce, it is a term used to describe the preservation of intermediary positions whereby consumers can access goods or information on the Internet that traditionally required the assistance of an intermediary such as a retailer, travel agent or a banker.

ANTIDIURETIC HORMONE, ADH (ARGININE VASOPRESSIN; VASOPRESSIN): A hormone produced in some mammals by specialised nerve cells in the hypothalamus, which is transported to, stored and secreted by the posterior pituitary gland. Its function is to regulate water re-absorption by the kidneys, thus reducing water loss in the urine. It also has various functions in the brain and is also released in response to a decrease in blood volume. An absence or low levels release of antidiuretic hormone causes diabetes insipidus.

ANTIDOTE: A substance that counteracts the effects of a poison or disease.

ANTIELECTRON (POSITRON): The antiparticle or the antimatter counterpart of the electron. It has the same mass and the same properties as the electron, except for its electrical charge, which is positive (+1e) and a spin of $1/2$. Like electrons, they appear as decay products of many heavier particles. When a low-energy antielectron collides with a low-energy electron, annihilation occurs, producing two 511 keV gamma rays travelling in opposite directions. Antielectrons can combine with other antiparticles to create atoms of antimatter.

ANTIFEEDANT: 1. A natural substance that occurs in certain plants such as neem tree, which adversely affects insects or other animals which feed on them. 2. A synthetic substance that acts as a repellent and stops or inhibits feeding by a pests, especially insects.

ANTIFERROMAGNETIC MATERIAL: Material with a very low magnetic susceptibility that increases with temperature up to a certain temperature, called the **Néel Temperature** or **Magnetic Ordering**

Temperature (the temperature above which an antiferromagnetic material becomes paramagnetic). Above the Néel temperature, the material becomes paramagnetic. Examples include hematite, metals such as chromium, alloys such as iron manganese (FeMn) and oxides such as nickel oxide (NiO).

ANTIFERROMAGNETISM: The magnetic moments of atoms or molecules, usually related to the spins of electrons, aligned in a regular pattern with neighbouring spins (on different sublattices) pointing in opposite directions. Adjacent ions behave as tiny magnets (in this case manganese ions, Mn^{2+}) spontaneously align themselves at relatively low temperatures and vanishing at and above a certain temperature, (the Néel temperature) into opposite or antiparallel, arrangements throughout the material so that it exhibits almost no gross external magnetism.

ANTIFLUORITE STRUCTURE: A crystal structure for ionic compounds of the type M_2X , which is the same as the fluorite (CaF_2) structure except that the positions of anions and cations are reversed, i.e. the cations occupy the F sites and the anions occupy the Ca^{2+} sites. Examples of the antifluorite structure are given by K_2O and K_2S .

ANTIFOAMING AGENT: A substance that inhibits the formation of bubbles in a liquid during its agitation by reducing its surface tension. Examples of substances used are polyamides, polysiloxanes and silicones. Antifoaming agents are employed in processes such as electroplating, paper-making and in water for boilers.

ANTIFREEZE: A substance, usually a liquid of ethylene glycol (ethanediol) that is added to a water cooling system such as a radiator to prevent it from freezing during a cold weather.

ANTIFREEZE MOLECULE: Any substance produced as a specialised adaptation by certain fishes, insects, plants and bacteria in order to prevent freezing of their tissues or body fluids when subject to subzero environmental temperatures. Examples are glycerol or dimethyl sulfoxide in insects, glycerol or trehalose in nematodes and rotifers and proteins in Antarctic fishes.

ANTIGEN: A substance (protein in nature) that the body regards as foreign and which stimulates the immune system to produce a set of specific antibodies. Antigens are usually proteins and are commonly found on the surface of infectious organisms such as some bacteria and viruses, pollen grains, transfused blood cells and organ transplants. They may be formed in or introduced into the body. The presence of an antigen triggers the production of an antibody, which reacts specifically with the antigen and either neutralises it, causes it to destroy itself or attracts leukocytes to carry out the destruction.

ANTIGEN-PRESENTING CELLS: A heterogeneous group of immunocompetent cells that mediate cellular immune response by processing and presenting antigens to the T-cells receptor. Examples include macrophages, dendritic cells, Langerhans cells and B-lymphocytes.

ANTIGENIC DETERMINANT: The site on an antigen to which an antibody binds, forming an antigen-antibody complex.

ANTIGENIC VARIATION: The ability of a pathogenic agent particularly virus, to alter the antigens on its outer surface.

ANTIGRAVITY: A hypothetical force that is free from the force of gravity and acts in a direction opposite to that of normal gravity. It may also refer to any means of reducing or cancelling a gravitational field, through some form of shielding or otherwise.

ANTIHISTAMINE: Any of a group of substances that counteracts the effects of histamine. It is used in the treatment of allergies, allergic rhinitis (hay fever), urticaria (hives), pruritus (itching) and insect bites and stings. It works by blocking the effects as of histamine, a chemical released during allergic reactions. They are sometimes used in cough and cold remedies because they dry up a runny nose and suppress the nerve centres in the brain that trigger the cough reflex. They are also used in antiemetic drugs, because they suppress the vomiting reflex.

A

ANTIHYDROGEN: The antimatter counterpart of hydrogen. The antihydrogen atom is made of one antiproton and one antielectron or positron. It is stable as the hydrogen atom, but because it is antimatter, it very soon annihilates and disappears. When an antihydrogen atom comes into contact with ordinary matter, its constituents quickly annihilate. The positron, which is an elementary particle, annihilates with an electron, with their mass-energy being released as gamma rays.

ANTIKNOCK: A substance added to fuel to reduce knocking characteristics and enhance smooth and efficient combustion in combustion engines. Examples are methanol, ethanol and tetraethyl lead, which have been replaced by methylcyclopentadienyl manganese (MMT) due to the toxicity of lead.

ANTILEPTON: An antiparticle of a lepton having equal mass but opposite sign and different properties. Antileptons interfere with subspace communications.

ANTILITHS: Herbs that prevent the formation or removal of urinary calculi and stones in the urinary system (kidneys and the bladder). Such herbs include comfrey leaves, silver birch and celery.

ANTILOGARITHM (ANTIOLOG): The inverse function of the logarithm defined such that $\text{antilog}_b(\log_b n) = \text{antilog}_b z = b^z = n$. That is, if the log in base b of a number n , (i.e. $\log_b n$), is z , then the antilogarithm in base b of z is therefore $b^z = n$. For example, if the log in base 10 of 2, (i.e. $\log_{10} 2$), is 0.3010, then the antilog in base 10 of 0.3010: $(\text{antilog}_{10}(0.3010) = \text{antilog}_{10}(\log_{10} 2) = 100.3010 = 2$. Also, if the log in base 2 of 8, (i.e. $\log_2 8$), is 3, then the antilog in base 2 of 3, i.e. $\text{antilog}_2(\log_2 8) = \text{antilog}_2 3 = 2^3 = 8$. In general, if logarithm $_a b = c$, then logarithm $_a c = b$. If no base is specified, it is usually taken to be 10.

ANTIEMETIC HERBS: Herbs that relieve or prevent nausea and vomiting. They include cloves, ginger, raspberry, lemongrass, lemons, and wild sage. Antiemetics work in the reverse way of emetics which are drugs or substances which induce nausea and vomiting.

ANTIMAGIC SQUARE:

An $n \times n$ square arrangement of the numbers 1 to n^2 such that the totals of the n rows and n columns and two long diagonals form a sequence of $(2n + 2)$ consecutive integers. The smallest antimagic squares have order 4. An example of a 4×4 antimagic square is shown in the diagram. Note that the numbers in every row add up to the same number.

1	3	4	2
4	2	1	3
2	4	3	1
3	1	2	4

A 4 × 4 Anti Magic Square

ANTIMATTER: A substance composed of elementary particles having the mass and electric charge of ordinary matter such as electrons and protons but for which the charge and related magnetic properties are opposite in sign. The antimatter particles corresponding to electrons, protons and neutrons are called positrons (e^+), antiprotons (p) and antineutrons (n), which are collectively known as antiparticles. The electrical properties of antimatter being opposite to those of ordinary matter, the positron has a positive charge and the antiproton a negative charge; the antineutron, though electrically neutral, has a magnetic moment opposite in sign to that of the neutron. Matter and antimatter cannot coexist at close range for more than a small fraction of a second because they collide with and annihilate each other, releasing large quantities of energy in the form of gamma rays or elementary particles.

ANTIMESON: The antiparticle of a meson.

ANTIMETABOLITE: A substance, which is structurally similar to a metabolite and competes with it or replaces it and so prevents or reduces its in physiological reactions.

ANTIMICROBIAL: Capable of destroying or inhibiting the growth of microorganisms, especially pathogenic microorganisms. Examples of such agents or drugs include antibiotics, antifungals, antiprotozoals, and antivirals.

ANTIMONIC COMPOUNDS: A general term which refers to all compounds of antimony (Sb) in +5 oxidation state. For example, antimonichloride is antimony(V) chloride (SbCl_5).

ANTIMONOUS COMPOUNDS: Compounds of antimony in +3 oxidation state. For example, antimonous chloride is antimony (III) chloride (SbCl_3).

ANTIMONY: A brittle silvery white metal, with the chemical symbol Sb, atomic number 51, atomic weight 121.76, melting point 630.5 °C (903.65 K), boiling point 1587 °C (1860 K) and relative density 6.691 which is used especially in alloys to impart hardness and as a donor impurity in semiconductors.

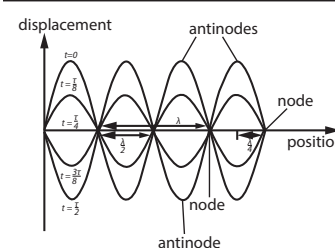
ANTIMONY-XENON LAMP (ARC): An intense source of ultraviolet, visible and near infra-red radiation produced by an electrical discharge in a mixture of antimony vapour and xenon under high pressure. Its output in the ultraviolet region is higher than that of the mercury-xenon arc.

ANTIMYCOTIC (FUNGICIDE): A substance used to kill a fungus or to inhibit its growth. An example is Bordeaux mixture, an antifungal agent consisting of a solution of copper sulfate and quicklime.

ANTINEUTRINO: A lepton, which is antiparticle of neutrinos. It is a neutral particle emitted in nuclear beta decay, where a neutron turns into a proton. It has a spin of $\frac{1}{2}$. Antineutrinos, like neutrinos, interact with other matter only through the gravitational and weak forces, making them very difficult to detect experimentally.

ANTINEUTRON: The antimatter counter particle of the neutron, having the same mass, average lifetime, spin and magnitude of magnetic moment as the neutron. It has no net electric charge, but has opposite baryon number: +1 for neutron and -1 for the antineutron. It was first produced in 1956 at the Bevatron particle accelerator at the University of California, Berkeley, by passing an antiproton beam through matter. Like any other antiparticle, it behaves like matter and can associate with other antiparticles, such as the positron and the antiproton, to form stable atoms.

ANTINODE: 1. The position in a standing wave pattern at which the amplitude of vibration is greatest. Antinodes form at those points where the component wave always arrive in phase. The formation of antinodes is an example of constructive interference. 2. Either of the two points on an orbit where a line in the orbit plane, perpendicular to the line of nodes and passing through the focus, intersects the orbit.



Nodes and antinodes of a standing wave

ANTINUTRIENTS: Chemicals produced by plants serving as a defence mechanism that inhibit the action of digestive enzymes in insects that attempt to eat the plants.

ANTIOXIDANT: Chemical additive which slows down oxidation reactions. They are widely used in manufacturing foods especially those containing fat, for example, thiols, ascorbic acid or polyphenols. Aromatic amines, phenols and aminophenols are used to delay loss of elasticity in rubber and gummy deposits in gasoline. In the body, antioxidants such as vitamins C and E and selenium may reduce oxidation caused by free radicals.

ANTIPARALLEL: 1. For a pair of lines, cutting two parallel lines in such a way that the interior opposite angles of the quadrilateral sum up to two right angles. 2. Two lines are said to be antiparallel with

respect to a transversal if they still form equal angles with the latter but in a reversed order. For a pair of directed lines, having the same direction but opposite sense, so that given any line segment AB, the directed lines AB and BA are antiparallel. 3. Of or relating to two adjacent molecules whose sequences of atoms or other elements run in opposite directions. For example, the two strands of DNA are antiparallel, with the head of one strand always lying against the tail of the other strand of DNA.

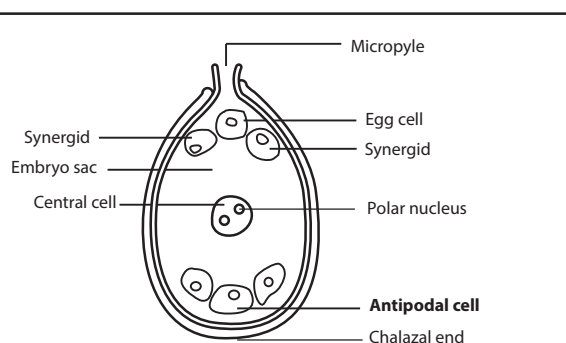
ANTIPARALLEL SPINS: Neighbouring spinning electrons in which the spins and hence the magnetic moments of the electrons are aligned in the opposite direction. Under some circumstances the interactions between magnetic moments in atoms favour parallel spins, while under other conditions such as antiferromagnetism they favour antiparallel spins.

ANTIPARALLEL VECTORS: Vectors directed along the same line in opposite direction, a pair of lines or vectors that are parallel but in the opposite direction.

ANTIPARTICLE: A subatomic particle that has the same mass, average life span, spin and magnitude of electric charge as another particle and equal but opposite value of some other property or properties. An example is an electron and a positron, which have the same mass but different charges.

ANTIPASMODICS: Herbs or agents that prevent or relieve muscular spasms or convulsions by suppressing muscle contraction. Such herbs include guava, neem, licorice, mistletoe, nutmeg oil, peppermint oil, wild lettuce and wild sage.

ANTIPODAL CELL (ANTIPODAL): Any of the three haploid cells in the embryo sac of angiosperms located at the chalazal end opposite to the micropyle. They have no known function and degenerate after fertilisation.



Antipodal cell (antipodal) of angiosperm

ANTIPODAL POINTS (ANTIPODES): Two points located on a sphere that are at the opposite ends of a diameter.

ANTIPORTER: A membrane protein that effects the simultaneous or sequential transport of two different types of substrates across a cell membrane in opposite directions, one into the cell, for example Na^+ and one out of the cell, for example Ca^{2+} .

ANTIPRISM: A semi-regular polyhedron constructed from two n -sided polygons and $2n$ triangles. The octahedron is a triangular antiprism and the simplest antiprism.

ANTIPROTON (NEGATRON): The antiparticle of the proton, having all of its features such as mass, average lifetime, spin, magnitude of magnetic moment, but with opposite electrical charge sign. While protons have positive charges, antiprotons have negative ones. Like any other antiparticle, it behaves like matter and can associate with other antiparticles, as the positron, to form stable atoms.

ANTIPSYCHOTIC DRUG (NEUROLEPTIC DRUG): Any of the class of drugs that acts on the nervous system by simultaneously

modifying agitation and neuronal activity. These drugs are usually prescribed for mental problems such as schizophrenia or bipolar disease.

ANTIPIRETTICS (FEBRIFUGE): Substances or agents that reduce fever and body temperature. Examples are analgesic drugs such as ibuprofen and paracetamol. They also include herbs such as guava leaves, dandelion, neem, olive, peppermint, and raspberry leaf.

ANTIRESISTANT: A substance, which is used as an additive to a pesticide formulation in order to reduce the resistance of insects to the pesticide.

ANTISENSE DNA: A single-stranded DNA molecule that can bind to a complementary base sequence in a particular messenger RNA (mRNA) molecule to prevent synthesis of the protein encoded by the mRNA.

ANTISENSE RNA (asRNA): A single stranded RNA molecule that is complementary to the messenger RNA (mRNA) strand. It is transcribed within the cell that is introduced to inhibit translation or anticodon complementary base pairing.

ANTISEPTIC: Any substance that kills, stops or inhibits the growth of disease causing microbes or micro-organisms, without damage to human skin tissue. It is non-toxic to cells of the body. Antiseptics are used to treat small wounds. Common antiseptics include hydrogen peroxide, salt and iodine solutions and ethanol.

ANTISERUM: A substance containing large numbers of antibodies to a particular antigen or an animal or human blood serum containing one or more antibodies that can provide immunity against a disease or counteract venom.

ANTISTATIC: Inhibiting or preventing electrostatic discharge, such as on textiles, phonograph records or paper products. In computer technology, an **antistatic floor** mat or ground mat is an anti-static device that safeguards an individual or piece of equipment such as a PC from an electrostatic discharge. Computer components that are sensitive to static electricity are motherboards, CPUs, expansion cards and memory devices. Antistatic mats are used under keyboards or mice and may also be placed on or under other equipment. They are particularly useful when working on computer components.

ANTISYMMETRIC RELATION: A binary relation $*$ is said to be antisymmetric on a set Z if, for all x and y in Z , whenever $x * y$ and $y * x$, then $x = y$.

ANTISYMMETRY PRINCIPLE (PAULI EXCLUSION PRINCIPLE): The quantum mechanical principle that electrons must be described by wavefunctions which are antisymmetric with respect to interchange of the coordinates (including spin) of a pair of electrons. A corollary of the principle is the Pauli exclusion principle. All particles with half-integral spin (fermions) are described by antisymmetric wavefunctions and all particles with zero or integral spin (bosons) are described by symmetric wavefunctions.

ANTITHETIC (HETEROMORPHIC): Describing a life cycle in which the alternating generations are morphologically and physiologically distinct, as in the bryophytes and pteridophytes.

ANTITRANSPIRANT: A compound sprayed on the stems and leaves of plants that reduces transpiration. It is often used to protect evergreens from dry winter winds. It reduces photosynthesis and affects various other metabolic processes.

ANTIVIRAL: 1. A term for a drug or other agent that inactivates viruses and is used to combat viral infections. 2. Of or relating to an antiviral and also eliminating or inhibiting the effectiveness of viruses.

ANTIVIRUS: A software used to prevent, detect and remove malware, including but not limited to computer viruses, computer worms, Trojan horses, spyware and adware. Computer security, including protection from social engineering techniques, is commonly

offered in products and services of antivirus software companies. No matter how useful antivirus software can be, it can sometimes have drawbacks. Antivirus software can impair a computer's performance. Inexperienced users may also have trouble understanding the prompts and decisions that antivirus software presents them with. An incorrect decision may lead to a security breach. If the antivirus software employs heuristic detection, success depends on achieving the right balance between false positives and false negatives. False positives can be as destructive as false negatives. Antivirus software generally runs at the highly trusted kernel level of the operating system, creating a potential avenue of attack.

ANTLER: Any of branched, bony, deciduous horns found on the head of animals such as a male deer, moose or elk.

ANTLERITE: A green mineral consisting of basic copper sulfate with the chemical formula $\text{Cu}_3[\text{OH}]_4\text{SO}_4$ and Mohs scale hardness of 3 to 3.5. It is a minor copper ore of widespread occurrence and formed by oxidation of primary copper minerals.

ANTON VAN LEEUWENHOEK (1632-1723): A Dutch maker of microscopes, who made pioneering discoveries concerning protozoa, red blood cells, capillary systems and the life cycles of insects.

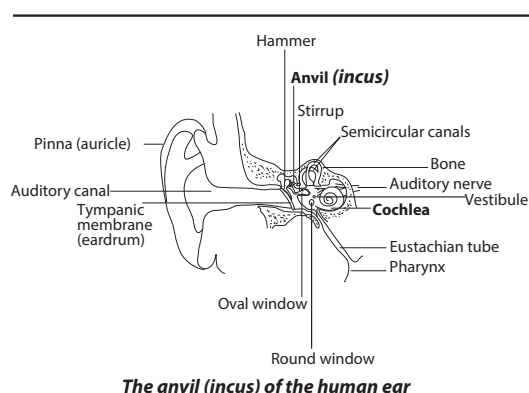
ANTORSE: In botany, it is a term used for an arrangement of hairs, in which the anther projects forwards, bent or is directed upwards.

ANUCLEATE: Any cell that does not contain a nucleus. Examples are matured erythrocytes and sieve tube element.

ANURA: An order of class Amphibia, which are tail-less as adults made up of frogs and toads. They live in a variety of environments from deserts, mountains to ponds and paddles. Frogs have smooth, moist skin, a broad body and long hind legs that make them excellent jumpers. Most frogs live in or near water bodies although some species live on trees. Examples are the red-eyed tree frog (*Agalychnis callidryas*) West African caecilian and (*Geotrypetes seraphini*). Unlike frogs, toads have a dry, warty skin, relatively short legs which are well adapted for dry environments. All adult forms are carnivorous, eating a wide variety of insects. Both frogs and toads return to water to reproduce, laying their eggs directly in water. The eggs are fertilised externally and hatch into swimming larval forms called tadpoles. Tadpoles live in water where they feed on minute algae. After considerable growth, the body of the tadpole abruptly changes into that of an adult. Tail, gills and lateral line system disappear and legs grow from the body. A sac-like bladder in the throat divides into sacs that become lungs. The pulmonary vein appears and the heart develops its internal wall and this process of change is known as metamorphosis.

ANUS: Posterior opening of the alimentary canal in most animals through which indigestible material or faeces pass out of the body.

ANVIL (INCUS): The middle of the three-ear ossicles, which is an anvil-shaped small bone in the mammalian middle ear that transmits sound vibrations from the malleus to the stapes.



ANYON: A type of particle that occurs only in two-dimensional systems, mathematically, which may play a role in the ceramics' superconductivity. It is a generalisation of the fermion and boson concept, obeying an unconventional form of quantum statistics, which is characterised by a parameter that can take on any of a continuum of values, just two of which represent Bose-Einstein and Fermi-Dirac statistics.

AOE (ATA OVER ETHERNET): An open standard protocol that allows direct network access to disk drives by client hosts such as web servers, mail servers or cluster servers.

AORTA: The main blood vessel in higher vertebrates through which oxygenated blood leaves the heart from the left ventricle. It branches to form many small arteries which in turn branch many times to supply oxygen to all living cells in the body except the lungs. It is the largest artery in the human body.

AORTIC ARCHES (ARTERIAL ARCHES): Six pairs of arteries in vertebrate embryos, which connect the cardiac aorticus with the paired dorsal aorta by going around the pharynx. They develop one after the other in the pharyngeal arches.

AORTIC BODY: The receptors and supporting cells located along the aortic arch.

AORTIC VALVE: A heart valve lying between the left ventricle and the aorta. It consists of three semilunar cusps and prevents blood returning to the ventricle from the aorta. It is the largest artery in the human body.

AORTOGRAPHY: An imaging procedure of the aorta using x-rays following the injection of a radiopaque substance. This enables the aorta and its branches to be seen on X-ray film.

APACHE (APACHE WEB SERVER): An open-source HTTP server for modern operating systems including UNIX, Microsoft Windows, Mac OS/X and Netware. Apache has been the most popular web server on the internet since April 1996.

APACHE INCUBATOR: Created in October 2002, it is the entry point for projects and software seeking to become part of the Apache Software Foundation's (ASF) efforts. The ASF is a non-profit organisation that oversees the development of Apache software. Membership is based on contribution to the foundation's open source projects.

APATITE: A common calcium phosphate mineral, with the chemical formula $\text{Ca}_5(\text{PO}_4)_3(\text{OH}, \text{F}, \text{Cl})$, which has a hexagonal structure and occurs widely in igneous rocks, such as pegmatite and in metamorphic rocks, such as marbles. It is a source of phosphorus for plants and used in the manufacture of fertilisers. The enamel of teeth is made up mainly of this mineral.

APE: Any of the four, usually tailless primates including gorilla, chimpanzee, gibbon and orang-utan.

APERIENTS: Agents or substances that induce bowel movement making an individual experience frequent bowel movements. They include herbs such as mahogany and guava.

APERIODIC TILING: A tiling made from the same basic elements or tiles that can cover an arbitrarily large surface without ever exactly repeating itself.

APERTURATE: A term used in botany to describe a pollen grain having one or more apertures or areas where the exine is either thinner or absent.

APERTURE: 1. The length of the refracting surface on which the incident rays can be refracted. 2. An adjustable opening for admitting light into a camera or the effective diameter of a lens or mirror. In photography, the aperture stop of a photographic lens can be adjusted to control the amount of light reaching the film or image sensor. In combination with variation of shutter speed, the aperture size will regulate the film's or image sensor's degree of exposure to light. A fast shutter speed will require a larger aperture to ensure

sufficient light exposure and a slow shutter speed will require a smaller aperture to avoid excessive exposure. In a telescope it is the effective diameter of an objective lens or primary mirror. The larger the aperture, the greater the ability to collect light and detect faint distant objects. A rough guide for estimating the magnification that an aperture can handle well is $2\times$ per mm. 3. An opening in certain kinds of mollusc shells, which is the main opening of the shell, where part of the body of the animal emerges for locomotion, feeding, etc. 4. In botany, aperture is very small spots, which appear on the exine or walls of pollen grains, where the wall is thinner and/or softer. The pollen tube is able to break through the very tough pollen wall for the pollen tube to transport the chromosomes to the egg deep down in the pistil. There are two different types of apertures: pores and colpi. If the apertures are colpi the pollen is termed **colpate** and if they are pores, the pollen is called **porate**. 5. A photo editing and management software program that was developed by Apple for the Mac OS X operating system, first released in 2005.

APERTURE SYNTHESIS: A method of combining signals from a collection of individual antennae or telescopes to provide an image with a resolution equivalent to that of a single telescope with a size roughly equal to the maximum distance between the individual antennae. In the simplest case, two antennas are used as a radio interferometer and the phase and amplitude of the combined radio signal are measured continuously. As the Earth rotates in the course of a day, one antenna is automatically carried around the other, effectively sweeping out a ring. On successive days the separation between the two antennas is changed, so that a large elliptical area is gradually covered. When the records are combined in a computer, it is possible to produce a radio map of the section of sky under observation with the detail resolved as if the telescope aperture were the size of the total area swept out. Astronomical interferometers are commonly used for high-resolution optical, infrared, submillimetre and radio astronomy observations.

APETALOUS: A term used to refer to a flower without petals.

APEX: 1. Generally, it is the top or highest part of something, especially one forming a point; the vertex. For example, the apex of a triangle or the apex of a hill. 2. In Astronomy, it is also known as **solar apex** and it is the point on the celestial sphere, lying in the constellation Hercules, towards which the Sun appears to move at a velocity of 20 kilometres per second relative to the nearest stars. 3. A narrowed or pointed end of an anatomical structure such as the lower pointed end of the heart situated in humans opposite the space between the cartilages of the fifth and sixth ribs on the left side. 4. The tip or the point furthest from the point of attachment such as the upper pointed end of a leaf.

APHASIA: Partial or total loss of ability to speak or understand spoken language. This is caused by damage to the brain.

APHELION: The point at which an object such as a planet or comet travelling in an elliptical orbit around the Sun is at its farthest point from the Sun. For example, the Earth is at aphelion on about the 3rd of July.

APHID (PLANT LOUSE): Any of various small, soft-bodied insects of the family *Aphididae* (greenfly, plant lice) of the insect order Hemiptera. Aphids are small plant bugs that have mouthparts specially adapted for sucking plant juices. Aphids are serious pests of crops all over the world as many species carry harmful viruses and fungi. Many species such as the blackfly (*Aphis fabae*) are pests on broad beans. Other species are chiefly important as vectors of plant virus diseases, such as cucumber mosaic virus, which is transmitted by several species of aphid including *Aphis gossypii*.

APHOTIC ZONE: A term describing those parts of the ocean or large deep lake that are not reached by sunlight. It is formally defined as the depths beyond which less than 1% of sunlight penetrates.

APHRODISIACS: Agents and substances that increases libido, or intensifies sexual desire as well as keep the body firm and tone up the

tissues. They act on the mind and stimulate the arousal of the mood and are used to help restore normal sexual potency and function. They include herbs such as mahogany, cloves ginger, ginseng, nutmeg oil, white lotus and wild asparagus.

APHYLLOPODIC: A plant having the lowermost leaves modified into small scales.

APHYLLOUS: Referring to or describing a plant without leaves.

APIACEAE (UMBELLIFERAE): A dicotyledonous aromatic family containing about 2500-3000 species in about 300 genera and commonly known as the carrot family. It occurs most commonly in temperate regions. Its members are distinguished by their characteristic inflorescence, the umbel. It has alternate leaves, widening at the base into a sheath that clasps the stem. Most umbellifers are herbaceous and have hollow internodes and often a characteristic odour. The flowers are usually compound and almost always concentrated in flat-topped umbels. The rays of the primary umbel give rise to a secondary umbel with the flower-bearing pedicels. The flowers have 5 petals, usually uneven and 5 stamens. The seeds and fruit form below where the petals and stamens originate. The fruit is dry and usually ridged and many such as caraway (*Carum carvi*) and coriander (*Coriandrum sativum*) are used as spices. Species used as herbs include parsley (*Petroselinum crispum*) and fennel (*Foeniculum vulgare*), while the carrot (*Daucus carota*) and parsnip (*Pastinaca sativa*) are important root crops.

APIARIST (BEEKEEPER): A person who keeps bees for such purposes as to produce honey, beeswax, pollen, royal jelly, pollinating fruits and vegetables and raising queens and bees for sale or for scientific purposes. A place with a number of hives or where bees are kept is called an **apiary**.

APICAL: Designating certain bonds in a bipyramidal or pyramidal structure. For example, in trigonal bipyramid the two bonds aligned through the central atom are the apical bonds and the groups attached to these bonds are in apical positions.

APICAL DOMINANCE: Inhibition of the growth of lateral buds in a plant by the presence of a growing apical bud. It is caused by the action of auxins and abscissic acid from the apical bud. This allows the plant to grow taller, increasing its exposure to light. Removal of the terminal bud releases some of the lateral buds from inhibition.

APICAL MERISTEM: The tissue of undifferentiated actively dividing cells found lying at the shoot tip or root tip of a plant in which cell divisions are continually occurring to produce new stem and root tissue, respectively and also causing increase in length. The new tissues produced are known collectively as the primary tissues of the plant.

APICAL PLACENTATION (PENDULOUS PLACENTATION; SUSPENDED PLACENTATION): A form of arrangement in flowering plants that causes one or few ovules to develop at the top of a simple or compound ovary.

APICOMPLEXA (SPOROZOA): A phylum of parasitic protists whose members may have a number of different animal hosts. They lack locomotory structures and have a complex life cycle that involves the alternation of asexual reproduction and sexual reproduction and the production of resistant spores. Examples are coccidian, gregarines, piroplasm, haemogregarines and malaras.

APICULUS (APICULATE): A structure or organ ending abruptly in a small, slender point as found in certain leaves.

APLANATIC LENS: A lens that reduces both spherical aberration and coma.

APLANETIC: Describing organisms in which there is no motile stage.

APLANOSPORE: A nonmotile asexual spore produced, for example, by fungi in the *Zygomycotina* and by certain algae in the *Chlorophyta* and *Chrysophyta*.

APNOEA (SLEEP APNOEA; APNEA): A disorder in which breathing stops for periods longer than 10 seconds during sleep. It can be

caused by failure of the automatic respiratory system to respond to elevated blood levels of carbon dioxide.

A

APO: In naming of organic compounds, it is an unitalicised prefix, preceded by a locant, used to indicate that the entire molecule beyond the carbon atom corresponding to that locant has been replaced by hydrogen atoms.

APOACTIVATOR: A regulatory protein that stimulates transcription from one or more genes in the presence of a coactivator molecule.

APOCARPOUS: A flower with carpels forming separate pistils, as found in buttercup.

APOCARPY: The condition in which the female reproductive organs (carpels) of a flower are made up of many free carpels not joined to each other. Examples exist in flamboyant and buttercup, which become fertilised to produce individual true, simple fruits.

APOCHROMATIC LENS: A lens or optical system, which is virtually free of chromatic aberration. It uses special optical designs and optical materials (SLD or ELD glass) to correct chromatic and spherical aberrations so that light of at least three different wavelengths is brought to focus at the same point, resulting in improved images which have greater contrast, sharpness and colour definition than a comparable non-APO type lens.

APOCRINE GLANDS: Sweat glands that are located primarily in the armpits and groin area. They are larger than the more widely distributed endocrine glands.

APOCYNTHION: The point in the orbit around the moon or a satellite launched from the Earth that is farthest from the Moon.

APODA: An order belonging to the class Amphibia, having a highly specialised group of tropical burrowing amphibians. They are legless worm-like creatures. They have very small eyes and are often blind. They eat worms and other soil invertebrates. The male of these vertebrates deposits sperm directly into the female, which bears live young ones. An example is *Caecilia pachynema*.

APODEME: A ridge-like ingrowth of the exoskeleton of an arthropod that supports the internal organs and provides attachment points for the muscles that enable movement of the limbs.

APOENZYME: The protein component of an enzyme that combines with the coenzyme to form the active enzyme. Examples of enzymes, which need both apoenzyme and cofactor for catalytic activity include alcohol dehydrogenase, which requires zinc ions; and kinases, which require magnesium or manganese ions.

APOGAMY: A phenomenon seen in certain ferns, fungi and algae in which the sporophyte develops directly from the gametophyte, without fertilisation. Apogamy often occurs when the gametophyte has been produced by apospory. The resulting sporophyte therefore has the same chromosome number as the gametophyte. If male gametes are produced they are redundant although they have been shown to be capable of functioning in certain species. It may also be induced by ageing or chemical agents. The term may also refer to the development of an unreduced diploid cell of the embryo sac into an embryo without fertilisation occurring or parthenogenesis.

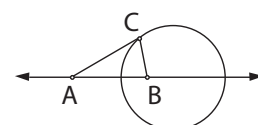
APOGEE: The point in the orbit of the Moon or an artificial earth satellite, at which it is furthest from the Earth. For example, at apogee, the Moon is 400,700 km from the Earth. The opposite of apogee is **perigee** or the point nearest to the Earth.

APOLLO ASTEROID: Member of a group of asteroids whose orbits cross that of the Earth. They are named after the first of their kind, Apollo, discovered in 1932 and then lost until 1973. Apollo asteroids are so small and faint that they are difficult to see except when close to Earth.

APOLLO-SOYUZ TEST PROJECT: Joint US-Soviet space mission in which an Apollo and a Soyuz craft docked while in orbit around the Earth on 17th July 1975. The crafts remained attached for two days and crew members were able to move from one craft to another through an airlock attached to the nose of the Apollo. It was the last Apollo mission, the first joint U.S./Soviet space flight and the last manned US space mission until the first Space Shuttle flight in April 1981.

APOLLONIUS OF PERGA (ABOUT 262 – 190 BC): Greek mathematician, who worked on conics. His work has persisted to modern times, which are now referred to as conics sections: the ellipse, hyperbola and parabola. He suggested the idea of epicycle motion of the planets.

APOLLONIUS' CIRCLE: A circle, named after the Greek mathematician, Apollonius of Perga (261 BC-190 BC) defined as the locus of a point *C* whose distance from a fixed point *A* is a multiple *r* of its distance from another fixed point *B*. If *r* is 1, then the locus is a line, the perpendicular bisector of the segment *AB*. If *r* is not equal to 1, then the locus is a circle.



APOLLONIUS' THEOREM: The equation that relates the length of a median in a triangle to the lengths of its sides. If *a* is the length of one side and *b* is the length of another, the third side is divided into two equal lengths *c* by a median of length *m*, then $a^2 + b^2 = 2m^2 + 2c^2$.

APOLUNE: The point in the path of a body such as a satellite orbiting the Moon, where the distance between the satellite and the Moon is farthest from the centre of the Moon.

APOMIXIS (AGAMOSPERMY): An asexual reproductive process in plants that superficially resembles normal sexual reproduction but in which there is no fusion of gametes.

APOMORPHY (DERIVED TRAIT): A specialised characteristic that is unique to a particular species and all its descendants and which can be used as a defining character for a species or group in phylogenetic terms. The possession of feathers in birds is an example.

APOPETALOUS (ACHOPETALOUS; POLYPETALOUS): Having a corolla composed of separate petals not united to each other.

APOPHYSIS: 1. In botany, it is a swollen region or enlargement at the base of the stalk or seta in certain mosses or on the cone scale of certain conifers. Its cells are rich in chloroplasts, there are numerous intercellular spaces and the epidermis contains stomata, similar in structure to those of vascular plants. The apophysis is thus an active photosynthetic region and helps to nourish the developing sporogonium. 2. In anatomy, it refers to any bony small projection, process or outgrowth, such as a tuberosity without an independent centre of ossification. It can also be defined as a natural swelling, projection or outgrowth of an organ or part, such as the process of a vertebra. 3. A windows application made in delphi for creating, editing and rendering fractal flames. Fractal flame is an extension on the IFS fractal.

APOPLAST: An interconnected system of cell walls in plants which allows the passage of water from one cell to another. The apoplast pathway is one route by which water taken up by the plant travels across the root cortex to the endodermis.

APOPTOSIS (PROGRAMMED CELL DEATH): The process of cell death that occurs naturally as part of the normal development, maintenance and renewal of tissues within an organism, characterised by DNA fragmentation, cytoplasmic shrinkage, nuclear condensation, membrane blebbing and cell death without lysis or damage to neighbouring cells. It is different from cell necrosis in which the cell death may be caused by toxic substance.

APOREPRESSOR: A regulatory protein, that combines with the corepressor (a substance that inhibits the expression of genes) to inhibit transcription of certain genes.

APOSEMATIC COLOURATION (APOSEMATISM; WARNING COLOURATION): The warning colouration markings that make a dangerous, poisonous or foul tasting animal particularly conspicuous and recognisable to a predator. Aposematic signals are beneficial for both the predator and prey, who both avoid potential harm.

APOSEPALOUS: In botany, it refers to a plant that has a calyx with separate or unlinked sepals; not connate.

AOSPORY: The development of the gametophyte from the sporophyte without meiosis and spore production, so that the gametophyte has the same number of chromosomes as the sporophyte. If fertile gametes are formed by such a gametophyte then a sporophyte with twice the original number of chromosomes is produced. Apospory can be induced artificially in certain ferns, such as the lady fern (*Athyrium filix-femina*), by pinning segments of frond to damp sand. Prothalli then arise from buds on the frond.

APOSTEMONOUS: A flower with separate stamens; or having stamens separate from each other.

APOTHECIUM: The disc- or cup-shaped ascocarp characteristic of discomycete fungi (except the Tuberales). The tips of the asci are freely exposed. Most lichens contain discomycete fungi and such lichen fungi also form apothecia.

APOTHEM: A line segment from the centre of a regular polygon perpendicular to the centre of a side.

APPARATUS: 1. Equipment that can be used, especially in scientific experiments. Examples are containers such as beakers, flasks, plastic bags and measuring instruments such as rulers and burettes. 2. A system of bodily organs. An example is regulatory apparatus.

APPARENT BRIGHTNESS: The energy received from a star per unit time per unit area of the Earth's surface.

APPARENT DEPTH: Depth that a transparent material such as water or glass appears to have when viewed from above. This is less than its real depth because of the refraction of light that takes place when light passes from the denser medium (i.e. water/glass) into a less dense medium.

APPARENT DIAMETER (ANGULAR DIAMETER: The angle that the actual diameter of an object makes in the sky. The Moon, with an actual diameter of 3,476 kilometres, has an angular diameter of 29' 21" to 33' 30", depending on its distance from Earth.

APPARENT EXPANSIVITY: The expansion of a liquid in relation to temperature, which can be observed directly. It is measured in a graduated container without taking into account the container's expansion.

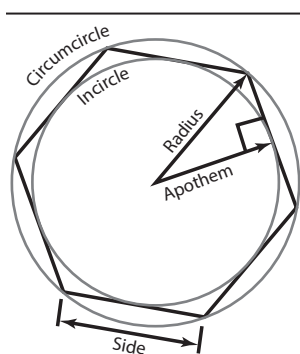
APPARENT HORIZON: 1. The boundary of a region in space-time in which the gravitational field is so strong that the cross-sectional area of an outgoing light pulse decreases. 2. Also known as **visible horizon**, it refers to the visible line of demarcation between land or the sea and the sky as viewed from any given point.

APPARENT MAGNITUDE: A numerical measure of the brightness of an astronomical object such as a star as seen by the ordinary eye from Earth, on a scale in which a lower number represents greater brightness. A decrease of one magnitude represents an increase in apparent brightness by a factor of 2.512.

APPARENT SPECIFIC GRAVITY: The ratio of the mass of a volume of the impermeable portion of a material at a stated temperature to the mass of an equal volume of distilled water at a stated temperature.

APPARENT VISCOSITY: The ratio of shear stress to shear rate in a fluid, calculated from measurements of forces and velocities as though the liquid were Newtonian (homogenous). It remains the same with changing shear strain rate at constant temperature and pressure. If the liquid is actually non-Newtonian, the apparent viscosity depends on the type and dimensions of the apparatus used.

APPARENT WEIGHT: Directly observed or measured weight that does not take into account external factors other than gravity, for



An apothem formed in an inner circle

example, the weight of an object in an accelerating elevator, distortion caused by the measuring instruments themselves or levitation (process by which an object is suspended by a physical force against gravity).

APPEARANCE ENERGY (APPEARANCE POTENTIAL): The ionisation of a molecule or atom by electron collision or photon absorption. In mass spectrometry it has often been reported as the voltage which corresponds to the minimum energy of the electrons in the ionising beam necessary for the production of a given fragment ion. In photoionisation it is the minimum energy of the quantum of light which produces ionisation of the absorbing molecule. According to IUPAC, the synonym, **appearance potential** is obsolete.

APPEARANCE TEMPERATURE: The temperature of the atomisation surface at which the analyte signal/noise (S/N) ratio reaches a value of 3 when the quantity of analyte in the atomiser is one hundred times the characteristic mass for peak high absorption.

APPEASEMENT: Presentation of special postures or displays by organisms to inhibit aggressive behaviour by the opponent of usually the same species. Threatening structures and markings are covered or turned away and vulnerable parts of the body may be exposed.

APPENDAGE: The attachment of a secondary part to the main structure.

APPENDICITIS: An inflammation of the appendix. The appendix is a small pouch attached to the beginning of the large intestine. Appendicitis may be caused by a blockage (most often by faecal material) of the inside of the appendix (the lumen). The blockage leads to increased pressure, impaired blood flow and inflammation. If left untreated, it can rupture, causing infection and even death. Appendicitis can also be caused by foreign bodies such as intrauterine device (birth control device inserted into the uterus to prevent implantation), traumatic abdominal injury or tumours.

APPENDICULAR: A term describing body parts that are associated with the limbs or the appendix.

APPENDICULAR SKELETON: The collective components of the vertebrate skeleton that are attached to the main supporting or axial skeleton. It is made up of paired appendages such as arms and legs together with the pelvic and pectoral girdles.

APPENDIX (VERMIFORM APPENDIX): An outgrowth of the caecum or a small tube-shaped bag of tissue attached to the intestine. It is associated with the digestion of cellulose and is very large in herbivores, but in humans, it is a vestigial organ, serving no digestive function.

APPLANATE: A flattened structure as found in the fruits of legumes.

APPLE : 1. *Malus domestica* is a deciduous tree of about 4.6 metres tall in the genus *Malus* of rose (Rosaceae) family and cultivated worldwide as a fruit tree. There are more than 7,500 known cultivars of apples. The leaves are dark green, simple oval with serrated margins and slightly downy underside. It produces white flowers with a pink tinge that gradually fades. The plant bears inflorescence consisting of a cyme with 4-6 flowers. The fruits mature in late summer or autumn with varieties of sizes. The skin of ripe apples is red, yellow, green, pink with others having bi- or tri-coloured varieties. It can be grown from seed or vegetatively propagated by grafting. The trees are large if grown from seed, but small if grafted onto roots. The fruits can be eaten raw, used in cooking or processed into non-alcoholic fresh drinks or hard cider (an alcoholic drink). 2. **Apple Inc.** is a well known US multinational technology company, founded by two friends, Steven Jobs and Stephen Wozniak and introduced in 1984 that designs, develops, and sells consumer electronics, computer software, and online services. Its headquarters is located in Cupertino, California, United States. The company's hardware products include the iPhone smartphone, the iPad tablet computer, the Mac personal computer, the iPod portable media player, the Apple Watch smartwatch, etc. Its software includes the macOS and iOS operating systems, the iTunes media player, the Safari web browser, etc.

A

APPLESCRIPT: A scripting language developed by Apple Inc. for the Macintosh operating system. It is commonly used for programming using common gateway interfaces with Macintosh-based Web servers.

APPLET: A small application program written in the Java programming language that is designed to run within another application. For example, a Sun Java applet is a file on an Internet web page that provides additional functionality not supported by HTML. It can be retrieved from a Web server and executed by a Web browser. A reference to an applet appears in the markup for a Web page, in the same way that a reference to a graphics file appears; a browser retrieves an applet in the same way that it retrieves a graphics file. For security reasons, applet access rights are limited in two ways: the applet cannot access the file system of the client upon which it is executing and the applet communication across the network is limited to the server from which it was downloaded.

APPLETON LAYER: A band containing ionised gases in the Earth's upper atmosphere, above the E-layer (formerly the Kennelly-Heaviside layer), which is 180–350 km. It can act as a reflector of radio signals from about 3 MHz to above 30 MHz. Its ionic composition varies with the sunspot cycle.

APPLIANCE: Instruments or devices fitted together to make a piece of equipment that performs a particular task. For example, a can opener is a mechanical appliance, a radio is an electronic appliance and a refrigerator is an electrical appliance.

APPLIANCE COMPUTING: A computing architecture that provides the end-user's work station (appliance) with software resources over the Internet. The software applications reside on a Web server rather than on the end-user's work station. The workstation (appliance) is a thin client which does not have a CD-ROM or floppy drive.

APPLICATION: 1. In logic, it is the process of determining the value of a function for a given argument. 2. The primitive binary function $(x, y) = x(y)$, which is equivalent to Lambda-conversion. 3. Also known as **application program** or **application software**, it is a term used in computing to denote the set of programs that is written and designed for a specific need or purpose and related documentation or a software program that runs on a computer. It meets the need of the various users of the computer system and contributes directly to the performance and execution of a specific task. Examples are Web browsers, e-mail programs, word processors, games, pay roll system and utilities.

APPLICATION BLACKLISTING: A network administration security practice used to block or prevent the execution of undesirable programs. Such applications can be employed to deny access to specific usernames, IP addresses, or specific items on the list.

APPLICATION CLUSTERING (SOFTWARE CLUSTERING): A method of connecting multiple computer servers to work together by maintaining the same information and collectively, they perform the same administrative tasks such as load balancing, determining node failures and assigning fail over duty.

APPLICATION COMPATIBILITY TOOLKIT (ACT): A set of freely downloadable program utilities and related documents from Microsoft for ensuring compatibility among application programs in Windows operating systems, especially in a large network environment. The Toolkit can also be used to diagnose and fix problems that may be related to compatibility. The Toolkit can be used for applications running in Windows 2000 with Service Pack 3 or later, Windows XP and Windows Server 2003. Microsoft says that the tools can be used to resolve over 200 symptoms of problems.

APPLICATION DESKTOP TOOLBAR: An element of the Microsoft Windows interface that can be provided by a Windows application. This toolbar operates similarly to the Windows 95 taskbar and provides

access to controls for specific functions related to the respective application.

APPLICATION FIREWALL: An enhanced firewall which offers additional protection by controlling the execution of files or handling of data by specific application.

APPLICATION LAYER: A layer which provides services directly to user applications to ensure that effective communication with another application program in a network is possible.

APPLICATION PROGRAM INTERFACE (API): It is a set of commands, functions, protocols, and objects that programmers can use to create software or interact with an external system.

APPLICATION SECURITY: The use of software, hardware and procedural methods to protect applications from external threats.

APPLICATION SERVER: A server program in a computer (in a distributed network) consisting of a server operating system (OS) and server hardware that works together to provide computing-intensive operations and services to the residing application.

APPLICATION SERVICE PROVIDER (ASP): A company or organisation that provides software applications to customers over the Internet. These Internet-based applications are also known as **software as a service (SaaS)** and are often made available on a subscription basis. This means ASP clients often pay a monthly fee to use the software, rather than purchasing a traditional software license. Some SaaS applications can be accessed via a web browser, while others operate over a proprietary secure port.

APPLICATION SPECIFIC INTEGRATED CIRCUIT (ASIC): An integrated circuit built for a specific application, such as a particular kind of transmission protocol or a hand-held computer.

APPLICATION WHITELISTING: A computer administration practice used to allow only pre-approved and specified programs to run in a computer system. This is to protect the computers and network from harmful applications.

APPLICATIONS PACKAGE: A term used in computing to denote the set of programs and related documentation used in a particular application.

APPLIED MATHEMATICS: A branch of mathematics concerned with mathematical methods that are typically used in fields outside of mathematics such as in science, engineering, business and industry. It also focuses on the formulation and study of mathematical models.

APPLIED POTENTIAL: The difference of potential measured between identical metallic leads to two electrodes of a cell. The applied potential is divided into two electrode potentials: ohmic potential drop through the solution and another ohmic potential drop through each electrode. Each of them is the difference of potential existing between the bulk of the solution and the interior of the conducting material of the electrode. Applied potential is denoted by the term voltage, which is not recommended by IUPAC.

APPLY: To use prior understanding, conceptual knowledge, or procedural skill to solve problems or to make sense of novel situations.

POSITION: The layers of cellulose to the inner surface of a plant cell wall, at its junction with the plasma membrane. It helps in thickening and strengthening the plant cell wall.

APPRESSED: Pressed close or flat against another organ but not united; or **in botany**, pressed close to or lying against, something for its whole length, as against a stem or as hairs on certain plant stems.

APPRESSORIUM: The swollen tip of a fungal germ-tube that attaches itself to the host before penetration takes place.

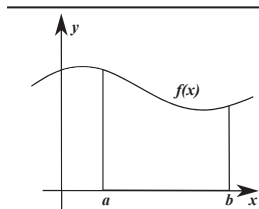
APPROPRIATE TECHNOLOGY: Simple or small scale machinery and tools that because they are cheap and easy to produce and maintain, may be of utmost use in the developing world.

APPROXIMATE: 1. To come or bring near or close; be almost the same. 2. In mathematics, almost correct or exact, close in value or amount but not precise; or to find an expression for (some quantity)

accurate to a specified degree; or an estimation of a number or rounding a number to its nearest place value. **Approximately equal to** with the symbol $\approx$ is an estimate or approximation to an exact value. For example, $\pi \approx 3.14$. 3. In botany, it refers to organs or structures borne close together, but not fused, for example, approximate leaves.

APPROXIMATE INTEGRATION:

Any of the various techniques for finding an approximate value for definite integral. There are many integrals that cannot be evaluated exactly and approximation techniques are used to estimate values for such integrals. Both analytical and numerical approximate integration techniques exist. Two examples of numerical approximate integration are Simpson's Rule and the Trapezium Rule.



Area under a curve $f(x)$ with limits $x=a$ and $x=b$.

Approximate Integration

APPROXIMATION: A rough value, one whose accuracy is not exact.

In mathematics and physics, an approximation result that is not necessarily exact, but is within limits of accuracy required for a given purpose.

APPROXIMATION TECHNIQUE: A mode of solving a mathematical problem or its physical application that does not give an exact solution, but that enables an approximate solution to be found.

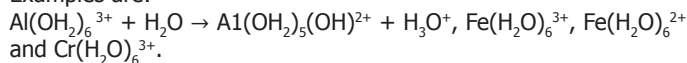
APPROXIMATION THEORY: The branch of numerical analysis which is relative to finding a function $f(x)$, which has approximately the same value as $F(x)$ over a given interval.

APROTIC (SOLVENT): A term used to describe a substance, especially a solvent that is non-protogenic or not capable of acting as a proton donor in a given situation. In other words, it is a polar solvent having a moderately high dielectric constant that does not contain replaceable hydrogen. With extremely strong Brønsted acids or bases, solvents that are normally aprotic may accept or lose a proton. For example, acetonitrile is in most instances an aprotic solvent, but it is protophilic in the presence of concentrated sulfuric acid and protogenic (capable of acting as a proton donor) in the presence of potassium *tert*-butoxide. Similar considerations apply to benzene, trichloromethane, etc.

APSE: 1. Also known as **apsidal point**, it is any point at which the direction of motion of a point moving round a closed curve is perpendicular to its radius vector. Every point of a circle is thus an apse and the apsidal points of an ellipse are the endpoints of its axes. 2. Also known as **apsides**, it is a point on an orbit at which the motion of the orbiting object is at right angle to the central radius vector of the orbit. The apsidal distance (distance from an apse to the centre of the motion equals the radius vector at its maximum or minimum value.

APTERYGOTA: A subclass of small primitive wingless insects in which metamorphosis is little or absent. Examples are the orders Thysanura and Diplura.

AQUA ACID (HYDRATED CATION): A type of acid in which the acidic hydrogen is on a water molecule coordinated to a metal ion. Examples are:



AQUA REGIA: A mixture of three parts of concentrated hydrochloric acid and one part of concentrated nitric acid by volume, which dissolves all metals except silver. Its ratio by volume is 3HCl:1HNO₃. It is used in the analysis of minerals and alloys and as a powerful cleaning agent.

AQUACULTURE: The cultivation of fish, shellfish or aquatic plants such as seaweed in natural or controlled marine or freshwater environments. An **aquaculturist** is a professional, who practices aquaculture.

AQUAGEL: 1. A hydrogel in which the network component is a colloidal network. 2. A clear, water-soluble, non-irritating lubricant, bacteriostatic and alcohol-free all-purpose lubricating gel used for medical procedures.

AQUALING (SCUBA): Underwater breathing apparatus worn by divers.

AQUAMARINE: 1. A transparent blue-green variety of the mineral beryl, which is a semiprecious gemstone. 2. Light blue-green or greenish blue.

AQUAPLANING: A phenomenon in which a vehicle such as a high-speed aircraft lands on a water-covered runway and the tyres cease to make direct contact with the runway surface, owing to the presence of a thin film of water; rendering braking relatively ineffective and causing the aircraft to skid.

AQUAPORIN: A protein, forming an integral part of the plasma membrane of red blood cells and proximal convoluted tubules of the kidney that is responsible for the high permeability of these cells to water. It accelerates the process of osmosis.

AQUARIUM: An artificial pond or glass tank, where live fish and other water creatures and plants are kept alive for study and exhibition. The plural is **Aquaria**.

AQUARIUS: A zodiacal constellation, which is a little south of the celestial equator near Pegasus.

AQUATIC: The term that describes organisms such as plants and animals that live in water or near water. It also refers to anything relating to water such as aquatic environment. A plant that is adapted to living in aquatic environments is called a **hydrophytic plant** or a **hydrophyte**. Many hydrophytes absorb water and gases over the whole surface and have no stomata, for example, the spiked water milfoil (*Myriophyllum spicatum*), which is completely submerged in water.

AQUATION: The incorporation of one or more integral molecules of water into another species with or without displacement of one or more other atoms or groups. For example, the incorporation of water into the inner ligand sphere of an inorganic complex is an aquation reaction.

AQUEDUCT: Any artificial channel or conduit for water. It is an elevated structure of stone, wood or iron built for conduction of water across a valley.

AQUEOUS: 1. Also known as **aqueous solution**, it is a term used to refer to a solution in water or a solution where the solvent is water. It is often abbreviated to aq in a formula, for example, NaCl_(aq). There are three main types of reactions that occur in aqueous solutions: ion-exchange reactions, redox reactions and precipitation reaction (sometimes also known as ion-exchange reactions). Ion-exchange reactions occur between two substances, each consisting of positively and negatively charged ions that involve an exchange of one or more ionic components. The reaction can be represented by the following equation: AB(aq) + CD (aq) → AD + CB; either AD or CB may be a solid or a gas. In redox reactions, electrons are exchanged, with an ion losing electrons and becoming more positive, while the other ion gains electrons and becomes more negative, causing an increase or decrease in oxidation numbers. In a precipitation reaction, a precipitate is formed when ions in solution react with each other to form an insoluble product. 2. In geology, it means formed from matter deposited by water, as in certain sedimentary rocks.

AQUEOUS AMMONIA (AMMONIA SOLUTION): The International Union of Pure and Applied Chemistry (IUPAC) name for ammonium hydroxide, chemical formula NH_{3(aq)}.

AQUEOUS HUMOUR: The clear fluid filling the space in the front of the eyeball between the lens and the cornea. It does not have its own blood supply, as it is not served by any blood vessels. Its functions include exerting enough pressure to maintain the round shape of the eye; providing nutrition such as protein, glucose, lactic acid and ascorbic acid for the cornea and lens; removing metabolic waste and carbon dioxide away from the cornea and lens; and to refract light.

AQUIFER: A body of underground rock or sediment containing sandstone, gravel, sand, or silt fractured limestone that contains water

and releases it in appreciable amounts that can be used as a water source. Aquifers must be both permeable and porous and the pores linked so that they can absorb water. Aquifers may be shallow or deep.

AQUO COMPLEX ION: A hydrated positive ion present in a crystal or in solution, for example, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$.

ARABAN (ARABINAN): A polysaccharide in which the major monosaccharide subunit is the pentose sugar arabinose. Arabans are found with pectic substances in mature primary cell walls and in hemicelluloses and gums.

ARABIC NUMERALS (FIGURES): Any of the characters 0 to 9 and so on. The term is used when distinguishing between values represented by those characters and the same values represented by another character set.

ARABIDOPSIS: A genus of flowering plants from the family *cruciferae*, which is widely used as a research tool in molecular genetics and developmental biology because it has a small and simple genome, over half of which codes for protein and can be easily cultured and have a life cycle of only 6-8 weeks.

ARABINOSE (PECTINOSE): A crystalline pentose sugar, with chemical formula $\text{C}_5\text{H}_{10}\text{O}_5$, density 1.585 g/cm^3 and melting point $164\text{-}165^\circ\text{C}$ ($437\text{-}438 \text{ K}$). It is derived from plant polysaccharides such as gums and used as a biological culture medium.

ARABIS MOSAIC: A viral disease that infects a wide range of horticultural crops, making the leaves stunted and the plants die.

ARABLE: A term describing a land suitable for the cultivation of crops. **Arable farming** is a type of agriculture in which the farmer ploughs the land, sow seeds and grow plants.

ARACHIDONIC ACID: A type of polyunsaturated fatty acid which plays an important role in the growth of animals. It acts as a precursor to several biologically active compounds including prostaglandins and plays an important role in membrane production and fat metabolism.

ARACHNID: A type of arthropod belonging to the class Arachnida. Their bodies are divided into two distinct regions, the cephalothorax and the abdomen, with four pairs of legs attached to the cephalothorax. Arachnids lack wings and antennae. They are joint-legged invertebrate animals in the subphylum Chelicerata. They comprise over 100,000 named species, including spiders, scorpions, ticks and mites. Even though they are found throughout the world, from ponds to rainforests to desert to attics, they prefer warmer climates. Most species are considered to be pests or are thought to be dangerous to humans because some spiders can kill humans (even though these occurrences are rare). Scorpions are also poisonous and ticks can spread deadly illnesses, such as Lyme disease.

ARACHNOID: 1. In anatomy, of or relating to a delicate membrane enclosing the spinal cord and brain or a delicate membrane that encloses the spinal cord and brain and lies between the pia mater and dura mater. 2. Of, relating to or resembling arachnids. 3. Covered with or consisting of thin, soft, entangled hairs or fibres resembling those of a cobweb or spider's web.

ARACHNOID MEMBRANE: A thin delicate membrane lying between the dura matter and pia matter which surrounds the brain and spinal cord.

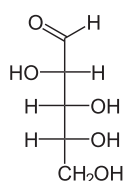


Fig I: Structure of d-arabinose

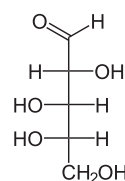
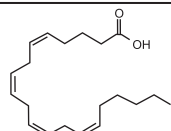


Fig II: Structure of l-arabinose



Chemical Structure of Arachidonic Acid

ARAGO'S SPOT (FRESNEL BRIGHT SPOT; POISSON SPOT):

A bright spot at the centre of the shadow of a circular opaque obstacle occurring as a consequence of the wave theory of light.

ARAGONITE: A form of anhydrous calcium carbonate, which are found in rocks. They occur as deposits in limestones, pearls and shells of molluscs. They are less stable than calcite, a more common form of calcium carbonate hence they recrystallise to form calcite over a period of time.

ARBITRARY CONSTANT: A non-numerical symbol which is not a variable in a generalised operation. For example, $y = ax^2 + bx + c$ is a general quadratic equation, where a , b and c are arbitrary constants. Also, $\int 3x^2 dx = x^3 + c$, where c is an arbitrary constant whose value can be determined by a boundary condition.

ARBOREAL (ARBORESCENT; ARBOREOUS): 1. Living in trees such as some species of animals. 2. Relating to, resembling or consisting of trees. 3. Having many trees; wooded.

ARBOREAL LOCOMOTION (ARBOREAL QUADRUPEDALISM):

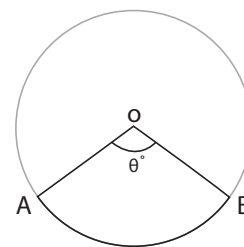
Travelling or movement of mammals on the branches, twigs and trunks of trees and woody shrubs. Most primates, many rodents, marsupials, carnivores, move about on arboreal substrates to forage, escape predators and acquire shelter. Arboreal supports are usually far enough from the ground that a slip or fall could cause serious injury or death or deprive the animal of a mate, food or energy. Arboreal animals frequently have elongated limbs that help them cross gaps, reach fruit or other resources and test the firmness of support ahead. Larger species may be restricted to larger-diameter branches that can support their weight, while smaller species may avoid competition by moving in the narrower branches.

ARBORETUM: An area, where a wide selection of all the woody plants such as trees, shrubs and vines may be grown in a particular climatic region for scientific, educational and recreational purposes.

ARBOVIRUS: One of a large heterogeneous group of RNA containing viruses that are transmitted from animals to man through the bite of mosquitoes and ticks.

ARBUSCULE: A branched fungal hyphae formed in the root of vascular plants by arbuscular mycorrhizal fungi that can penetrate plant cells. **Arbuscular mycorrhiza** is a symbiotic association formed between a fungus and the roots of a vascular plant.

ARC: 1. Also known as **arc of a circle**, it is a section of the circumference of a circle or a part of a curve that cuts at two specific points. For example, the endpoints of the diameter of a circle define an arc called a semicircle. An arc is named by its endpoints, for example, in the diagram shown, the arc is AB. If an arc of a circle of radius r subtends an angle θ° at the centre of the circle as in the diagram, then



Arc (AB) of a circle

the length of the arc AB is given by

$$\frac{2\pi r \theta^\circ}{360^\circ}. \text{ If } \theta \text{ is in radians, then the}$$

length of the arc is $r\theta$. The length of the arc of a curve from

$x = a$ to $x = b$ is given by $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$. 2. A stable electrical

discharge of high current density and low burning voltage between two or more electrodes.

ARC LAMP (ARC LIGHT): Electric light that uses the illumination of an electric arc maintained between two electrodes.

ARC LENGTH: The measure of the distance along the curved line making up an arc. It is longer than the chord. If an arc of a circle of radius r subtends an angle θ at the centre of a circle, then the length of the arc AB is given by $\frac{2\pi r \theta^\circ}{360^\circ}$. If θ is in radians, the length of an arc

is $r\theta$. The length of the arc of a curve from $r = a$ and $r = b$ is given

$$\text{by } \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

ARC-: A prefix for the inverse function of a given trigonometric or hyperbolic function. An example is arcsin, which can also be written as asin or $\sin^{-1}$.

ARC-COS (ARCCOS): The inverse of the cosine function.

ARC-COSEC (ARCCOSEC): The inverse of the cosecant function.

ARC-COSECANT (ARCCOSECANT; ARCCOSEC): An inverse of a cosecant function, usually denoted by cosec^{-1} or csc^{-1} and abbreviated as **acsc**. Its value for any argument is an angle in radians whose cosecant equals the given argument; that is, $y = \text{cosec}^{-1}x$ if and only if $x = \text{cosec}y$. It is defined for arguments less than -1 or greater than 1 and its principal values, often written as $\text{cosec}^{-1}y$, are by convention taken to lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ on a graph.

ARC-COSECH (ARCCOSECH): The inverse hyperbolic cosecant function, usually denoted by cosech^{-1} , csch^{-1} and abbreviated as **acsch** and defined for all non-zero arguments. For any argument the hyperbolic cosecant of its value equals the given argument; that is, $y = \text{cosech}^{-1}x$ if and only if $x = \text{cosech}y$. The derivative of

$\text{cosech}^{-1}x$ is $\frac{-1}{|x|\sqrt{1+x^2}}$, and an antiderivative or indefinite integral is

$$x\text{cosech}^{-1}x + \frac{x}{|x|} \sinh^{-1}x + c.$$

ARC-COSH (ARCCOSH): The inverse of the hyperbolic cosine function, so that $y = \cosh^{-1}x$ if and only if $x = \cosh y$. Its symbol is $\cosh^{-1}$, ch^{-1} or **acosh**. It is defined for arguments greater than or equal to 1, when $\cosh^{-1}x = \ln[x + \sqrt{(x^2 - 1)}]$; and its principal values, often written $\text{Cosh}^{-1}y$ are by convention taken to be the non-negative ones on a graph. The derivative of $\cosh^{-1}x$ is given by $\frac{1}{\sqrt{x^2 - 1}}$, and an

antiderivative or indefinite integral is $x\cosh^{-1}x - \sqrt{x^2 - 1} + c$.

ARC-COSINE (ARCCOSINE; ARCCOS): The inverse of the trigonometrical function cosine, with the symbol $\cos^{-1}$ and abbreviated as **arccos** or **acos**. It is the angle measured in radians that has a cosine equal to a given number, that is, $y = \cos^{-1}x$ if and only if $x = \cos y$. It is defined for arguments between -1 and 1 and its principal values, usually written as $\text{Cos}^{-1}y$, are by convention taken to be those between 0 and π on a graph. The derivative of $\cos^{-1}x$ is given by $\frac{-1}{\sqrt{1-x^2}}$, and

the antiderivative or indefinite integral is $x\cos^{-1}(x) - \sqrt{1-x^2} + c$.

ARC-COTANGENT (ARCCOTANGENT; ARCCOT): An inverse cotangent function, which is the radian measure of the arc on a circle corresponding to a given value of cotangent. Its symbol is $\cot^{-1}$ or cotan^{-1} and is abbreviated as **arccot**. It is defined for all real arguments and its principal values, usually written as $\text{Cotan}^{-1}y$ and $\text{Cot}^{-1}y$ and are by convention taken to be those strictly between 0 and π . The derivative of $\cotan^{-1}x$ is $\frac{-1}{\sqrt{1+x^2}}$, and an antiderivative or indefinite integral is

$$x\cotan^{-1}x + \frac{1}{2} \ln(1+x^2) + c.$$

ARC-COTANH (ARCCOTANH; ARC-COTH; ARCCOTH): The inverse hyperbolic cotangent function. It is defined for arguments less than -1 or greater than 1. An inverse hyperbolic cotangent of x can be denoted by $\text{arcoth } x$ or $\text{arch } x$ or $\text{coth}^{-1}x$: $\text{Coth}^{-1}x = \tan^{-1}\left(\frac{1}{x}\right)$. The

derivative of $\cotan^{-1}x$ is $\frac{1}{1-x^2}$, and the antiderivative or indefinite

integral is $x\cotan^{-1}x + \frac{1}{2} \ln(1-x^2) + c$.

ARC-COTH (ARCCOTH): An inverse hyperbolic cotangent. An inverse hyperbolic cotangent of x can be denoted by

$$\text{arcoth } x \text{ or } \text{arch } x \text{ or } \text{coth}^{-1}x. \text{Coth}^{-1}x = \tan^{-1}\left(\frac{1}{x}\right).$$

ARC-SECANT (ARCSECANT): The inverse of the secant function, often denoted as $\sec^{-1}$ or **asec** and abbreviated as **arsec** or **asec**. For any argument the value of the arc-secant is an angle in radians whose secant equals the given argument; that is, $y = \sec^{-1}$ if and only if $x = \sec y$. It is defined for arguments between -1 and 1 and its principal values are by convention taken to lie between 0 and π on a graph. The derivative of $\sec^{-1}x$ is given by $\frac{1}{x\sqrt{x^2-1}}$, and an

antiderivative or indefinite integral is $x\sec^{-1}x - \ln[x + \sqrt{(x^2 - 1)}] + c$.

ARC-SECH (ARCSECH): The inverse of the hyperbolic secant function, often denoted as sech^{-1} or **arsech**. For any argument the hyperbolic secant of its value equals the given argument; that is, $y = \text{sech}^{-1}$ if and only if $x = \text{sech}y$. It is defined for arguments between 0 and 1 and its principal values are by convention taken to

be the positive ones shown on a graph. The derivative of $\sec^{-1}x$ is given by $\frac{1}{x\sqrt{1-x^2}}$, and an antirivative or indefinite integral given by

$$x\text{sech}^{-1}x + \sin^{-1}x + c.$$

ARC-SECOND (ARCSECOND; SECOND OF ARC): An angular measurement. There are 60 arc-seconds in one arc-minute, 60 arc-minutes in one degree and 360 degrees in a full circle. One thousandth of an arc-second = 1 milliarc-second.

ARC-SINE (ARCSINE): The inverse of the trigonometrical sine function, denoted by $\sin^{-1}$ or **asin**, with abbreviations **arcsin** or **asin**. Its value for any argument is an angle in radians whose sine equals the given argument; that is, $y = \sin^{-1}x$ if and only if $x = \sin y$. It is defined for arguments between -1 and 1 and its principal values are by convention taken to be those between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ on a graph.

The derivative of $\sin^{-1}x$ is given by $\frac{1}{\sqrt{1-x^2}}$, and an antiderivative or indefinite integral given by $x\text{sech}^{-1}x + \sqrt{1-x^2} + c$.

ARC-SINH (ARCSINH): The inverse of the hyperbolic sine function, usually denoted by $\sinh^{-1}$, **sh**⁻¹ and abbreviated as **asinh**, so that $y = \sinh^{-1}x$ if and only if $x = \sinh y$, defined by all real x by $\sinh^{-1}x = \ln[x + \sqrt{(x^2 + 1)}]$. The derivative of $\sinh^{-1}x$ is given by $\frac{1}{\sqrt{x^2+1}}$, and an antiderivative or indefinite integral

given by $x \sinh^{-1}x - \sqrt{(1+x^2)} + c$.

ARC-TANGENT (ARCTANGENT): The inverse of the trigonometrical function tangent, usually written as $\tan^{-1}$ with **arctan** or **atn** as symbol and abbreviation. Its value for any argument is an angle in radians whose tangent equals the given argument; that is, $y = \tan^{-1}x$ if and only if $x = \tan y$. It is defined for all real arguments and its principal values by convention taken to be those strictly between $-\pi/2$ and $\pi/2$. The derivative of $\tan^{-1}x$ is given by $\frac{1}{\sqrt{1+x^2}}$, and an antiderivative or indefinite integral given

$$\text{by } x\tan^{-1}x - \frac{1}{2} \ln(1+x^2) + c.$$

ARC-TANH: The inverse of the hyperbolic tangent function, often written **tanh**⁻¹ or **th**⁻¹ with the symbol **atanh**, so that for any argument

the hyperbolic tangent of its value equals the given argument; that is, $y = \tanh^{-1}x$ if and only if $x = \tanh y$. It is defined for arguments between -1 and 1 on a graph. The derivative of $\tanh^{-1}x$ is given by

$$\frac{1}{\sqrt{1-x^2}}, \text{ and an antiderivative or indefinite integral given by } x \tanh^{-1}x + \frac{1}{2} \ln(1-x^2) + c.$$

ARCH: 1. A strong, curved structure usually built of steel, reinforced concrete or timber that may take many different forms and that provides both an opening and a support for overlying material. Uses of arches include as a support for bridges, railroad tracks and as part of a building, where it provides a large open space unobstructed by columns. 2. An organ or structure having a curved or bowl-like appearance, especially either of two arched sections of the bony structure of the foot.

ARCHAEA: A domain of prokaryotic organisms or single-celled organisms without nuclei that can survive in extreme environments. Archaea contain the archaeobacteria, including the methanogens which produce methane.

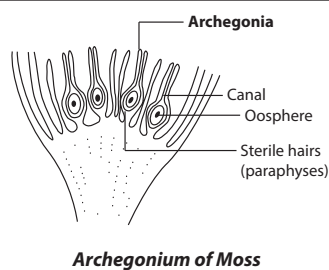
ARCHAEAN (ARCHEAN; ARCHAEZOIC): The earliest era of geological time, the first part of the Precambrian, from the formation of Earth up to about 2,500 million years ago. During the Archaean Eon a large percentage of the Earth's continental crust formed, plate tectonics began, very warm climates and oceans existed and life appeared on Earth in the form of unicellular organisms.

ARCHAEBACTERIA: A term referring to three groups of bacteria: extreme halophiles, methanobacteria and sulfur-dependent extreme thermophiles, whose DNA differs significantly from that of other bacteria. All are strict anaerobes and cannot function when oxygen is present.

ARCHAEOLOGY: The study of history based on the examination of physical remains. Principal activities include preliminary field or site surveys, excavation, classification, dating and interpretation of finds.

ARCHAEZOA (ARCHEZOA): Primitive eukaryotic kingdom, with a nucleus and rod shaped chromosome but no mitochondria or plastid. Archaezoa is composed of two kingdoms of protists, Kingdom Diplomadida, which includes the intestinal parasite *Giardia lamblia* (a protist) and Kingdom Parabasala. It is believed to be the intermediate stage between prokaryotes and eukaryotes. It was proposed by Thomas (Tom) Cavalier-Smith (1942-), Professor of Evolutionary Biology at Oxford University. The kingdom Archezoa has now been abandoned, because they are now known to have developed from mitochondriate ancestors and trees based on other genes do not support their basal placement.

ARCHEGONIUM: The female organ of seedless plants such as ferns and mosses that bears the female gamete (egg/oosphere).



ARCHENTERONS

(GASTROCOEL): The primitive gut that forms during gastrulation in the developing blastula and which develops into the digestive tract of an animal.

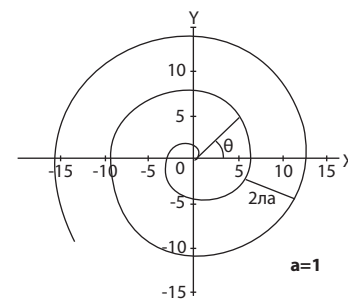
ARCHESPORIUM: 1. The tissue that gives rise to the spore mother cells. 2. The basic structural and functional unit of all organisms; they may exist as independent units of life (as in monads) or may form colonies or tissues as in higher plants and animals.

ARCHIE: An indexing program that can be used to search for files available on one or more File Transfer Protocol (FTP) servers and then indexes them in one large file.

ARCHIMEDEAN PROPERTY: 1. The order axiom for the real line which states that if a and b are real numbers such that $a < b/n$ for all

natural numbers n then $a \leq 0$ or equivalently, that for any positive a and b there is a positive integer n such that $a < nb$. An infinitesimal is non-Archimedean as it is less than any positive non-zero number. 2. The corresponding property of a partial order on an ordered vector space. This fails in the lexical order on Euclidean 2-space.

ARCHIMEDEAN SPIRAL: A curve whose equation in polar coordinates is $r = a\theta$, where ($a > 0$) is a constant. Equivalently, (r, θ) can be represented by the equation $r = a + b/\theta$ with real numbers a and b . It is the locus of points corresponding to the locations over time of a point moving away from a fixed point with a constant speed along a line which rotates with constant angular velocity. Changing the parameter a will turn the spiral, while b controls the distance between successive turnings.



Archimedean Spiral

In the figure shown, the Archimedean spiral is described by the polar equation $r = a\theta$, where a is a parameter determining the density of spiral turns. The separation distance between successive turnings in the Archimedean spiral is constant and equal to $2\pi a$. The following are the general formulas for converting the polar coordinates (r, θ) to Cartesian ones (x, y) : $x = r \cos \theta$ and $y = r \sin \theta$.

ARCHIMEDES (287-212 BC): A Greek mathematician who worked on levers and other aspects of mechanics. He formulated the Archimedes' principle and in hydrostatics, he devised a pump called the Archimedian Screw. His method of successive approximations allowed him to determine the value of π (pi), which is a constant in Euclidean plane geometry, defined as the ratio of a circle's circumference C to its diameter d : $\pi = C/d$, which equals to approximately 3.142. The ratio C/d is constant, regardless of a circle's size. For example, if a circle has twice the diameter d of another circle it will also have twice the circumference C , preserving the ratio C/d .

ARCHIMEDES' CONSTANT (PI): The ratio of a circle's circumference C to its diameter (d), expressed as: C/d , which equals to $22/7$ or approximately 3.142.

ARCHIMEDES' METHOD: The method of estimating π by computing the area or perimeter of inscribed and circumscribed polygons for a unit circle using more and more sides. Archimedes estimated π to be between $3\frac{10}{71}$ and $3\frac{1}{7}$, that is $3\frac{10}{71} < \pi < 3\frac{1}{7}$. from computations with polygons with 96 sides. This gives 3.14 ...; being accurate to two decimal places.

ARCHIMEDES' NUMBER: The dimensionless number, which is the ratio of gravitational force to viscous force. It characterises the motion of liquids due to density differences.

ARCHIMEDES' PRINCIPLE (ARCHIMEDES' LAW): A law of physics named after its discoverer, Archimedes of Syracuse that concerns the upward force or upthrust on any object fully or partly submerged in a fluid. It states that the buoyant force on an object is equal to the weight of the fluid it displaces. Mathematically, (buoyant force) $F_b = m_f g$... , where m_f is the mass of the fluid displaced by the body. Archimedes' principle can be expressed in other words that an object in a fluid such as water experiences an upthrust from the fluid which is equal to the weight of the fluid that the body displaces. The upward force is called the upthrust or buoyant force, expressed mathematically as:

Upthrust = weight in air - weight in liquid. The resultant force on the

body is the difference between the upthrust and the body's weight. This difference is zero for a floating body, i.e. upthrust and weight are equal and opposite. Hot air displaces a balloon, producing upthrust. This upthrust is balanced by the weight of the balloon. A hot air balloon rises when the upthrust on it is greater than its weight. A diver may attach lead weight to his suit if he wants to reach a greater depth. His weight will then be greater than the upthrust on him. For a given body to float in a given fluid depends on its weight and volume or the relative density (weight per unit volume). This is compared to that of the fluid in order to determine its buoyant force. If the body is less dense than the fluid, it will float. If the body is denser than the fluid, it will sink. The principle applies to both floating and submerged bodies and to all fluids, i.e., liquids and gases. Archimedes' principle is an important and underlying concept in the field of fluid mechanics.

ARCHIMEDES' SCREW: One of the earliest kinds of pump, thought to have been invented by Archimedes. It consists of a spiral revolving inside a close fitting cylinder. It is used, for example, to raise water for irrigation.

ARCHIPELAGO: An area of sea containing a group of islands. The islands of an archipelago are usually volcanic in origin and they sometimes represent the tops of peaks in areas around continental margins flooded by the sea.

ARCHIVE: In computer science, it refers to a single file that contains multiple files and/or folders. Archives may be created by several different file archiving utilities and can be saved in one of several different formats. They may also be compressed to reduce the file size or encrypted for security purposes. **Archival data** is data that is not subject to change and not directly accessible to the user of a computer system. It is maintained for long-term storage and record keeping purposes. Examples are research, medical records and historical data.

ARC MINUTE (ARC MINUTE; MINUTE OF ARC): A unit for measuring small angles, used in geometry, surveying, map making and astronomy. An arcminute, with the symbol ' is one-sixtieth of a degree in the sky and an arc second, with the symbol " is one-sixtieth of an arc minute. The number of square arcminutes in a complete sphere is given by $4\pi(\frac{10800}{\pi})^2 = \frac{466560000}{\pi}$ or approximately 148,510,660.498 square arcminutes.

ARCOGRAPH (CYCLOGRAPH): An instrument for drawing arcs of large diameter circles, whose centres are inconveniently or inaccessibly located, therefore does not use a central point.

ARCTIC CIRCLE: Imaginary line that encircles the North Pole at latitude 66°32' N. Within this line there is at least one day in the summer during which the Sun never sets and at least one day in winter during which the Sun never rises.

ARCTURUS (ALPHA BOOTIS): Brightest star in the constellation Bootes, which is approximately 36 light-years from Earth. It is the fourth brightest star in the sky.

ARCUATE: A structure curved into an arch, like a bow.

AREA: A measure of the extent of a surface. It is also a quantity expressing the amount of a surface enclosed in a plane region bounded by a closed curve.

AREA MODEL: 1. A model for mathematics problems in which the length and width of a rectangle represent the factors, and the area of the rectangle represents the product. 2. A model showing fractions as parts of a whole. The whole is a region, such as a circle or a rectangle, representing the whole or entire unit.

AREA OF SURFACE OF REVOLUTION: The area of a surface obtained as a result of rotating, through one revolution about the

		80	4
10		800	40
2		160	8

Area Model (defn. 1)

x -axis, the arc of the curve $y = f(x)$ between $x = a$ and $x = b$ is given

by $2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$, where $y = f(x)$ is a graph of the function $f(x)$ such that $f(x)$ is continuous on the interval $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$.

AREA UNDER A CURVE: The area enclosed by a curve can be found by finding the definite integral between two limits or points. To find the area enclosed under the curve $y = f(x)$ the lines $x = a$ and $x = b$, the equation of the curve $y = f(x)$ is integrated between the limits between the limits $x = a$ and $x = b$, thus the required area is

$A = \int_a^b f(x) dx$. If the area is

between the curve $y = f(x)$ and the

lines $y = a$ and $y = b$, the required area is $A = \int_a^b x dy$. If the area is enclosed between two curves y_a and y_b , then the required area is

$A = \int_{x_1}^{x_2} (y_a - y_b) dx$, where y_a is the curve bounding the area above

and y_b the curve bounding the area below; x_1 and x_2 are the points of intersection of the two curves.

ARECACEAE (PALMAE): A monocotyledonous family of mainly tropical and subtropical woody shrubs, vines or trees comprised of about 200 genera and 2,500 species. The plant usually consists of an unbranched trunk of the same diameter from base to top; with a crown of long and large palmate or pinnate leaves at the apex. When branching does occur, such as in *Hyphaene* (dour palm), it is dichotomous or forks into two branches. Palm inflorescences are large and may contain thousands of flowers in a panicle or spike. In some palms, such as *Metroxylon sagu* (sago palm), the inflorescence is terminal and the plant dies after flowering. The fruits are usually one-seeded drupes or berries, which may attain a considerable size. Some commercially important palms include coconut (*Cocos nucifera*), date (*Phoenix dactylifera*) and oil palm (*Elaeis guineensis*).

ARECIBO: A site in Puerto Rico, where the world's longest single dish radio telescope, 350m in diameter is located.

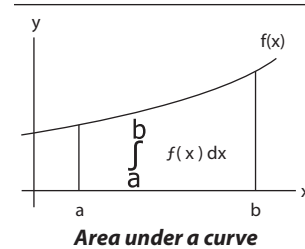
ARECIDAE: One of the four subclasses of the monocotyledons containing both herbaceous and arborescent (the palms) plants. Members of the *Arecidae* are characterised by numerous, small flowers usually unisexual and grouped into an inflorescence subtended by a spathe. They also have broad, petiolate leaves without typical parallel venation.

AREIC: 1. Attribute to a physical quantity obtained by division by area. **Areic charge** is the charge on a surface divided by the surface area. It is measured in coulombs per square metre. 2. In geography, it applies to areas of permeable rocks, such as limestone, which lack surface drainage or without streams or rivers.

ARENACEOUS SOILS (SANDY SOILS): Soil with a high amount of sand particles or soil which contains a considerable amount of quartz-sand or is largely made up of sandy particles.

ARENE COMPLEX: A complex compound involving an aromatic compound and a metal atom achieved through the use of pi-electrons. For example, in ferrocene, $(C_5H_5)_2Fe$, Iron (II) is sandwiched between two molecules of cyclopentadienyl ions.

ARENES (AROMATIC HYDROCARBONS): A member of the class of hydrocarbons, of which benzene is the first member, consisting of assemblages of cyclic conjugated carbon atoms and characterised by large resonance energies. In other words, they are a class of hydrocarbons having a closed ring system of carbon atoms with delocalised π -electrons. They resist addition reactions and common oxidising agents. They can be monocyclic (MAH) or polycyclic (PAH). Some examples are toluene and naphthalene



ARENICOLOUS: In biology, it refers to organisms that grow, live or burrow in sand. An example is the marine fungi that grow on the sandy beaches along the coast of southern Africa.

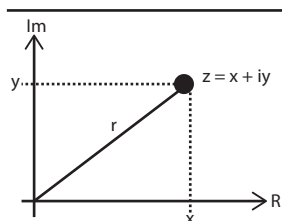
AREOLA (AREOLE): A small well defined area between the veinlet of a leaf or the region of a cactus bearing the spine or flowers.

AREOLAR CONNECTIVE TISSUE: Protein fibres made up of elastin, collagen, fibroblast, mast cells, macrophages and fat cells that are jelly-like and found throughout the skin and the body connecting organs and tissues.

AREOLE: 1. A sunken cushion representing a condensed lateral shoot from which spines, branches and flowers arise in cacti. Areoles may occur either singly on tubercles, for example, in *Mammillaria zeilmanniana* or in rows along raised ridges. 2. Any area outlined on a surface, for example, a segment of leaf lamina surrounded by veins. A surface divided into areoles, for example, lichen thalli split by cracks into roughly hexagonal areas, is described as areolate.

ARG.: Abbreviation and symbol for the argument of a complex number. The principal value is usually taken to be in the range $-\pi < \theta \leq \pi$; and denoted Arg .

ARGAND DIAGRAM (GAUSSIAN PLANE): Named after Jean-Robert Argand (1768-1822), Swiss-born mathematician, it is a two-dimensional method for representing complex numbers by Cartesian coordinates x, y . Along the x -axis or horizontal axis are plotted the real numbers and along the y -axis or the vertical axis the non-real or imaginary numbers. The number $x + iy$ is represented by the point (x, y) or by the corresponding vector. If the polar coordinates of (x, y) are (r, θ) , r is the modulus and θ the argument of $x + iy$.



An Argand diagram indicating the imaginary axis (y-axis) and real axis (x-axis).

ARGAND PLANE (COMPLEX PLANE)

PLANE: The geometric representation of the complex numbers as identified with the infinite two-dimensional space defined by the real axis and the orthogonal imaginary axis of the Argand diagram.

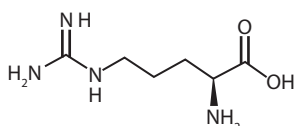
ARGAND, JEAN ROBERT (1768-1822): Swiss-born mathematician, who was one of several people, including Gauss, who invented a geometrical representation for complex numbers. He also discovered how to represent complex numbers on the Cartesian plane. Argand is also renowned for delivering a proof of the fundamental theorem of algebra. The word "Argand diagram" was named in his honour.

ARGENTITE: A sulfide ore of silver with chemical formula Ag_2S . It crystallises in the cubic system. It is grey-black in colour but appears shiny when freshly cut.

ARGENTOUS COMPOUNDS: Compounds of silver in its lower (+1) oxidation state. For example, argentous chloride is silver (I) chloride.

ARGILLACEOUS: 1. Plants which usually grow on clay soils. 2. It refers to sedimentary rocks and deposits composed of very fine-grained material, such as clay, shale, etc.

ARGININE: A basic amino acid with the chemical formula $\text{C}_6\text{H}_{14}\text{N}_4\text{O}_2$. It is obtained from the hydrolysis or digestion of plant and animal protein. It is important in histone proteins, which are rich in this amino acid.



Structure of Arginine

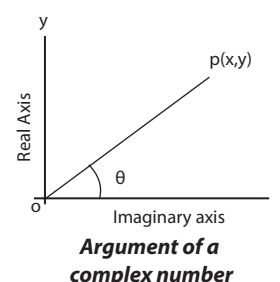
ARGON: A colourless, odourless gas with chemical symbol Ar , atomic number 18, relative atomic mass 39.95, melting point -189.3°C (83.85 K) and boiling point -185.9°C (87.25 K). It is the most abundant noble gas. By volume it makes up 0.9% of the atmosphere from which it is commercially obtained by fractionation. Under normal conditions, it

has little tendency to react. It is used to fill electric light bulbs; and fluorescent tubes; to protect metals from oxidation; to provide inert atmosphere for welding stainless steel, titanium and other metals.

ARGUMENT: 1. Argument of a function, also known as an **independent variable** is a specific value of the independent variable of a function of x or the variable, term or expression on which a function operates. For example, the argument of $\sin(2A)$ is $2A$, the argument of e^{x-5} is $x-5$ and the argument of $f(x)$ is x . 2. In logic, a process or instance of inductive or deductive reasoning that purports to show its conclusion to be true. 3. In logic, an argument is a sequence of sentences, called **premises** that leads to a resulting sentence called **conclusion**. 4. In mathematics, also known as **amplitude, anomaly** or **azimuth** of a complex number, it is the angle that describes the direction of a complex number when it is plotted in a complex plane using polar coordinates. It is measured in radians as an angle in standard position. 5. A term used in computing to denote the value on which a function operates. For example, if the argument 16 is operated on by the function square root, the answer 4 is produced.

ARGUMENT OF A COMPLEX NUMBER (COMPLEX ARGUMENT):

The angle describing the direction of a complex number on the complex plane. The numeric value is given by the angle in radians and is positive if measured counter-clockwise. For a complex number in polar form $r(\cos \theta + i \sin \theta)$, the argument is θ . Algebraically, it is a complex number z of the form $z = x + yi$, where x and y are real numbers and i is the imaginary unit, with the property $i^2 = -1$. The real number x is called the real part of the complex number. It is denoted by $\text{Re}(z)$. The real number y is the imaginary part. It is denoted by $\text{Im}(z)$. In



the diagram shown, line OP is called **modulus z** , where $|z| = \sqrt{x^2 + y^2}$.

The argument of $z = x + yi$ is given by $\arg z = \tan^{-1}\left(\frac{y}{x}\right)$.

ARHIZOUS: A structure without root.

ARIANE: A series of European civilian launch vehicles built by the European Space Agency. The launch site is Kourou in French Guinea.

ARID: Lacking water, especially having insufficient rainfall to support vegetation. An arid region or area has less than 250 mm precipitation per year and is very dry with little vegetation.

ARIES: 1. In astronomy, it is a zodiacal and autumnal constellation in the northern hemisphere between Pisces and Taurus, near Auriga. 2. In astrology, it is the first sign of the zodiac governing the period March 21 – April 19. It is represented as the legendary ram whose golden fleece was sought by Jason and Argonauts.

ARIL: Accessory seed cover other than a pericarp often derived from the stalk of the ovule. It may be fleshy and sometimes brightly coloured, woody or hairy.

ARISTARCHUS OF SAMOS (ABOUT 270 BC): A Greek astronomer, who first verifies that the Earth rotates around the Sun. He represented astronomy mathematically and used geometrical procedures to calculate the relative sizes of the Sun and the Moon and their relative distances from the Earth.

ARISTATE: A term used to describe a structure, such as a glume or lemma; bearing an awn.

ARISTOTELIAN LOGIC: The logical theories of Aristotle, especially as developed in the Middle Ages, concerned mainly with the principle of the syllogism; as opposed to modern, symbolic or mathematical logic.

ARISTOTLE (384-322 BC): Greek philosopher, who attended Plato's Academy in Athens. He made important contributions to mathematics through his work on deductive logic. Aristotle studied almost every subject possible at the time such as anatomy, astronomy, embryology, geography, geology, meteorology, physics and zoology and made significant contributions to most of them.

ARITHMETIC: A branch of science concerned with the study of numbers and their properties. The fundamental operations of arithmetic are addition, subtraction, multiplication and division.

ARITHMETIC FACTS: The addition facts (whole-number addends 9 or less); their inverse subtraction facts; multiplication facts (whole-number factors 9 or less); and their inverse division facts, except there is no division by zero. They are as follows:

100 addition facts: $0 + 0 = 0$ through $9 + 9 = 18$;

100 subtraction facts: $0 - 0 = 0$ through $18 - 9 = 9$;

100 multiplication facts: $0 \times 0 = 0$ through $9 \times 9 = 81$;

90 division facts: $0/1 = 0$ through $81/9 = 9$.

ARITHMETIC FUNCTION: Any function defined on the natural numbers or integers and dependent upon those properties of the integer itself as a number, in contrast to functions that are defined for other values (real numbers, complex numbers or even other functions) and that involve various operations from algebra and calculus. For example, the Euler phi function is an arithmetic function.

ARITHMETIC LOGIC UNIT (ALU): A digital circuit, which is part of a computer processor (CPU) that is used to perform arithmetic and logic operations.

ARITHMETIC MEAN: 1. The average of a set of numbers or quantities obtained by dividing their sum by the number of terms. For example, the arithmetic mean of 3, 4 and 8 is 5. 2. Also known as **average set of data**, it is a measure of central tendency, to characterise the centre of a frequency distribution of a quantitative variable by considering all the observations with the same weight according to each. It is calculated by summing the observations and dividing by the number of observations. For example, for the set $x_1, x_2, \dots, x_n$, the arithmetic mean is given by $\frac{1}{n} \sum_{i=1}^n x_i$.

ARITHMETIC OPERATOR: Any symbol that takes numerical values as their operands and return a single numerical value. Examples are addition (+), subtraction (-), multiplication (*), and division (/). They are used in common arithmetic and most computer languages contain a set of such operators that can be used within equations to perform a number of types of sequential calculation.

ARITHMETIC PROGRESSION (LINEAR SEQUENCE): A number pattern which increases or decreases by the same amount known as the **common difference** each time. It is generated by writing down the terms in a definite order, for example, increasing or decreasing by the same amount, such as 4, 8, 12, 16 with an increase of 4. The terms are connected by the formula $U_n = a + (n - 1)d$, where a is the first term, d is the common difference and n is the number of terms.

ARITHMETIC SERIES: The sum of an arithmetic sequence or the indicated sum of a sequence of terms, with each term being added to those that precede it. The sum of an arithmetic series is found by multiplying the number of terms by the average of the first and last terms. Given that the first term is a_1 , the common difference is d and the number of terms is n , then the following formula can be used to find the sum of terms in an arithmetic sequence:

$$S_n = \frac{n}{2}[a + u_n] \text{ or } S_n = \frac{n}{2}[2a + (n-1)d].$$

ARITHMETIC-GEOMETRIC MEAN INEQUALITY: The inequality that the arithmetic mean of a set of numbers is always greater than the geometric mean of the numbers. The set of numbers can be equal if and only if every number in the list is the same.

ARITHMETIC-GEOMETRIC MEAN ITERATION: The two term iteration in which one repeatedly computes both the arithmetic mean

and the geometric mean of two positive numbers: $a_0 = a, b_0 = b$ and $a_{n+1} = \frac{1}{2}(a_n + b_n); b_{n+1} = \sqrt{a_n b_n}$.

ARITY: The number of arguments of a function or relation.

ARK: 1. In agriculture, it is a mobile poultry house with wire or slatted floors used to house chickens at night and in bad weather. 2. In agriculture, it is a corrugated metal or wooden hut, used to shelter sows kept outdoors in fields.

ARKOSE: A pale grey to pink sedimentary rock composed of sand-size fragments containing about 30% feldspar. It forms in marine and freshwater sediments and continental deposits, usually from the breakdown of granite.

ARM: 1. In biology, it is the upper limb of the human body, from the shoulder to the hand. 2. In mathematics, it refers to any of the two lines forming an angle.

ARM-CELL: A cell with incomplete septate (cross wall) extending inwards. An example is found among the leaf mesophyll cells of some members of the grass family.

ARMA MODEL (BOX-JENKINS MODELS): Autoregressive moving average models used in time series analysis. The autoregressive part is denoted AR and consists of a finite linear combination of previous observations. The moving average part, MA, consists of a finite linear combination of the previous values for a white noise (a sequence of mutually independent and identically distributed random variables).

ARMATURE: 1. Any moving part in an electrical machine in which a voltage is induced by a magnetic field, especially the rotating coils in an electric motor or generator and the ferromagnetic bar attracted by an electromagnet in a relay. An armature may also be part of a dynamo that rotates in a magnetic field to produce an electric current, rotating coil(s) in an electric motor or the moving plate of a bell or relay. 2. A piece of soft iron used to connect the two poles of a magnet or electro-magnet, in order to complete the circuit or to receive and apply the magnetic force. In the ordinary horseshoe magnet, it serves to prevent the dissipation of the magnetic force. 3. The moving part in an electromechanical device like a loudspeaker or a buzzer. 4. In biology, it is the protective covering, structure or organ of an animal or a plant, such as teeth, claws, spines, thorns, prickles, barbs or the shell of a turtle. 5. A framework serving as a supporting core for the material that is used to make a sculpture.

ARMoured CABLE: A cable which is shielded by wires and insulating materials to provide mechanical protection for use outside or when laid below the surface of the ground. The armour is placed below the sheath of the cable to provide protection for the conductor and insulating layers. Aluminium wire armour and steel wire armour are the protection most often used.

ARMPIT (AXILLA; OXTER; UNDERARM): The area on the human body directly under the joint where the arm connects to the shoulder.

ARMSTRONG NUMBER (NARCISSISTIC NUMBER; PERFECT NUMBER): An n -digit number, which is equal to the sum of its digits raised to the n th power. For instance, 371 is narcissistic because $3^3 + 7^3 + 1^3 = 371$ and 9474 is narcissistic because $9^4 + 4^4 + 7^4 + 4^4 = 9474$. The smallest narcissistic number of more than one digit is $153 = 1^3 + 5^3 + 3^3$. The largest narcissistic number (in base 10) is 11513221901876399256509559797 971522401, which is the sum of the 39th powers of its digits.

ARNDT-EISERT SYNTHESIS: A method of converting a carboxylic acid into the next higher homologue acid or one of its derivatives. It is a popular method of producing beta-amino-acids from alpha-amino-acids. Diazomethane is used to insert a CH_2 group.

AROMATIC: 1. Aromatic compounds are organic chemical compounds containing one or more planar rings, each of which contains usually three conjugated double bonds and $(4n + 2)$ delocalised pi-electrons, where n is a small integer, characteristic of the benzene series or naphthalene and related organic groups. They undergo substitution reactions more readily than addition reactions. 2. Having an aroma such as onions, leeks, garlic and scallions; fragrant or sweet-smelling.

AROMATIC HYDROCARBONS (ARENES): A member of the class of hydrocarbons, of which benzene is the first member, consisting of assemblages of cyclic conjugated carbon atoms and characterised by large resonance energies. In other words, they are a class of hydrocarbons having a closed ring system of carbon atoms with delocalised π -electrons. They resist addition reactions and common oxidising agents. They can be monocyclic (MAH) or polycyclic (PAH).

AROMATICITY: A state in which three (3) or more carbon atoms present in an unsaturated organic compound joined in a ring that exhibit delocalisation of π -electrons as in benzene and its compounds. In other words, it is a term used to describe the cyclic delocalisation of a π -electron system of a conjugated compound. A molecule or ion with such delocalised electrons is represented by several contributing structures called resonance structures or canonical forms. When a molecule cannot be completely represented by a single structure but its chemical and physical properties can be described by two or more different structures, then the true structure is said to be resonance hybrid, which is the overall combination of all canonical forms. It is more stable than all canonical forms and shows all the characteristics of the molecule. This phenomenon is called resonance effect. All C-C bonds are equivalent and the ring is flat or almost flat. The total number of delocalised π -electrons must obey the Hückel Rule, $4n + 2$, where n takes the values 0, 1, 2, 3,...

ARONSZAJN TREE: An uncountable tree such that all branches and levels are countable.

AROUSAL: Excitatory responsiveness of an organism to a particular stimulus usually controlled by the reticular activating system of the brain and varies between sleep and full alertness.

ARPANET (ADVANCED RESEARCH PROJECTS AGENCY NETWORK): The early packet switching network that became the basis for the present day Internet. It was based on a concept first published in 1967 which was developed under the direction of the US Advanced Research Projects Agency in 1969. In the 1970s, it became possible to develop and expand the networks.

ARRANGEMENT: 1. A permutation (in ordered sequence) or combination (not in ordered sequence) of a set of objects. This is given by ${}^n P_r = \frac{n!}{(n-r)!}$. 2. In algebra, any ordered sequence of elements.

For example, 3,1,2 is an arrangement, whereas the permutation $2,6,8 \rightarrow 8,6,2$ is an operation upon arrangements.

ARRAY: 1. An orderly arrangement of numbers or objects, often in rows, columns or a matrix that is most commonly used as a visual tool for demonstrating multiplication and division. 2. An n -dimensional ordered set of data items identified by a single name and one or more indices, so that each element of the set is individually addressable. Examples are matrices and vectors. 3. In computer science, it is a group of related data values grouped together as one value. It is used in computer programs to organise data so that a related set of values can be easily sorted or searched.

ARRECTOR PILI: Small muscles in the dermis of the skin that are attached to the bases of hair follicles. Contraction in them in response to cold or fear pulls the hair into vertical position, thus trapping an insulating layer of air around the body and causing goose flesh in human.

ARRHENIUS EQUATION: A mathematical relation, named after Svante Arrhenius (1859-1927), which expresses the exponential relationship between the rate constant and the activation energy of a reaction at a particular temperature. It is expressed as $k = A \exp(-E_a / RT)$, where k is the rate constant of a given reaction and E_a the activation energy. A is a constant for a given reaction, called the pre-exponential factor. Often the equation is written in logarithmic form $\ln k = \ln A - E_a / RT$. A graph of $\ln k$ against $1/T$ is a straight line with a gradient $-E_a / R$ and an intercept on the $\ln k$ axis of $\ln A$.

ARRHENOTOKY: A sexual dimorphism arising from the hatching of

fertilised eggs into females and unfertilised eggs into males as occurs in honey bees and wasps.

ARRHYTHMIA: A defect in the normal rhythm of the heart that may be slower or faster than normal or just irregular. It is usually caused by reduced blood flow within the heart or a defect in the sinoatrial node or pacemaker of the heart.

ARROW (MORPHISM): In mathematics, the elements of an arbitrary category that play the role of mappings of one set into another.

ARROWROOT: A starch yielding tuber plant, *Maranta arundinacea*, in the tropics, which is particularly used in the preparation of a particularly pure form of starch, which contains only a trace of protein (0.2%) and is free from vitamins. It is used to thicken sauces and in bland, low-salt and protein-restricted diets.

ARSANES: The saturated hydrides of trivalent arsenic, having the general formula $As_n H_{n+2}$. Individual members having an unbranched arsenic chain are named arsane, diarsane, triarsane, etc.

ARSENIC: A brittle, greyish-white metalloid or semi-metallic element, with the chemical symbol As, atomic number 33, atomic weight 74.92160, melting point 817°C (1 090.15 K) at 28 atmospheres pressure, sublimation point 613°C (886.15 K). It consists of very poisonous compounds, especially the oxide (As_2O_3). It was discovered in 1250 and is a member of Group 15 (VA) of the Periodic Table, it has three allotropic forms: yellow, black and grey and found widely distributed in nature. However, the main sources are realgar and arsenopyrite. It is used in the manufacture of semi-conductors in electronics, glass manufacturing to remove green colour caused by impurities of iron compounds, hardening of lead and in the manufacture of military poison gases. Arsenic also means relating to or containing arsenic, especially with a valence of 5.

ARSENIC(III) OXIDE (ARSENIC TRIOXIDE; ARSENIUS OXIDE; WHITE ARSENIC; DIARSENIC TRIOXIDE; ARSENOUS ANHYDRIDE): A white or colourless solid inorganic compound, with the chemical formula As_2O_3 , molecular weight 197.84, melting point 312.3°C (585.45 K), which exists in three forms. The commonest has cubic or octahedral crystals with a relative density of 3.87 and sublimes at 193°C (466.15 K). It is soluble in water, ethanol and alkali solutions. It occurs naturally as arsenolite.

ARSENIC(V) OXIDE (ARSENIC OXIDE; ARSENIC PENTOXIDE; DIARSENIC PENTAOXIDE): A white amorphous deliquescent solid, with the chemical formula As_2O_5 , relative density 4.32, decomposes at 315°C (588.15 K) and soluble in water and ethanol. It cannot be obtained by direct combination of arsenic and oxygen. It is usually prepared by the reaction of arsenic with nitric acid followed by dehydration of the arsenic acid thus formed.

ARSENIDES: Compounds obtained from arsines by replacing one or more hydrogen atoms by a metal.

ARSENIUS ACID (ARSENIC(III) ACID; ARSENOUS ACID): A tribasic acid, with the chemical formula H_3AsO_3 from which arsenites are derived. An example is an aqueous solution of arsenic (III) oxide (arsenious oxide, white arsenic) with the chemical formula As_4O_6 .

ARSENITE (ARSENATE(III) SALT): A salt or an ester of arsenious acid, with the chemical formula H_3AsO_3 . It is the arsenic oxoanion where arsenic has oxidation state +3. Examples of arsenites include sodium arsenite which contains a polymeric linear anion, $[AsO_2^-]_n$ and silver arsenite, Ag_3AsO_3 , which contains the trigonal, AsO_3^{3-} anion. The arsenite ion is sometimes called ortho-arsenite.

ARSENOLITE: A mineral form of arsenic(III) oxide, As_4O_6 . It is formed as an oxidation product of arsenic sulfides. Commonly found as small octahedra it is white, but impurities of realgar or orpiment may give it a pink or yellow hue.

ARSINE (ARSENIC TRIHYDRIDE; TRIHYDRIDOARSENIC; ARSENOUS HYDRIDE; HYDROGEN ARSENIDE): A colourless and highly poisonous gas, with the chemical formula AsH_3 , melting point -111.2°C (162 K) and boiling point -62.5°C (211 K). It is soluble in water, chloroform and benzene. It is used for doping gas

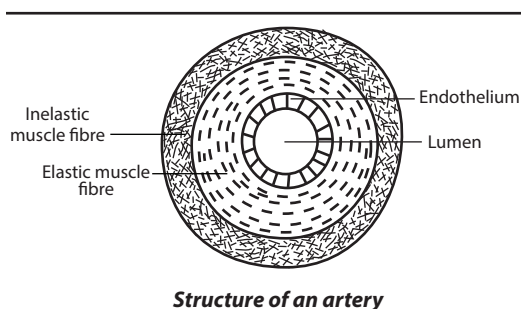
in the process of semi-conductors in the micro electronics industry, in organic synthesis and in forensic science.

ARTERIOGRAPHY: An imaging technique for examining blood vessels in which an x-ray visible dye called contrast material or radiopaque is injected into the bloodstream. This dye allows the arteries to be imaged with x rays. The technique is particularly useful in detecting atherosclerosis.

ARTERIOLE: A smaller artery usually linking an artery to a capillary and found close to organs but distant from the aorta. Example is the afferent arteriole in the kidney that connects to the renal artery.

ARTERIOSCLEROSIS: A disease of the circulatory system manifested in the arteries as a result of high cholesterol deposits causing pressure build up in the arterial wall, thickening and loss of elasticity. The main risk factors and causes of arteriosclerosis include hypertension, diabetes mellitus and smoking. It can be avoided by taking regular vigorous aerobic exercise.

ARTERY: A blood vessel that carries blood away from the heart to the rest of the body. It has thick muscular elastic walls that can withstand considerable pressure. Arteries carry oxygenated blood, except the pulmonary arteries which convey deoxygenated blood from the heart to the lungs.



ARTESIAN BASIN: A structural basin in the crust of the earth within which a zone of water-bearing rock is confined between impermeable beds.

ARTESIAN WELL: A well that is supplied with water rising under its own pressure from an underground water-saturated rock layer or aquifer.

ARTHRITIS: Inflammation of the joints, characterised by pain, stiffness, swelling and restricted motion that worsens with age. There are three main types of arthritis: rheumatoid arthritis; osteoarthritis; and septic arthritis. Rheumatoid arthritis and osteoarthritis are the commonest type of arthritis. Rheumatoid arthritis is an autoimmune disorder that occurs, when the body's immune system attacks the synovium (the lining of the joints). Osteoarthritis is caused by the natural breakdown of cartilage tissue. Other types of arthritis may be caused by gout, infection, crystal deposition, injury and diseases, such as psoriasis (a disease that causes cells to build up rapidly on the surface of the skin) and lupus (a chronic inflammatory disease that occurs when the body's immune system attacks the tissues and organs).

ARTHROLEPTIDAE: A family in the order Anura which includes squeakers and cricket frogs. They have smooth skin without the large warts seen in toads and many other types of frogs. Some are reddish, greenish brown, brown and others are almost black in colour; toes are without webs between them, but ending in large pads in some species. Most of them have thin front and hind legs. Few are burrowers with heavy legs, thick shovel-like bumps or tubercles on the heels of their hind feet for digging.

ARTHROPODS: A group of invertebrates including insects, arachnids, myriapods and crustaceans with hard chitinous or calcareous outer covering or exoskeleton, a segmented body and several paired jointed appendages. They comprise over one million species and they inhabit marine, freshwater and terrestrial habitats worldwide.

ARTHROSCOPE: An endoscope for examining the interior of a joint and for performing diagnostic and therapeutic procedures within the joint. It consists of a rigid telescope fitted with a lens system and fibre-optic illumination that is inserted into a joint through a small incision to generate a magnified picture of the joint interior that can be viewed on a video screen. The diameter of arthroscopes ranges from 2 mm for small joints to 5 mm for larger joints.

ARTHROSPORE (OIDIUM): A spore that forms by the organised fragmentation of a hypha as seen, for example, in *Endomyces*.

ARTICLE: A constricted joint separating a section of a fruit from others as found in *Desmodium*.

ARTICULATE: In biology, it refers to jointed or having joints where separation may occur naturally. **Articulation** is the attachment of two bones usually by means of a joint. The thighbone, for example, articulates with the pelvic girdle.

ARTICULATING SURFACE: The surface of a bone that rubs against or glides over the surface of another in a slightly movable joint. It is covered with a protective layer of cartilage.

ARTIFICIAL CLASSIFICATION: In biology, the grouping of organisms into groups that do not take the true evolutionary relationships into account. An example is the grouping together of plants according to the number and situation of their stamens, styles, and stigmas rather than their evolutionary relationships.

ARTIFICIAL CHROMOSOME: A chromosome that is used as a vector for cloning large segments of DNA because it has features of a true chromosome.

ARTIFICIAL INSEMINATION (A.I.): The introduction of living sperms collected from a male into a female's uterus or cervix at proper time under most hygienic conditions to facilitate pregnancy through in vivo fertilisation by means other than sexual intercourse. It is used for selected breeding of animals, including dairy cattle and pigs and in humans in some cases of impotence and infertility. Where donor sperm is supplied by a sperm bank, it will always be quarantined and frozen, and will need to be thawed before use. When an ovum is released, semen is introduced into the woman's vagina, uterus or cervix, depending on the method being used. Sperm is occasionally inserted twice within a 'treatment cycle'. Common methods used in A.I. include the following: (i) **Spectrum method** in which spectrum is placed in the vagina of the cow, which provides passage outside to the site of insemination, then an inseminating tube is passed through a speculum (instrument used to keep the vagina walls open so that the cervix is more easily visible) and semen is deposited at the cervix. (ii) **Recto-vaginal method** of insemination through the vulva to the vagina and cervix and observed with the hand in the rectum that the A.I. gun loaded with semen straw reaches the cervix, then the semen is deposited by injecting the gun, and after depositing the semen the gun is removed. (iii) **Vaginal method** in which a hand is passed through the vagina and the inseminating tube is guided by hand to the site of insemination and semen is deposited. In this method, there is a risk of contamination and injury of female genitalia.

ARTIFICIAL INTELLIGENCE (AI): A branch of computer science concerned with creating computer programmes that can perform tasks thought to require human intelligence. Current AI research covers areas such as planning, language understanding, pattern recognition, knowledge representation, agent design and machine learning.

ARTIFICIAL MANURE: 1. Organic residues, or a mixture of organic residues and soil, that have been piled, moistened, and allowed to decompose. 2. Also called synthetic fertilisers, man-made combinations of chemicals and inorganic substances such as nitrogen, phosphorus, potassium, calcium, magnesium and other elements in different ratios.

ARTIFICIAL NEURAL NETWORKS: A computational simulation of the functioning of human neurons, for example, the electrical messages they produce between input such as from the eyes or nerve

endings in the hand, processing by the brain and the final output from the brain such as reacting to light or from sensing touch or heat.

ARTIFICIAL PROPAGATION: The deliberate, directed reproduction of plants using plant cells, tissues or organs to produce new plants. Methods used include cutting, layering, grafting, tissue culture and division of the roots and of the tubers. Artificial propagation is often used to reproduce a plant with desirable characteristics, create better plants, hasten fruit production or prevent a species from becoming extinct. Since artificial propagation does not rely on seeds, it allows new plants to be produced quickly.

ARTIFICIAL REEFS: A man-made, underwater structure that promotes marine life in areas with a generally featureless bottom. It also controls erosion, block ship passage, block the use of trawling nets, or improve surfing. Shipwrecks also serve as artificial reefs when preserved on the sea floor.

ARTIFICIAL RESPIRATION (KISS OF LIFE): Any procedure used to supply air or oxygen to the lungs of a suffocating person to restore normal breathing or a method of resuscitating a person who suffers from asphyxia or lack of oxygen. It assists or stimulates respiration, a metabolic process to have the overall exchange of gases in the body by pulmonary ventilation, external respiration and internal respiration. The procedure takes many forms, but generally entails providing air for a person who is not breathing or is not making sufficient respiratory effort on their own. However, it must be used on a patient with a beating heart or as part of cardiopulmonary resuscitation to achieve the internal respiration. Some common methods used include the following: (i) **Mouth to mouth**, with the rescuer making a seal between his or her mouth and the patient's mouth and blowing air into the patient's body; (ii) **Mouth to nose**, in which the rescuer may need to form a seal with the patient's nose; and (iii) **Mechanical resuscitator**, which uses an electric unit designed to breathe for patients. All methods, however, require good airway management to perform, which ensures that the method is effective.

ARTIFICIAL SATELLITE: A man-made object that moves in an orbit around the Earth, the Moon, the Sun or a Planet. It is launched from the Earth's surface by a rocket and once placed in orbit, maintains its motion mainly by the gravitational influence of one celestial body. Satellites can be divided into five main types: research, communications, weather, navigational and applications. Research satellites measure fundamental properties such as magnetic fields, the flux of cosmic rays and micrometeorites and properties of celestial objects of outer space. Communication satellites relay information, telephone, TV and radio signals around the world. Weather satellites or meteorological satellites provide continuous, up-to-date, information about large-scale atmospheric conditions such as cloud cover and temperature profiles. Navigation satellites are designed to aid the navigation of sea and air traffic. An example is the Global Positioning System (GPS), which uses a web of 24 Navstar satellites in 12-hour orbits to determine position. Applications satellites are designed to test ways of improving satellite technology. Satellites have also been used for a number of military purposes. Presently, more than 1,000 artificial satellites are in orbit.

ARTIFICIAL SELECTION: Selective breeding of a plant or animal that exhibits the particular characteristics that a breeder wishes to develop. In plants desirable features may include resistance to disease, high yield, attractive appearance, improvement of production, etc.

ARTIFICIAL SPECIATION: The creation of new species by people. This is achieved through laboratory experiments, where scientists mostly research insects like fruit flies and in animal husbandry. For example, laboratory experiments with fruit flies have shown that it is possible to induce reproductive isolation between laboratory populations grown in different habitat types and food substrates. Also domestic sheep can no longer interbreed with their wild ancestor, *Ovis orientalis*.

ARTIFICIAL VARIABLES: In linear programming, variables that are temporarily introduced to ease a computation but are disposed of during the computation. The artificial variables must be driven to

zero to obtain a basic solution to original constraints. One way of doing this is to solve an auxiliary linear program where the objective is to minimise the sum of the artificial variables.

ARTIFICIALLY ACQUIRED PASSIVE IMMUNITY: A type of short-term immunity consisting of antibodies produced by another organism or by in vitro method that is introduced to the body through intravenous or intramuscular means.

ARTIN'S CONJECTURE ON PRIMITIVE ROOTS: A quantitative form of the conjecture that every non-square integer is a primitive root of infinitely many primes.

ARTINIAN MODULE: In abstract algebra, a module that satisfies the descending chain condition on its poset of submodules. They are for modules what Artinian rings are for rings and a ring is Artinian if and only if it is an Artinian module over itself (with left or right multiplication). Every Artinian module is also a Noetherian module, but not necessarily vice versa; for example, the integers are a Noetherian but not an Artinian $\mathbb{Z}$ -module. It is named after Emil Artin (1898-1962), German-born American algebraist and group theorist.

ARTINIAN RING: A ring that is regarded as a left or right R -module is an Artinian module.

ARTINITE: A mineral form of basic magnesium carbonate, with the chemical formula $\text{MgCO}_3 \cdot \text{Mg}(\text{OH})_2 \cdot 3\text{H}_2\text{O}$, Mohs hardness 2.5, relative density 2, occurring in low-temperature hydrothermal veins and in serpentinized ultramafic rocks.

ARTIODACTYLA: An order of hoofed mammals comprising the even toed ungulates, in which the third and fourth digits are equally developed and bear the weight of the body. The order contains the families Antilocapridae (prohorn antelope), Bovidae (antelope, cattle, bison, buffalo, goats, sheep), Camelidae (camels, guanacos, llamas, alpacas, vicuñas), Cervidae (moose, elk), Giraffidae (giraffe, okapi) and Tragulidae (mouse deer).

ARYL GROUP: A group of atoms remaining after a hydrogen atom is removed from the aromatic system, for example, phenyl group, C_6H_5- , derived from benzene; tolyl group, $\text{CH}_3\text{C}_6\text{H}_4-$, derived from toluene; and xylol group, $(\text{CH}_3)_2\text{C}_6\text{H}_3-$, derived from xylene.

ARYNE: A compound such as benzyne, formed from an arene by removing two adjacent hydrogen atoms to convert a double bond into a triple bond.

ASA SYSTEM: A numbering system derived by the American Standard Association, (now ANSI) used in photography for rating the spread of films. It has now been superseded by the International Organisation for Standardization (ISO) system.

ASAPHIDAE: A family of order Anura which includes Tailed frogs. The two known species are *Ascaphus truei* and *Ascaphus montanus*. They range from British Columbia south to Mendocino County, California, the Rocky Mountains of Idaho and Montana and adjacent Washington and Oregon in or near clear, rocky, swift-moving streams that flow through forests. The tailed frogs get their name from the copulatory organ of the male, which resembles a short tail, but is really an expanded portion of the cloaca. These species are among the very few frogs with internal fertilisation; the copulatory organ is used to transfer sperm to the female. The tympana is absent. They are not known to have an advertisement call and are considered as silent frogs.

ASBESTOS: Any of several naturally occurring fibrous silicate minerals of the amphibole or serpentine groups that are resistant to heat, chemicals and friction. Formerly it was widely used for fireproofing, electrical insulation, building materials, brake linings and chemical filters. Asbestos can cause lung diseases such as **asbestosis** for people chronically exposed to asbestos, if breathed into the lungs and may give rise to mesothelioma and is associated with an increased risk of lung cancer, particularly in smokers.

ASCARIASIS: Infestation of the intestine by common roundworms or related nematode worms (ascarids).

ASCARID: A parasitic nematode worm, such as the common roundworm.

ASCARIS LUMBRIOCOIDES: A large common roundworm that is parasitic in the intestines of humans and some other animals such as dogs and pigs. It is the the largest and most common parasitic worm in humans. Males are 2-4 mm in diameter and 15-31 cm long and the females are 3-6 mm wide and 20-49 cm long. The male's posterior end is curved ventrally and has a bluntly pointed tail. If contaminated food or water is taken, the eggs hatch in the upper part of the small intestine. The larvae go to the villi, liver, heart, lungs, trachea and mouth, where they are swallowed again into the stomach and intestine. Here the larvae become adult worms, where they absorb the digested food of the host and mate and produce eggs. It is responsible for the disease ascariasis with symptoms of restlessness, fever and sometimes diarrhoea in humans. Symptoms may include loss of appetite, anaemia, swollen limbs or cheeks, pneumonia, nausea and vomiting.

ASCENDING (ASSURGENT; ASCENDENT): In biology, it is a growth, which goes obliquely upwards and is usually curved.

ASCENDING CHAIN CONDITION: The condition on a ring that every ascending sequence of left or right ideals has only a finite number of distinct members. In other words, a partially ordered set (poset) P is said to satisfy the ascending chain condition if every strictly ascending sequence of elements eventually terminates.

ASCENDING NODE: The point at which an orbiting body rises through the plane of the ecliptic travelling northward; or a point, where the moon crosses to the north of the ecliptic. The opposite point in the orbit, where the body moves from north to south across the reference plane, is the descending node. The longitude of the ascending node is an orbital element.

ASCII (AMERICAN STANDARD CODE FOR INFORMATION INTERCHANGE): The most common format for text files in computers and on the Internet, in which numbers are assigned to letters, digits and punctuations. It encodes 128 specified characters into 7-bit binary integers. The characters encoded are numbers 0 to 9, lowercase letters a to z, uppercase letters A to Z, basic punctuation symbols, control codes that originated with Teletype machines and space. It was developed by the American National Standards Institute (ANSI)

ASCOCARP: The fruiting body of fungi belonging to the phylum Ascomycota producing spores known as ascus. An example is the spore of yeast. Ascus may be flask-shaped (perithecium), small opening (ostiole), cup-shaped (apothecium) or closed sphere (cleistothecium).

ASCOGENOUS HYPHAE: The hyphae, in fungi of the Ascomycota that grows from the ascogonium after it has been fertilised by the antheridium and ultimately gives rise to the asci (ascus).

ASCOGONIUM: The large coiled multinucleate female gametangium or reproductive organ of ascomycetous fungi. It is fertilised either by antheridial contents or by spermatia from which ascogenous hyphae develops.

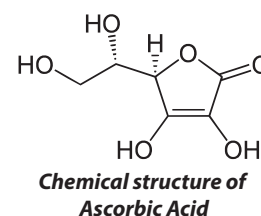
ASCOLI'S THEOREM: The result that a family of functions that is point-wise bounded and on a compact space is totally bounded in the supremum norm. Every sequence in such a family will contain a norm convergent subsequence. In the complex case, this is known as the **Arzela-Ascoli theorem**. It is named after the Italian analyst Giulio Ascoli (1843-96).

ASCOMYCETES: A class of fungi in which sexual reproduction produces haploid ascospores, eight of which are characteristically produced within a spherical or cylindrical cell, the ascus. The asci are usually contained in a fruit-body called ascocarp.

ASCOMYCOTA (SAC FUNGI): A subdivision of the *Eumycota*, class *Ascomycetes*, whose members reproduce sexually by ascospores inside a sac-like cell called an ascus (asci) during sexual reproduction. It

contains about 16500 species in almost 2000 genera. Its members include yeast, some species of edible fungi, etc.

ASCORBIC ACID (VITAMIN C): A white or light yellow solid organic compound with antioxidant properties found in fresh fruits and vegetables. Its chemical formula is $C_6H_8O_6$, melting point 190-192 °C, 463-465 K and density of 1.65 g/cm³. It is prepared industrially from glucose in a method based on the Reichstein process. It acts as a cofactor in the hydroxylation of proline to hydroxy-proline. Overcooking of vegetables reduces their vitamin C content. Lack of it causes skin sores or ulcers, mouth and gum problems, scurvy, etc.



ASCOSPORE: One of usually eight sexually produced fungal spore formed within an ascus. The two nuclei of the dikaryotic ascus initially fuse and undergo meiosis to produce four haploid daughter nuclei. These then undergo one mitotic division resulting in eight haploid nuclei. The cytoplasm of the ascus is then isolated around each nucleus to form the ascospores. They grow well under a variety of conditions including growing indoors on damp materials. Many are known to be plant pathogens.

ASCUS: A membranous, often club-shaped specialised cell in fungi of the phylum Ascomycota which normally contains eight ascospores, produced by a meiotic cell division followed, in most species, by a mitotic cell division.

ASEPALOUS: It refers to a plant without sepals or calyx.

ASEPSIS: The practice of ensuring that bacteria are excluded from open sites during surgery, wound dressing, blood sampling and other medical procedures to prevent infection.

ASEPTIC: A term describing equipment or a place that is free from pathogens and other harmful microbes. **Aseptic techniques** are procedures performed under strict sterile conditions and commonly performed, for example, in microbiology, (e.g., transferring cultures, inoculating media, isolation of pure cultures, etc.), surgical operating theatres, clinics, outpatient care centres, etc.

ASEXUAL REPRODUCTION: A mode of propagation, in which a part or parts of only one parent plant is used; or reproduction that does not involve the fusion of sex cells. Asexual reproduction may be achieved by any of the following: (i) **budding**, which is growing out and eventual splitting off of a new individual, as in Hydra; (ii) **binary fission**, which is the simple division of a single individual, as seen in bacteria or protozoa; (iii) by **spore formation** as seen in fungi or; (iv) **vegetative reproduction** in the higher plants in which new plants are formed by rooting stem or leaf cuttings, bulbs or by layering a stem or dividing roots.

ASH: 1. The solid residue which remains after the combustion of a fuel such as coal. Ash consists largely of treated mineral matter, but it may contain some products of the incomplete combustion of the fuel as well. 2. Any of various chiefly deciduous ornamental timber trees of the genus *Fraxinus*, having opposite, pinnately compound leaves, clusters of small flowers and one-seeded winged fruits. 3. A pulverized particulate matter ejected by volcanic eruption. 4. The mineral residue of incinerated organic matter, used as an additive in pet foods.

ASHING: An analytical process in which the chemical material being analysed is oven-heated to leave only noncombustible ash.

ASIC: In computing, it is the acronym for **Application Specific Integrated Circuit**, which is an integrated circuit built for a specific application, such as a particular kind of transmission protocol or a hand-held computer.

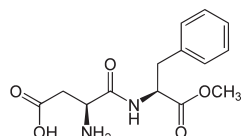
ASP: In computing, it is the acronym for **Active Server Pages** (an HTML (Hypertext Markup Language) page that includes one or more scripts (small embedded programs) that are processed on a Microsoft

Web server before the page is sent to the user) or **Application Service Provider** (a company or organisation that provides software applications to customers over the Internet).

ASPARAGINE: An uncharged polar amino acid with the chemical formula $C_4H_8N_2O_3$. It is found in many proteins and present in large amounts in some plants, such as asparagus, that is easily hydrolysed to aspartic acid. Because ammonia is extremely toxic to living cells, it is used to synthesise glutamine and asparagine, in which form it can be stored for later use in amino acid synthesis.

ASPARTAME (N-(L- α -ASPARTYL)-L-PHENYLALANINE; 1-METHYL ESTER): A non-carbohydrate sweetener used in food

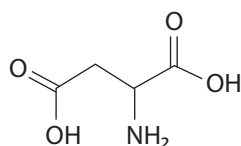
under various trade names such as Naturesweet, Canderel and Equal. Aspartame consists of two amino acids, aspartic acid forming about 40% by weight and phenylalanine forming about 50% combined with a methyl ester. Its molecular formula is $C_{14}H_{18}N_2O_5$ and molecular weight is 294.31. It is about 200 times as sweet as sugar and unlike saccharine, has no after taste. Aspartame loses its sweetness when heated and it is usually not suitable for baking. Aspartame is favoured as a sweetener by those on low carbohydrate diet such as diabetics, some athletes and actors/actresses and all other people who need to watch their weights. Drinks such as Diet Coca Cola, Coca Cola Zero, Diet Pepsi, some chewing gums such as Wrigleys Airwaves and Orbit and some other store-bought foods such as cereals contain aspartame. Some other artificial non-caloric sweeteners include saccharin, sucralose, neotame and acesulfame potassium. Some other natural product sugar substitutes include sorbitol or xylitol.



Chemical structure of Aspartame

ASPARTIC ACID (2-AMINOBUTANEDIOIC ACID; AMINOSUCCINIC ACID, ASPARAGIC ACID, ASPARGINIC ACID): An acidic amino acid with the formula $C_4H_7NO_4$,

density 1.7 g/cm^3 , melting point 270°C (543 K) and boiling point of 324°C , 597 K (decomposes). It is formed by a transamination, in which the amino group of glutamic acid is transferred



Chemical Structure of Aspartic Acid

to oxaloacetic acid so forming aspartic acid. Breakdown is by the reverse reaction, followed by further oxidation in the TCA cycle. Aspartate is an important precursor of nitrogenous compounds. The amino acids leucine, isoleucine, threonine, methionine and lysine are all derived from aspartic acid and it is necessary for synthesis of purines, pyrimidines and porphyrins. Aspartate also acts as an amino-group donor in several reactions.

ASPECT: The apparent position of any of the planets or the Moon relative to the Sun, as seen from Earth. The four main aspects are conjunction, greatest elongation, opposition and quadrature.

ASPECT DIVERSITY: It is the measure of the different physical appearances that are found in a group of species living in a common habitat and are hunted by other animals that use visual hunting skill to identify and kill their prey.

ASPECT EXPERIMENT: A two-photon correlation experiment conducted by French physicist, Alain Aspect and his colleagues in 1982 to investigate the validity of the entanglement effect in quantum mechanics by using some kind of Bell inequality. Bell's Theorem states that a Bell inequality must be obeyed under any local hidden variable theory but can in certain circumstances be violated under quantum mechanics. The experiment involves producing photons from a source

of excited calcium ions. A Bell test experiment is one designed to test whether or not the real world obeys a Bell inequality.

ASPECT RATIO: An image projection attribute that describes the proportional relationship between the width of an object and its height. It is commonly expressed as two numbers separated by a colon, as in 16:9. For example, movies which are usually short with a wide angle lens have an aspect ratio that is typically 16.9, which means that the width of the image area is almost twice its height.

ASPERGILLALES (PLECTASCALES; EUROTIALES): An order of the Plectomycetes containing about 150 species (80 genera) of various moulds having a closed ascocarp (cleistothecium) with the asci scattered rather than gathered in a hymenium.

ASPERGILLOSIS: A disease affecting mucous membranes, lungs and sometimes bones that is caused by infection with the fungus *Aspergillus*. Symptoms vary depending on the type of infection and where it is in the body. Common symptoms include coughing up blood or mucus and a high temperature.

ASPERITY: A tiny hair-like prickle or projection of an epidermal cell.

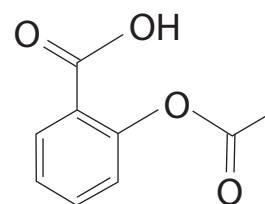
ASPEROUS: Rough to touch, scabrous. An example is a plant having a rough surface due to a covering of short stiff hairs.

ASPHALT (BITUMEN): A black or brown petroleum-like material that has a consistency varying from viscous liquid to glassy solid. It is obtained either as a residue from the distillation of petroleum or from natural deposits. Asphalt consists of compounds of hydrogen and carbon with minor proportions of nitrogen, sulfur and oxygen. It softens when heated and is elastic under certain conditions. It is used principally in road surfacing. Asphalt is also used for roofs, coatings, floor tilings and waterproofing and in industrial products.

ASPHYXIA: A condition due to the deficiency of oxygen and excess of carbon dioxide in the blood and body tissues. Symptoms of asphyxia vary but may include light-headedness, nausea and gasping, followed by unconsciousness and death. Causes of asphyxia include drowning, suffocation or strangling; inhalation of toxic gases; obstruction of the larynx, trachea or bronchi, which can occur in severe cases of asthma or diphtheria.

ASPIRATOR: 1. Any device for removal of fluids, gases or solids from a cavity by vacuum. Examples are squeeze bulb, pump, Venturi, etc. 2. Also called a **pooter**, it is a device used for drawing out small insects from the crevices of walls by suction.

ASPIRIN (ACETALSALICYLIC ACID): A naturally occurring carboxylic acid, with the chemical formula $CH_3COOC_6H_4COOH$, density 1.40 g/cm^3 , melting point 135°C (408 K) and boiling point 140°C (413 K) and decouples. Aspirin is part of a group of medications called nonsteroidal anti-inflammatory drugs (NSAIDs). It is commonly used in tablet form to relieve pain and reduce fever and inflammation and also as an antiplatelet agent by inhibiting the production of thromboxane. It is also used long-term, at low doses, to help prevent heart attacks, strokes and thrombosis or blood clot formation in people at high risk of developing blood clots. Low doses of aspirin may be given immediately after a heart attack to reduce the risk of another heart attack or of the death of cardiac tissue. The undesirable side-effects of aspirin taken by mouth include gastrointestinal ulcers, stomach bleeding and tinnitus, especially in higher doses.



Chemical Structure of Aspirin

ASS: Any of the several hoofed mammals (*Equus asinus*) of the genus *Equus* belonging to the order Perissodactyla. It is generally smaller than the horse but with longer ears. It is used mainly as a beast of burden.

ASSAY: 1. The determination of the quantity of a given substance present in a sample. It usually refers to determining the purity of precious metals. 2. In metallurgy, also known as **assaying**, it refers

to the amount of metal present in a sample of material such as ores and alloys; especially the proportion of gold, silver or platinum present. 3. In pharmacology, it is the estimation of the strength of a drug or substance by comparing its effects in test animals to a reference standard.

ASSEMBLER: A program in computing that translates a program written in an assembly language into a complete machine code program that can be executed by a computer. Each instruction in the assembly language is translated into only one machine code instruction.

ASSEMBLY LANGUAGE: A low-level computer programming language closely related to a computer's internal codes. It consists, chiefly of a set of codes, which are translated by a program called an assembler, into machine code for the computer's Central Processing Unit (CPU) to execute directly.

ASSESSMENT INSTRUMENT: Particular methods developed that are typically used in the evaluation of ability, achievement, interests, personality, or some other factor. For example, a school can use assessment instrument to gather information about student achievements. Such information may include the following: (i) knowledge (remembering previously learned information), comprehension (demonstrating an understanding of the facts), (iii) application (applying knowledge to actual situations), (iv) analysis (breaking down objects or ideas into simpler parts and find evidence to support generalisations), (v) synthesis (compiling component ideas into a new whole or propose alternative solutions), and evaluation (making and defend judgments based on internal evidence or external criteria).

ASSIGNMENT: 1. In logic, a function that associates specific elements of domain with each free variable in a formal calculus. 2. In computing, a statement in a program that allocates a value to a variable usually written as in $x := y + z$. If the new value is a function of the previous value of that same variable the notation is still used; thus $x := x + 1$ is an instruction to increase the value of x by one.

ASSIGNMENT PROBLEM: One of a class of combinatorial optimisation problems in the branch of optimisation or operations research in mathematics derived from the problem of matching individuals and tasks, often in order to maximise job satisfaction or some other measure of appropriateness.

ASSIMILATION (BIOLOGICAL ASSIMILATION; BIO-ASSIMILATION): A combination of two processes to supply animal cells with nutrients or tissues. Assimilation occurs in every cell of the body to help develop new cells. The first process is absorption of vitamins, minerals and other chemicals from food within the gastrointestinal tract. In humans this is achieved by chemical breakdown by enzymes and acids and physical breakdown by oral mastication and stomach churning. The second process is the chemical alteration of substances in the bloodstream by the liver or cellular secretions resulting in the absorbed food reaching the cells via the liver. Examples of assimilation in organisms are: (i) **Photosynthesis** in which carbon dioxide and water are transformed into a number of organic molecules in plant cells. (ii) **Nitrogen fixation** from the soil into molecules by symbiotic bacteria which live in the roots of certain plants, such as Leguminosae. (iii) The absorption of nutrients into the body after digestion in the intestine and its transformation in biological tissues and fluids.

ASSIMILATORY NITRATE REDUCTION: One of the biological pathways in nitrate reduction that takes a nutrient in the soil, moves it into the cell and uses it for biosynthesis of macromolecules. In other words, it is a biological process in which a highly oxidised form of inorganic nitrogen is reduced to ammonia by plants, fungi and certain bacteria that cannot fix nitrogen gas.

ASSOCIATE NUMBERS: Any pair of elements of an integral domain that are unit multiples of each other, such as $3 + i$ and $3i - 1$ in the Gaussian integers.

ASSOCIATED IONS: Species formed by the collision of dissolved ions of opposite charges that live for a short time.

ASSOCIATION: 1. An ecological phenomenon in which two or more species occur in closer proximity to one another than would be expected on the basis of chance. 2. In chemistry, it is the combination of molecules of one substance with those of another to form chemical species that are held together by forces weaker than normal chemical bonds. 3. In mathematics, it is also known as **correlation**, which is the degree of relationship between two sets of information. If one set of data increases at the same time as the other, the relationship is said to be positive or direct and if one set of data increases as the other decreases the relationship is said to be negative or inverse. Mathematically, a correlation is expressed by a correlation coefficient that ranges from -1 (never occur together), through 0 (absolutely independent), to 1 (always occur together).

ASSOCIATION FOR COMPUTING MACHINERY, ACM: A US organisation made up of computer professionals of all types. It hosts the computing industry's leading Digital Library and serves its global members and the computing profession with journals and magazines, conferences, workshops, electronic forums and Learning Centre. Its monthly journal, the Communications of the Association for Computing Machinery, is peer-reviewed and serves its global members and the computing profession with journals and magazines, conferences, workshops, electronic forums and Learning Centre. Its subsidiary special interest groups or SIGGRAPH (graphics) and SIGCHI (human-computer interaction), focus on areas such as graphics and human-computer interaction. The equivalent UK organisation is the British Computer Society (BCS).

ASSOCIATION CENTRE: The part of the brain that deals with the interpretation and meaning of primary sensory impulses and links them with other parts of the brain such as the motor and memory areas.

ASSOCIATION REACTION (ASSOCIATIVE COMBINATION): The reaction of a (slow moving) ion with a neutral species wherein the reactants combine to form a single ionised species.

ASSOCIATIVE: 1. Of a binary operation, having the property that the bracketing of its arguments may be disregarded since $(a \bullet b) \bullet c = a \bullet (b \bullet c)$, where $\bullet$ is the operator. 2. Of an algebraic structure, possessing an associative operator. For example, in $a + (b + c) = (a + b) + c$, the operation indicated by $+$ is associative.

ASSOCIATIVE DINITROGEN FIXATION: An enhanced rate of dinitrogen fixation, brought about by a close relationship between free-living diazotrophic organisms and a higher plant.

ASSOCIATIVE IONISATION: This occurs when two excited gaseous atoms or molecular moieties interact and the sum of their internal energies is sufficient to produce a single, additive ionic product.

ASSOCIATIVE LAW: One of the laws of algebra, it is a mathematical law stating that the value of an expression is independent of the grouping of the symbols or terms in the expression under operation. For example, the associative law of addition states that numbers may be added in any order, for example, $(x + y) + z = x + (y + z)$, $(a + b) + c = a + (b + c) = (a + c) + b$ and for multiplication $(xy)z = x(yz)$, $x, y, z \in R$. But for subtraction and division, the law does not apply. For example, $(3 + 5) + 2 = 3 + (5 + 2) = (3 + 2) + 5 = 10$ and $(3 \times 5) \times 2 = 3 \times (5 \times 2) = (3 \times 2) \times 5 = 30$ but $(7 - 3) - 5 \neq 7 - (3 - 5) \Rightarrow -1 \neq 9$, $(16 \div 8) \div 4 \neq 16 \div (8 \div 2) \Rightarrow \frac{1}{2} \neq 8$,

A

ASSOCIATIVE OPERATION: An operation in mathematics in which the outcome is independent on the grouping of the numbers or symbols concerned. If $*$ is a binary operation defined on the set of real numbers R , then the operation $*$ is associative if $(x * y) * z = x * (y * z)$.

ASSOCIATIVE SYMBIOSIS: Interaction between two dissimilar organisms or biological systems, which is normally mutually beneficial.

ASSORTATIVE MATING: Selective mating in a population between individuals that are genetically related or have similar characteristics. For example, in humans, a person may choose a mate according to religious, cultural or ethnic preferences, professional interests or physical traits. If similar phenotypes breed together it is called **positive assortive mating** and leads to **inbreeding**. If dissimilar phenotypes breed together it is called **negative assortive mating** and leads to **outbreeding**. If sufficiently consistent, assortative mating can theoretically result in the evolution of new species without geographical isolation.

ASSUMPTION: A statement that is presumed to be true on a particular circumstance. Any conclusions reached are dependent on the validity of the assumption. In other words, it is a proposed explanation that is still conjectural and subject to experimentation. It contrasts to well-thought out propositions that are factual.

ASSURGENT (ASCENDING; ASCENDENT): In biology, it is a growth, which goes obliquely upwards and is usually curved.

ASTABLE: In electronics, it describes a transistor switching circuit in which the output is not stable in either the on or the off state, but continually switches between the high and low states by itself, producing a square wave. When connected to a LED, an astable timer gives a continuously flashing light.

ASTATIC GALVANOMETER: A sensitive form of a moving-magnet galvanometer, which is independent of the Earth's magnetic field. It has two magnetic needles parallel to each other, with opposite polarities. The needle assembly is suspended by a silk thread and has no net magnetic dipole moment. The lower needle is inside the current sensing coils and is deflected by the magnetic field created by the passing current. It is used for measuring small currents.

ASTATINE: A non-metallic unstable radioactive element with the symbol At, atomic number 85, melting point 302 °C (575.15 K) and boiling point 337 °C (610.15 K). It was discovered in 1940. Its most stable and longest lived isotope has a mass number of 210 and a half-life of 8.3 hours. It is the heaviest in the halogen group. Very rare in nature, it is obtained by bombardment of bismuth with alpha particles. It is used in medicine as a tracer element and in cancer treatment. Like iodine, it is known to accumulate in the thyroid gland.

ASTEMONOUS (ANANDROUS): A term used for flowers that do not have stamens.

ASTER: 1. Any of various plants of the genus *Aster* in the composite family, having radiate flower heads with white, pink or violet rays and a usually yellow disk. 2. A star-shaped structure formed by fibrils radiating from a centriole. Asters become conspicuous at the poles of the spindle as cell division commences in some primitive plant cells.

ASTERACEAE (COMPOSITAE; SUNFLOWER FAMILY): The largest family of flowering plants in the major group Angiosperms (Flowering plants) of about 12,000 species worldwide, consisting of numerous economic plants. Asteraceae was formerly known as Compositae. Most members are herbaceous, but a significant number are also shrubs, vines and trees. The stem may be solid or hollow and it may be prickly. The roots may be short and near the surface, as in the Daisy or there may be a long taproot, as in the Dandelion or Thistle. The leaves are alternate, opposite or less commonly whorled and range from simple to pinnately or palmately compound; with no stipules. They are characterised by composite flower type in the form of capitula surrounded by involucre bracts. The inflorescence, called a capitulum or head consists of several small flowers, called florets, that are crowded together, sessile, on a receptacle. This inflorescence

is often mistaken for a single flower. The flowers are of two basic types: those with tubular actinomorphic corollas and those with strap-shaped or radiate zygomorphic corollas, often within the same head. Either type may be bisexual or unisexual. The fruit often possesses a ring of hairs, the pappus, to aid dispersal. Compositae is important primarily for its many garden ornamentals, such as ageratum, asters, chrysanthemums, cosmos, dahlias, marigolds (*Tagetes*) and zinnias. Some genera include noxious weeds, such as dandelion (*Taraxacum*), ragweed (*Ambrosia*) and thistle (*Carduus*, *Cirsium* and others). Several other members have economic importance as food crops. Artichokes (*Cynara*), lettuce (*Lactuca*), endive (*Cichorium*) and salsify (*Tragopogon*) are commonly eaten as vegetables and the edible seeds of safflower (*Carthamus*) and sunflower (*Helianthus*) are used in the production of cooking oils.

ASTERIDAE (SYMPETALAE): A subclass of the dicotyledons containing mostly herbaceous plants having flowers with fused petals. They have few stamens and usually only two carpels and the seeds are surrounded by only one integument. Examples are milkweeds, forget-me-nots, honeysuckles and plantains.

ASTEROID (MINOR PLANET; PLANETOID): Any small planet with a diameter of 10 metres up to 1,000 km that orbits the Sun or another star (usually between Mars and Jupiter). The largest ones are spherical in shape, and the smaller ones are irregular.

ASTEROID BELT: A region between the orbits of Mars and Jupiter where the majority of asteroids in the Solar System are to be found. The main asteroid belt extends from 2.15 to 3.3 astronomical units (255 to 600 million km) from the Sun – between Mars (1.5 AU) and Jupiter (5.2 AU) – and may contain over a million objects bigger than 1 km across, the largest being Ceres (1,003 km), Pallas (608 km) and Vesta (538 km). Within the belt, the distribution of asteroids is non-uniform, with concentrations in asteroid groups and families and also relatively empty zones known as Kirkwood gaps.

ASTEROID IMPACTS: A hypothesis that links certain mass extinction events with the impact of a comet or asteroid, most notably the mass extinction 65 million years that caused the disappearance of dinosaurs and many other reptilian groups. Asteroid impacts early in earth history also contributed to the formation of the atmosphere and oceans.

ASTEROIDEA (SEA STARS; STARFISH): A class of echinoderm of the phylum *Echinodermata*, which includes the starfish; and are important members of many marine benthic communities. They are found throughout the world's oceans and are familiar along shorelines, where sometimes large numbers may aggregate on the rocks. They also live among coral reefs. They are star-shaped with five or multiple of five arms, although some species have more. There is no clear distinction between the arms and the central body. The aboral or upper surface may be smooth, granular or spiny and is covered with overlapping plates. Most starfish are predators, feeding on sessile or slow-moving prey such as molluscs and barnacles.

ASTHENOSPHERE: The solid, plastic layer of the mantle beneath the lithosphere and the upper mantle below with depths of between 80 and 200 km below the earth's surface. It is made of mantle rock and has so much pressure and heat that the rocks become partially plastic and can flow like a liquid. Because it flows very slowly, it allows tectonic plates to move on top of it.

ASTHMA: A chronic respiratory disorder characterised by wheezing and coughing due to spasmodic contraction of the bronchi. There is no single cause of asthma, but certain things such as genetics and the environment may increase the likelihood of developing it. Almost all asthma sufferers have allergies. Allergic reactions triggered by antibodies in the blood often lead to the airway inflammation that is associated with asthma. All asthma sufferers have allergies. Depending on individuals, allergies include tobacco smoke, dust, grass, pollens, animal fur or feathers, mould, mites, weather conditions, etc.

ASTIGMATISM: A visual defect in which the lens has different focal lengths for rays of different orientations, resulting in a distortion of the image. Rays of light from horizontal and vertical lines in a plane on the object are not focused to the same plane on the edges of the image.

ASTON, FRANCIS WILLIAM (1877-1945): British chemist and physicist, who worked with J.J. Thompson at Cambridge University. He was awarded a Nobel Prize for Chemistry for designing the Mass Spectrograph.

ASTRAL: 1. Of or emanating from stars. 2. In biology, of, relating to or shaped like the mitotic aster; star-shaped.

ASTROBIOLOGY (EXOBIOLGY; BIOASTRONOMY): The study of life throughout the universe, which includes the origin, evolution, distribution and future of life. It seeks to understand the origin and evolution of life on earth, to determine if life exists elsewhere in the universe and to predict the future of life on earth and in the rest of the universe.

ASTROBLEME: An ancient circular crater in the Earth's crust created by the impact of a meteorite as a result of explosions of the meteorite on impact and characteristically gives rise to shutter cones in the adjacent rocks.

ASTROCHEMISTRY: The study of molecules in interstellar space. Interstellar molecules are usually detected by their spectra in the radio, microwave or infrared regions of the electromagnetic spectrum. Over 140 different molecules have been detected.

ASTROCYTE: A star-shaped glial cell found in the central nervous system (CNS). Astrocytes have many long arms or processes and are the largest and most numerous glial cells in the CNS; they outnumber neurons 10 to 1.

ASTRODYNAMICS: The science concerned with all aspects of the motion of satellites, rockets and spacecraft. It involves the practical application of celestial mechanics, astrobballistics, propulsion theory and allied fields.

ASTROID: A hypocycloid (the path of a point on a circle rolling inside another circle) with the diameter of the fixed circle four times the diameter of the rolling circle. It has four cusps.

ASTROLABE: An ancient navigational instrument used to determine the altitude of the Sun or other celestial bodies, now replaced by the sextant. It usually consists of a flat disc with a sighting rod that could be pivoted to point at the sun or brightest star.

ASTROLOGY: The study of the position of the stars and movement of planets in the belief that it influences human affairs and can be used to predict the destiny of humans. It originated in Mesopotamia in modern-day Iraq around 3000 BC.

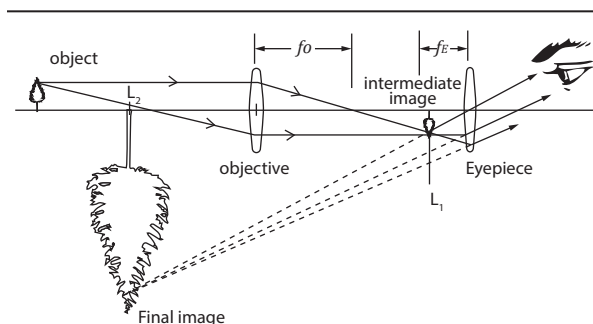
ASTROMETRY: The branch of astronomy concerned with the measurement of the precise positions of the celestial bodies in space. The information obtained by astrometric measurements provides information on the kinematics and physical origin of our Solar System and galaxy, the Milky Way.

ASTRONAUT (COSMONAUT): A person trained to pilot, navigate or otherwise participate as a crew member of a spacecraft. The term commonly refers to those participating in U.S. space missions. **Cosmonaut** is the Russian equivalent and **taikonaut** is the Chinese equivalent.

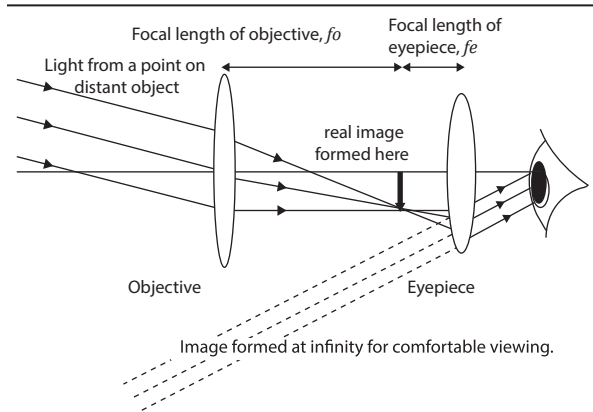
ASTRONAUTICS: The application of scientific principles, engineering techniques or technology to space travel. Astronautics encompasses both manned and unmanned missions.

ASTRONOMICAL CLOCK: A highly accurate clock that keeps either sidereal time (time measured relative to the fixed stars) or Universal Time (mean solar time for Greenwich meridian or longitude 0°).

ASTRONOMICAL TELESCOPE: A balloon-borne telescope designed for gathering light and viewing astronomical objects. It consists of two converging lenses, as the objective lens and the eyepiece lens. The objective lens has long focal length whereas the eyepiece lens has short focal length. This ensures that the telescope has a high magnifying power. The objective lens collects parallel rays from a distant object and forms a real inverted and diminished image L_1 at the principal focus F_0 of the lens. The eyepiece then acts as a magnifying lens and forms a magnified virtual image of L_1 at L_2 . When the principal focus of the objective lens is arranged to coincide with that of the eyepiece lens, the final image is formed at infinity. The telescope is used to view distant objects such as stars and planets.



Astronomical Telescope with image formed at near point



Astronomical telescope with final image formed at infinity

ASTRONOMICAL UNIT (AU): A unit of distance used in astronomy and defined by the International Astronomical Union as the length for which the Gaussian gravitational constant (k) takes the value 0.017,202,098,95, when the units of measurement are the astronomical units of length, mass and time. It is equal to about 149,597,870.7 kilometres, which is the mean distance between the Sun and the Earth. Light travels this distance in approximately 8.3 minutes. It is a derived constant and used to indicate distances within the solar system.

ASTRONOMY: The scientific study of the universe beyond the Earth's atmosphere or the scientific study of celestial bodies and all other objects in the universe. These include the Sun, Moon, Stars and the Planets.

ASTROPHYSICS: The scientific study of the properties and structures of cosmic objects, including the universe as a whole. It is the study of physics and astronomy that deals principally with the physical processes of celestial bodies and phenomena. Some concerns of astrophysics are how much light the stars give off and the size, mass and temperature of planets and stars.

ASTROSCLEREID (ASTEROSCLEREID; STAR SCLEREID): A relatively short sclereid cell that tends to be radiately branched but is otherwise quite variable, occurring in the leaves of many plants and

in the petioles of Camellia. Astroclereids are usually present singly or in small groups and are often found in the mesophyll of leaves, where they act as a strengthening agent.

ASTROSEISMOLOGY: The study of the internal structure and evolution of stars through the interpretation of their pulsation.

ASTROSOCIOLOGY: The study of astrosocial phenomena or social and cultural patterns related to space. It examines human behaviour related to space and how these activities influence a particular society as well as human societies in the international community.

ASTYLOCARPELLOUS: A flower without a style and a stipe.

ASTYLOCARPEPODIC: A flower without a style but with a stipe.

ASTYLOUS: A flower lacking a style.

ASYMMETRIC: 1. Not symmetric, so that two sides are not identical on both sides of a central line or the property of objects being invariant to transformation for example reflection. 2. Of a structure or organ, not divisible into two equal halves, as in some leaves. 3. In logic, of a relation never holding between a pair of elements in one order when it holds between them in the other; i.e ordered pairs of elements (x, y) and (y, x) can never both have the relation

ASYMMETRIC CARBON ATOM: A carbon atom that is covalently bonded to four different groups with no two of them alike. Compounds containing such a carbon atom are capable of existing in two optically active forms which are distinguished by being respectively levorotatory and dextrorotatory and also in some cases by having enantiomorphous crystal forms. The asymmetric carbon atom is the one attached to two carbon atoms, an oxygen atom and a hydrogen atom. For example, malic acid has four carbon atoms but just one of them is asymmetric. An example of asymmetric carbon atom is the simplest sugar glyceraldehydes, which is an aldotriose with three carbon atoms. An aldotetrose with four carbon atoms is also a good example of compounds with two asymmetric carbon atoms.

ASYMMETRIC COMMUNICATIONS: In telecommunications, the term refers to any system in which the data speed or quantity differs in one direction as compared with the other direction averaged over time.

ASYMMETRIC CRYPTOGRAPHY (PUBLIC-KEY CRYPTOGRAPHY): A cryptography in which a pair of keys is used to encrypt and decrypt a message so that it arrives securely. Initially, a network user receives a public and private key pair from a certificate authority. Any other user who wants to send an encrypted message can get the intended recipient's public key from a public directory. They use this key to encrypt the message and they send it to the recipient. When the recipient gets the message, they decrypt it with their private key, which no one else should have access to.

ASYMMETRIC DIGITAL SUBSCRIBER LINE (ADSL): A form of Digital Subscriber Line technology that enables faster data transmission or broadband communication via the internet by using pre-existing copper telephone lines, while allowing simultaneous use of line for normal telephone calls by utilising a DSL filter or splitter. This is achieved by utilising frequencies that are not used by a voice telephone call. ADSL can generally only be distributed over short distances of between 4 km to 8 km but can reach over 8 km if the originally laid wire gauge allows for further distribution. It is generally used to connect a home to an ISP.

ASYMMETRIC FILM: A film bounded by two different bulk phases. When the bulk phases are identical the film is described as **symmetric**.

ASYMMETRIC INDUCTION: A synthesis of a particular enantiomer or diastereoisomer of a chiral molecule using regioselective and stereoselective reagents.

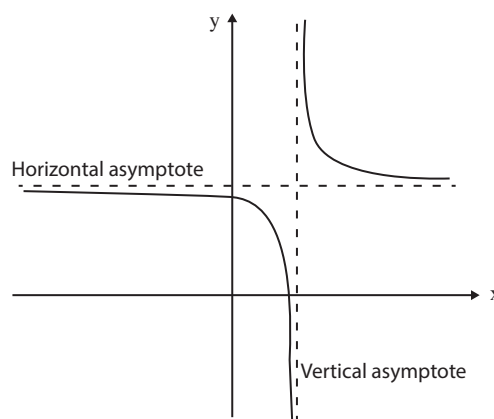
ASYMMETRIC MEMBRANE: A membrane constituted of two or more structural planes of non-identical morphologies.

ASYMMETRIC RELATION: A binary relation $*$ on a set S is asymmetric, if, for all x, y in S , whenever $x * y$, then $y * x$ does not hold. For example, the relation $<$ on the set of integers is asymmetric.

ASYMMETRIC TRANSFORMATION (DERACEMIZATION): The conversion of a racemate into a pure enantiomer or into a mixture in which one enantiomer is present in excess or of a diastereoisomeric mixture into a single diastereoisomer or into a mixture in which one diastereoisomer predominates.

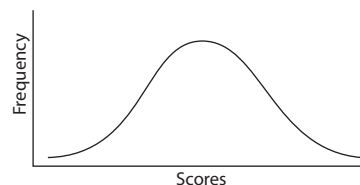
ASYMMETRICAL: 1. Having parts which fail to correspond to one another in shape, size or arrangement; lacking symmetry. 2. Of or relating to a carbon atom having four different atoms or structural groups attached to it, resulting in an unbalanced spatial arrangement of atoms in a molecule.

ASYMPTOTE: A line or a curve that approaches a given curve arbitrarily closely. In other words, it is a straight line whose perpendicular distance from a curve decreases to zero as the distance from the origin increases without limit. For example, a curve in the form $y = R(x)$, where $R(x)$ is a rational function of x and its asymptotes are at the point where the denominator disappears or where the function ceases to exist. A function $y = 1/(x - 1)$ does not exist at $x = 1$, therefore, the line $x = 1$ is an asymptote to the curve $y = 1/(x - 1)$. A line, $x = a$ to a curve is a vertical asymptote, the line $y = b$ to a curve is a horizontal asymptote and the line $y = mx + b$ to a curve is an oblique asymptote. The graph of $y = 1/(x - 1)$ with the asymptote $x = 1$ is a tangent to the curve at infinity.



Asymptote

ASYMPTOTIC: 1. Of a function, or series, approaching a given value, the asymptote, closely, as the independent variable or some expression containing a variable, approaches a limit or tends to infinity. Graphically, the perpendicular distance between a curve and its asymptote tends to zero as the distance from the origin tends to a limit or infinity.



Normal Curve

For example, a normal curve, also known as normal distribution curve or Gaussian curve is asymptotic as shown as shown in the diagram. 2. Of a formula, becoming increasingly exact as a variable approaches a limit, usually infinity. 3. Of a pair of functions, approaching each other infinitely closely as their arguments tend to a certain value, normally infinity. This is usually expressed as $f(x)$ is asymptotic to $h(x)$

at u if $\lim_{x \rightarrow u} \frac{f(x)}{h(x)} = 1$.

ASYMPTOTIC DENSITY (NATURAL DENSITY): A method for measuring the size of natural numbers. A subset A of positive integers has asymptotic density α , where $0 \leq \alpha \leq 1$, if the proportion of elements of A among all natural numbers from 1 to n is asymptotic to α as n tends to infinity.

ASYMPTOTIC DIRECTION: At a point on a surface, any direction in which normal curvature vanishes, the contact of the tangent plane being of at least third order.

ASYMPTOTIC EQUIPARTITION PROPERTY: A general property used extensively in information theory concerning the output samples of a stochastic data source. Roughly speaking, it says the typical output samples of a stochastic data source are all equally likely.

ASYMPTOTIC FREEDOM: The property of the theory of quantum chromodynamics which predicts that the intrinsic strength of the force between quarks decreases as they approach each other.

ASYMPTOTIC FUNCTION: Two functions are said to be asymptotic if they approach each other infinitely closely as their argument tends to a certain value, normally infinity. This is usually expressed as $f(x)$ is asymptotic to $h(x)$ at u if $\lim_{x \rightarrow u} (f(x)/h(x)) = 1$.

ASYMPTOTIC NORMALITY: The distribution of statistics is said to be asymptotically normal if, as the sample size increases, the distribution of the statistics approaches a normal distribution.

ASYMPTOTIC SERIES: A series formed by the expansion of a function in a certain form, such that the error resulting from terminating the series at a certain term tends to zero more rapidly than a certain other, as the independent variable x tends to infinity. An asymptotic series expansion is not necessarily a convergent

series. For example, a series of the form $a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_n}{x^n} + \dots$

is an asymptotic series of the function $f(x)$ if there exists a number N such that for all $n \neq N$ the quantity $x_n[f(x) - S_n(x)]$ approaches zero as x approaches infinity, where $S_n(x)$ is the sum of the first n terms in the series.

ASYNCHRONOUS (ASYNCR): 1. Pertaining to two or more processes that do not depend upon the occurrence of specific events such as common timing signals. 2. Without regular time relationship; unexpected or unpredictable with respect to the execution of program instructions.

ASYNCHRONOUS COMMUNICATION: A method of communication supported by the operating system that allows an exchange of data with a remote device, using either a start-stop line or an X.25 line. Asynchronous communication includes the file transfer support and the interactive terminal facility support.

ASYNCHRONOUS TRANSFER MODE (ATM): A networking technology that transfers data in packets or cells of a fixed size, using 53-byte cells (5 bytes for the address header and 48 bytes for the data). These extremely small cells can be processed through switches fast enough to maintain data transfer speeds of over 600 mbps. The technology was designed for the high-speed transmission of all forms of media from basic graphics to full-motion video.

ASYNCHRONOUS TRANSMISSION (START/STOP TRANSMISSION): A data transmission where the communication can start and stop at any time. Data sent through an asynchronous transmission contains a start bit and stop bit, helping the receiving end know when it has received all of its data.

ASYSTOLE: Inability of the heart to beat, which may be due to cardiac arrest.

AT REST: The state of an object when it is not in motion relative to a particular frame of reference or another object. An object at rest will remain at rest until a force makes it move due to inertia.

ATACTOSTELE: A stele, typical of monocotyledon stems, in which the vascular bundles are arranged more or less irregularly in the ground tissue.

ATANASOFF-BERRY COMPUTER (ABC): One of the first electronic digital computers and the first machine to use vacuum tubes. It was developed in 1937 by Prof. John Vincent Atanasoff (1903 - 1995) and his student, Cliff Berry in 1937 at the Iowa State College (now Iowa State University). It incorporated binary arithmetic, regenerative memory, parallel processing and separation of memory and computing functions.

ATAPI: Acronym for **Advanced Technology Attachment Packet Interface**, which is a type of hardware connected between hard

disks, CD-ROMs and tape drives and other computer peripherals to a PC, to support the additional commands needed for controlling a CD-ROM player or tape backup so that the computer can use the IDE interface and controllers to control these relatively newer device types.

ATAVISM: 1. A term used in genetics to denote the reappearance of a characteristic not apparent in the immediately preceding generations. 2. A term used in psychology to denote the manifestation of primitive forms of behaviour.

ATAXIA: The lack of coordination between muscles resulting in unsteady gait, difficulty in fine movements and speech disorder. Ataxia can be brought on by an injury, infection or degenerative disease of the central nervous system, for example, syphilis, encephalitis, brain tumour or multiple sclerosis.

ATAXITE: A rare variety of iron meteorite (designated type D) made almost entirely of taenite, a solid solution of iron and 27% to 65% nickel.

ATHERMALIZATION: The design of an instrument so that temperature changes do not affect what is being measured. For example, good displacement transducers are athermalized.

ATHEROSCLEROSIS: A disease of the arteries characterised by the deposition of fatty materials, cholesterol and other substances in and on the artery walls resulting in the hardening and narrowing of the artery walls and formation of plaque, which can restrict blood flow to the tissues and organs. If the plaque is not ruptured, it can cause coronary artery disease resulting in angina. If the plaque is suddenly ruptured causing blood to clot inside an artery at the site of the rupture, it can cause the heart muscle to die, called heart attack or myocardial infarction. It can also cause cerebrovascular disease resulting in strokes and brain injury.

ATHLETE'S FOOT: A contagious fungal infection of the feet, caused by a fungus, *Tinea pedis* resulting in inflammation, itching, blisters, cracking and scaling, especially between the toes. Treatment consists of foot hygiene, dusting powder, certain carboxylic acids and antifungal antibiotics.

ATIAH, SIR MICHAEL FRANCIS (1929 -): A British mathematician who contributed massively to the field of topology, geometry, analysis, the transcendental theory of algebraic varieties, differential operators and quantum field theory. He won the Fields Medal award in 1966 and the Abel prize in 2004.

ATLAS: 1. A book containing maps and vital statistics relating to geographic regions. 2. The first cervical vertebra in a mammal. It is a ring-like bone on which the skull sits. It permits, by its form, nodding movements of the head. 3. A collection of charts that cover a manifold. It is said to be a C^r -atlas if each pair of charts has a C^r -overlap. 4. One of the innermost of Saturn's moons. It was discovered by Richard Terrile in 1980 from Voyager 1 photos and is also known as Saturn XV. It has a diameter of $46 \times 38 \times 19$ km, a semi major axis of 137,670 km and an orbital period of 0.602 day (14 hr 27 mins).

ATM: 1. In computing, it is the acronym for **asynchronous transfer mode**, a networking technology that transfers data in packets or cells of a fixed size, using 53-byte cells. 2. Acronym for **automated teller machine**, a computerised outlet that allows bank customers to access accounts independently without the help of a bank representative to perform functions such as to withdraw cash, make deposits, and pay bills by the use of a magnetically encoded plastic card with a code or pin number (personal identification number).

ATMOLYSIS: The separation of a mixture of gases by means of their different rates of diffusion through a porous septum or diaphragm.

ATMOMETER: An instrument that measures the rate of water evaporation from a non-living wet surface, such as a porous pot.

ATMOSPHERE: 1. A mixture of gases which surrounds a planet, sun or moon inside a closed space held in place by the gravity of the body. The Earth's atmosphere is composed of approximately 20% oxygen, 78% nitrogen and 2% carbon dioxide and other gases, that covers the surface of the Earth. The Earth's atmosphere consists

of the **troposphere**, which includes the planetary boundary layer or peplosphere as the lowest layer; the **stratosphere**, includes the ozone layer, the mesosphere; the **thermosphere**, contains the ionosphere, the exosphere and the magnetosphere. 2. A unit of pressure, with the symbol atm, which is equal to 760 mmHg, or 1.0132×10^5 pascals (newton per square metre) or 1013.25 millibars and defined as the amount of pressure that will support a column of mercury 760 millimetres high at 0°C (273.15 K) under standard gravity. For practical purposes it has been replaced by the bar, which is 10^5 Pa.

ATMOSPHERIC CHERENKOV TELESCOPES (ACTs): Telescopes that are located on the ground to detect very-high-energy gamma-ray photons in the photon energy range of 50 GeV to 50 TeV from space. These gamma rays interact with atoms in Earth's atmosphere, producing Cherenkov Light.

ATMOSPHERIC CIRCULATION (GENERAL CIRCULATION): The large-scale movement of air together with ocean circulation by which heat is distributed on the surface of the Earth. It is caused by temperature differences in the atmosphere and the rotation of the Earth, which transfers heat from warm zones (the tropics) to cooler zones (toward the poles). The poleward circulation due to heat transference, mostly in the form of convection, gives rise to large-scale eddies such as cyclones and anticyclones, low-pressure troughs and high-pressure ridges. The eddies also take part in the longitudinal atmospheric circulation around the Earth, with the Earth's rotation maintaining easterly winds toward the equator and westerlies toward the poles. Narrow jet streams blow swiftly over middle latitudes, usually horizontally east in the stratosphere and move farther toward the poles during the summer.

ATMOSPHERIC DIFFUSION: The exchange of fluid parcels between regions in the atmosphere in the apparently random motions of a scale too small to be treated by equations of motion.

ATMOSPHERIC DISPERSION MODELLING: The mathematical simulation of how air pollutants disperse in the ambient atmosphere. It is performed with computer programs that solve the mathematical equations and algorithms, which simulate the pollutant dispersion. The dispersion models are used to estimate or to predict the downwind concentration of air pollutants or toxins emitted from sources such as industrial plants, vehicular traffic or accidental chemical releases.

ATMOSPHERIC POLLUTION (AIR POLLUTION): The introduction into the Earth's atmosphere of harmful by-products such as carbon dioxide, carbon monoxide, lead, nitrogen oxides, smoke and sulfur dioxide resulting from human activities that may adversely affect humans or other living organisms. Although there are many sources of air pollution, the most common are emissions from the burning of hydrocarbons or fossil fuels such as coal and oil products. Atmospheric pollution is a serious global problem and is especially problematic in large urban areas.

ATMOSPHERIC PRESSURE: The pressure exerted by the weight of the column of air above it at any point on the Earth's surface, which decreases with height. The pressure varies both with altitude and weather patterns. The standard is an average atmospheric pressure at sea level and is defined as 1 atmosphere on Earth, equal to 760 millimetres of mercury (760 Torr) and 101,325 Pascals. The standard that is due to the weight of air value at sea level in SI unit is 101325 Pa above it. It is measured with a barometer.

ATMOSPHERIC REFRACTION: The refraction of light rays passing through the Earth's atmosphere, due to variations in the density and temperature which produce corresponding variations in its refractive index. Atmospheric refraction gives rise to a shift in the apparent direction of a celestial object; the effect increases the observed altitude of an object and is greatest (just over half a degree) for objects on the horizon.

ATMOSPHERIC WINDOW: A range of electromagnetic wavelengths to which Earth's atmosphere is largely or partially transparent. All

spectral regions are affected to some extent by absorption in the atmosphere but there are two nearly transparent ranges, the optical window and the radio window and several narrow, partial infrared windows.

ATOLL: A coral reef, which encircles a shallow central lagoon. It may be continuous or move often, broken into closely spaced islets. It is mostly found in the Pacific Ocean and the sizes may vary to more than a hundred kilometres. The largest coral atoll in the world is the Kwajalein Atoll, in the Marshall Islands, which is 120km across. The reef and lagoon area in some of the atolls of the Maldives are larger than 3,000 km sq. For example, Saya de Malha Bank in the Western India Ocean is 35,000 km².

ATOM: 1. The smallest particle of an element that can exist and has the element's properties or the smallest unit of matter that can take part in a chemical reaction and that cannot be broken down chemically into anything simpler. It is a basic unit of matter that is made up of a dense central nucleus consisting of small dense positively charged protons and neutrons surrounded by negatively charged electrons. The nucleus of hydrogen-1 is the only stable nuclide with no neutrons. The electrons of an atom are bound to the nucleus by the electromagnetic force and a group of atoms can remain bound to each other to form a molecule. An atom containing an equal number of protons and electrons is electrically neutral, otherwise it has a positive charge if there are fewer electrons (electron deficiency) or negative charge if there are more electrons (electron excess). An atom whether positively or negatively charged is known as an **ion**. An atom is classified according to the number of protons and neutrons in its nucleus, with the number of protons determining the chemical element and the number of neutrons determining the isotope of the element. 2. A set, often a point, in a measure space with strictly positive measure and such that any subset of the set either has equal or zero measure. 3. A minimal non-zero element in a Boolean algebra.

ATOMATE (ATOMIFEROUS): In botany, possessing sessile or sub-sessile glands; or sprinkled with small particles; or having a surface covered in tiny shiny particles; powdered.

ATOMIC: 1. Extremely small; minute. 2. Of, using or characterised by the energy that is produced when atoms are split apart, for example, atomic bomb. 3. Of a substance, consisting of uncombined atoms rather than molecules. 4. In logic, of a sentence, formula, etc., having no internal structure at the appropriate level of analysis. In predicate calculus, *Fa* is an atomic sentence and *Fx* an atomic predicate. 5. Also called **purely atomic**, a measurable set which has positive measure and contains no set of smaller but positive measure.

ATOMIC ABSORPTION SPECTROSCOPY (AAS): An analytical technique used to determine the concentration of a specific metal element in a sample. The sample is converted into atoms by heating and the free atoms are made to absorb electromagnetic radiation of a particular wavelength. The amount of radiation absorbed is proportional to the concentration of the absorbing atoms of the element and can be used to determine the concentration of the element present in the sample.

ATOMIC BOMB (ATOM BOMB; A-BOMB; FISSION BOMB): A type of bomb whose destructive force is provided by nuclear fission, nuclear fusion or both. The design of nuclear weapons allows them to release huge amount of energy in a small space in a short time. This provides a far-reaching blast and a fireball of temperature greater than a million degrees celsius, creating immense explosion and fire damage over very large areas. The first atomic bomb was produced at a laboratory in Los Alamos, New Mexico in the U.S. and successfully tested on July 16, 1945. The only atomic bombs used in war were dropped by the U.S. during the Second World War on Hiroshima in Japan on 6 August 1945 with an estimated equivalent explosive force of 12,500 tons of TNT, killing and maiming some 140,000 civilians. Three days later, a more powerful one was dropped on Nagasaki, also in Japan.

ATOMIC CHARGE: The electric charge of an ion, equal to the number of electrons the atom has gained or lost in its ionisation multiplied by the charge on one electron.

International Atomic Weight/Mass Table Of Elements

Element	Symbol	Atomic Number	Atomic W/Mass	Element	Symbol	Atomic Number	Atomic W/Mass
Actinium	Ac	89	227.02779*	Neodymium	Nd	60	144.24
Aluminium	Al	13	26.98154	Neon	Ne	10	20.179
Americium	Am	95	243.061139*	Neptunium	Np	93	237.0482
Antimony	Sb	51	121.75	Nickel	Ni	28	58.71
Argon	Ar	18	39.948	Niobium	Nb	41	92.9064
Arsenic	As	33	74.9216	Nitrogen	N	7	14.0067
Astatine	At	85	209.98704*	Nobelium	No	102	255.093*
Barium	Ba	56	137.33	Osmium	Os	76	190.2
Berkelium	Bk	97	247.07032*	Oxygen	O	8	15.9994
Beryllium	Be	4	9.01218	Palladium	Pd	46	106.4
Bismuth	Bi	83	208.9808	Phosphorus	P	15	30.97376
Boron	B	5	10.81	Platinum	Pt	78	195.09
Bromine	Br	35	79.904	Plutonium	Pu	94	244.06424*
Cadmium	Cd	48	112.41	Polonium	Po	84	208.98244*
Calcium	Ca	20	40.08	Potassium	K	19	39.0983
Californium	Cf	98	251.07961*	Praseodymium	Pr	59	140.9077
Carbon	C	6	12.011	Promethium	Pm	61	144.91279*
Cerium	Ce	58	140.12	Protactinium	Pa	91	231.0359*
Cesium	Cs	55	132.9054	Radium	Ra	88	226.0254
Chlorine	Cl	17	35.453	Radon	Rn	86	222*
Chromium	Cr	24	51.996	Rhenium	Re	75	186.2
Cobalt	Co	27	58.9332	Rhodium	Rh	45	102.9055
Copper	Cu	29	63.546	Rubidium	Rb	37	85.4678
Cunum	Cm	96	247.07038*	Ruthenium	Ru	44	101.07
Dysprosium	Dy	66	162.50	Samarium	Sm	62	150.4
Einsteinium	Es	99	254.08805*	Scandium	Sc	21	44.9559
Erbium	Er	68	167.26	Selenium	Se	34	78.96
Europium	Eu	63	151.96	Silicon	Si	14	28.0855
Fermium	Fm	100	257.09515*	Silver	Ag	47	107.868
Fluorine	F	9	18.998403	Sodium	Na	11	22.9898
Francium	Fr	87	223.01976*	Strontium	Sr	38	87.62
Gadolinium	Gd	64	157.25	Sulphur	S	16	32.06
Gallium	Ga	31	69.737	Tantalum	Ta	73	180.9479
Germanium	Ge	32	72.59	Techneium	Tc	43	96.9062*
Gold	Au	79	196.9665	Tellurium	Te	52	127.60
Hafnium	Hf	72	178.49	Terbium	Tb	65	158.9254
Helium	He	2	4.00260	Thallium	Tl	81	204.37
Holmium	Ho	67	164.9304	Thorium	Th	90	232.0381
Hydrogen	H	1	1.0079	Thulium	Tm	69	168.9342
Indium	In	49	114.82	Tin	Sn	50	118.69
Iodine	I	53	126.9045	Titanium	Ti	22	47.90
Iridium	Ir	77	192.22	Tungsten	W	74	183.85
Iron	Fe	26	55.847	Uranium	U	92	238.029
Krypton	Kr	36	83.80	Vanadium	V	23	50.9415
Lanthanum	La	57	138.9055	Xenon	Xe	54	131.30
Lawrencium	Lr	103	256.099*	Ytterbium	Yb	70	173.04
Lead	Pb	82	207.2	Yttrium	Y	39	88.9059
Lithium	Li	3	6.941	Zinc	Zn	30	65.38
Lutetium	Lu	71	174.967	Zirconium	Zr	40	91.22
Magnesium	Mg	12	24.305	Element 104†		104	257*
Manganese	Mn	25	54.9380	Element 105†		105	260*
Mendelevium	Md	101	258*	Element 106†		106	263*
Mercury	Hg	80	200.59	Element 107†		107	258*
Molybdenum	Mo	42	95.94				

* The mass of the isotope with the longest known half-life

† Names for elements 104-107 have not yet been approved by IUPAC. The USSR has proposed Kurchatovium (Ku) for element 104, and Bohrium (Bh) for element 105.

The United States has proposed Ruthetordium (Rt) for element 104, and Hahnium (Ha) for element 105.

ATOMIC CLOCK: An apparatus for measuring or standardising time that is based on periodic phenomena within atoms or molecules.

ATOMIC DECAY (RADIOACTIVE DECAY): The disintegration of a nucleus that occurs spontaneously or as a result of electron capture. One or more different nuclei are formed and usually particles and gamma rays are emitted.

ATOMIC EMISSION SPECTROSCOPY: A chemical analytical technique in which a sample is vapourised in a flame, plasma, arc or spark and the excited atoms present are detected by their emission of electromagnetic radiation at characteristic wavelengths. The intensity of the radiation emitted is proportional to the concentration of the atoms.

ATOMIC ENERGY (NUCLEAR ENERGY): The energy stored in the nucleus of an atom and released through fission, fusion or radioactivity.

ATOMIC FLUORESCENCE: A combined process of photon absorption by an atom followed by spontaneous photon emission.

ATOMIC FORCE MICROSCOPY (AFM): A type of microscopy in which a small probe, consisting of a tiny chip diamond, is held on a spring-loaded cantilever in contact with the surface of the sample. The probe is moved slowly across the surface and the tracking force between the tip and the surface is monitored. The probe is raised and lowered so as to keep this force constant and a profile of the surface is produced. Scanning the probe over the sample gives a computer-generated contour map of the surface.

ATOMIC LASER: A gas laser which is pumped using energy transfer from other atoms or molecules. Examples are the helium-neon (He Ne) and the copper vapour laser.

ATOMIC MASS: The rest mass of an atom in its ground state or the weighted average mass of the isotopes of an element. Its symbol is m_a . The commonly used unit is the unified atomic mass unit (amu). Atomic masses are usually determined by mass spectrography. It works by bombarding gaseous atoms with fast-moving electrons which knock out an electron from the atom. For example, the atomic mass of carbon is 12.011 and that of hydrogen is 1.0079. (Refer to page 109 for International Atomic Mass/Weight of Elements).

ATOMIC MASS CONSTANT: One twelfth (1/12th) of the mass of a carbon-12 atom in its nuclear and electronic ground state, $m_u = 1.6605402 (10) \times 10^{-27}$ kg. It is equal to the unified atomic mass unit.

ATOMIC MASS UNIT (AMU; DALTON): A unit used to express the mass of individual atoms. It is expressed as the mass equal to 1/12th of the mass of a carbon-12 isotope. It has a value of $1.6605402 (10) \times 10^{-27}$ kg. A mass expressed on this standard is called a relative atomic mass (r.a.m.), with the symbol A_r . The symbol for amu is u and that for dalton is Da. One dalton is approximately equal to the mass of one proton or one neutron.

ATOMIC NUMBER (PROTON NUMBER): The number of protons in the nucleus of an atom, which is identical to the charge number of the nucleus. It is determined experimentally equal to the number of electrons orbiting the neutral atom. Its symbol is Z , which is written as a subscript before an element's symbol, for example, ${}_{92}\text{U}$. All atoms in an element have the same atomic number. Every element has a unique atomic number.

ATOMIC ORBITAL: 1. A mathematical function that describes the wave-like behaviour of either one electron or a pair of electrons in an atom. This function can be used to calculate the probability of finding any electron of an atom in any specific region around the atom's nucleus. The term may thus refer directly to the physical region defined by the function where the electron is likely to be; or a region in space around the nucleus of an atom, where there is a high probability of finding a specified electron. Orbitals of the same principal quantum number belong to the same subshell. 2. In physics, it is the region in which an electron may be found around a single atom.

ATOMIC PILE: An early type of nuclear reactor whose core consisted of layers of graphite block interspersed with uranium, designed to create a sustained fission reaction.

ATOMIC RADIATION (NUCLEAR RADIATION): Energy given out by disintegrating atoms during radioactive decay. The energy may be in the form of fast moving particles known as alpha particles and beta particles or in the waveform of high energy electromagnetic waves known as gamma radiation. Prolonged exposure to atomic radiation can lead to radiation sicknesses.

ATOMIC RADIUS: A measure of the radius of an atom, typically the total distance from an atom's nucleus to the outermost orbital of electron.

ATOMIC SIZE (ATOMIC RADIUS): A length equal to half the distance between the nuclei of two covalently bonded atoms or the size of an atom expressed as the radius in angstroms or other unit of length. For metallic elements, it is half the experimentally determined distance between the nuclei of the nearest neighbour in the metal. The size of the atom in metallic elements is always greater than the size of its ion. For non-metallic elements the reverse is true.

ATOMIC SPECTRUM: The range of characteristic frequencies of electromagnetic radiation that is readily absorbed and emitted by an atom. It could be absorption or emission spectrum.

ATOMIC STORAGE (ATOMIC MEMORY): A nanotechnology data storage system that uses single atoms as computer bits. IBM researchers used a technology called spin resonance to enable atoms to be written and read independently at a width of just one nanometre, a billionth of a metre. Current hard disk drive can store a bit of data on around 100,000 atoms.

ATOMIC STRUCTURE: The arrangement of the parts of an atom, which consists of a nucleus surrounded by a cloud of electrons arranged in orbits describable in terms of quantum mechanics.

ATOMIC SYMBOL: An internationally agreed one to three-letter code used to represent the atom in chemical formulae. For example, the atomic symbol for lithium is Li.

ATOMIC THEORY: A theory of the nature of matter. It states that matter is composed of discrete units called atoms, as opposed to the obsolete notion that matter could be divided into any arbitrarily small quantity.

ATOMIC TIME: Time as given by atomic clocks, regulated by natural resonance frequencies of particular atoms, displaying a continuous count of seconds.

ATOMIC UNITS (AU; A.U.): A system of natural units designed to simplify the form of the fundamental equations of quantum mechanics by eliminating from them fundamental constants. The atomic unit of length is the Bohr radius, a_0 ; masses are specified in atomic mass unit, amu and electron mass unit, m_e . The advantage of atomic units is that if all calculations are directly expressed in such units, the results do not vary with any revision of the numerical values of the fundamental constants.

ATOMIC VOLUME: The relative atomic mass of an element divided by its density; which is the volume occupied by 1 gram-atom of an element in the solid state.

ATOMIC WEIGHT: Former name for **relative atomic mass**, defined as weighted average of the masses of the constituent isotopes of a chemical element or the ratio of the average mass of a chemical element's atoms to 1/12th the mass of an atom of the carbon-12 isotope. Its symbol is A_r . Measuring the masses of atoms in grams, for example, would imply dealing with extremely small numbers, thus, the mass of an atom is compared with that of an atom of carbon-12. In calculating the relative atomic mass of an element with isotopes, the relative mass and proportion of each is taken into account. For example, naturally occurring chlorine consists of atoms of relative isotopic masses 35 (75%) and 37 (25%). Its relative atomic mass is 35.5. That is $A_r = \left(\frac{75}{100} \times 35\right) + \left(\frac{25}{100} \times 37\right) = 35.5$. The relative masses of atoms are measured using an instrument called

a mass spectrometer, invented by the English physicist Francis William Aston (1877-1945). The cations formed are brought down on to a detector in turn according to their mass-to-charge ratio. The instrument provides a measure of the relative mass (compared to ^{12}C) and the relative number of each isotope. Through the use of this instrument he discovered the existence of isotopes of elements. Relative atomic masses have no units.

ATOMICITY: The number of atoms in a molecule of an element. It is denoted by the letter Z. For example, oxygen (O_2) has an atomicity of 2, ozone (O_3) an atomicity of 3 and benzene (C_6H_6) an atomicity of 12.

ATOMISE (ATOMIZE): To reduce to atoms or fine particles.

ATOMISER (ATOMIZER): Any system which is capable of converting the analyte into atomic vapour. Alternatively, a device such as spinning disk generator or vibrating orifice generator that produces a spray of fine droplets of liquid from a bulk liquid.

ATOMISM: The belief that all matter consists of invisible, indestructible material particles.

ATP (ADENOSINE TRIPHOSPHATE) SYNTHETASE: An enzyme complex occurring in the inner membrane of the mitochondrion and also in the thylakoid membranes of chloroplasts that catalyses the formation of adenosine triphosphate (ATP) from adenosine diphosphate (ADP) and inorganic phosphate.

ATPase: A family of enzymes that brings about the hydrolysis of adenosine triphosphate (ATP) into adenosine diphosphate (ADP) with a release of energy. There are different forms of membrane-associated ATPases to meet specific demands of cells. They include the following: (i) The F-Adenosine triphosphate ATPases (phosphorylation factor, also known as H⁺-transporting Adenosine triphosphate ATPases or F(0)F(1)-Adenosine triphosphate ATPases, V-Adenosine triphosphate ATPases or Vacuole Adenosine triphosphate (ATP)ases, A-Adenosine triphosphate ATPases or Archaea Adenosine triphosphate ATPases, P-Adenosine triphosphate ATPases also known as E1-E2 Adenosine triphosphate ATPases and E-Adenosine triphosphate ATPases or Extracellular Adenosine triphosphate ATPases.

ATTOSECOND: A term usually used in photon research to represent a million trillionth (10^{-18}) of a second.

ATRIAL NATRIURETIC PEPTIDE (ANP; ATRIAL NATRIURETIC FACTOR (ANF); ATRIAL NATRIURETIC HORMONE; ATRIOPEPTIN): A peptide hormone, produced by certain cells in the wall of the atrium of the heart in response to high blood pressure, which promotes the excretion of sodium ion in the urine. It acts to reduce the water, sodium and adipose loads on the circulatory system to reduce blood pressure.

ATRIOVENTRICULAR NODE (AVN; AV NODE): A specialised group of cardiac muscle fibres situated in the fibrous ring between the right atrium and ventricle of the heart. It receives heartbeat impulses from the sinoatrial node and directs them to the walls of the ventricles.

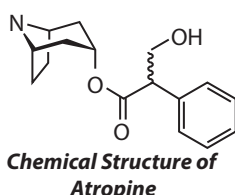
ATRIOVENTRICULAR VALVE (AV): The valve between each auricle and ventricle of the heart. It maintains a unidirectional or a one way blood flow through the heart.

ATRIUM (AURICLE): Either of the two chambers in the upper part of the mammalian heart that receives blood from the veins and forces it by powerful muscular contraction into the ventricle. The atria (plural of atrium) have thinner walls than the ventricles.

ATRO-: A prefix meaning dark or blackish.

ATROPHY: The degeneration, withering or wasting away of a cell organ, tissue or part of the body through lack of nourishment or use.

ATROPINE: A poisonous crystalline alkaloid, with the chemical formula $\text{C}_{17}\text{H}_{23}\text{NO}_3$. It is isolated from deadly nightshade (*Atropa belladonna*). It inhibits muscarinic acetylcholine receptors and also causes dilation of the pupil and hence used extensively in eye surgery.



ATROPISOMERS: Conformers that have highly restricted rotation about a single bond and can consequently be separated as distinct species. An example is the case of substituted biphenyls with large groups at the ortho positions of the rings.

ATROPOUS (ORTHOTROPOUS): A term used to describe a form of ovule orientation in the ovary in which the ovule develops in an upright position from the placenta so that the micropyle is at the end opposite the stalk. This arrangement may be found in some members of the Polygonaceae.

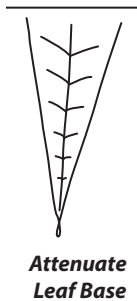
ATTACHMENT: 1. A transformation by which one molecular entity, called the substrate is converted into another by the formation of only one two-centre bond between the substrate and another molecular entity and which involves no other changes in connectivity in the substrate. The product of an attachment may also be the adduct of the two reactants, but not all adducts can be represented as the products of an attachment. 2. In telecommunications, an attachment is a file sent with an e-mail message. It can be a picture, a word document, a movie, a sound file, an excel document or any other file that requires another program to open it.

ATTAINABLE SET (REACHABLE SET): In a directed graph, the set of points that can be reached from a given vertex V.

ATTENTION-DEFICIT HYPERACTIVITY DISORDER, ADHD: A chronic psychiatric condition occurring in young children characterised by hyperactivity, disruptive behaviour, easily distracted, inability to concentrate or pay attention, learning difficulties and underachievement. It most commonly affects boys.

ATTENUANCE FILTER (ATTENUATOR; NEUTRAL-DENSITY FILTER): An optical filter, which reduces the radiant power of a light beam by a constant factor over all wavelengths within its operating range.

ATTENUATE: 1. To weaken or become weak; reduce in size, strength, density or value. 2. To make or become thin or fine. 3. Diluted, weakened, slender or reduced. 4. In botany, tapering gradually to a narrow tip or base. 5. To make something such as a pathogenic bacterium, virus, etc less virulent, as by culture in special media or exposure to heat. 6. In electronics, to decrease the amplitude of an electronic signal, whether digital or analogue.



ATTENUATED TOTAL REFLECTANCE SPECTROSCOPY (ATRS): A variation of infrared spectroscopy in which the IR source is reflected from the sample and absorption occurs only in the surface layer. It is used in forensic science for analysis of thin layers like paint.

ATTENUATION: 1. A term used in medicine to refer to a process of reducing the disease producing ability of a micro organism by chemical treatment, heating, drying, irradiation, etc. 2. A reduction in density, force or intensity of a radiation as it passes through matter. It is the weakening of a beam of particle or of electromagnetic radiation by any material placed in its path. Examples are the attenuation of sunlight by dark glasses; the attenuation of x-rays by lead; and the attenuation of light and sound when they pass through seawater. 3. Any reduction in the strength of a signal, whether digital or analogue. Attenuation affects the propagation of waves and signals in electrical circuits, in optical fibres, as well as in radio waves. It is a natural consequence of signal transmission over long distances. It is measured in decibels (dBs). In telecommunication, attenuation occurs on networks for several reasons such as distance, radio interference or physical obstructions like walls and wire size (on wired networks, thinner wires suffer from higher (more) attenuation than thicker wires).

A

ATTENUATION COEFFICIENT: Ratio of the natural log of 2 to the half-value thickness of material. It is a constant used to calculate the intensity of X-rays given any thickness of material. It shows how easily a material can be penetrated by a beam of light, sound, particles, or other energy or matter. Its symbol is μ . A large attenuation coefficient means that the beam is quickly weakened as it passes through the medium, and a small attenuation coefficient means that the medium is relatively transparent to the beam. The SI unit of attenuation coefficient is the reciprocal metre (m^{-1}).

ATTENUATION-TO-CROSSTALK RATIO (HEADROOM; ACR): The difference, expressed as a figure in decibels (dB), between the signal attenuation produced by a wire or cable transmission medium and the near-end crosstalk (NEXT).

ATTERBERG LIMITS LIQUID LIMITS: The limits of soil consistency for the classification of fine-grained soils, as first defined by Swedish scientist, Albert Atterberg. The method is based on the determination of the water content at distinct transitions between different states of soil consistency. These transitions are defined as shrinkage limit, plastic limit, and liquid limit, defined as follows: liquid limit is the boundary between the liquid and plastic states; plastic limit is the boundary between the plastic and semi-solid states; and shrinkage limit is the boundary between the semi-solid and solid states, or the water content where further loss of moisture will not result in any more volume reduction. The liquid limit, also known as the upper plastic limit, is the water content at which soil changes from the liquid state to a plastic state, or the minimum moisture content at which a soil flows upon application of very small shear force. The plastic limit can also be defined as the moisture content where the thread breaks apart at a diameter of 3.2 mm. The values for these limits are dependent on various soil parameters such as particle size, specific surface area of the particles that are able to attract water molecules.

ATTO-: A prefix used in the metric system to denote 10^{-18} . Its symbol is *a*.

ATTRACTION: In physics, it is an electric or magnetic force that tends to make objects, which are oppositely charged move towards each other, as for example, between the north and south poles of a magnet.

ATTRACTOR: 1. A set of physical properties toward which a system tends to evolve, regardless of the starting conditions of the system. 2. In mathematics, in a dynamical system, a value or set of values toward which variables tend to evolve for a wide variety of starting conditions of the system. It is defined formally as a closed set E such that for a given mapping f , $f(E)$ is contained in E and for all x in some E , the distance from $f^m(x)$ to E tends to zero as m tends to infinity. It is usually required that the orbit of f is dense in E for some value of x . If the attractor, E , is a fractal set, it is said to be **strange**. For example, in the Julia sets, $z \rightarrow z^2 - \mu$ makes E strange for some values of μ .

ATTRIBUTE: 1. A characteristic or property of an object such as length, weight, capacity, time, sides, shape, or colour that helps to identify it. 2. A characteristic that identifies and describes a managed object. The characteristic can be determined and possibly changed, through operations on the managed object. 3. Information within a managed object that is visible at the object boundary. An attribute has a type, which indicates the range of information given by the attribute and a value, which is within that range. 4. A basic property of an experimental unit whose value cannot be changed by the experimenter.

ATTRITION: A form of coastal or river erosion, when the bed load is eroded by itself. As rocks are transported downstream along a riverbed, the regular impacts between them cause them to be broken up into smaller fragments, making them rounder and smoother.

AUDIBLE RING: In a telephone system, it is the ringing tone that is returned from a called party's phone to indicate to the caller that the desired party's phone is active and ringing.

AUDIO CD (AUDIO COMPACT DISC): A digitally encoded data such as voice or music on an optical disc that is played back by a laser.

AUDIO CODEC: A digital electronic device or software that is integrated into Intel chips for encoding or decoding a digital data stream or signal. Examples include MP3, Windows Media Audio (WMA), Dolby Digital and DTS. The standard found in most computers is Audio Codec 97 (AC97).

AUDIO EDITING SOFTWARE: A software which allows users to create, listen and manipulate sound data.

AUDIO FREQUENCY (AUDIBLE FREQUENCY BANDS; AUDIBLE FREQUENCY): Sounds with frequencies from about 20Hz to 20,000Hz that can be detected by the normal human ear. In humans, the upper limit becomes inaudible with old age. The state or quality of being heard is **audibility**.

AUDIO FREQUENCY AMPLIFIER (A.F.): An amplifier that amplifies signals in the approximate range of 10 Hz to 20 Hz. 2. In electronics, audio frequency (AF) is an alternating current (AC) having a frequency such that, if applied to a transducer such as a loudspeaker or headset, will produce acoustic waves within the range of human hearing (20 Hz to 20,000 Hz). All telephone circuits operate with AF signals in a restricted range of approximately 200 Hz to 3000 Hz. A telephone-line modem is an AF device that converts binary digital data into analogue signals that can be transmitted over the telephone circuit and also converts incoming AF signals into binary digital data.

AUDIO INPUT DEVICE: A computer device which is used to send audio signals such as music and voice into a computer. Examples are CD-ROM players, microphone, speech recognition and voice recognition.

AUDIO VIDEO INTERLEAVE (AVI): The standard file format used for video files under Microsoft Windows Resource Interchange File Format (RIFF) specification and ends with an .avi extension. The maximum size of an AVI file is 2 Gig. AVI is used for multiple encoding methods, such as AVI video compression methods (video codecs) including DivX, Xvid, Cinepak, Indeo, DV and Motion JPEG, as well as uncompressed RGB and YUV2. AVI audio includes compressed MP3 and Microsoft's ADPCM (MS ADPCM), as well as uncompressed PCM. AVI files require a special player that may be included with the Web browser or may require downloading.

AUDIO-ONE: Digital recording and editing software developed for the dance, music and audio businesses. It includes a waveform recorder that allows signal manipulation and editing.

AUDIOBLOG: A blog with audio clips such as Advanced Audio Coding (AAC), MP3 or other audio recordings as its central part. An example is podcast.

AUDIOGRAPHIC TELECONFERENCING: A type of electronically transmitted communication using audio, graphics, figures and printed text with the computer screen shared by more than one site. It is useful for participants which are at different geographic locations to interact with one another or each other.

AUDIOMETER: An instrument that generates a sound of known frequency and intensity used for measuring the hearing ability of an individual and for assessing hearing impairment or hearing loss.

AUDIT TRAIL: In telecommunications, it is a record of computer operations; showing both completed and attempted accesses and services, the changes that have been made to a database or file or data forming a logical path linking a sequence of events. The term is taken from accountancy but used widely in computing to check many aspects of computer security and to trace the transactions that have affected the contents of a record.

AUDITORY: Relating to the ear; capable of hearing; producing sound.

AUDITORY CANAL (EAR CANAL; EXTERNAL AUDITORY MEATUS): The tube extending from the outer ear and opening into the eardrum. In humans, the ear canal is about 2.5 cm long. **Auditory** refers to the organs of hearing or the process of hearing.

AUDITORY NERVE (VESTIBULOCOCHLEAR NERVE; ACOUSTIC NERVE): The eighth cranial nerve that transmits information from the inner ear to the hindbrain, which it enters between the pons and medulla. It serves two distinct functions: The acoustic portion carries sensations of sound and pitch from the organ of hearing, the cochlea; these are relayed to the auditory part of the cerebral cortex; The vestibular portion carries postural sensation from the semicircular canals of the inner ear to the cerebellum. It provides information about position.

AUDITORY OSSICLES (EAR OSSICLES): The three tiny bones in the middle ear of vertebrates, which connect the eardrum to the inner ear. They transmit vibrations of the eardrum caused by sound waves in the air to the fluid of the inner ear via the oval window. Three ossicles, the malleus (hammer), incus (anvil) and stapes (stirrup), are present in the mammalian ear. They form a lever system, diminishing the force of sound waves and increasing the force on the inner ear. A single ossicle occurs in the ear of birds, amphibians and reptiles.

AUDITORY PATTERN: A set of sounds that are arranged following a rule to determine similarities and differences in patterns of sounds.

AUDITORY PROCESSING DISORDER, APD (CENTRAL AUDITORY PROCESSING DISORDER, CAPD): A neurological acquired disorder that affects auditory analysis of mostly children. Children with this condition cannot process what they hear in the same way as other kids do because their ears and brain do not fully coordinate.

AUFBAU PRINCIPLE (AUFBAU RULE; BUILDING-UP PRINCIPLE): A rule used to determine how electrons organise into shells and subshells around the atomic nucleus. It states that as much as possible, electrons will enter the orbital with the lowest energy first. It is used to determine the electron configuration of an atom, molecule or ion. An Aufbau principle uses arrows to represent electrons. The electrons are filled into the orbitals in order of decreasing stability, taking account of the Pauli Exclusion Principle. As they are added, they assume their most stable conditions, called electron orbitals, with respect to the nucleus and those electrons already there. In the notation used to show electronic configuration, a number is used to designate the principal shell and the letters *s*, *p*, *d* and *f* to identify the subshells.

AUGEND: A number or quantity to which another, the addend, is added to form a sum, for example in $x + y$, x is the 'augend'.

AUGER EFFECT: The ejection of an electron from an atom without the emission of x-rays or gamma rays. It may be as a result of the de-excitation of an excited electron within the atom. This type of transition occurs in the x-ray region of the emission spectrum.

AUGER ELECTRON SPECTROSCOPY: Any technique in which a specimen is bombarded with keV-energy electrons or X-rays and the energy distribution of the electrons produced through radiationless de-excitation of the atoms in the sample (Auger electrons) is recorded. The derivative curve may also be recorded.

AUGER (SOIL AUGER): A tool having a variety of heads attached to a shaft that is used for drilling or boring into the soil to obtain a sample of soil for observation and analysis.

AUGMENTATION: The enlargement of a set of equations or a matrix in linear programming, matrix theory or control theory.

AUGMENTED EUCLIDEAN GEOMETRY: A Euclidean geometry to which have been added a notional line at infinity on which parallel lines meet and unless the geometry is real, complex points that have complex coordinates in any system of Cartesian coordinates. It can be represented by all non-zero complex (or real) triples, where two triples represent the same point if they are complex multiples of one another.

AUGMENTED LAGRANGIAN: Any of a number of combinations of a penalty function term, usually quadratic, with a Lagrangian, in order to produce algorithms that exploit the best features of both.

AUGMENTED MATRIX: The coefficient matrix that has an extra column containing the constant terms, this extra column being separated by a vertical line. It is created by joining the coefficient of the matrix to the numbers on the right hand side of the equations. The new column is set apart by a dotted vertical line. For example, for a system of linear equations $2x - 3y = 8$ and $4x + 5y = 1$, the

augmented matrix is given as: $\left(\begin{array}{cc|c} 2 & -3 & 8 \\ 4 & 5 & 1 \end{array} \right)$. The solution of the

augmented matrix can be obtained by transforming it to reduced echelon form by elementary row reduction operations.

AUGMENTED REALITY, AR: The use of digital information such as sound, video and graphics to superimpose or overlay on video or other real-life representations, thus providing a composite view. In other words, it is putting new information into an existing picture. An example is a video of a car engine with the mechanical drawings superimposed or overlaid.

AURELIA (JELLY FISH): A common widely distributed marine organism. Large specimen may reach 30 cm in diameter. The polyp form, called a scyphistoma, is small, sessile and lives attached to rocks and other submerged objects in shallow coastal waters. It is radially symmetrical in four parts. The mouth is square in shape having four long oral arms arising from the corners of the square. Along the arms are short oral tentacles that enable it to capture food such as small planktonic animals, which are then moved towards the mouth along the ciliated, groove on the oral side each arm.

AURIC (GOLD(III); Au³⁺): Trivalent gold or containing gold with a valence of three.

AURICLE 1. Also known as pinna, it is the external flap of the ear. It consists of a thin plate of yellow fibrocartilage covered with skin. It is fixed in position by certain ligaments and possesses two sets of weak muscles – an extrinsic group, passing to the cartilage from the epicranial aponeurosis and the mastoid process and an intrinsic group, attached to the cartilage alone. 2. Also known as **auricle of the atrium** or **atrial appendage**, it is a small conical upper chamber located at each atrium of the heart. The left auricle collects oxygenated blood as it leaves the lungs and moves the blood into the left ventricle. The right auricle collects deoxygenated blood from the bloodstream and moves it into the heart's right ventricle. 3. **Auricle of a leaf** is a small earlike projection from the base of a leaf or petal, as shown in the diagram. Auricles are seen at the base of the leaf blade in grasses (Gramineae).



AURICULAR: 1. Of or pertaining to the outer ear, or to something else that is ear-shaped, such as the atrium of the heart. 2. A set of feathers that are found near a bird's ear openings.

AURICULATE-CLASPING: Ear-shaped lobes at the base of a leaf, encircling the stem.

AURONES: One of the two groups of anthochlor flower pigments. They are probably formed by oxidation of chalcones. Aurones are found in many members of the Compositae, especially the genus Coreopsis.

AURORA: The luminous natural electrical phenomenon that appears in the middle and high latitudes of the Earth's magnetic poles in the Northern and Southern Hemispheres characterised by bands or streamers of reddish or greenish light in the sky. The effect is caused by the interaction of solar charged particles with atoms in the upper atmosphere. In the Northern Hemisphere, it is called **northern lights** or **aurora borealis**; and in the Southern Hemisphere, it is called **Southern Lights** or **aurora australis**.

AUROUS (GOLD(I); Au⁺): Monovalent gold or containing gold with a valence of one.

AUSTENITE GAMMA PHASE IRON: A metallic non-magnetic solid solution of carbon and iron that exists in steel above the critical temperature of 723 °C (996.15 K). It was named after Sir William Chandler Roberts-Austen (1843–1902). As it cools, this structure either breaks down into a mixture of ferrite and cementite usually in the structural forms pearlite or bainite or undergoes a slight lattice distortion known as martensitic transformation. Because of its face-centred cubic (FCC) structure it holds a high proportion of carbon in solution.

AUSTRALOPITHECUS: A genus of fossil primates that lived in Africa 4-2 million years ago, coexisting for some of this time with early forms of human. They walked erect and had teeth like those of modern human.

AUTACOID (AUTOCOID): A physiologically active substance such as histamine and hormone, that acts locally to regulate the activities of certain cells within a given tissue. It is produced in one part of an organism and transported by the blood, lymph or sap to another part of the organism.

AUTAPOMORPHY: Any derived character state possessed only by members of one particular taxon. Autapomorphies distinguish a taxon from other related taxa. The number of autapomorphies possessed by one taxon as compared to another taxon assumed to be derived from the same common ancestor, provides a measure of anagenesis.

AUTECOLOGY: The branch of ecology that deals with the biological relationship between an individual organism or an individual species and its environment.

AUTHENTICATED E-MAIL (CERTIFIED E-MAIL; STAMPED E-MAIL): A system in which senders of commercial e-mail messages pay a small fee to ensure that their messages will bypass spam filters to reach intended recipients.

AUTHENTICATION, AUTHORIZATION AND ACCOUNTING (AAA): A system of management and security protocols for controlling a client's activities on a secure network which enables the network server (NAS) to identify users who want to access the secure network (**Authentication**); to grant access to the secure network resources (**Authorization**); and to collect data regarding what network resources are accessed, enforce policies, etc., (**Accounting**).

AUTHOR: In taxonomy, the person who published the first valid name of a taxon. The name of the author or a recognised abbreviation of it, should follow the name of the taxon for it to be complete.

AUTHORING: Development of interactive multimedia, especially a hypertext or multimedia application for presentations or for electronic publishing. It includes organising the necessary audio, video, graphics and text files and other interactive programs and formatting them for display. **Authoring program** or **tool** is any software that allows developers to create multimedia presentations or World Wide Web pages visually like a desktop publishing program. Examples of such tools for creating the World Wide Web include Microsoft FrontPage, Macromedia Dreamweaver and Adobe PageMill.

AUTHORIZATION: In computing, it is a security mechanism that gives or denies access to a particular computer system, including programs, files, services, data and application features. The conditions are defined by the system administrator for the system upon which users may be allowed or denied access to the system and may include authentication or identification, conditions of use, privileges of use such as access to file directories, hours of access, amount of allocated storage space, etc.

AUTISM (CLASSICAL AUTISM): A developmental disorder that appears in children by the age of three and affects the normal functioning of the brain. Autism is variable in expression but normally characterised and diagnosed by three distinctive behaviours: (i) difficulties with social interaction; (ii) problems with verbal and nonverbal communication; (iii) repetitive behaviours or narrow, obsessive interests rather than with flexible and imaginative ones.

AUTO NUMBER IDENTIFICATION (ANI): The service that provides the receiver of a telephone call with the source number; determined by the service provider.

AUTOCAD: A computer-aided design (CAD) software package, published by the US company, AutoDesk and used for 2-D and 3-D design and drafting by professionals such as engineers, architects and designers.

AUTOCATALYSIS (AUTOCATALYTIC REACTION): A chemical reaction in which a product (or a reaction intermediate) also functions as a catalyst; or form of catalysis in which one of the products of a reaction serves as a catalyst for the reaction. In such a reaction the observed rate of reaction is often found to increase with time from its initial value. For example, when an oxalate reacts with acidified potassium permanganate solution the manganese(II) ions resulting from the reduction of the permanganate catalyse the reaction. Before the emergence of protein-based enzymes, which are the principle biological catalysts today, it is believed that a substance, such as RNA, may have served as both a replicator and an autocatalyser in the first terrestrial life-forms.

AUTOCHTHONOUS: 1. In geology, it refers to rocks, minerals, etc., which are formed in the place where found. An example is coal. 2. Of an organism, native to the place in which it is found. 3. In pathology, found in the part of the body in which it originates. An example is cancerous lesion.

AUTOClave: A strong pressurised steel vessel that uses superheated steam to carry out sterilisation of materials and equipment such as surgical instruments. It is similar in principle to a pressure cooker.

AUTOCORRELATION (SERIAL CORRELATION): A measure of the correlation of a particular time series delayed by k lags (the distance between the observations that are so correlated). It is obtained by dividing the covariance between two observations, separated by k lags, of a time series (auto-covariance) by the standard deviation y_t and y_{t-k} . If the autocorrelation is calculated for all values of k the autocorrelation function is obtained. For a time series that does not change over time, the autocorrelation function decreases exponentially to zero.

AUTOCOVARANCE: The condition occurring when successive items in a sequence are correlated so that their covariance is not zero and not independent. It describes the strength of linear relationship between two random variables. This is comparable to autocorrelation but it can be used to measure the covariance between the variables.

AUTODIPLOID: A gamete containing a diploid rather than haploid number of chromosomes. Such gametes may arise either through faulty meiosis or from tetraploid tissues. They may fuse with another such gamete to give a tetraploid zygote or may develop parthenogenetically. The term is also applied to diploid plants that have been obtained by doubling the chromosomes of a haploid plant. Such plants are completely homozygous.

AUTOECIOUS: A term for rust fungus that passes its life cycle in association with only one host plant.

AUTOEXEC.BAT: In computing, it is an abbreviation for **automatically executed batch**, which is a file in the MS-DOS operating system that is automatically run when the computer is booted. Batch indicates that this is a file containing a sequence of commands entered from a file rather than interactively by a user.

AUTOGAMY: A type of reproduction common to ciliated protozoans of the genus *Paramecium* in which only a single isolated individual produces young ones.

AUTOGENESIS (ABIOTENESIS; BIOPOESIS): A hypothesis that some organisms originated from abiotic or inanimate material or the study of how life on Earth could have arisen from inanimate matter.

AUTOGENIC: That, which relates to or is caused by a change in an individual organism's environment due to some biotic factors such as species diversity or a change in soil structure or composition. On the other hand, if the change is due to abiotic factors such as flooding,

drought, or earthquakes, it is called **allogenic**. Both (autogenic and allogenic) can bring about a change to the organisms in that environment called **autogenic succession** or **allogenic succession**.

AUTOGENIC INHIBITION: Reflex inhibition of a motor unit in response to excessive tension in the muscle fibres it supplies. Muscle tension is monitored by the Golgi tendon organs. Autogenic inhibition is a protective mechanism, preventing muscles from exerting more force than the bones and tendons can tolerate.

AUTOGENICS: A system developed in the 1900s by German physician Johannes Schultz to facilitate mental control of biological and physiological functions, generally considered to be involuntary. It can be used in inducing relaxation, managing pain, assisting healing processes, reducing sleeping disorders such as insomnia and relieving psychosomatic disorders. It can also be used to overcome addictions such as smoking, change unwanted behaviours such as nail biting and resolve phobias such as acrophobia (abnormal fear of heights). It is regarded as a precursor of biofeedback.

AUTOGENOUS: In medicine, it refers to anything that is generated from the body such as vaccines prepared from bacteria or other materials that are obtained from the affected person. For example, autogenous blood transfusion involves the collection of blood from a patient, prior to undergoing surgery, for use during and after the operation. It is a valuable procedure for operations that may require large transfusions or where a person has a rare blood group.

AUTOGYRO: A type of aircraft that supports itself in the air with a rotary wing or rotor similar to that of a helicopter but unpowered. It was invented by the Spanish aeronautical engineer, Juan de la Cierva and first flown successfully in 1923.

AUTOIMMUNITY (AUTOIMMUNE DISEASE): A type of disorder in the body's immune system causing disease conditions in which antibodies produced to fight antigens tend to destroy the body's own tissues. Examples of such diseases include rheumatoid, arthritis, insulin-dependent diabetes mellitus and systemic lupus erythematosus (SLE), an inflammation of the connective tissue occurring most often in young women.

AUTOIONISATION (PRE-IONISATION): The spontaneous ionisation of excited atoms, molecules or fragments of an ion. Atoms can autoionise when either two or more valence electrons are excited or one or more inner-shell electrons are missing, called the Auger effect. Molecules, in addition, can experience vibrational autoionisation.

AUTOLOADER: Any tool or resource that helps to automatically load or handle data or files.

AUTOLYSINS: An antibody that in the presence of complement is capable of destroying its own cells and tissues.

AUTOLYSIS (SELF DIGESTION): A self-defence of cells that is executed by lysosomes by secreting enzymes for digesting the entire cell, cell organelle or tissues invaded by foreign materials.

AUTOMATA THEORY: A theoretical branch of computer science concerned with mathematical study and modelling of certain abstract machines and their capacity to solve various types of problem by means of the algorithms available to them. In other words, it is the study of abstract machines and automata and computational problems. It develops methods by which computer scientists can describe and analyse the dynamic behaviour of discrete systems, in which signals are sampled periodically.

AUTOMATED FINGERPRINT IDENTIFICATION SYSTEM (AFIS): A biometric identification (ID) methodology that uses digital imaging technology to obtain, store and analyse fingerprint data.

AUTOMATED SPEECH RECOGNITION (ASR): A technology that allows users of information systems to speak entries rather than punching on a keypad. ASR is used primarily to provide information and to forward telephone calls.

AUTOMATED TEST EQUIPMENT: It is a computer-controlled equipment that performs tests on different devices referred to as devices under test (DUT) for functionality and performance.

AUTOMATIC CALL DISTRIBUTOR (ACD): A telephone facility that manages incoming calls and handles them based on the number called and an associated database of handling instructions. Many companies offering sales and service support use ACDs to validate callers, make outgoing responses or calls, forward calls to the right party, allow callers to record messages, gather usage statistics, balance the use of phone lines and provide other services.

AUTOMATIC CALLBACK: A telephone system that permits a caller who gets a busy signal to instruct the system to establish the connection when the line becomes available. The system does this by retaining both the called and calling numbers, redialling the called number periodically and alerting the caller when the connection has been made.

AUTOMATIC CALLING UNIT (ACU): A dialing device that permits a computer to automatically dial calls over a network.

AUTOMATIC FALLBACK: In computing, a feature allowing two communicating modems to drop to a slower speed if they experience data corruption, such as line noise. Modem speeds are typically rated according to one or another CCITT standard (known as a V number). All modems rated for a specific standard are backwards compatible.

AUTOMATIC HYPHENATION: A feature of some word processors which inserts hyphens into words which would otherwise extend beyond the right hand margin of the page.

AUTOMATIC IDENTIFICATION AND DATA CAPTURE (AIDC): A broad category of technologies used to collect information from an individual, object, image or sound without manual data entry. AIDC systems are used to manage inventory, delivery, assets, security and documents. Sectors that use AIDC systems include distribution, manufacturing, transportation, medicine, government and retail, among many others.

AUTOMATIC NERVOUS SYSTEM (AUTONOMIC NERVOUS SYSTEM; AUTONOMIC SYSTEM): The parts of the nervous system responsible for controlling or regulating involuntary bodily actions (or actions which are not consciously directed) such as heartbeat, digestive processes, blood flow, breathing, pupillary response, urination, defecation and other metabolism, emotional responses, electrolyte balance, sexual response, body temperature, etc. It operates by receiving information from the other parts of the body and the environment and it is divided into the sympathetic, parasympathetic and enteric nervous systems. The sympathetic nervous system mobilizes the body for quick (stimulate) response while the parasympathetic nervous system will act slowly to reverse, lower or inhibit the action. The enteric nervous system is confined to the gastrointestinal tract.

AUTOMATIC PILOT (AUTOPILOT OR GYROPILOT): A central device that keeps an aeroplane flying automatically on a given course at a given height and speed.

AUTOMATIC TELLER MACHINE (ATM): A computerised machine made available by banks and building societies to enable account holders to withdraw cash automatically using cash card from their account.

AUTOMATICITY: 1. The ability to recall or generate an answer quickly and effortlessly from memory without conscious thought or through force of habit. 2. In medicine, the ability of the heart muscles to contract independently of each other when separated or in unison when reattached in whole or in groups. Also, the capacity of a cell to generate an action potential spontaneously without an external stimulus.

A

AUTOMATION: The use of self-regulating machines, control systems and information technologies to increase productivity in the production of goods and delivery of services or to operate systems such as washing machines, refrigerated or air-conditioning systems, aircraft/ship systems, etc. The fundamental constituents of an automated process are a power source, a feedback control mechanism and a programmable command structure. By using feedback, sensors check a system's operations and send signals to a computer or other devices that automatically regulates a process.

AUTOMATISM: Unconscious actions that are performed without awareness, which one may not remember. It may occur in some neurological and psychiatric disorders such as narcolepsy (the sudden recurrent uncontrollable compulsion to sleep) and schizophrenia. It is also seen in sleepwalking.

AUTOMOBILE: A self-propelled passenger land vehicle that usually has four wheels and an internal-combustion engine, used for transport. They include motorcars for personal or commercial operations such as taxis; buses, which carry large numbers of commercial passengers; and medium- and heavy-duty trucks, which carry heavy or bulky loads of freight or other goods and materials.

AUTOMORPHIC FUNCTION: An analytic function, f , on a domain D , such that for a group of Möbius transformations, T , $T(z)$ is in D and $f(T(z)) = f(z)$ for all z in D .

AUTOMORPHIC NUMBER (AUTOMORPH): A number n , whose square ends with n . For example, 5 is automorphic because $5^2 = 25$, which ends with 5. A number n is called **trimorphic**, if n^3 ends in n . An example is 49, since $49^3 = 117649$. Not all trimorphic numbers are automorphic. A number n is called **tri-automorphic** if $3n^2$ ends in n . An example is 667 because 3×667^2 is 1334667, ends with 667.

AUTOMORPHISM: An isomorphism from a set onto itself. It is, in some sense, a symmetry of the object and a way of mapping the object to itself while preserving all of its structure. For example, in set theory, an automorphism of a set X is an arbitrary permutation of the elements of X . The automorphism group of X is also called the symmetric group on X .

AUTONOMIC MOVEMENT: A plant movement that results from internal growth changes and is independent of changes in the external environment. Autonomic movements include the beating of cilia and flagella, protoplasmic streaming, circumnutation and the movement of chromosomes during nuclear division.

AUTONOMIC NERVOUS SYSTEM: Part of the nervous system of mammals that controls involuntary actions by supplying stimulation along the motor nerves to the smooth and cardiac muscles and to glands of the body under the control of mainly the medulla oblongata and the hypothalamus.

AUTONOMOUS: 1. A differential equation or a system of ordinary differential equations is said to be autonomous if it does not explicitly have an independent variable in the equation $\frac{dy}{dt} = f(y)$. For example,

the equation $\frac{dx}{dt} = x$ is autonomous, but $\frac{dx}{dt} = t$ is not. 2. Of an ordinary differential equation of the second order, able to be reduced to a first-order equation by writing $\frac{du}{dt} = p$, $\frac{d^2u}{dt^2} = p \frac{dp}{du}$, and regarding u as the independent variable, where u is the dependent variable and t the independent variable of the given equation.

AUTOPHAGY: A process utilised by certain cells to digest worn out or superfluous cell organelles through the action of enzymes originating within the same cell. An autophagic vacuole forms around a portion of cytoplasm containing one or more unwanted organelles. Hydrolytic enzymes are secreted into the vacuole by the surrounding cytoplasm and digestion occurs followed by the subsequent reabsorption of the breakdown products. An organelle called lysosome also releases enzymes to digest unwanted cell organelles within a cell.

AUTOPILOT (AUTOMATIC PILOT): A navigation mechanism, which uses electronic, hydraulic and mechanical control system to maintain a preset course without assistance from a human being. The autopilot monitors the course and speed of the aircraft and corrects any deviations from the flight plan. Systems range from simple wing-levellers to computer-operated units consisting of a gyroscope; an electric servomechanism; and an accelerometer, which measures the acceleration of the aircraft. Example of autopilot includes self-steering gear for ships or auto guidance of spacecraft and missiles.

AUTOPOEISIS: A theory of what life is, developed by the Chilean scientists Humberto Maturana and Francisco Varela. It is based on the way living systems address and engage with the domains in which they operate and to define and explain the nature of living systems. The eukaryotic cell, for example, is made of various biochemical components such as nucleic acids and proteins and is organised into bounded structures such as the cell nucleus, various organelles, a cell membrane and cytoskeleton. These structures, based on an external flow of molecules and energy, produce the components which, in turn, continue to maintain the organised bounded structure that gives rise to these components (not unlike a wave propagating through a medium).

AUTOPOISONING (SELF-POISONING): A phenomenon in which a product of a reaction may cause poisoning or inhibition.

AUTOPOLYPLOID: A polyploid organism in which the multiple sets of chromosomes are all derived from the same species. An example is the doubling of chromosome number during mitosis.

AUTOPROLYSIS: A proton (hydron) transfer reaction between two identical molecules, usually a solvent, one acting as a Brønsted acid and the other as a Brønsted base. For example, two water molecules ($H_2O + H_2O$) self-ionises to hydroxide ion (OH^-) and hydronium ion (H_3O^+) as follows: $H_2O + H_2O \rightleftharpoons OH^- + H_3O^+$.

AUTOPSY: The medical examination of a dead body in order to establish the cause and circumstances of death.

AUTORADIOGRAPHY: A technique for detecting the distribution of radioactive isotope cells or section of tissue that has been treated or injected with a radioactively labelled isotope or that has absorbed or ingested such an isotope. An **autoradiograph** is the image recorded on a photographic film or plate and shows the location of the radioactive substance in the form of concentrated dark patches.

AUTOREGRESSIVE MODEL: A model that is used in time series analysis. In the model x_t observation, at time t is a linear function of the previous values of the series. The first order of autoregressive model is given by $x_t = \phi x_{t-1} + a_t$, where a_t is a random disturbance and ϕ is a parameter of the model. The corresponding model of order p is given by: $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + a_t$.

AUTOREGRESSIVE MOVING-AVERAGE MODEL: The combination of autoregressive model and a moving-average model in time series. The general model of order p, q which is denoted by $ARMA(p, q)$ is $X_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + a_t - \theta_1 a_{t-1} - \dots - \theta_q a_{t-q}$, where $\phi_1, \phi_2, \dots, \phi_p$ and $\theta_1, \theta_2, \dots, \theta_q$ are the parameters of the model and $a_t, a_{t-1}, a_{t-2}, \dots$ are white noise sequences.

AUTOREGULATION: The process by which a gene regulates its own expression, especially the maintenance by an organ or tissue of a constant blood flow despite changes in arterial pressure; and the adjustment of blood flow through an organ in accordance with its metabolic needs.

AUTORESPONDER: A program or script on a mail server that automatically replies to e-mails. Though it is run from the mail server, an autoresponder can usually be set up by the user through a Web-based interface. For example, a company might set up an autoresponder for their support e-mail address to let users know they have received their support requests.

AUTOSAVE (AUTORECOVER; AUTO-SAVE): In computer science, it is a feature in an application that automatically saves data at regular intervals so that only a small amount of information is lost if the computer system fails. The frequency of how often the program automatically saves can be set by the user. Normally, many programs

perform this task every 5 to 10 minutes if a file running is being edited. Autosave is not an option for manually saving regularly.

AUTOSOMAL DOMINANT: A trait or disorder in which an affected individual possesses one copy of a mutant allele and one normal allele. In other words, it is a gene on one of the non-sex chromosomes that is always expressed, even if only one copy is present. It is a pattern of inheritance characteristic of some genetic diseases. Examples of such diseases are Huntington's disease, neurofibromatosis, achondroplasia (in some cases) and polycystic kidney disease. Individuals with autosomal dominant diseases have a 50-50 chance of passing the mutant allele and hence the disorder onto their children.

AUTOSOMAL LINKAGE (LINKAGE): Inheritance of the genes on a given autosomal chromosome (any chromosome except for the sex chromosomes; humans have 44 autosomal chromosomes), implying that the genes are on the same chromosome and all genes present on homologous chromosomes (pair of matching chromosomes in an organism, with one being inherited from each parent) belong to the same linkage groups (all genes on a single chromosome, inherited as a group, moving and acting as a unit, rather than independently, during cell division).

AUTOSOMAL RECESSIVE: A trait or disorder in which an affected individual possesses two copies of a mutant allele. Examples of autosomal recessive diseases include cystic fibrosis, Fanconi anaemia, sickle cell disease and Tay-Sachs disease.

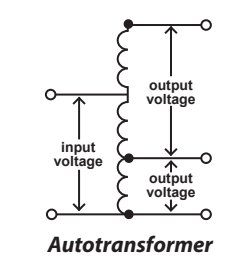
AUTOSOME: Any pair of numbered chromosomes that does not play a principal role in sex determination but rather bears genes of the body. Humans have 22 pairs of autosomes (a total of 46 chromosomes) and one pair of sex chromosomes (the X and Y). In botany the term is used for plant chromosomes when discussing dioecious species that have cytologically identifiable sex chromosomes.

AUTOSUM: A function used in Microsoft Excel spreadsheet program to find the sum of various numbers, which inserts a formula in the selected cell that adds the numbers in the column above it.

AUTOSTYLIC JAW SUSPENSION: A type of jaw suspension seen in some sharks, lungfish and tetrapods, in which the upper jaw is connected directly to the cranium.

AUTOTOMY: The shedding by an animal of part of its body, such as the tail of certain lizards or the claw of a lobster followed by the regeneration of the lost part. Autotomy is achieved by the contraction of muscles at specialised regions in the body. It serves as protective mechanism if the animal is damaged. It is common as a method of sexual reproduction in *Polychaeta* worms, in which both new head and tail regions may be regenerated.

AUTOTRANSFORMER: A transformer that has a single winding, which is tapped at different points to provide both the primary and secondary circuits. It has many uses and applications including to start induction motors, to regulate the voltage of transmission lines and to transform voltages when the primary to secondary ratio is close to unity. Its advantages include being smaller, lighter and cheaper than typical dual-winding transformers. Some of its disadvantages include the following: (i) it does not provide electrical isolation, so that a failure of the insulation of the windings can result in full input voltage being applied to the output. (ii) It cannot safely be used for stepping down higher voltages to much lower voltages suitable for smaller loads. (iii) A break in the part of the winding that is used as both primary and secondary will result in the transformer acting as an inductor in series with the load, which under light load conditions may result in near full input voltage being applied to the output. (iv) If the secondary circuit suffers a short-circuit condition, the resulting



primary current would be much larger than an equivalent double wound transformer due to the increased flux linkage damaging the autotransformer.

AUTOTROPH: Any living organism that synthesises organic substances from inorganic molecules by using light or chemical energy. Autotrophic nutrition or holophytic nutrition is characteristic of organisms possessing chlorophyll (eg. Cynobacteria, Euglena, Spirogyra/green algae, green plants) by trapping light and combining inorganic substances, carbon dioxide and water to produce food by photosynthesis. Autotrophs are the primary producers in all food chains since the materials they synthesise are the energy sources of all other organisms. Autotrophs produce their own sugars, lipids and amino acids using carbon dioxide as a source of carbon and ammonia or nitrates as a source of nitrogen. Organisms such as the green plants, certain algae and the pigmented sulfur bacteria that use light energy to synthesise organic compounds are called **photosynthetic autotrophs**. Organisms including the iron bacteria, the nitrifying bacteria and the nonpigmented sulfur bacteria that oxidise such compounds as hydrogen sulfide (H_2S) to obtain energy are called **chemosynthetic autotrophs** or **chemotrophs**. **Heterotrophs** are organisms that must obtain their energy from organic compounds. **Autotrophic** means capable of synthesising organic substances from inorganic molecules by using light or chemical energy.

AUTOTROPHIC NUTRITION: Autotrophic Nutrition is the process by which organisms produce their food from simple inorganic substances as water, carbon dioxide and mineral salts in the presence of sunlight. Plants undergo such a process by a method called photosynthesis.

AUTOTROPHIC NITRIFICATION: The combined nitrification action of two autotrophic organisms, one converting ammonium to nitrite and the other oxidising nitrite to nitrate.

AUTUMN (FALL): The third season in the year after summer and before winter. In the Northern Hemisphere, it is from September to December and in the Southern Hemisphere from March to June. During this season, the weather begins to get colder; the days get shorter; deciduous trees and shrubs lose their leaves and become dormant; some animals begin to prepare for the winter by storing food or fat on their bodies; etc. Autumnal means characteristic of, or occurring in autumn, for example organisms or flowers appearing in autumn.

AUTUMNAL EQUINOX: The first day of autumn, which occurs about 22 September in the Northern Hemisphere and 20 March in the Southern Hemisphere. At this period, the Sun crosses the plane of Earth's equator and day and night are approximately equal in duration all over the earth (about 12 hours each).

AUXANOMETER: An instrument used for studying and measuring growth in shoot length. It is made up of a scale and a lever attached to the shoot apex and works on a clockwise mechanism so that an increase in shoot length will cause a dip in the lever that can be recorded on the scale.

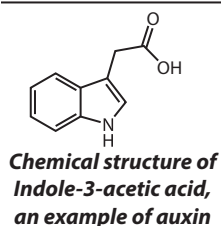
AUXILIARY ELECTRODE (COUNTER ELECTRODE): An electrode used in a three electrode electrochemical cell for voltammetric analysis or other reactions in which an electrical current is expected to flow. The other two electrodes are: (i) an **indicator electrode** also known as **test electrode** or **working electrode**, at which the cell reaction takes place, (ii) a **reference electrode**, which establishes the electrical potential against which other potentials may be measured. The auxiliary electrode functions as a cathode whenever the working electrode is operating as an anode and vice versa. The auxiliary electrode often has a surface area much larger than that of the working electrode to ensure that the half-reaction occurring at the auxiliary electrode can occur fast enough so as not to limit the process at the working electrode.

AUXILIARY EQUATION: Any simpler equation that assists the solution of a given more difficult one, usually obtained by taking transforms; especially an equation of the same form as a given

differential equation but with seal variables replacing the derivatives. For example, the auxiliary equation

$$\frac{d^2x}{dy^2} + b\frac{dx}{dy} + cy = 0 \text{ is } D^2 + bD + c = 0.$$

AUXIN: A class of plant growth hormones that promote stem and root growth and development. They influence many aspects of plant growth and development, including cell enlargement, inhibition of development of axillary buds, tropisms and the initiation of roots. Some of the most important auxins include: indole-3-acetic acid, 4-chloroindole-3-acetic acid, Indole-3-butyric acid and 2-phenylacetic acid.



AUXIN-BINDING PROTEIN (ABP): A docking protein forming an activated complex with auxins that stimulates the action of auxins in cells for cell growth and expansion.

AUXOCHROME: A group of atoms in a molecule that have the ability to increase the wavelength of absorption of the compound by extending the conjugation of the multiple bonds by delocalising its lone pair of electrons. This usually manifests itself in colour changes of the compound or intensifies the existing colour of the compound. An example is -OH, -NH₂, the aldehydes.

AUXOSPORE: The protoplast of a diatom usually resulting from the union of two smaller cells or their contents before formation of a siliceous cell wall occurs. Auxospores are usually formed following a sequence of vegetative cell divisions in which the daughter cells become progressively smaller. Auxospores may also form after sexual fusion or by apogamy or parthenogenesis.

AUXOTROPH: A mutant organism, such as a strain of bacteria, that has lost the ability to synthesise certain substances required for its growth and requires a supplement in its diet to replace the substance it has lost the ability to make. For example, certain mutants of the mould *Neurospora crassa* need the amino acid arginine in the culture medium to grow normally.

AVAILABLE NUTRIENT: Any element or compound in the soil that is present in the correct chemical form and ratio that can be readily absorbed and assimilated by growing plants. Available nutrients may be derived from the soil or the atmosphere.

AVAILABLE SOIL WATER HOLDING CAPACITY: The amount of soil water that is stored within the rooting zone and available for plant use, or the amount of water volume between field capacity (the maximum amount of water the soil can hold) and wilting point (where the plant can no longer extract water from the soil). When water enters a soil profile, some will be stored within the rooting zone for plant use, some will evaporate and some will drain away from the plant root zone. Plant available water is the difference between field capacity and the wilting point measured over 100 cm or maximum rooting depth. Beyond the wilting point there is still water in the soil profile, however it is contained in pores that are too small for the plant roots to access. Soil texture, soil structure and plant rooting depth are the crucial factors in determining the amount of water available for plants to access. Soil that is made up of smaller particle sizes, such as silt and clay, has larger surface area and therefore easier for the soil to hold onto water so it has a higher water holding capacity. The water holding capacity for sandy soil is low because it has large particle sizes which results in smaller surface area. Soil organic matter (SOM) is another factor that can help increase water holding capacity.

AVAILABLE WATER: The portion of water in a soil that can be readily absorbed by plant roots. In other words, it is the amount of water available, stored, or released between field capacity and the permanent wilting point water contents. It is considered to be the water held in the soil against a pressure of up to approximately 15 bars.

AVAILABLE SPACE THEORY: The theory that the origin of new leaf primordia at a stem apex is governed physically by the amount of space between existing primordia. It is postulated that a new primordium only arises at a point where a certain minimum amount of space has become available. This is usually between the penultimately formed primordium and the one formed previous to that. The pattern that results from such restrictions is the same as would occur if inhibitors were released by older primordia.

AVALANCHE: 1. Increase in the number of ions or electrons, usually within a medium exposed to an applied electromagnetic field, due to collisions of the ions or electrons with the medium or a shower of ionised particles created by a single ionisation, as a result of the original electron and ion being accelerated in an electric field. 2. A large mass of snow, ice, earth, rock or other material falling swiftly down a mountainside or over a precipice.

AVATAR: In computer science, it refers to a character that represents an online user. Avatars are commonly used in multiplayer gaming, online communities and Web forums.

AVENA CURVATURE TEST: A quantitative method used to correlate the degree of curvature obtained in a plant part, after the application of an external stimulus, with the amount of endogenous growth substance (auxin) required to bring it about. The tips of Avena (oat) coleoptiles are used as a standard.

AVERAGE: 1. A value that stands fairly for all the values of a set. The three types of averages are the mean, median and mode. The **mean** is adding up all values in the set and dividing by the number of values. The **median** is the value at the middle when members of the set are arranged in order of magnitude. The **mode** is the most common value in the set. However, the usual sense of averages is the mean. 2. The geometric average of a set of n numbers is the n th root of their product.

AVERAGE ACCELERATION: A change in velocity over an interval of time divided by the time interval.

AVERAGE DEVIATION (MEAN ABSOLUTE DEVIATION): It is the arithmetic mean of the absolute deviations. Its symbol is μ . Statistically, it is defined as

$$\mu = \frac{\sum_{i=1}^N |X_i - \bar{X}|}{N} = \frac{|X_1 - \bar{X}| + |X_2 - \bar{X}| + \dots + |X_N - \bar{X}|}{N}, \text{ where } X_1, X_2, \dots, X_N \text{ are}$$

the sample points and $\bar{X}$ is the sample mean based on $X_1, X_2, \dots, X_N$.

AVERAGE RATE OF FLOW: The ratio of the mass of a drop, at the instant when it is detached from the tip of the capillary, to the drop time; the average value of the instantaneous rate of flow over the entire life of the drop.

AVERAGE SET OF DATA (ARITHMETIC MEAN): A measure of central tendency, to characterise the centre of a frequency distribution of a quantitative variable by considering all the observations with the same weight accorded to each. It is calculated by summing the observations and dividing by the number of observations. For example,

for the set $x_1, x_2, \dots, x_n$, the arithmetic mean is given by $\frac{1}{n} \sum_{i=1}^n x_i$.

AVERAGE SPEED: A change in distance with respect to time. It is the overall rate at which an object moves. It can be calculated by dividing the total distance by total time.

AVERAGE VELOCITY: A change in displacement with respect to time.

AVERSION THERAPY: A type of behaviour therapy designed to inhibit the occurrence of certain undesirable habits such as drug addiction, alcoholism, smoking and gambling by making the individual associate the bad behaviour with unpleasant or uncomfortable stimulus, such as a weak electric shock. It is hoped that by repeating this treatment, the individual will develop an aversion to the undesired behaviour. It is most commonly used in treating drug and alcohol addictions.

EVERY, OSWALD THEODORE (1877-1955): A Canadian bacteriologist who became famous for his identification of DNA as the hereditary material in cells of *Pneumococcus* bacteria, contrary to the previous idea that protein was the main hereditary material. This discovery paved way for James Watson and Francis Crick, then at Cambridge University to discover the structure and true chemical basis of heredity.

AVES (BIRDS): A class of vertebrates with feathers, wings and a beak. They evolved from reptilian ancestors in the Jurassic period. Modern birds still have scaly legs, like reptiles.

AVIAN: Pertaining to poultry or birds. For example, **avian flu** is a type of influenza caused by the H_3N_2 strain of Influenza A. It is usually characterised by fever and respiratory problems. It affects birds, including domestic chickens and is capable of infecting humans. Symptoms in humans include fever, sore throat, cough, headache and muscle aches.

AVIARY: A large cage or building for keeping birds such as in a zoo.

AVIATION: The science or practice of flying aircraft or development and operation of aircrafts. Aviation includes general aviation, air transport aviation and military aviation. General aviation comprises all aviation not included in military or air-transport aviation. Military aviation includes all forms of aviation in military activities and air-transport aviation is primarily the operation of commercial airlines essentially as a public utility for the movement of persons and commodities.

AVIDIN: A glycoprotein component of egg white that binds strongly to the vitamin biotin, making it unavailable for metabolism. An abundance of avid in the diet can result in a deficiency of biotin.

AVIDITY: 1. The total binding strength of an antibody molecule to its target site(s) or epitopes, on the antigen. It is determined by both the number of binding sites of a particular antibody that are attached to epitopes and the strength of binding each individual site. 2. The dissociation-dependent strength of an acid or base. 3. Degree of affinity.

AVIONICS: The study and development of electronic circuits and devices as used in either aeronautics or astronautics. These include navigation instruments; control instruments that aid in steering and controlling the craft; power indicators such as tachometers, torque meters and exhaust pressure ratio indicators; and performance indicators such as altimeters, vertical velocity indicators and velocity gauges.

AVOCADO TREE (ALLIGATOR PEAR; *Persea americana*): A fast-growing subtropical tree which grows to 9 - 20 metres high with a trunk diameter of 30-60 centimetres. It is native to Mexico and widely cultivated in some tropical and subtropical regions. Botanically, cultivated avocado is classified into three groups: West Indian Avocado (*Persea americana* variety *americana*), the Mexican Avocado (*Persea americana* variety *drymifolia*) and the Guatemalan Avocado (*Persea nubigena* variety *guatemalensis*). The tree has quite a dense foliage and a spreading growth form. The roots are highly competitive and tend to suppress other plants nearby. It has alternate leaf arrangement and the leaves are glossy, elliptic and dark green with paler veins. It produces inconspicuous flowers which are greenish-yellow, borne in large clusters of 200-300 each, with only 2 - 3 fruits developing from each cluster. There are two types of flowers: A and B. Type A flowers are receptive to pollen in the morning, then release pollen the following afternoon, while type B flowers release pollen in the morning and are receptive to pollen the following afternoon. Production is best with cross-pollination between types A and B. The flowers attract bees and hoverflies. Reproduction is by seed, layering and grafting. The fruit produced by the plant, called avocado pear is pear-shaped with a large central seed. The fruit is low in calories, high in vitamin A, carotenoids and potassium, high in monounsaturated fat that gives

high-density lipoprotein (HDL), the good cholesterol. It can be eaten fresh by itself or accompanied by other foods such as pastries or can be used to prepare salad. Oil can be extracted from the seeds and used in the cosmetic industry.

AVOGADRO, AMEDEO (1776-1856): Italian physicist and chemist, who proposed Avogadro's Hypothesis, which states that equal volumes of gases under equal conditions of temperature and pressure contain equal numbers of molecules. From Avogadro's law the converse follows that equal numbers of molecules of any gases under identical conditions occupy equal volumes. Therefore, under identical physical conditions the gram-molecular weights of all gases occupy equal volumes.

AVOGADRO'S CONSTANT (AVOGADRO'S NUMBER): The number of units (atoms, molecules or ions) that is contained in one mole of any substance which is equal to $6.022140857 \times 10^{23}$. It can be used to calculate the number of elementary entities in any given number of moles. Avogadro's number is determined by calculating the spacing of the atoms in a crystalline solid through x-ray methods and combining this data with the measured volume of one mole of the solid to obtain the number of molecules per molar volume.

AVOGADRO'S LAW: It states that at the same temperature and pressure equal volumes of all gases contain the same number of molecules. From Avogadro's law the converse follows that equal numbers of molecules of any gases under identical conditions occupy equal volumes.

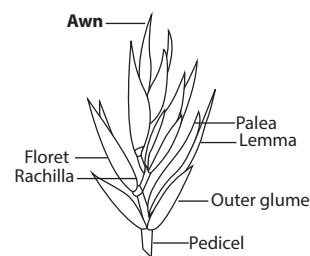
AVOGADRO'S NUMBER: The number of particles in a mole of any substance, which is equal to $6.022 \times 10^{23} \text{ mol}^{-1}$.

AVOIRDUPOIS: A system of units of mass based on the pound, with the symbol lb (1 lb = 0.45 kg). It consists of 16 ounces or 7,000 grains.

AWK: In computer science, it is a scripting language suitable for the analysis of text and log files, using its "match and act" method of processing. The program is a sequence of patterns and actions that tell you what to look for in the input data and what to do when it is found. It searches a set of files for lines matched by any of the patterns; when a matching line is found, the corresponding action is performed. It is used to perform tasks such as tallying information from text files and creating reports from the results; translating files from one format to another; creating small databases; and performing mathematical operations on files of numeric data. Named after its inventors Aho, Weinberg and Kernighan, it has been a part of the Unix environment since its beginning.

AWL-SHAPED: Sharply-pointed, linear and narrowly triangular, like an awl. It is used in botany especially for plant organs such as a leaf, spine and scale.

AWN: In botany, a stiff bristle that projects from the tip of a plant part or organ. The glumes and lemmas of grasses (Gramineae) commonly possess awns, as do some fruits and less commonly leaves. Cereals and grasses having ears or spikes covered in awns are termed **awned** or **bearded**. An individual organ bearing an awn is termed **aristate**. An awn may act to bury a fruit in the soil by uncoiling in damp conditions and is so doing pushing the fruit into the ground.



Awn of a spikelet

AX SPECTRUM: A general pattern of the nuclear magnetic spectrum of a molecule AX , where A and X are both spin- $1/2$ nuclei. If X is spin a , the spin-spin interaction between the nuclei of A and X results in one line in the spectrum of A being shifted by $J/2$ from the frequency of precession it has in the absence of coupling. If X is spin a , the change in frequency of precession of A is $-J/2$. Thus single lines in the resonances of both A and X are split

into doublets, with splitting J . The A resonance in molecules of the type A_nX is also a doublet with the splitting J .

A

AXENIC CULTURE: The growth of organisms of a single species in the absence of contaminating living organisms or living cells of any other species, done in an environment that has been previously sterilised. It is widely used in microbiology to determine the basic growth requirements or degree of inhabitation by antibiotics or other chemicals, of a particular species.

AXIAL (AXILE): 1. Positioned on the axis. 2. A type of placentation in which ovules are attached to the axis

AXIAL FILAMENT: The organ of motility found in spirochetes (type of bacteria).

AXIAL PERIOD: The time taken for an object to make one complete rotation on its axis. The axial period of a planet is its "day".

AXIAL ROTATION: The spin of an object around its axis.

AXIAL SKELETON: The central longitudinal part of the vertebrate skeleton including the skull, the vertebral column and the rib cage. In the head and trunk of the human body, it is composed of five parts: the human skull, the ossicles of the middle ear, the hyoid bone of the throat, the rib cage and the vertebral column consisting of the 80 bones.

AXIAL VECTOR (PSEUDO-VECTOR): A vector that does not reverse its sign when the coordinate system is changed to a new system by a reflection in the origin ($x'_i = -x_i$).

AXIL: The angle between a leaf petiole or stalk or a branch and the stem from which it grows. Axillary or lateral buds develop in the axil of a leaf and its presence distinguishes a leaf from a leaflet.

AXILLA (OXTER; ARMPIT; UNDERARM): The area on the human body directly under the joint where the arm connects to the shoulder.

AXILLARY: Arising or positioned in an axil like a flower or bud. Axillary bud is a bud that develops in the axil of a leaf and its presence distinguishes a leaf from a leaflet.

AXILLARY ARTERY: The chief artery of the upper limb that supplies the arm, axilla (armpit) and chest wall. It is a continuation of the subclavian artery in the armpit, becoming the brachial artery in the arm, with superior thoracic, thoracoacromial, lateral thoracic, subscapular and posterior and superior circumflex humeral branches.

AXILLARY VEIN: The axillary vein has the same extent as the axillary artery. It is the continuation of the basilic and brachial veins that runs from the lower border of the teres major muscle to the outer border of the first rib where it becomes the subclavian vein.

AXILROD-TELLER FORMULA: An expression for three-body interactions in intermolecular forces, named after B.M. Axilrod and E. Teller, who discovered it in 1943. These dipole dispersion terms are of importance in the third virial coefficient. At low temperatures, the Axilrod-Teller terms can have a comparable size to two-body interactions, making them of importance to such properties of liquids like pressure and surface tension.

AXIOM (POSTULATE): A statement in logic or mathematics that is assumed to be true without a proof and upon which theorems are proved by using logical deduction.

AXIOM OF CHOICE (ZERMELO'S AXIOM): The axiom of set theory stating that from every family of disjoint sets, a set can be constructed containing exactly one element of each of the given family of sets.

AXIOM OF INFINITY: The axiom of set theory that specifies an algorithm for constructing infinitely many distinct sets.

AXIOMATIC METHOD: A method of mathematical reasoning based on logical deduction from assumptions known as axioms. The method is fundamental to the philosophy of mathematics. In an axiomatic system, certain undefined terms are taken and described by a set of axioms. Other, often surprising, relationships or theorems are then deduced by logical reasoning.

AXIOMATIC PROBABILITY: The study of probability in terms of probability measure. . Consider the event A, the probability that the event A will occur is given by $P(A)$, which gives the precise measure of the chance that A will occur. The probability axioms are given by (i) for any event A, $P(A) \geq 0$, (ii) $P(A) = 1$, (iii) If $A_1, A_2, A_3, \dots$, is an infinite collection of disjoint events, then $P(A_1 \cup A_2 \cup A_3, \dots) = \sum_{i=1}^{\infty} P(A_i)$.

AXIOMATIC SET THEORY: The presentation of set theory as a formal set of uninterpreted axioms and rules of inference, rather than as the formalisation of an antecedent body of knowledge. In technology, axiomatic set theory is one of several approaches to set theory, consisting of a formal language for talking about sets and a collection of axioms describing how they behave. There are many different axiomatisations for set theory. Each takes a slightly different approach to the problem of finding a theory that captures as much as possible of the intuitive idea of what a set is, while avoiding the paradoxes that result from accepting all of it, the most famous being Russell's paradox.

AXIOMATIC SYSTEM: In logic, any logical system with a set of explicitly stated axioms from which theorems are derived by transformation rules.

AXION: A hypothetical elementary particle having no charge or spin and small mass. It was postulated by the Peccei-Quinn theory in 1977 to the existence of certain symmetries of the strong nuclear force. Axions have not been detected experimentally, although it has been possible to put limits on some astrophysical phenomena such as the cooling of stars. It has also been suggested that they account for some or all of the missing matter in the universe.

AXIS: 1. One of the set of reference lines used to locate points on a graph or in a coordinate system its perpendicular or angular distance from them. In a Cartesian coordinate system, the number of axes equals the dimension of the space and all axes are mutually perpendicular. The axes are usually labelled $x, y, z, \dots$ 2. **Axis of rotation** is a straight line through all the fixed points of a rotating rigid body around which all other points of the body move in circles. 3. The main stem, central part or central column of an inflorescence of a plant on which other plant parts, such as branches or leaflets, are arranged.

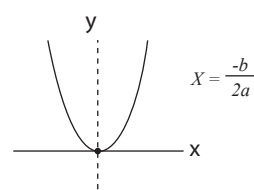
AXIS OF A COORDINATE GRID: Either of the two number lines used to form a coordinate grid. A coordinate grid has two perpendicular lines, or axes, labelled like number lines. The horizontal axis is called the x -axis. The vertical axis is called the y -axis. The point where the x -axis and y -axis intersect is called the origin.

AXIS OF ROTATION: A line about which a solid figure such as Earth rotates. For example, the Earth's axis is an imaginary line that passes through the North and South Poles of the planet about which it rotates once everyday (approximately 24 hours).

AXIS OF SYMMETRY (LINE OF SYMMETRY; REFLECTION SYMMETRY): A line through a plane

or a 2-dimensional (2-D) shape so that each side of this line is a mirror image of each other. It is a line that divides a figure into two symmetrical parts, so that when the figure is rotated about this line, it does not topple over. For example, the letter **W** has one axis of symmetry, continuous; the equilateral triangle has three axis of symmetry; and a circle has an infinite axes of symmetry. The graph of the function

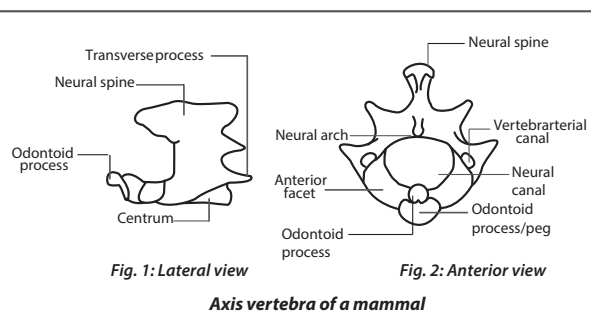
$$y = ax^2 + bx + c$$



A parabola with its line of symmetry and its equation

$y = ax^2 + bx + c$ is a parabola, given in the diagram, has a line of symmetry which passes through the vertex and has the equation $x = \frac{-b}{2a}$.

AXIS VERTEBRA: The second cervical vertebra in a mammal which articulates with the first cervical vertebra (atlas). It is located below the atlas at the top of the backbone. In reptiles, birds and mammals, the articulation between the two permits side-to-side movements of the head.



AXLE: A thin rod or shaft attached to the centre of a wheel. A wheel and axle together makes a simple machine and a small effort at the wheel can move a large load attached to the axle.

AXON: The long thread-like part of a neuron or nerve cell that conducts electrochemical impulses away from the cell body to other cells or towards an effector organ such as a muscle. It is the main part of the nerve fibre and is covered in an insulating coat. The dendrite transmits impulses received from other cells at synapses to the cell body.

AXONEME: A bundle of fibrils that form the central core of a cilium consisting of two central microtubules surrounded by nine other pairs of microtubules. Alternatively, it is the core of an undulipodium, which is the slender cell extension in all organisms except bacteria used for locomotion and feeding.

AZALEA: A flowering shrub of the rhododendron family, consisting of nine pairs of microtubules surrounding a central pair.

AZEOTROPE (AZEOTROPIC MIXTURE; CONSTANT-BOILING MIXTURE): A solution of two or more liquids which on distillation behaves as a pure liquid. Its boiling point is invariable at a given pressure and the composition of the vapour phase is the same as that of the liquid.

AZEOTROPIC DISTILLATION: A distillation technique employed in separating azeotropic mixtures by the addition of a third compound called the entrainer, which associates with one of the compounds of the mixture to form a third azeotrope but not the other components of the original azeotrope and makes it possible to distil this new azeotrope and separate it from the original mixture. For example, benzene (entrainer) can be added to an ethanol/water mixture to form a new, lower boiling point azeotrope.

AZIDES: Compounds containing the ion N_3^- or the group $-N_3$.

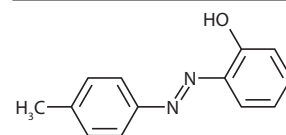
AZIDOTHYMININE, AZT (ZIDOVUDINE; 3'-AZIDO-3'-DEOXYTHYMININE; RETROVIR): A drug, with the chemical formula $C_{10}H_{13}N_5O_4$ used to treat patients infected with the Human Immunodeficiency Virus (HIV), which causes Acquired Immune Deficiency Syndrome (AIDS). AZT affects HIV's ability to reproduce by inhibiting the transcription of RNA to DNA but does not stop it entirely. It is particularly effective in preventing transmission of HIV from infected pregnant woman to the foetus. HIV may become AZT-resistant over time; therefore it is, usually used in conjunction with other NRTIs and anti-viral drugs.

AZIMUTH: 1. In astronomy, it is the arc of the horizon or the angular distance measured clockwise between a celestial object and the northern point of the horizon as seen by the observer. In other words, it is measured from the north point of the horizon to the intersection with the horizon of the vertical circle passing through a celestial body. In navigation, it is the angular distance usually measured clockwise from the north point of the horizon to the intersection with the horizon of the vertical circle passing through a celestial body. 2. In polar coordinates, it is also known as **amplitude**, **argument** or **anomaly**, which is the angle between the positive real axis and the vector representing a given complex number in the Argand diagram. If the point (x, y) , representing the number $x + iy$, has polar coordinates (r, θ) , its amplitude is θ , i.e., $\text{amp}(x + iy) = \arctan(y/x)$ and $r(\cos\theta + i\sin\theta)$ equals the given number.

AZIMUTHAL QUANTUM NUMBER (ORBITAL ANGULAR MOMENTUM QUANTUM NUMBER; SECOND QUANTUM NUMBER): One of the four quantum numbers of an atomic orbital that determines its orbital angular momentum and describes the shape of the orbital. In other words, it is one of the four quantum numbers that determines the magnitude of the angular momentum of the electron around the nucleus. It also describes the shape of the orbital. It is denoted as l and takes the values $0, 1, 2, \dots, n-1$, where n is the principal quantum number. Azimuthal quantum numbers $0, 1, 2$ and 3 , correspond to the s, p, d and f orbitals, respectively.

AZINE: An organic heterocyclic compound containing a six-membered ring formed from carbon and nitrogen atoms. Pyridine is an example containing one nitrogen atom (C_5H_5N). Diazines have two nitrogen atoms in the ring, for example, $C_4H_4N_2$ and isomers exist depending on the relative positions of the nitrogen atoms. Triazines contain three nitrogen atoms.

AZO COMPOUND: An organic compound containing at least one of two doubly bonded nitrogen atoms with each atom linking two other groups through single bonds: $R-N=N-R'$, in which R and R' can be either aryl or alkyl, for example, $PhN=NPh$, azobenzene. The more stable derivatives contain two aryl groups. The $N=N$ group is called an **azo group**. An example of an azo compound is azo dye, which is a synthetic dye containing the azo group of two nitrogen atoms ($-N=N$) connecting aromatic ring compounds. Azo dyes are usually red, brown or yellow and make up about half the dyes produced. They are manufactured from aromatic amines.



Chemical Structure of Azo Dye

AZONAL SOIL: A major soil group without distinct genetic horizons or having no zonal organization or structure because, for example, they are of recent formation or due to their locations such as on very steep slopes which prohibit profile development. Examples are some mountain, alluvial, marine, glacial, windblown, and volcanic soils.

AZOTOBACTER: A nitrogen-fixing bacterium belonging to a group found in soil, which contributes to soil fertility.

AZULENE: 1. A blue crystalline compound, with the chemical formula $C_{10}H_8$, melting point 99°C (372 K) and boiling point 242°C (515 K). It contains a five-membered ring and has aromatic properties. It converts to naphthalene when heated. 2. Any of a number of blue oils that can be produced by distilling or heating essential oils from plants.

AZURITE (BICYCLO[5.3.0]DECAPENTAENE): A secondary mineral consisting of hydrated basic copper carbonate, in monoclinic crystalline form, with the chemical formula $Cu_3(OH)_2(CO_3)_2$. It is a minor ore of copper and generally formed in the upper zone of copper ore deposits and often occurs with malachite. Its intense azure blue colour made it formerly important as a pigment. It is used as a gemstone.

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B

5-BROMOURACIL (5-BU): An analogue of the naturally occurring pyrimidine thymine (5-methyl uracil), in which bromine replaces the methyl group on carbon five of the pyrimidine ring. During chromosome replication the analogue may be incorporated into DNA in place of thymine, but it is less specific in its pairing properties.

6-BENZYLAMINOPURINE (BAP, BENZYL-ADENINE): A synthetic cytokinin, with the molecular formula $C_{12}H_{11}N_5$ and molar mass of 225.2492 g/mol. It is a plant growth regulator and stimulates cell division and elicits plant growth and development responses, setting blossoms and stimulating fruit richness.

B: In the hexadecimal notation, it represents the number 11.

B FORM: The most common form of duplex DNA, containing a right handed helix and about 10-10.5 nucleotides per turn of the helix axis. It carries the genetic information in the cell and is capable of self-replication and synthesis of RNA.

B HORIZON: The third layer of soil below A horizon or topsoil and generally referred to as subsoil. The B horizon is characterised by an accumulation of soluble or suspended organic material, clay, iron or aluminium.

B*-ALGEBRA: A Banach algebra that satisfies the condition $\|x^{**}\| = \|x\|^2$, for all x in the given B*-algebra. It automatically implies that the *-involution is isometric, i.e., $\|x^{**}\| = \|x\|^2$. Hence $\|xx^{**}\| = \|x^{**}\| = \|x\|^2$, and therefore, a B*-algebra is also a C*-algebra. Conversely, the C*-condition implies the B*-condition.

B-CELL ANTIGEN RECEPTOR (BCR): A type of surface immunoglobulin, located on the outer surface of B-cell (a type of lymphocyte) that activates it (B-cell) when a specific antigen present in the body binds to its receptor.

B-CELL (B-LYMPHOCYTE): A type of white blood cell present in quantity in blood and lymph and known for its strong immune function by binding to antigen to facilitate its removal.

B-CHROMOSOME (ACCESSORY CHROMOSOME; SUPERNUMERARY CHROMOSOME): Any of a number of small chromosomes containing a centromere occasionally seen in tissue culture of some organisms in addition to the fixed number of stable chromosomes characteristic of the species (the A-chromosomes). B-chromosomes are composed largely of heterochromatin and in many cases appear to have little effect. However, when present in large numbers they may be deleterious.

B-FIELD: A three dimensional space in which the magnitude of the resultant magnetic field strength, B at any point is mostly non-zero. The zero values of B are found at neutral point.

B-SCAN ULTRASONOGRAPHY (B-SCAN; BRIGHTNESS SCAN): A diagnostic test used to examine the tissues of the eye in cross-section, using a high-frequency ultrasound technique. It is used on the outside of the closed eyelid to view the eye, especially

if there are problems (such as edema or severe cataracts) that prevents examination of the eye normally. B-scan can help view accurately other eye structures like the lens, choroid, sclera, vitreous and retina.

B-TREE: In computer science, it is an information storage and retrieval in the form of a tree in which all terminal nodes (all functional devices that are accessible through the network) are the same distance from the base, and all non-terminal nodes have between n and $2n$ subtrees or pointers (where n is an integer). B-tree keeps data sorted and allows searches, sequential access, insertions, and deletions in logarithmic time.

B-Z REACTION (BELOUSOV-ZHABOTINSKII REACTION): A chemical reaction that shows a periodic colour change between magenta and blue with a period of about one minute. It occurs with a mixture of sulfuric acid, potassium bromate(V), cerium sulfate and propanedioic acid.

BABBAGE, CHARLES (1792 – 1871): British mathematician, who invented the mechanical general-purpose computer, called the analytical engine, which was not built. He had earlier designed another one, called the difference engine, which was finally built in 1991 based on his original design. His analytical engine has a number of features, which is essential in modern design of computers.

BABBAGE'S ENGINE: The mechanical general-purpose computer, called the analytical engine, designed in the 1830s by English mathematician, Charles Babbage (1792-1871) and inventor, which was intended to automatically compute mathematical tables, but was never built due to lack of funds and other problems. He had earlier designed another one, called the difference engine, which was finally built in 1991 based on his original design.

BABBIT METAL: Any of a group of related alloys used to reduce friction in bearings. It may be an alloy of tin, lead, copper and antimony.

BABO'S LAW: A law stating that the vapour pressure of a liquid is decreased when a solute is added and is proportional to the amount of solute dissolved.

BABOON: A large African or Arabian monkey of the family Cercopithecidae, especially of the genus *Papio* or *Chaeropithecus* with an elongated, doglike muzzle, a short tail and bare calluses on the buttocks. It weighs between 14 – 40 kg and with a body length of about 50 – 115 cm long, excluding the tail.

BACCATE: A structure or organ that is soft or berry-like.

BACCIFORM: Having the shape of a berry.

BACILLARIOPHYTA: Any of various microscopic one-celled marine or freshwater algae comprising the diatoms of the class Bacillariophyceae, having cell walls of silica consisting of two interlocking symmetrical valves. Asexual reproduction is by fission and auxospore formation. There are two classes, the radially symmetrical or centric diatoms and the bilaterally symmetrical or pennate diatoms. The cell walls are extremely resistant to decay and form deep deposits of diatomaceous earth or kieselguhr in lake and ocean beds.

BACILLE CALMETTE-GUERIN (BACILLUS CALMETTE-GUÉRIN; BCG): A vaccine that is prepared from a strain of the attenuated (weakened) live bovine tuberculosis bacillus, *Mycobacterium bovis*, that has lost its virulence in humans and used to effectively immunize against tuberculosis (TB). It is also used in the treatment of diseases such as leprosy, certain types of cancer and type I diabetes. It is named after Camille Guérin (1872-1961) and Albert Calmette (1863-1933), French bacteriologists who produced the first BCG strain of the bacteria in 1921 at the Pasteur Institute in Paris.

BACILLIFORM: Having the shape of a rod or club.

BACILLUS: A member of a group of rod-shaped Gram-positive eubacteria belonging to the family Bacillaceae that occur everywhere